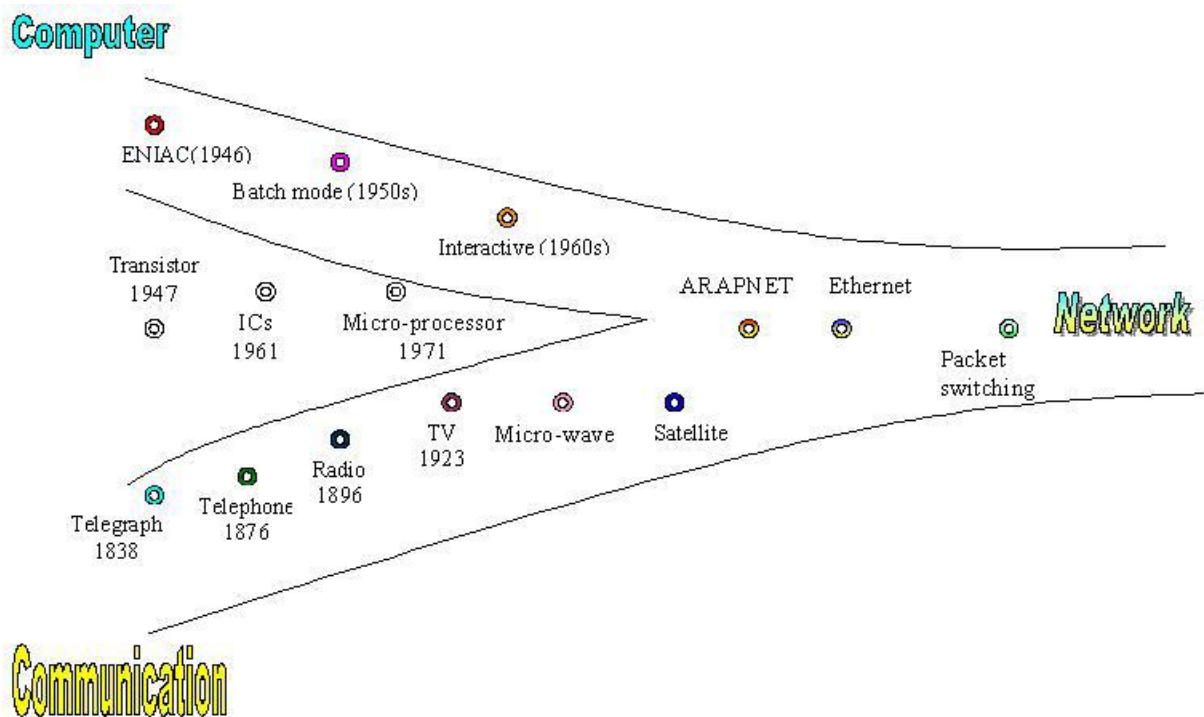


## Chapter 1

### Introduction to computer Network

The concept of Network is not new. In simple terms it means an interconnected set of some objects. For decades we are familiar with the Radio, Television, railway, Highway, Bank and other types of networks. In recent years, the network that is making significant impact in our day-to-day life is the Computer network. By computer network we mean an interconnected set of autonomous computers. The term autonomous implies that the computers can function independent of others. However, these computers can exchange information with each other through the communication network system. Computer networks have emerged as a result of the convergence of two technologies of this century- Computer and Communication as shown in Fig. 1.1. The consequence of this revolutionary merger is the emergence of an integrated system that transmit all types of data and information. There is no fundamental difference between data communications and data processing and there are no fundamental differences among data, voice and video communications.



*Figure 1.1: Evolution of Network*

#### 1. Definition:

A **computer network** is a set of connected computers. Computers on a network are called nodes. The connection between computers can be done via cabling, most commonly the Ethernet cable, or wirelessly through radio waves. Connected computers can share resources, like access to the Internet, printers, file servers, and others. A network is a multipurpose connection, which allows a single computer to do more.

A computer network, or data network, is a digital telecommunications network which allows nodes to share resources. In computer networks, computing devices exchange data with each other using connections (data links) between nodes. These data links are

Established over cable media such as wires or optic cables, or wireless media such as Wi-Fi.

Network computer devices that originate, route and terminate the data are called network nodes. Nodes can include hosts such as personal computers, phones, servers as well as networking hardware. Two such devices can be said to be networked together when one device is able to exchange information with the other device, whether or not they have a direct connection to each other. In most cases, application-specific communications protocols are layered (i.e. carried as payload) over other more general communications protocols. This formidable collection of information technology requires skilled network management to keep it all running reliably.

Computer networks support an enormous number of applications and services such as access to the World Wide Web, digital video, digital audio, shared use of application and storage servers, printers, and fax machines, and use of email and instant messaging applications as well as many others. Computer networks differ in the transmission medium used to carry their signals, communications protocols to organize network traffic, the network's size, topology, traffic control mechanism and organizational intent. The best-known computer network is the Internet.

### **Advantages of Computer Networking**

#### **1. It enhances communication and availability of information.**

Networking, especially with full access to the web, allows ways of communication that would simply be impossible before it was developed. Instant messaging can now allow users to talk in real time and send files to other people wherever they are in the world, which is a huge boon for businesses. Also, it allows access to a vast amount of useful information, including traditional reference materials and timely facts, such as news and current events.

#### **2. It allows for more convenient resource sharing.**

This benefit is very important, particularly for larger companies that really need to produce huge numbers of resources to be shared to all the people. Since the technology involves computer-based work, it is assured that the resources they wanted to get across would be completely shared by connecting to a computer network which their audience is also using.

#### **3. It makes file sharing easier.**

Computer networking allows easier accessibility for people to share their files, which greatly helps them with saving more time and effort, since they could do file sharing more accordingly and effectively.

#### **4. It is highly flexible.**

This technology is known to be very flexible, as it gives users the opportunity to explore everything about essential things, such as software without affecting their functionality. Plus, people will have the accessibility to all information they need to get and share.

**5. It is an inexpensive system.**

Installing networking software on your device would not cost too much, as you are assured that it lasts and can effectively share information to your peers. Also, there is no need to change the software regularly, as mostly it is not required to do so.

**6. It increases cost efficiency.**

With computer networking, you can use a lot of software products available on the market which can just be stored or installed in your system or server, and can then be used by various workstations.

**7. It boosts storage capacity.**

Since you are going to share information, files and resources to other people, you have to ensure all data and content are properly stored in the system. With this networking technology, you can do all of this without any hassle, while having all the space you need for storage.

**Disadvantages of Computer Networking**

**1. It lacks independence.**

Computer networking involves a process that is operated using computers, so people will be relying more of computer work, instead of exerting an effort for their tasks at hand. Aside from this, they will be dependent on the main file server, which means that, if it breaks down, the system would become useless, making users idle.

**2. It poses security difficulties.**

Because there would be a huge number of people who would be using a computer network to get and share some of their files and resources, a certain user's security would be always at risk. There might even be illegal activities that would occur, which you need to be careful about and aware of.

**3. It lacks robustness.**

As previously stated, if a computer network's main server breaks down, the entire system would become useless. Also, if it has a bridging device or a central linking server that fails, the entire network would also come to a standstill. To deal with these problems, huge networks should have a powerful computer to serve as file server to make setting up and maintaining the network easier.

**4. It allows for more presence of computer viruses and malware.**

There would be instances that stored files are corrupt due to computer viruses. Thus, network administrators should conduct regular check-ups on the system, and the stored files at the same time.

**5. Its light policing usage promotes negative acts.**

It has been observed that providing users with internet connectivity has fostered undesirable behavior among them. Considering that the web is a minefield of distractions—online games, humor sites and even porn sites—workers could be tempted during their work hours. The huge network of machines could also encourage them to engage in illicit practices, such as instant messaging and file sharing, instead of working on work-related matters. While many organizations draw up certain policies on this, they have proven difficult to enforce and even engendered resentment from employees.

**6. It requires an efficient handler.**

For a computer network to work efficiently and optimally, it requires high technical skills and know-how of its operations and administration. A person just having basic skills cannot do this job. Take note that the responsibility to handle such a system is high, as allotting permissions and passwords can be daunting. Similarly, network configuration and connection is very tedious and cannot be done by an average technician who does not have advanced knowledge.

**7. It requires an expensive set-up.**

Though computer networks are said to be an inexpensive system when it is already running, its initial set up cost can still be high depending on the number of computers to be connected. Expensive devices, such as routers, switches, hubs, etc., can add up to the cost. Aside from these, it would also need network interface cards (NICs) for workstations in case they are not built in.

## **Applications**

In a short period of time computer networks have become an indispensable part of business, industry, entertainment as well as a common-man's life. These applications have changed tremendously from time and the motivation for building these networks are all essentially economic and technological.

Initially, computer network was developed for defense purpose, to have a secure communication network that can even withstand a nuclear attack. After a decade or so, companies, in various fields, started using computer networks for keeping track of inventories, monitor productivity, communication between their different branches offices located at different locations. For example, Railways started using computer networks by connecting their nationwide reservation counters to provide the facility of reservation and enquiry from anywhere across the country.

And now after almost two decades, computer networks have entered a new dimension; they are now an integral part of the society and people. In 1990s, computer network started delivering services to private individuals at home. These services and motivation for using them are quite different. Some of the services are access to remote information, person-person communication, and interactive entertainment. So, some of the applications of computer networks that we can see around us today are as follows:

**Marketing and sales:** Computer networks are used extensively in both marketing and sales organizations. Marketing professionals use them to collect, exchange, and analyze data related to customer needs and product development cycles. Sales application includes teleshopping, which uses order-entry computers or telephones connected to order processing network, and online-reservation services for hotels, airlines and so on.

**Financial services:** Today's financial services are totally depended on computer networks. Application includes credit history searches, foreign exchange and investment services, and electronic fund transfer, which allow user to transfer money without going into a bank (an automated teller machine is an example of electronic fund transfer, automatic pay-check is another).

**Manufacturing:** Computer networks are used in many aspects of manufacturing including manufacturing process itself. Two of them that use network to provide essential services are computer-aided design (CAD) and computer-assisted manufacturing (CAM), both of which allow multiple users to work on a project simultaneously.

**Directory services:** Directory services allow list of files to be stored in central location to speed worldwide search operations.

**Information services:** A Network information service includes bulletin boards and data banks. A World Wide Web site offering technical specification for a new product is an information service.

**Electronic data interchange (EDI):** EDI allows business information, including documents such as purchase orders and invoices, to be transferred without using paper.

**Electronic mail:** probably it's the most widely used computer network application.

**Teleconferencing:** Teleconferencing allows conference to occur without the participants being in the same place. Applications include simple text conferencing (where participants communicate through their normal keyboards and monitor) and video conferencing where participants can even see as well as talk to other fellow participants. Different types of equipment's are used for video conferencing depending on what quality of the motion you want to capture (whether you want just to see the face of other fellow participants or do you want to see the exact facial expression).

**Voice over IP:** Computer networks are also used to provide voice communication. This kind of voice communication is pretty cheap as compared to the normal telephonic conversation.

**Video on demand:** Future services provided by the cable television networks may include video on request where a person can request for a particular movie or any clip at any time he wish to see.

**Summary:** The main area of applications can be broadly classified into following categories:

- Scientific and Technical Computing
- Client Server Model, Distributed Processing
- Parallel Processing, Communication Media
- Commercial
- Advertisement, Telemarketing, Teleconferencing

- Worldwide Financial Services
- Network for the People (this is the most widely used application nowadays)
- Telemedicine,
- Distance Education
- Access to Remote Information
- Person-to-Person Communication
- Interactive Entertainment

## Networks Model

### Network Technologies

There is no generally accepted taxonomy into which all computer networks fit, but two dimensions stand out as important: Transmission Technology and Scale. The classifications based on these two basic approaches are considered in this section.

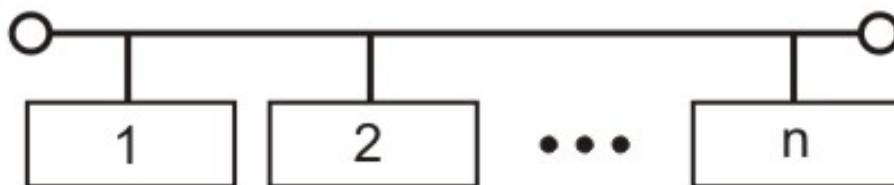
#### Classification Based on Transmission Technology

Computer networks can be broadly categorized into two types based on transmission technologies:

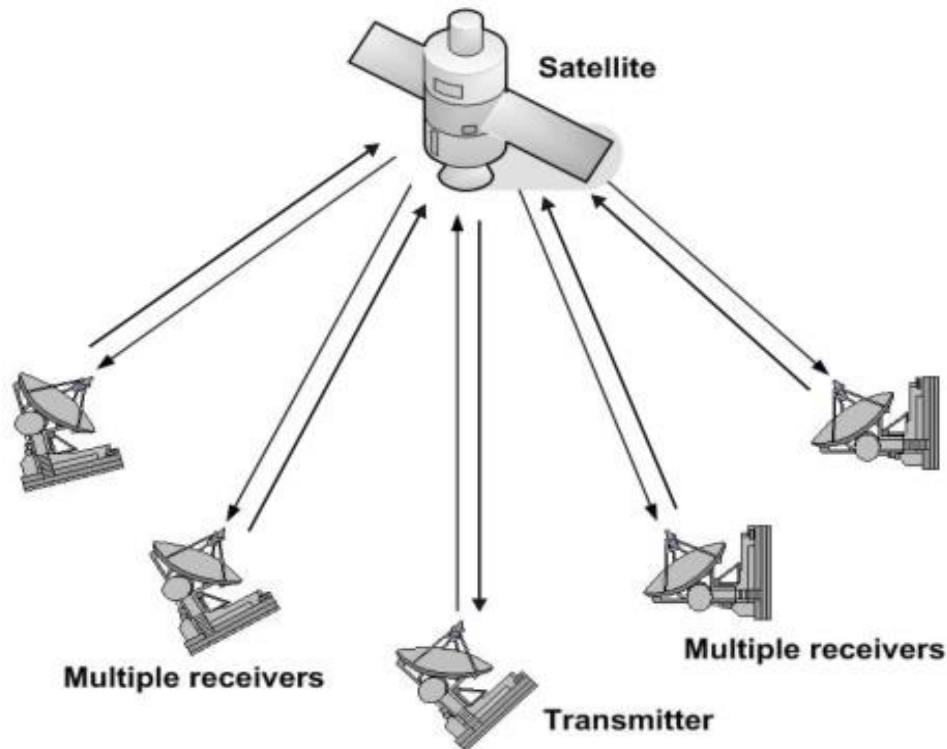
- Broadcast networks
- Point-to-point networks

#### Broadcast Networks

Broadcast network have a single communication channel that is shared by all the machines on the network as shown in Figs.1.2 and 1.3. All the machines on the network receive short messages, called packets in certain contexts, sent by any machine. An address field within the packet specifies the intended recipient. Upon receiving a packet, machine checks the address field. If packet is intended for itself, it processes the packet; if packet is not intended for itself it is simply ignored. This system generally also allows possibility of addressing the packet to all destinations (all nodes on the network). When such a packet is transmitted and received by all the machines on the network. This mode of operation is known as **Broadcast Mode**. Some Broadcast systems also supports transmission to a sub-set of machines, something known as **Multicasting**.



*Figure 1.2: Broadcast network based on shared bus*



*Figure 1.3: Broadcast network based on satellite communication*

### **Point-to-Point Networks**

A network based on point-to-point communication is shown in Fig. 1.4. The end devices that wish to communicate are called *stations*. The switching devices are called *nodes*. Some Nodes connect to other nodes and some to attached stations. It uses FDM or TDM for node-to-node communication. There may exist multiple paths between a source-destination pair for better network reliability. The switching nodes are not concerned with the contents of data. Their purpose is to provide a switching facility that will move data from node to node until they reach the destination. As a general rule (although there are many exceptions), smaller, geographically localized networks tend to use broadcasting, whereas larger networks normally use are point-to-point communication.

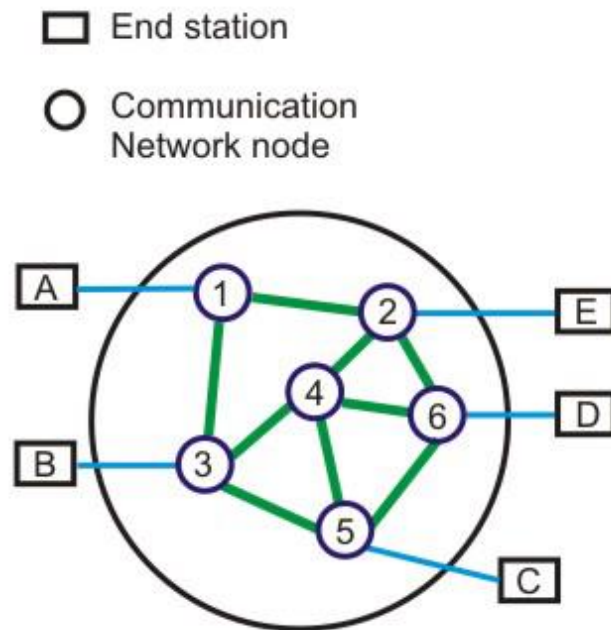


Figure 1.4: Communication based point to point protocol

### Classification based on Scale

Alternative criteria for classifying networks are their scale. They are divided into Local Area (LAN), Metropolitan Area Network (MAN) and Wide Area Networks (WAN).

#### Local Area Network (LAN)

LAN is usually privately owned and links the devices in a single office, building or campus of up to few kilometers in size. These are used to share resources (may be hardware or software resources) and to exchange information. LANs are distinguished from other kinds of networks by three categories their size, transmission technology and topology.

LANs are restricted in size, which means that their worst-case transmission time is bounded and known in advance. Hence this is more reliable as compared to MAN and WAN. Knowing this bound makes it possible to use certain kinds of design that would not otherwise be possible. It also simplifies network management.

LAN typically used transmission technology consisting of single cable to which all machines are connected. Traditional LANs run at speeds of 10 to 100 Mbps (but now much higher speeds can be achieved). The most common LAN topologies are bus, ring and star. A typical LAN is shown in Fig. 1.5.

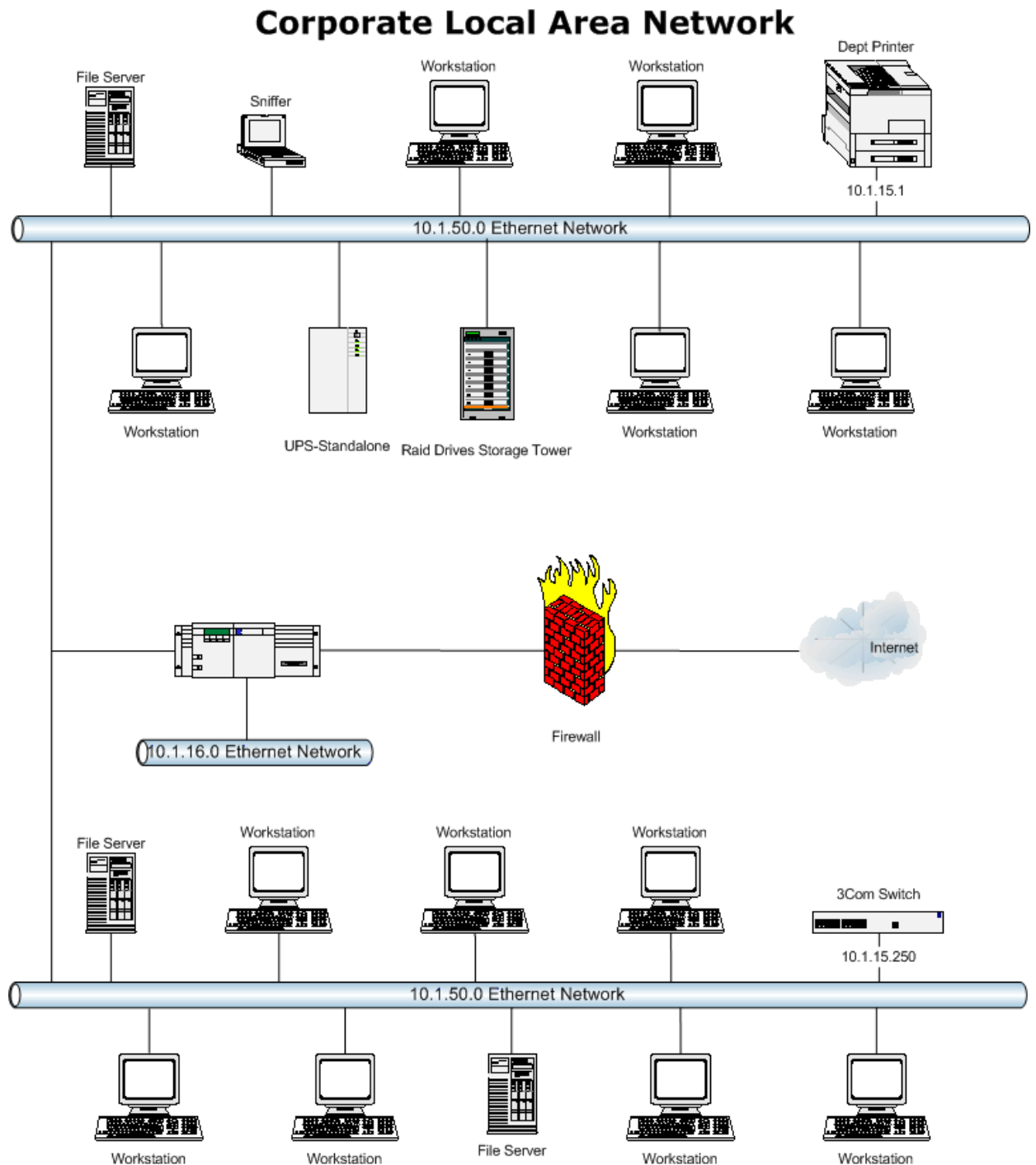
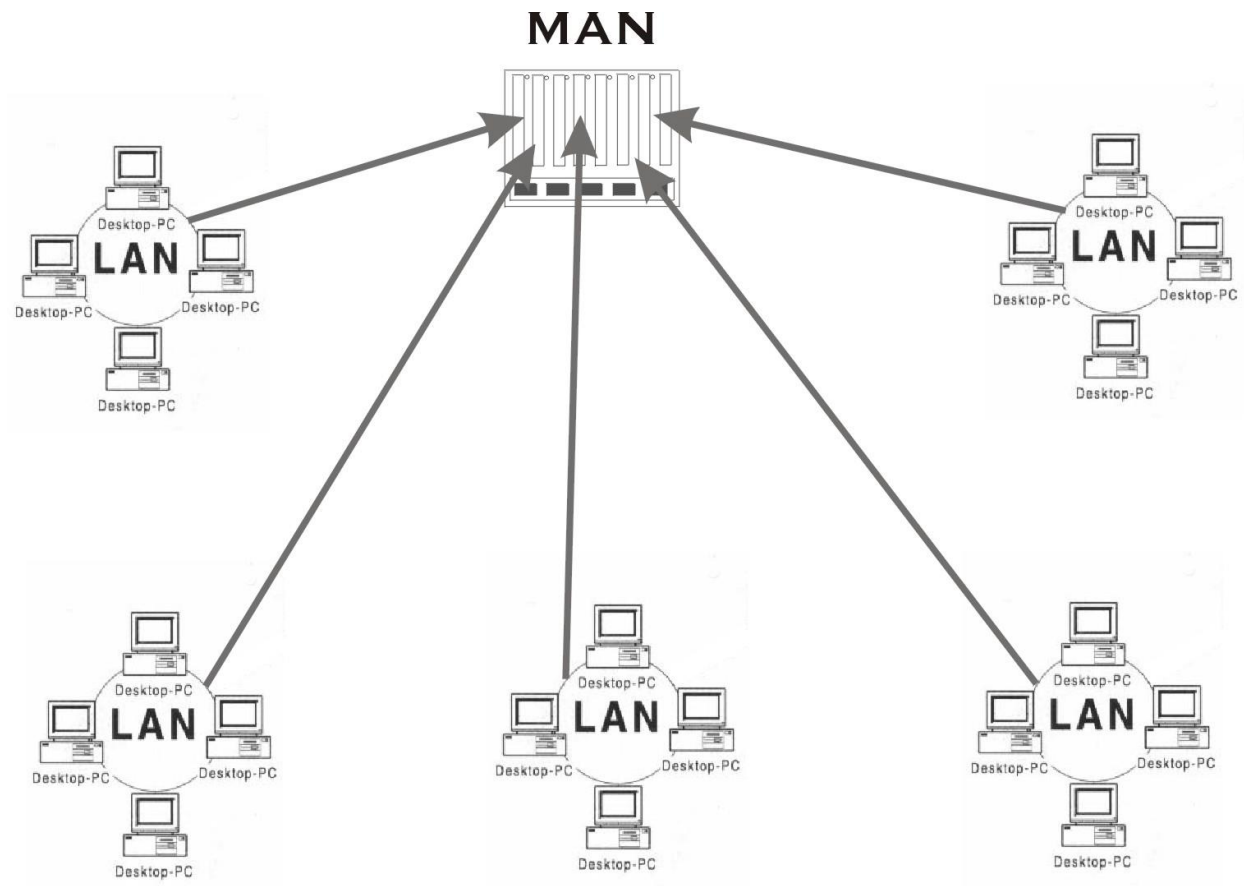


Figure 5: Local Area Network

### Metropolitan Area Networks (MAN)

**MAN is designed to extend over the entire city.** It may be a single network as a cable TV network or it may be means of connecting a number of LANs into a larger network so that resources may be shared as shown in Fig. 1.6. For example, a company can use a MAN to connect the LANs in all its offices in a city. MAN is wholly owned and operated by a private company or may be a service provided by a public company. The main reason for

distinguishing MANs as a special category is that a standard has been adopted for them. It is DQDB (Distributed Queue Dual Bus) or IEEE 802.6.



*Figure 1.6 Metropolitan Area Network*

### **Wide Area Network (WAN)**

WAN provides long-distance transmission of data, voice, image and information over large geographical areas that may comprise a country, continent or even the whole world. In contrast to LANs, WANs may utilize public, leased or private communication devices, usually in combinations, and can therefore span an unlimited number of miles as shown in Fig. 1.7. A WAN that is wholly owned and used by a single company is often referred to as *enterprise network*.

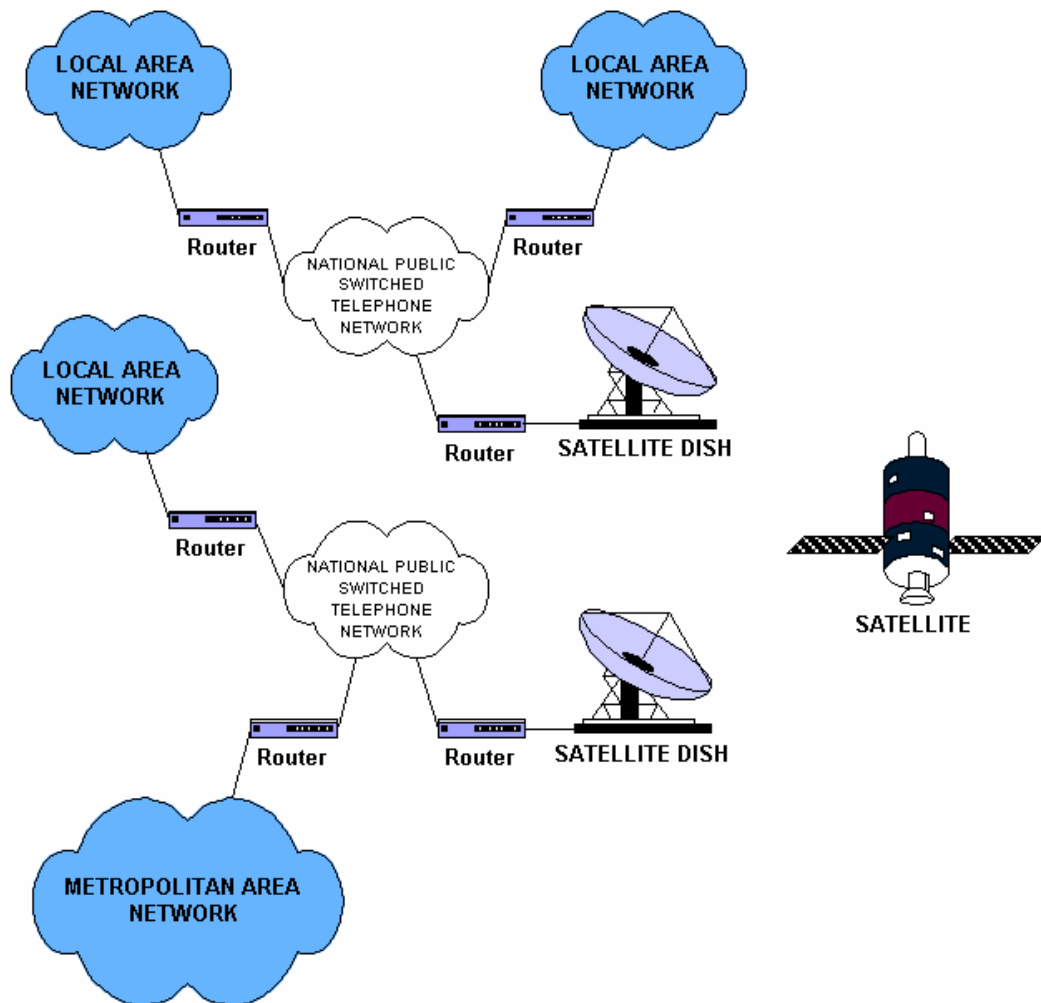


Figure 1.7: Wide Area Network

### Personal Area Networks

A personal area network (PAN) is a computer network for interconnecting **devices centered on an individual person's workspace**. A PAN provides data transmission amongst devices such as computers, smartphones, tablets and personal digital assistants. PANs can be used for communication amongst the personal devices themselves, or for connecting to a higher level network and the Internet (an uplink) where one master device takes up the role as gateway. A PAN may be carried over wired computer buses such as USB.

A wireless personal area network (WPAN) is a low-powered PAN carried over a short-distance wireless network technology such as IrDA, Wireless USB, Bluetooth and ZigBee. The reach of a WPAN varies from a few centimeters to a few meters.

### Global Area Network

- A global area network (GAN) refers to a network composed of different interconnected networks that cover an unlimited geographical area. The term is loosely synonymous with Internet, which is considered a global area network.
- Unlike local area networks (LAN) and wide area networks (WAN), GANs cover a large geographical area.
- Because a GAN is used to support mobile communication across a number of wireless LANs, the key challenge for any GAN is transferring user communications from one local coverage area to the next.
- The most sought-after GAN type is a broadband GAN. The broadband GAN is a global satellite Internet network that uses portable terminals for telephony. The terminals connect laptop computers located in remote areas to broadband Internet.

### Campus Area Network

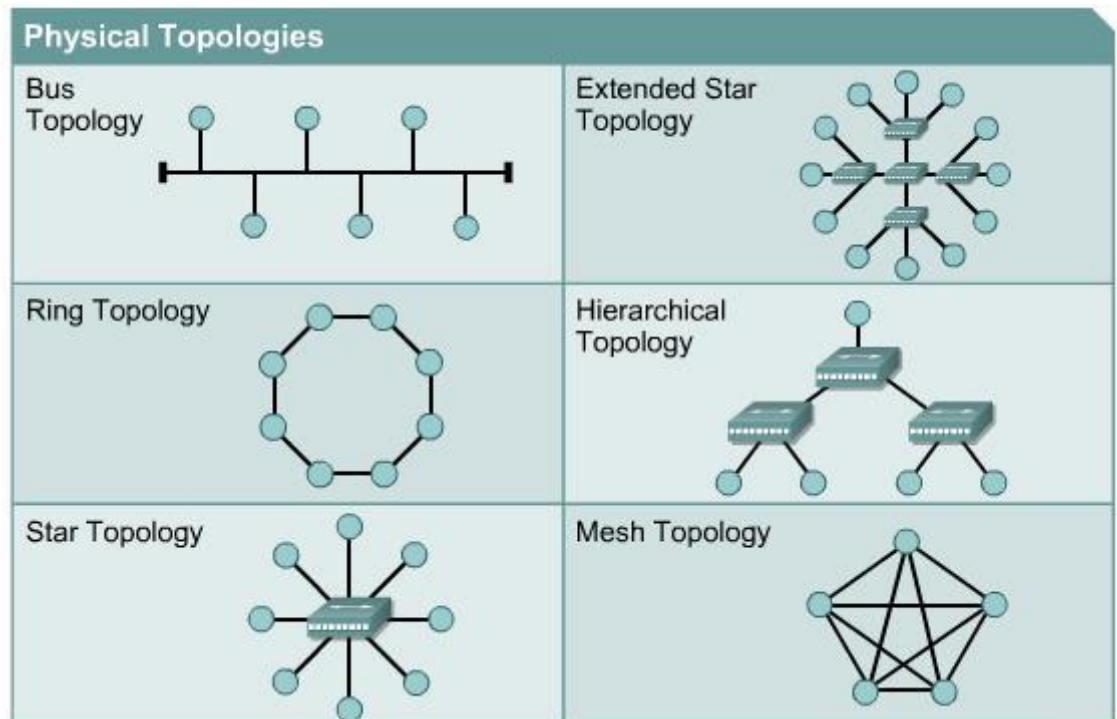
A campus network, campus area network, corporate area network or CAN is a computer network made up of an interconnection of local area networks (LANs) within a limited geographical area. The networking equipment's (switches, routers) and transmission media (optical fiber, copper plant, Cat5 cabling etc.) are almost entirely owned by the campus tenant / owner: an enterprise, university, government etc.

CAN benefits are as follows:

- Cost-effective
- Wireless, versus cable
- Multi departmental network access
- Single shared data transfer rate (DTR)

### Network Topology

- Network topology is an arrangement of the various elements like links, nodes etc. of biological network.
- It is topological structure of network, depicted as *physical* and *logically*.
- Physical topology refers to placement of network various component including device location and cable installation.
- Logical topology shows how data flows within a network, regardless of its physical design.



- **Bus**

In local area networks where bus topology is used, each node is connected to a single cable, by the help of interface connectors. This central cable is the backbone of the network and is known as the bus (thus the name). A signal from the source travels in both directions to all machines connected on the bus cable until it finds the intended recipient. If the machine address does not match the intended address for the data, the machine ignores the data. Alternatively, if the data matches the machine address, the data is accepted. Because the bus topology consists of only one wire, it is rather inexpensive to implement when compared to other topologies. However, the low cost of implementing the technology is offset by the high cost of managing the network. Additionally, because only one cable is utilized, it can be the single point of failure. In this topology data being transferred may be accessed by any node.

- **Linear bus**

The type of network topology in which all of the nodes of the network that are connected to a common transmission medium which has exactly two endpoints (this is the 'bus', which is also commonly referred to as the backbone, or trunk) – all data that is transmitted in between nodes in the network is transmitted over this common transmission medium and is able to be received by all nodes in the network simultaneously.

- **Distributed bus**

The type of network topology in which all of the nodes of the network are connected to a common transmission medium which has more than two endpoints that are created by adding branches to the main section of the transmission medium – the physical distributed bus topology functions in exactly the same fashion as the physical linear bus topology (i.e., all nodes share a common transmission medium).

- **Star network topology**

In local area networks with a star topology, each network host is connected to a central hub with a point-to-point connection. So it can be said that every computer is indirectly connected to every other node with the help of the hub. In star topology, every node (computer workstation or any other peripheral) is connected to a central node called hub, router or switch. The switch is the server and the peripherals are the clients. The network does not necessarily have to resemble a star to be classified as a star network, but all of the nodes on the network must be connected to one central device. All traffic that traverses the network passes through the central hub. The hub acts as a signal repeater. The star topology is considered the easiest topology to design and implement. An advantage of the star topology is the simplicity of adding additional nodes. The primary disadvantage of the star topology is that the hub represents a single point of failure. Since all peripheral communication must flow through the central hub, the aggregate central bandwidth forms a network bottleneck for large clusters.

- **Extended star**

The extended star network topology extend a physical star topology by one or more repeaters between the central node and the peripheral (or 'spoke') nodes. The repeaters are used to extend the maximum transmission distance of the physical layer, the point-to-point distance between the central node and the peripheral nodes. Repeaters permit to reach a greater transmission distance beyond the transmitting power of the central node. The use of repeaters can also overcome limitations from the standard upon which the physical layer is based.

A physical extended star topology in which repeaters are replaced with hubs or switches is a type of hybrid network topology and is referred to as a physical hierarchical star topology, although some texts make no distinction between the two topologies.

A physical hierarchical star topology can also be referred as a tier-star topology, this topology differ from a tree topology in the way start networks are connected together. A tier-star topology use central node, however a tree topology use central bus and can also be referred as star-bus network.

- **Distributed Star**

A type of network topology that is composed of individual networks that are based upon the physical star topology connected in a linear fashion – i.e., 'daisy-chained' – with no central or top level connection point (e.g., two or more 'stacked' hubs, along with their associated star connected nodes or 'spokes').

- **Ring network topology**

A ring topology is a bus topology in a closed loop. Data travels around the ring in one direction. When one node sends data to another, the data passes through each intermediate node on the ring until it reaches its destination. The intermediate nodes repeat (re transmit) the data to keep the signal strong.[5] Every node is a peer; there is no hierarchical relationship of clients and servers. If one node is unable to re transmit data, it severs communication between the nodes before and after it in the bus.

Advantages:

- When the load on the network increases, its performance is better than bus topology.
- There is no need of network server to control the connectivity between workstations.

Disadvantages:

- Aggregate network bandwidth is bottlenecked by the weakest link between two nodes.

- **Mesh network topology**

The value of fully meshed networks is proportional to the exponent of the number of subscribers, assuming that communicating groups of any two endpoints, up to and including all the endpoints, is approximated by Reed's Law.

➤ Fully connected network

In a fully connected network, all nodes are interconnected. (In graph theory this is called a complete graph.) The simplest fully connected network is a two-node network. A fully connected network doesn't need to use packet switching or broadcasting. However, since the number of connections grows quadratic ally with the number of nodes: This kind of topology does not trip and affect other nodes in the network this makes it impractical for large networks.

➤ Partially connected mesh topology

In a partially connected network, certain nodes are connected to exactly one other node; but some nodes are connected to two or more other nodes with a point-to-point link. This makes it possible to make use of some of the redundancy of mesh topology that is physically fully connected, without the expense and complexity required for a connection between every node in the network.

➤ Hybrid

Hybrid topology is also known as hybrid network. Hybrid networks combine two or more topologies in such a way that the resulting network does not exhibit one of the standard topologies (e.g., bus, star, ring, etc.). For example, a tree network (or star-bus network) is a hybrid topology in which star networks are interconnected via bus networks. However, a tree network connected to another tree network is still topologically a tree network, not a distinct network type. A hybrid topology is always produced when two different basic network topologies are connected.

- **Daisy chain**

Except for star-based networks, the easiest way to add more computers into a network is by daisy-chaining, or connecting each computer in series to the next. If a message is intended for a computer partway down the line, each system bounces it along in sequence until it reaches the destination. A daisy-chained network can take two basic forms: linear and ring.

A linear topology puts a two-way link between one computer and the next. However, this was expensive in the early days of computing, since each computer (except for the ones at each end) required two receivers and two transmitters.

By connecting the computers at each end, a ring topology can be formed. An advantage of the ring is that the number of transmitters and receivers can be cut in half, since a message will eventually loop all of the way around. When a node sends a message, the message is processed by each computer in the ring. If the ring breaks at a particular link then the transmission can be sent via the reverse path thereby ensuring that all nodes are always connected in the case of a single failure.

## Chapter 2

### Reference Model

#### Protocols and Standards

- Protocol suites are collections of protocols that enable network communication between hosts.
- A protocol is a formal description of a set of rules and conventions that govern a particular aspect of how devices on a network communicate. Protocols determine the format, timing, sequencing, and error control in data communication.
- Without protocols, the computer cannot make or rebuild the stream of incoming bits from another computer into the original format.
- Protocols control all aspects of data communication, which include the following:
  - How the physical network is built
  - How computers connect to the network
  - How the data is formatted for transmission
  - How that data is sent
  - How to deal with errors
- These network rules are created and maintained by many different organizations and committees. Included in these groups are the Institute of Electrical and Electronic Engineers (IEEE), American National Standards Institute (ANSI), Telecommunications Industry Association (TIA), Electronic Industries Alliance (EIA) and the International Telecommunications Union (ITU).

#### Layered Architectures

Network architectures define the standards and techniques for designing and building communication systems for computers and other devices. In the past, vendors developed their own architectures and required that other vendors conform to this architecture if they wanted to develop compatible hardware and software. There are proprietary network architectures such as IBM's SNA (Systems Network Architecture) and there are open architectures like the OSI (Open Systems Interconnection) model defined by the International Organization for Standardization. The previous strategy, where the computer network is designed with the hardware as the main concern and software is afterthought, no longer works. Network software is now highly *structured*.

To reduce the design complexity, most of the networks are organized as a series of **layers** or **levels**, each one build upon one below it. The basic idea of a layered architecture is *to divide the design into small pieces*. Each layer adds to the services provided by the lower layers in such a manner that the highest layer is provided a full set of services to manage communications and run the applications. The benefits of the layered models are modularity and clear interfaces, i.e. open architecture and comparability between the different providers' components.

A basic principle is to ensure independence of layers by defining services provided by each layer to the next higher layer without defining how the services are to be performed. This permits changes in a layer without affecting other layers. Prior to the use of layered protocol architectures, simple changes such as adding one terminal type to the list of those supported by an architecture often required changes to essentially all communications software at a site. The number of layers, functions and contents of each layer differ from network to network. However

in all networks, the purpose of each layer is to offer certain services to higher layers, shielding those layers from the details of how the services are actually implemented.

The basic elements of a layered model are services, protocols and interfaces. A *service* is a set of actions that a layer offers to another (higher) layer. *Protocol* is a set of rules that a layer uses to exchange information with a peer entity. These rules concern both the contents and the order of the messages used. Between the layers service interfaces are defined. The messages from one layer to another are sent through those interfaces.

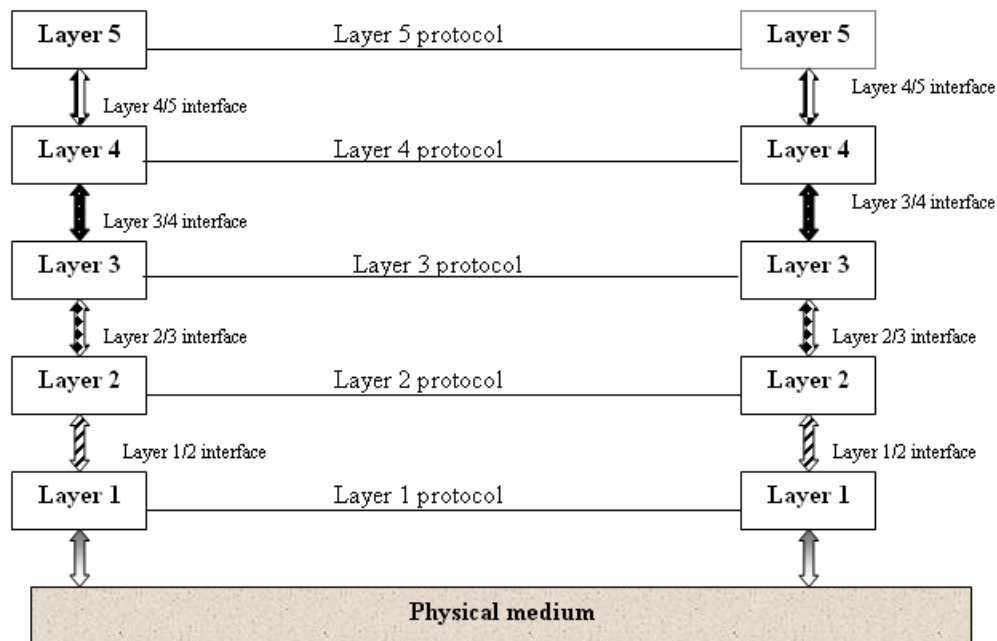


Figure 2.8 Basic Five Layer Architecture

Between each pair of adjacent layers there is an interface. The interface defines which primitives operations and services the lower layer offers to the upper layer adjacent to it. When network designer decides how many layers to include in the network and what each layer should do, one of the main considerations is defining clean interfaces between adjacent layers. Doing so, in turns requires that each layer should perform well-defined functions. In addition to minimize the amount of information passed between layers, clean-cut interface also makes it simpler to replace the implementation of one layer with a completely different implementation, because all what is required of new implementation is that it offers same set of services to its upstairs neighbor as the old implementation (that is what a layer provides and how to use that service from it is more important than knowing how exactly it implements it).

### Why Layered architecture?

1. To make the design process easy by breaking unmanageable tasks into several smaller and manageable tasks (by divide-and-conquer approach).
2. Modularity and clear interfaces, so as to provide comparability between the different providers' components.

3. Ensure independence of layers, so that implementation of each layer can be changed or modified without affecting other layers.
4. Each layer can be analyzed and tested independently of all other layers.

## Open System Interconnection Reference Model

The Open System Interconnection (OSI) reference model describes how information from a software application in one computer moves through a network medium to a software application in another computer. The OSI reference model is a conceptual model composed of seven layers, each specifying particular network functions. The model was developed by the International Organization for Standardization (ISO) in 1984, and it is now considered the primary architectural model for inter-computer communications. The OSI model divides the tasks involved with moving information between networked computers into seven smaller, more manageable task groups. A task or group of tasks is then assigned to each of the seven OSI layers. Each layer is reasonably self-contained so that the tasks assigned to each layer can be implemented independently. This enables the solutions offered by one layer to be updated without adversely affecting the other layers.

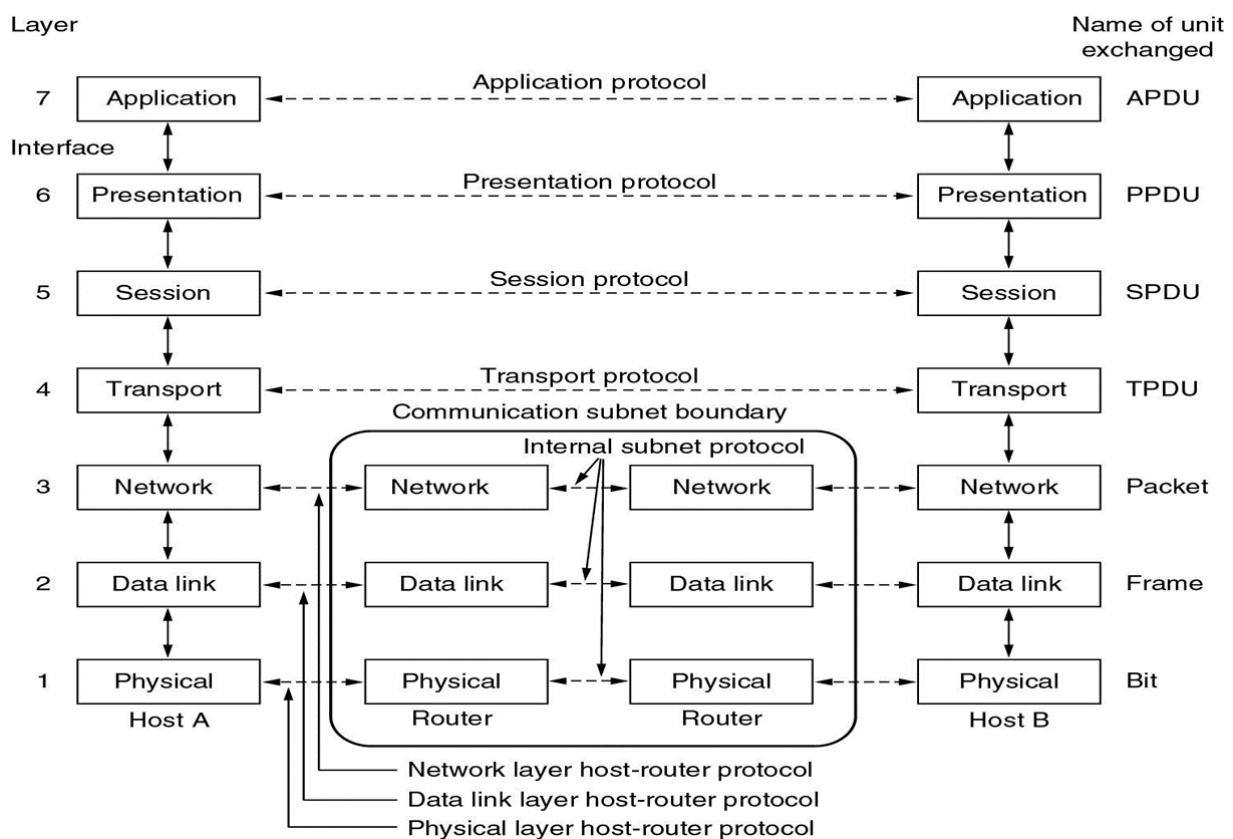


Figure 9.2 OSI layer

The OSI Reference Model includes seven layers:

**7. Application Layer:** Provides Applications with access to network services.

**6. Presentation Layer:** Determines the format used to exchange data among networked computers.

**5. Session Layer:** Allows two applications to establish, use and disconnect a connection between them called a session. Provides for name recognition and additional functions like security, which are needed to allow applications to communicate over the network.

**4. Transport Layer:** Ensures that data is delivered error free, in sequence and with no loss, duplications or corruption. This layer also repackages data by assembling long messages into lots of smaller messages for sending, and repackaging the smaller messages into the original larger message at the receiving end.

**3. Network Layer:** This is responsible for addressing messages and data so they are sent to the correct destination, and for translating logical addresses and names (like a machine name FLAME) into physical addresses. This layer is also responsible for finding a path through the network to the destination computer.

**2. Data-Link Layer:** This layer takes the data frames or messages from the Network Layer and provides for their actual transmission. At the receiving computer, this layer receives the incoming data and sends it to the network layer for handling. The Data-Link Layer also provides error-free delivery of data between the two computers by using the physical layer. It does this by packaging the data from the Network Layer into a frame, which includes error detection information. At the receiving computer, the Data-Link Layer reads the incoming frame, and generates its own error detection information based on the received frames data. After receiving the entire frame, it then compares its error detection value with that of the incoming frames, and if they match, the frame has been received correctly.

**1. Physical Layer:** Controls the transmission of the actual data onto the network cable. It defines the electrical signals, line states and encoding of the data and the connector types used. An example is 10BaseT.

## Functions of the OSI Layers

Functions of different layers of the OSI model are presented in this section.

### 1. Physical Layer

The physical layer is concerned with transmission of raw bits over a communication channel. It specifies the mechanical, electrical and procedural network interface specifications and the physical transmission of bit streams over a transmission medium connecting two pieces of communication equipment. In simple terms, the physical layer decides the following:

- Number of pins and functions of each pin of the network connector (Mechanical)
- Signal Level, Data rate (Electrical)
- Whether simultaneous transmission in both directions
- Establishing and breaking of connection
- Deals with physical transmission

There exist a variety of physical layer protocols such as RS-232C, Rs-449 standards developed by Electronics Industries Association (EIA).

## **2. Data Link Layer**

- The goal of the data link layer is to provide reliable, efficient communication between adjacent machines connected by a single communication channel. Specifically:
- Group the physical layer bit stream into units called frames. Note that frames are nothing more than ``packets" or ``messages". By convention, we shall use the term ``frames" when discussing DLL packets.
- Sender calculates the checksum and sends checksum together with data. The checksum allows the receiver to determine when a frame has been damaged in transit or received correctly.
- Receiver re-computes the checksum and compares it with the received value. If they differ, an error has occurred and the frame is discarded.
- Error control protocol returns a positive or negative acknowledgment to the sender. A positive acknowledgment indicates the frame was received without errors, while a negative acknowledgment indicates the opposite.
- Flow control prevents a fast sender from overwhelming a slower receiver. For example, a supercomputer can easily generate data faster than a PC can consume it.
- In general, data link layer provides service to the network layer. The network layer wants to be able to send packets to its neighbors without worrying about the details of getting it there in one piece.

### **Design Issues**

Below are the some of the important design issues of the data link layer:

#### **a) Reliable Delivery:**

Frames are delivered to the receiver reliably and in the same order as generated by the sender. Connection state keeps track of sending order and which frames require retransmission. For example, receiver state includes which frames have been received, which ones have not, etc.

#### **b) Best Effort:**

The receiver does not return acknowledgments to the sender, so the sender has no way of knowing if a frame has been successfully delivered.

When would such a service be appropriate?

1. When higher layers can recover from errors with little loss in performance. That is, when errors are so infrequent that there is little to be gained by the data link layer performing the recovery. It is just as easy to have higher layers deal with occasional loss of packet.
2. For real-time applications requiring ``better never than late" semantics. Old data may be worse than no data.

#### **c) Acknowledged Delivery**

The receiver returns an acknowledgment frame to the sender indicating that a data frame was properly received. This sits somewhere between the other two in that the sender keeps connection state, but may not necessarily retransmit unacknowledged frames. Likewise, the receiver may hand over received packets to higher layer in the order in which they arrive, regardless of the original sending order. Typically, each frame is assigned a unique sequence number, which the receiver returns in an acknowledgment frame to indicate which frame the ACK refers to. The sender must retransmit unacknowledged (e.g., lost or damaged) frames.

#### **d) Framing**

The DLL translates the physical layer's raw bit stream into discrete units (messages) called frames. How can the receiver detect frame boundaries? Various techniques are used for this: Length Count, Bit Stuffing, and Character stuffing.

#### **e) Error Control**

Error control is concerned with insuring that all frames are eventually delivered (possibly in order) to a destination. To achieve this, three items are required: Acknowledgements, Timers, and Sequence Numbers.

#### **f) Flow Control**

Flow control deals with throttling the speed of the sender to match that of the receiver. Usually, this is a dynamic process, as the receiving speed depends on such changing factors as the load, and availability of buffer space.

### **Link Management**

In some cases, the data link layer service must be "opened" before use:

- The data link layer uses open operations for allocating buffer space, control blocks, agreeing on the maximum message size, etc.
- Synchronize and initialize send and receive sequence numbers with its peer at the other end of the communications channel.

### **Error Detection and Correction**

In data communication, error may occur because of various reasons including attenuation, noise. Moreover, error usually occurs as bursts rather than independent, single bit errors. For example, a burst of lightning will affect a set of bits for a short time after the lightning strike. Detecting and correcting errors requires redundancy (i.e., sending additional information along with the data).

There are two types of attacks against errors:

- **Error Detecting Codes:** Include enough redundancy bits to detect errors and use ACKs and retransmissions to recover from the errors. Example: parity encoding.
- **Error Correcting Codes:** Include enough redundancy to detect and correct errors. Examples: CRC checksum, MD5.

## **Network Layer**

The basic purpose of the network layer is to provide an end-to-end communication capability in contrast to machine-to-machine communication provided by the data link layer. This end-to-end is performed using two basic approaches known as connection-oriented or connectionless network-layer services.

### **Four issues:**

1. Interface between the host and the network (the network layer is typically the boundary between the host and subnet)
2. Routing
3. Congestion and deadlock
4. Internetworking (A path may traverse different network technologies (e.g., Ethernet, point-to-point links, etc.)

### ➤ **Network Layer Interface**

There are two basic approaches used for sending packets, which is a group of bits that includes data plus source and destination addresses, from node to node called virtual circuit and datagram methods. These are also referred to as connection-oriented and connectionless network-layer services. In virtual circuit approach, a route, which consists of logical connection, is first established between two users. During this establishment phase, the two users not only agree to set up a connection between them but also decide upon the quality of service to be associated with the connection. The well-known virtual-circuit protocol is the ISO and CCITT X.25 specification. The datagram is a self-contained message unit, which contains sufficient information for routing from the source node to the destination node without dependence on previous message interchanges between them. In contrast to the virtual-circuit method, where a fixed path is explicitly set up before message transmission, sequentially transmitted messages can follow completely different paths. The datagram method is analogous to the postal system and the virtual-circuit method is analogous to the telephone system.

### ➤ **Overview of Other Network Layer Issues:**

The network layer is responsible for routing packets from the source to destination. The routing algorithm is the piece of software that decides where a packet goes next (e.g., which output line, or which node on a broadcast channel).

For connectionless networks, the routing decision is made for each datagram. For connection-oriented networks, the decision is made once, at circuit setup time.

### ➤ **Routing Issues:**

The routing algorithm must deal with the following issues:

- Correctness and simplicity: networks are never taken down; individual parts (e.g., links, routers) may fail, but the whole network should not.
- Stability: if a link or router fails, how much time elapses before the remaining routers recognize the topology change? (Some never do.)

- Fairness and optimality: an inherently intractable problem. Definition of optimality usually doesn't consider fairness. Do we want to maximize channel usage? Minimize average delay?

When we look at routing in detail, we'll consider both adaptive--those that take current traffic and topology into consideration--and non-adaptive algorithms.

➤ **Congestion**

The network layer also must deal with congestion:

- When more packets enter an area than can be processed, delays increase and performance decreases. If the situation continues, the subnet may have no alternative but to discard packets.
- If the delay increases, the sender may (incorrectly) retransmit, making a bad situation even worse.
- Overall, performance degrades because the network is using (wasting) resources processing packets that eventually get discarded.

➤ **Internetworking**

Finally, when we consider internetworking -- connecting different network technologies together -- one finds the same problems, only worse:

- Packets may travel through many different networks
- Each network may have a different frame format
- Some networks may be connectionless, other connection oriented

➤ **Routing**

Routing is concerned with the question: Which line should router J use when forwarding a packet to router K?

There are two types of algorithms:

- **Adaptive algorithms** use such dynamic information as current topology, load, delay, etc. to select routes.
- In **non-adaptive algorithms**, routes never change once initial routes have been selected. Also called static routing.

Obviously, adaptive algorithms are more interesting, as non-adaptive algorithms don't even make an attempt to handle failed links.

### 3. Transport Layer

The transport level provides end-to-end communication between processes executing on different machines. Although the services provided by a transport protocol are similar to those provided by a data link layer protocol, there are several important differences between the transport and lower layers:

1. **User Oriented.**

Application programmers interact directly with the transport layer, and from the programmers perspective, the transport layer is the ``network''. Thus, the transport layer

should be oriented more towards user services than simply reflect what the underlying layers happen to provide. (Similar to the beautification principle in operating systems.)

## **2. *Negotiation of Quality and Type of Services.***

The user and transport protocol may need to negotiate as to the quality or type of service to be provided. Examples? A user may want to negotiate such options as: throughput, delay, protection, priority, reliability, etc.

## **3. *Guarantee Service***

The transport layer may have to overcome service deficiencies of the lower layers (e.g. providing reliable service over an unreliable network layer).

## **4. *Addressing becomes a significant issue.***

That is, now the user must deal with it; before it was buried in lower levels.

Two solutions:

- Use well-known addresses that rarely if ever change, allowing programs to "wire in" addresses. For what types of service does this work? While this works for services that are well established (e.g., mail, or telnet), it doesn't allow a user to easily experiment with new services.
- Use a name server. Servers register services with the name server, which clients contact to find the transport address of a given service.

In both cases, we need a mechanism for mapping high-level service names into low-level encoding that can be used within packet headers of the network protocols. In its general form, the problem is quite complex. One simplification is to break the problem into two parts: have transport addresses be a combination of machine address and local process on that machine.

## **5. *Storage capacity of the subnet.***

Assumptions valid at the data link layer do not necessarily hold at the transport layer. Specifically, the subnet may buffer messages for a potentially long time, and an "old" packet may arrive at a destination at unexpected times.

## **6. *We need a dynamic flow control mechanism.***

The data link layer solution of reallocating buffers is inappropriate because a machine may have hundreds of connections sharing a single physical link. In addition, appropriate settings for the flow control parameters depend on the communicating end points (e.g., Cray supercomputers vs. PCs), not on the protocol used.

Don't send data unless there is room. Also, the network layer/data link layer solution of simply not acknowledging frames for which the receiver has no space is unacceptable. Why? In the data link case, the line is not being used for anything else; thus retransmissions are inexpensive. At the transport level, end-to-end retransmissions are needed, which wastes resources by sending the same packet over the same links multiple times. If the receiver has no buffer space, the sender should be prevented from sending data.

### 7. Deal with congestion control.

In connectionless Internets, transport protocols must exercise congestion control. When the network becomes congested, they must reduce rate at which they insert packets into the subnet, because the subnet has no way to prevent itself from becoming overloaded.

### 8. Connection establishment.

Transport level protocols go through three phases: establishing, using, and terminating a connection. For data gram-oriented protocols, opening a connection simply allocates and initializes data structures in the operating system kernel.

Connection oriented protocols often exchanges messages that negotiate options with the remote peer at the time a connection are opened. Establishing a connection may be tricky because of the possibility of old or duplicate packets.

Finally, although not as difficult as establishing a connection, terminating a connection presents subtleties too. For instance, both ends of the connection must be sure that all the data in their queues have been delivered to the remote application.

## 4. Session Layer

This layer allows users on different machines to establish session between them. A session allows ordinary data transport but it also provides enhanced services useful in some applications. A session may be used to allow a user to log into a remote time sharing machine or to transfer a file between two machines. Some of the session related services are:

**1. This layer manages Dialogue Control.** Session can allow traffic to go in both direction at the same time, or in only one direction at one time.

**2. Token management.** For some protocols, it is required that both sides don't attempt same operation at the same time. To manage these activities, the session layer provides tokens that can be exchanged. Only one side that is holding token can perform the critical operation. This concept can be seen as entering into a critical section in operating system using semaphores.

**3. Synchronization.** Consider the problem that might occur when trying to transfer a 4-hour file transfer with a 2-hour mean time between crashes. After each transfer was aborted, the whole transfer has to start again and again would probably fail. To eliminate this problem, Session layer provides a way to insert checkpoints into data streams, so that after a crash, only the data transferred after the last checkpoint have to be repeated.

## 6. Presentation Layer

This layer is concerned with Syntax and Semantics of the information transmitted, unlike other layers, which are interested in moving data reliably from one machine to other. Few of the services that Presentation layer provides are:

1. Encoding data in a standard agreed upon way.
2. It manages the abstract data structures and converts from representation used inside computer to network standard representation and back.

## 7. Application Layer

The application layer consists of what most users think of as programs. The application does the actual work at hand. Although each application is different, some applications are so useful that they have become standardized. The Internet has defined standards for:

- **File transfer (FTP):** Connect to a remote machine and send or fetch an arbitrary file. FTP deals with authentication, listing a directory contents, ASCII or binary files, etc.
- **Remote login (telnet):** A remote terminal protocol that allows a user at one site to establish a TCP connection to another site, and then pass keystrokes from the local host to the remote host.
- **Mail (SMTP):** Allow a mail delivery agent on a local machine to connect to a mail delivery agent on a remote machine and deliver mail.
- **News (NNTP):** Allows communication between a news server and a news client.
- **Web (HTTP):** Base protocol for communication on the World Wide Web.

## TCP/IP (Transmission Control Protocol/Internet Protocol)

TCP/IP, or the Transmission Control Protocol/Internet Protocol, is a suite of communication protocols used to interconnect network devices on the internet. TCP/IP can also be used as a communications protocol in a private network (an intranet or an extranet).

The entire internet protocol suite -- a set of rules and procedures -- is commonly referred to as TCP/IP, though others are included in the suite.

TCP/IP specifies how data is exchanged over the internet by providing end-to-end communications that identify how it should be broken into packets, addressed, transmitted, routed and received at the destination. TCP/IP requires little central management, and it is designed to make networks reliable, with the ability to recover automatically from the failure of any device on the network.

The two main protocols in the internet protocol suite serve specific functions. TCP defines how applications can create channels of communication across a network. It also manages how a message is assembled into smaller packets before they are then transmitted over the internet and reassembled in the right order at the destination address.

IP defines how to address and route each packet to make sure it reaches the right destination. Each gateway computer on the network checks this IP address to determine where to forward the message.

## The history of TCP/IP

The Defense Advanced Research Projects Agency (DARPA), the research branch of the U.S. Department of Defense, created the TCP/IP model in the 1970s for use in ARPANET, a

wide area network that preceded the internet. TCP/IP was originally designed for the UNIX operating system, and it has been built into all of the operating systems that came after it.

The TCP/IP model and its related protocols are now maintained by the Internet Engineering Task Force.

### How TCP/IP works

TCP/IP uses the client/server model of communication in which a user or machine (a client) is provided a service (like sending a webpage) by another computer (a server) in the network.

Collectively, the TCP/IP suite of protocols is classified as stateless, which means each client request is considered new because it is unrelated to previous requests. Being stateless frees up network paths so they can be used continuously.

The transport layer itself, however, is stateful. It transmits a single message, and its connection remains in place until all the packets in a message have been received and reassembled at the destination.

### TCP/IP model layers

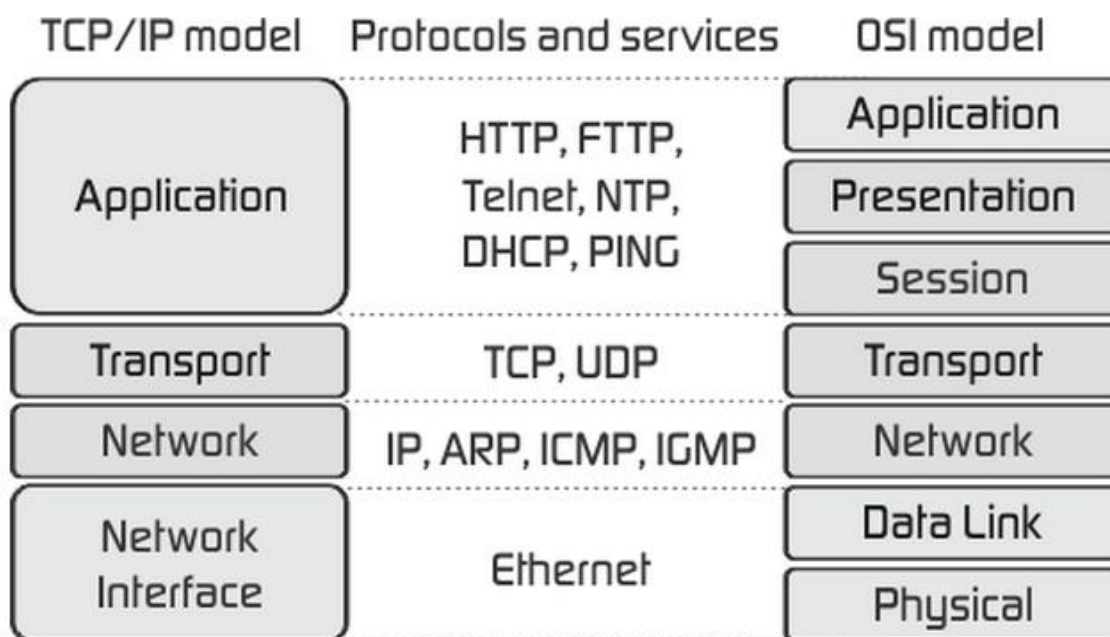


Figure 2.10 TCP/IP Model

TCP/IP functionality is divided into four layers, each of which include specific protocols.

The **application layer** provides applications with standardized data exchange. Its protocols include the Hypertext Transfer Protocol (HTTP), File Transfer Protocol (FTP), Post Office Protocol 3 (POP3), Simple Mail Transfer Protocol (SMTP) and Simple Network Management Protocol (SNMP).

The **transport layer** is responsible for maintaining end-to-end communications across the network. TCP handles communications between hosts and provides flow control, multiplexing and reliability. The transport protocols include TCP and User Datagram Protocol (UDP), which is sometimes used instead of TCP for special purposes.

The **network layer**, also called the internet layer, deals with packets and connects independent networks to transport the packets across network boundaries. The network layer protocols are the IP and the Internet Control Message Protocol (ICMP), which is used for error reporting.

The **physical layer** consists of protocols that operate only on a link -- the network component that interconnects nodes or hosts in the network. The protocols in this layer include Ethernet for local area networks (LANs) and the Address Resolution Protocol (ARP).

### **Advantages of TCP/IP**

TCP/IP is nonproprietary and, as a result, is not controlled by any single company. Therefore, the internet protocol suite can be modified easily. It is compatible with all operating systems, so it can communicate with any other system. The internet protocol suite is also compatible with all types of computer hardware and networks.

### **OSI vs TCP IP Model**

- TCP/IP is a communication protocol that allows for connections of hosts to the internet.
- OSI, on the other hand, is a communication gateway between the network and the end users. TCP/IP refers to Transmission Control Protocol used in and by the applications on the internet. This protocol can borrow its roots from the Department of Defense, which developed it to allow different devices to be connected to the internet.
- OSI, on the other hand, refers to Open Systems Interconnection, a communication gateway developed by the International Organization for Standardization (ISO).

### ***Just what differences are there among the two?***

- First off is the model of implementation on which each is developed. TCP/IP comes from the implementation of the OSI model, which led innovation in the field.
- OSI, on the other hand, was developed as a reference model that could be employed online. The model upon which TCP/IP is developed, on the other hand, points toward a model that revolves around the internet. The model around which OSI was developed upon is a theoretical model and not the internet.
- There are four levels or layers upon which TCP is developed. These layers include the Link Layer, the Internet Layer, Application Layer and the Transport Layer.
- The OSI gateway, on the other hand, is developed upon a seven-layer model. The seven layers include Physical Layer, DataLink Layer, Network Layer, Transport Layer, Session Layer, Presentation Layer and, last but not least, Application Layer.

### ***When it comes to general reliability,***

- TCP/IP is considered to be a more reliable option as opposed to OSI model.
- The OSI model is, in most cases, referred to as a reference tool, being the older of the two models.

- OSI is also known for its strict protocol and boundaries. This is not the case with TCP/IP. It allows for a loosening of the rules, provided the general guidelines are met.

***On the approach that the two implement,***

- TCP/IP is seen to implement a horizontal approach while the OSI model is shown to implement a vertical approach.
- It is also important to note that TCP/IP combines the session layer and presentation too in the application layer.
- OSI, on the other side, seems to take a different approach to the presentation, having different session and presentation layers altogether.
- It is also imperative to note the design followed when protocols were being designed. In TCP/IP, the protocols were first designed and then the model was developed.
- In OSI, the model development came first and then the protocol development came in second.

***When it comes to the communications,***

- TCP/IP supports only connectionless communication emanating from the network layer.
- OSI, on the other hand, seems to do quite well, supporting both connectionless and connection-oriented communication within the network layer.
- Last but not least is the protocol dependency of the two.
  - TCP/IP is a protocol dependent model, whereas
  - OSI is a protocol independent standard.

**Summary of OSI vs TCP**

- **TCP** refers to Transmission Control Protocol.
- **OSI** refers to Open Systems Interconnection.
- Model **TCP/IP** is developed on points toward a model the internet.
- **TCP/IP** has 4 layers.
- **OSI** has 7 layers.
- **TCP/IP** more reliable than **OSI**
- **OSI** has strict boundaries; **TCP/IP** does not have very strict boundaries.
- **TCP/IP** follow a horizontal approach.
- **OSI** follows a vertical approach.
- In the application layer, **TCP/IP** uses both session and presentation layer.
- **OSI** uses different session and presentation layers.
- **TCP/IP** developed protocols then model.
- **OSI** developed model then protocol.
- **TCP/IP** offers support for connectionless communication within the network layer.
- In the network layer, **OSI** supports both connectionless and connection-oriented communication.
- **TCP/IP** is protocol dependent.
- **OSI** is protocol independent.

### Similarities between TCP/IP & OSI models

- They share similar architecture. - Both of the models share a similar architecture. This can be illustrated by the fact that both of them are constructed with layers.
- They share a common application layer. - Both of the models share a common "application layer". However in practice this layer includes different services depending upon each model.
- Both models have comparable transport and network layers. - This can be illustrated by the fact that whatever functions are performed between the presentation and network layer of the OSI model similar functions are performed at the Transport layer of the TCP/IP model.
- Knowledge of both models is required by networking professionals.
- Both models assume that packets are switched. Basically this means that individual packets may take differing paths in order to reach the same destination.
- Both models are based on layered protocols. In both models, the transport service can provide a reliable end-to-end byte stream.

### Network Devices (Hub, Repeater, Bridge, Switch, Router, Gateways and Brouter)

#### 1. Repeater

A repeater operates at the physical layer. Its job is to regenerate the signal over the same network before the signal becomes too weak or corrupted so as to extend the length to which the signal can be transmitted over the same network. An important point to be noted about repeaters is that they do not amplify the signal. When the signal becomes weak, they copy the signal bit by bit and regenerate it at the original strength. It is a 2 port device.

#### 2. Hub

A hub is basically a multiport repeater. A hub connects multiple wires coming from different branches, for example, the connector in star topology which connects different stations. Hubs cannot filter data, so data packets are sent to all connected devices. In other words, collision domain of all hosts connected through Hub remains one. Also, they do not have intelligence to find out best path for data packets which leads to inefficiencies and wastage.

#### Types of Hub

**Active Hub:** - These are the hubs which have their own power supply and can clean, boost and relay the signal along the network. It serves both as a repeater as well as wiring center. These are used to extend maximum distance between nodes.

**Passive Hub:** - These are the hubs which collect wiring from nodes and power supply from active hub. These hubs relay signals onto the network without cleaning and boosting them and can't be used to extend distance between nodes.

#### 3. Bridge

A bridge operates at data link layer. A bridge is a repeater, with add on functionality of filtering content by reading the MAC addresses of source and destination. It is also used for interconnecting two LANs working on the same protocol. It has a single input and single output port, thus making it a 2 port device.

### Types of Bridges

- **Transparent Bridges:** - These are the bridge in which the stations are completely unaware of the bridge's existence i.e. whether or not a bridge is added or deleted from the network, reconfiguration of the stations is unnecessary. These bridges makes use of two processes i.e. bridge forwarding and bridge learning.
- **Source Routing Bridges:** - In these bridges, routing operation is performed by source station and the frame specifies which route to follow. The host can discover frame by sending a special frame called discovery frame, which spreads through the entire network using all possible paths to destination.

### 4. Switch

A switch is a multi-port bridge with a buffer and a design that can boost its efficiency (large number of ports imply less traffic) and performance. Switch is data link layer device. Switch can perform error checking before forwarding data that makes it very efficient as it does not forward packets that have errors and forward good packets selectively to correct port only. In other words, switch divides collision domain of hosts, but broadcast domain remains same.

### 5. Routers

A router is a device like a switch that routes data packets based on their IP addresses. Router is mainly a Network Layer device. Routers normally connect LANs and WANs together and have a dynamically updating routing table based on which they make decisions on routing the data packets. Router divide broadcast domains of hosts connected through it.

### 6. Gateway

A gateway, as the name suggests, is a passage to connect two networks together that may work upon different networking models. They basically works as the messenger agents that take data from one system, interpret it, and transfer it to another system. Gateways are also called protocol converters and can operate at any network layer. Gateways are generally more complex than switch or router.

### 7. Brouter

It is also known as bridging router is a device which combines features of both bridge and router. It can work either at data link layer or at network layer. Working as router, it is capable of routing packets across networks and working as bridge, it is capable of filtering local area network traffic.

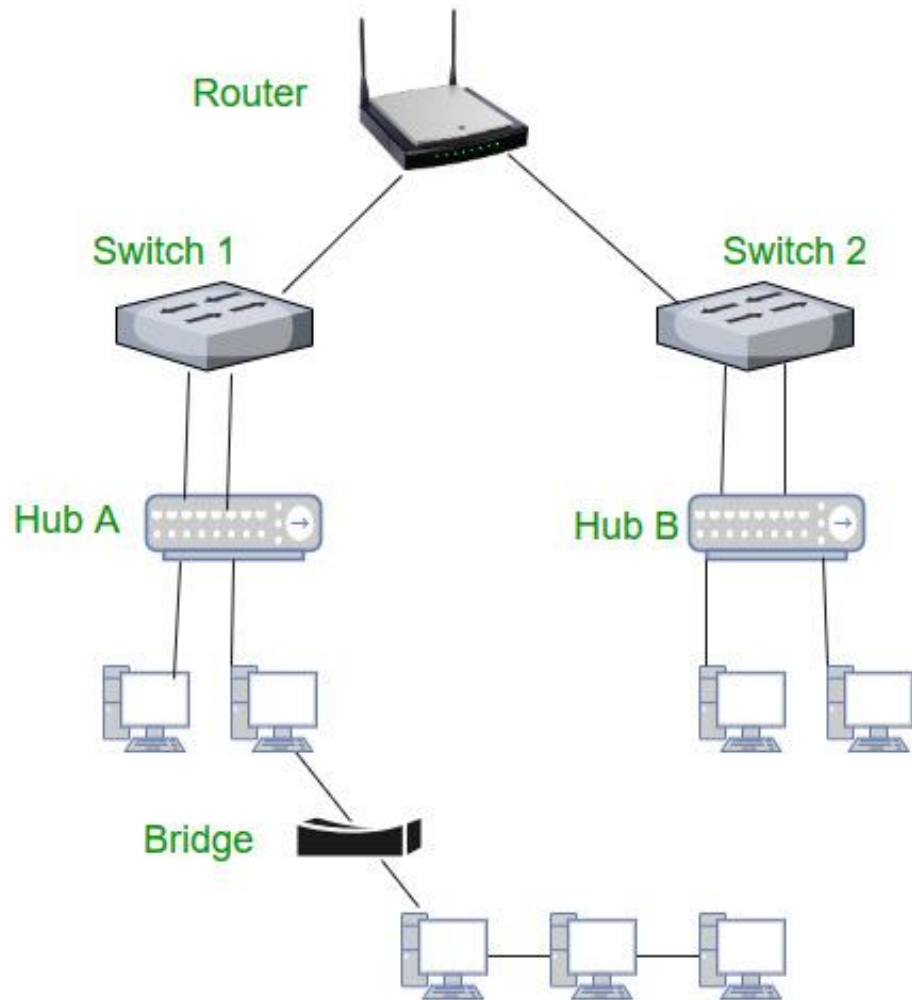


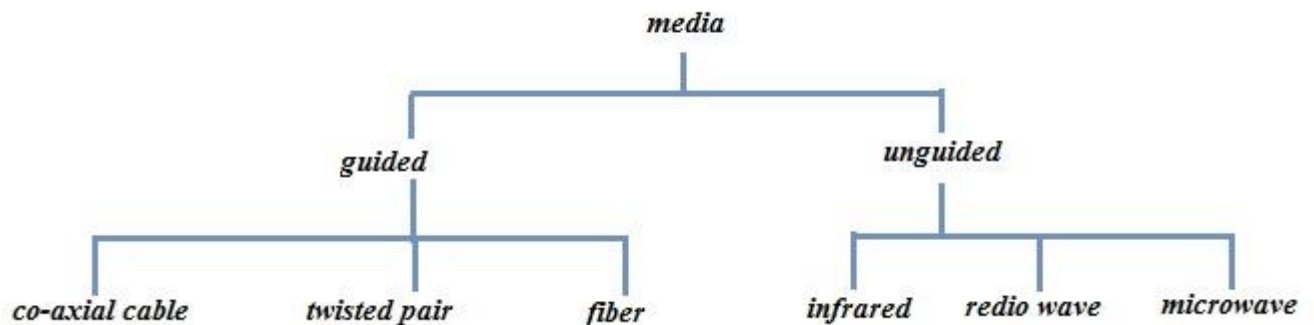
Figure 2.11 Network Hardware

## Chapter 3

### Physical Media

#### Transmission Media

In data communication terminology, a transmission medium is a physical path between the transmitter and the receiver i.e. it is the channel through which data is sent from one place to another. Transmission Media is broadly classified into the following types:



#### Guided Media:

It is also referred to as Wired or Bounded transmission media. Signals being transmitted are directed and confined in a narrow pathway by using physical links.

Features:

- High Speed
- Secure
- Used for comparatively shorter distances

There are 3 major types of Guided Media:

#### **(i) Twisted Pair Cable**

It consists of 2 separately insulated conductor wires wound about each other. Generally, several such pairs are bundled together in a protective sheath. They are the most widely used Transmission Media.

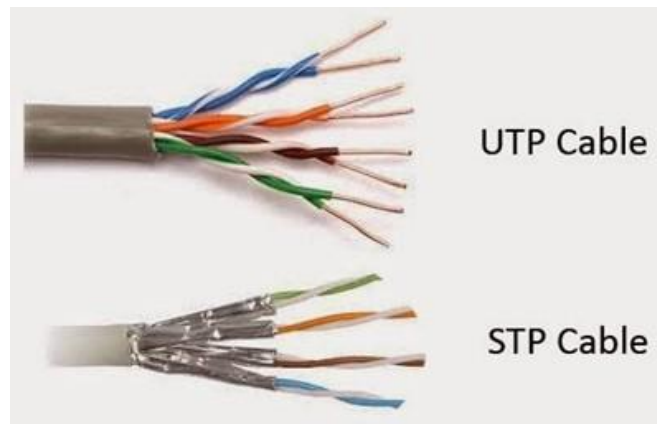


Figure 3.12 Twisted Cable (CAT5)

In twisted pair technology, two copper wires are strung between two points:

- The two wires are typically "twisted" together in a helix to reduce interference between the two conductors as shown in Fig.3.1 Twisting decreases the cross-talk interference between adjacent pairs in a cable. Typically, a number of pairs are bundled together into a cable by wrapping them in a tough protective sheath.
- Can carry both analog and digital signals. Actually, they carry only analog signals. However, the "analog" signals can very closely correspond to the square waves representing bits, so we often think of them as carrying digital data.
- Data rates of several Mbps common.
- Spans distances of several kilometers.
- Data rate determined by wire thickness and length. In addition, shielding to eliminate interference from other wires impacts signal-to-noise ratio, and ultimately, the data rate.
- Good, low-cost communication. Indeed, many sites already have twisted pair installed in offices -- existing phone lines!

**Twisted Pair is of two types:**

➤ **Unshielded Twisted Pair (UTP):**

This type of cable has the ability to block interference and does not depend on a physical shield for this purpose. It is used for telephonic applications.

Advantages:

- Least expensive
- Easy to install
- High speed capacity

Disadvantages:

- Susceptible to external interference
- Lower capacity and performance in comparison to STP
- Short distance transmission due to attenuation

➤ **Shielded Twisted Pair (STP):**

This type of cable consists of a special jacket to block external interference. It is used in fast-data-rate Ethernet and in voice and data channels of telephone lines.

Advantages:

- Better performance at a higher data rate in comparison to UTP
- Eliminates crosstalk
- Comparatively faster

Disadvantages:

- Comparatively difficult to install and manufacture
- More expensive
- Bulky

### (ii) Coaxial Cable

It has an outer plastic covering containing 2 parallel conductors each having a separate insulated protection cover. Coaxial cable transmits information in two modes: Baseband mode (dedicated cable bandwidth) and Broadband mode (cable bandwidth is split into separate ranges). Cable TVs and analog television networks widely use Coaxial cables.

Advantages:

- High Bandwidth
- Better noise Immunity
- Easy to install and expand
- Inexpensive

Disadvantages:

- Single cable failure can disrupt the entire network



Figure 3.13 Coaxial Cable

### (iii) Optical Fiber Cable

It uses the concept of reflection of light through a core made up of glass or plastic. The core is surrounded by a less dense glass or plastic covering called the cladding. It is used for transmission of large volumes of data.

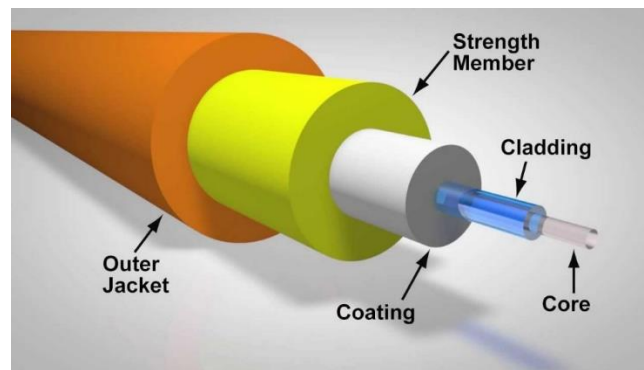


Figure 14.3: Optical Fiber

Three components are required:

- **Fiber medium:** Current technology carries light pulses for tremendous distances (e.g., 100s of kilometers) with virtually no signal loss.
- **Light source:** typically a Light Emitting Diode (LED) or laser diode. Running current through the material generates a pulse of light.
- **A photo diode light detector, which converts light pulses into electrical signals.**

Advantages:

- Very high data rate, low error rate. 1000 Mbps (1 Gbps) over distances of kilometers common.
- Error rates are so low they are almost negligible.
- Difficult to tap, which makes it hard for unauthorized taps as well. This is responsible for higher reliability of this medium.
- Light weight.
- Not susceptible to electrical interference (lightning) or corrosion (rust).
- Less signal attenuation.
- Greater repeater distance than coax.

Disadvantages:

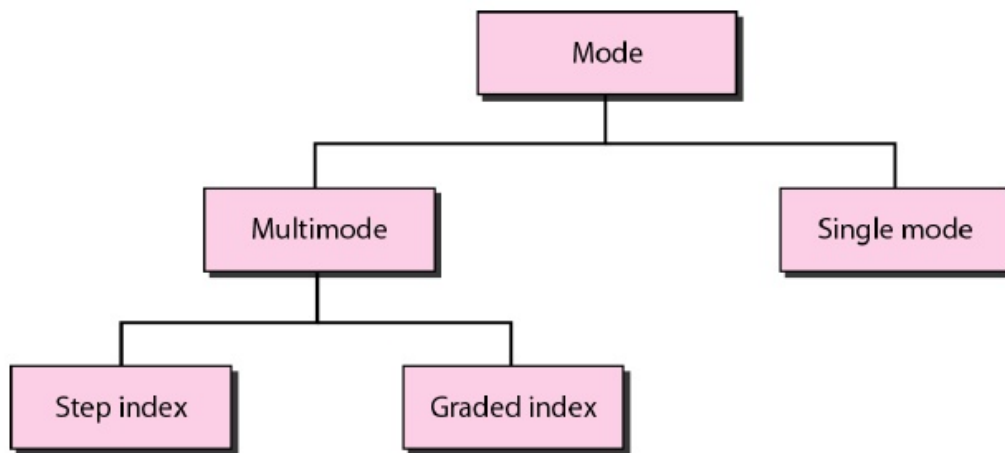
- Difficult to install and maintain
- High cost
- Fragile

### Propagation Mode

There are 2 types of propagation mode in fiber optics cable which are multi-mode and single-mode. These provide different performance with respect to both attenuation and time dispersion. The single-mode fiber optic cable provides the better performance at a higher cost.

The number of modes in a fiber optic cable depends upon the dimensions of the cable and the variation of the indices of refraction of both core and cladding across the cross section. There

are three principal possibilities which are multi-mode step index, single-mode step index and multi-mode graded index.



- **Single-mode Step Index**

The diameter of the core is fairly small relative to the cladding. Typically, the cladding is ten times thicker than the core. Comparing the output pulse and the input pulse note that there is little attenuation and time dispersion.

Single mode propagation exists only above a certain specific wavelength called the cutoff wavelength. Single-mode fiber optic cable is fabricated from glass. Because of the thickness of the core, plastic cannot be used to fabricate single-mode fiber optic cable.

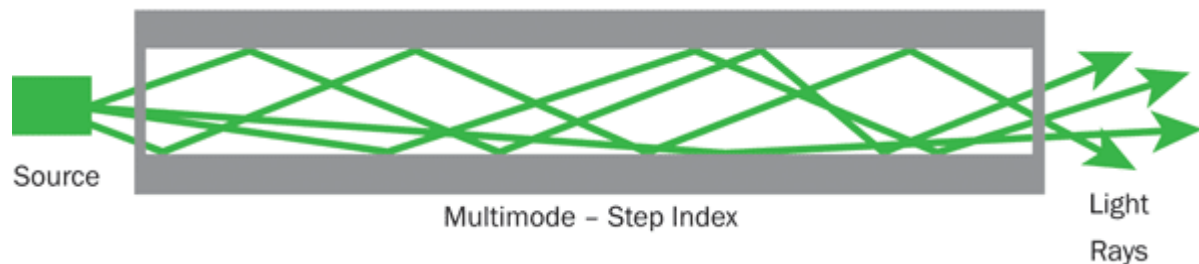
Less time dispersion of course means higher bandwidth and this is in the 50 to 100 GHz/ km range. However, single mode fiber optic cable is also the most costly in the premises environment. For this reason, it has been used more with Wide Area Networks than with premises data communications. It is attractive more for link lengths go all the way up to 100 km. Nonetheless, single-mode fiber optic cable has been getting increased attention as Local Area Networks have been extended to greater distances over corporate campuses.



- **Multi-mode Step Index**

The diameter of the core is fairly large relative to the cladding. Note that the output pulse is significantly attenuated relative to the input pulse. It also suffers significant time dispersion. The higher order modes, the bouncing rays, tend to leak into the cladding as they propagate down the fiber optic cable. They lose some of their energy into heat. This results in an attenuated output

signal. Consequently, they do not all reach the right end of the fiber optic cable at the same time. When the output pulse is constructed from these separate ray components the result is time dispersion.

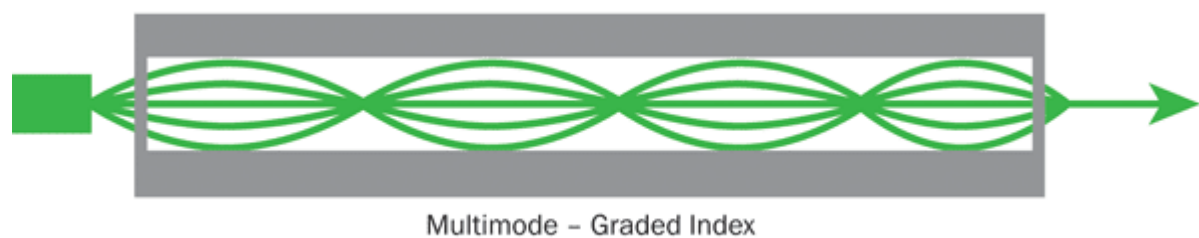


Fiber optic cable that exhibits multi-mode propagation with a step index profile is thereby characterized as having higher attenuation and more time dispersion than the other propagation candidates have. However, it is also the least costly and in the premises environment the most widely used. It is especially attractive for link lengths up to 5 km. usually, it has a core diameter that ranges from 100 microns to 970 microns. It can be fabricated either from glass, plastic or PCS.

- **Multi-mode Graded Index**

There is no sharp discontinuity in the indices of refraction between core and cladding. The core here is much larger than in the single-mode step index. When comparing the output pulse and the input pulse, note that there is some attenuation and time dispersion, but not nearly as great as with multi-mode step index fiber optic cable.

Fiber optic cable that exhibits multi-mode propagation with a graded index profile is thereby characterized as having attenuation and time dispersion properties somewhere between the other two candidates. Likewise its cost is somewhere between the other two candidates. This type of fiber optic cable is extremely popular in premise data communications applications.



### Unguided Media:

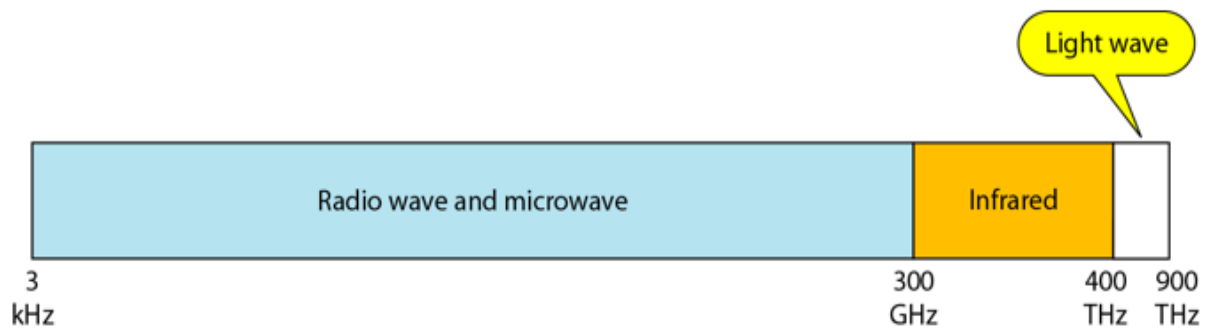
It is also referred to as Wireless or Unbounded transmission media. No physical medium is required for the transmission of electromagnetic signals.

Features:

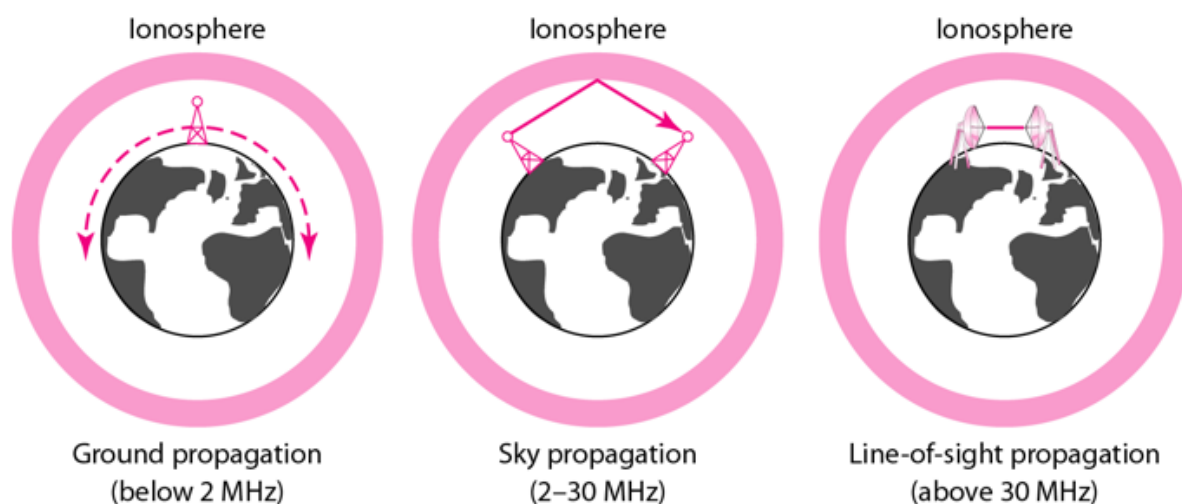
- Signal is broadcasted through air
- Less Secure
- Used for larger distances

Unguided medium transport electromagnetic waves without using a physical conductor. This type of communication is often referred to as wireless communication. Signals are normally broadcast through free space and thus are available to anyone who has a device capable of receiving them.

The below figure shows the part of the electromagnetic spectrum, ranging from 3 kHz to 900 THz, used for wireless communication.



Unguided signals can travel from the source to the destination in several ways: Ground propagation, Sky propagation and Line-of-sight propagation as shown in below figure.



### Propagation Modes

- **Ground Propagation:** In this, radio waves travel through the lowest portion of the atmosphere, hugging the Earth. These low-frequency signals emanate in all directions from the transmitting antenna and follow the curvature of the planet.
- **Sky Propagation:** In this, higher-frequency radio waves radiate upward into the ionosphere where they are reflected back to Earth. This type of transmission allows for greater distances with lower output power.
- **Line-of-sight Propagation:** in this type, very high-frequency signals are transmitted in straight lines directly from antenna to antenna.

We can divide wireless transmission into three broad groups:

- Radio waves
- Micro waves
- Infrared waves

### **Radio Waves**

Electromagnetic waves ranging in frequencies between 3 KHz and 1 GHz are normally called radio waves.

Radio waves are omnidirectional. When an antenna transmits radio waves, they are propagated in all directions. This means that the sending and receiving antennas do not have to be aligned. A sending antenna send waves that can be received by any receiving antenna. The omnidirectional property has disadvantage, too. The radio waves transmitted by one antenna are susceptible to interference by another antenna that may send signal using the same frequency or band.

Radio waves, particularly with those of low and medium frequencies, can penetrate walls. This characteristic can be both an advantage and a disadvantage. It is an advantage because, an AM radio can receive signals inside a building. It is a disadvantage because we cannot isolate a communication to just inside or outside a building.

### **Omnidirectional Antenna for Radio Waves**

Radio waves use omnidirectional antennas that send out signals in all directions.



*Figure3.15 Omni Directional Antenna*

## Applications of Radio Waves

- The omnidirectional characteristics of radio waves make them useful for multicasting in which there is one sender but many receivers.
- AM and FM radio, television, maritime radio, cordless phones, and paging are examples of multicasting.

## Micro Waves

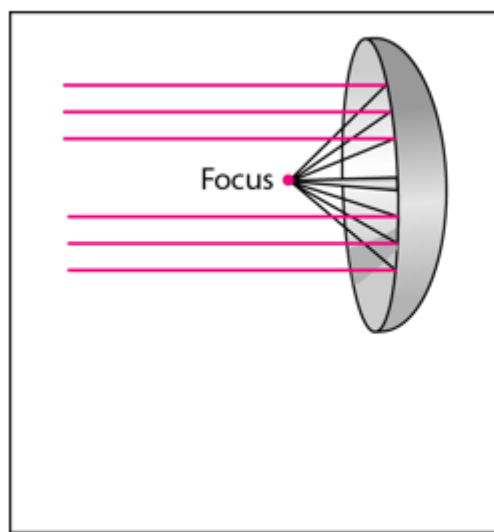
Electromagnetic waves having frequencies between 1 and 300 GHz are called micro waves. Micro waves are unidirectional. When an antenna transmits microwaves, they can be narrowly focused. This means that the sending and receiving antennas need to be aligned. The unidirectional property has an obvious advantage. A pair of antennas can be aligned without interfering with another pair of aligned antennas.

The following describes some characteristics of microwaves propagation:

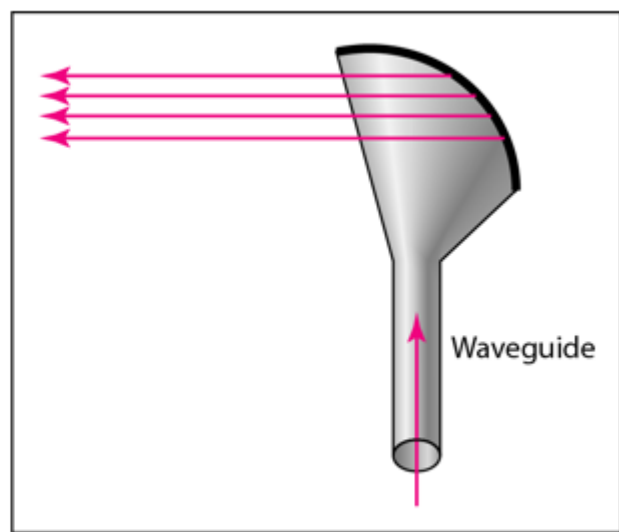
- Microwave propagation is line-of-sight. Since the towers with the mounted antennas need to be in direct sight of each other, towers that are far apart need to be very tall.
- Very high-frequency microwaves cannot penetrate walls. This characteristic can be a disadvantage if receivers are inside the buildings.
- The microwave band is relatively wide, almost 299 GHz. Therefore, wider sub-bands can be assigned and a high data rate is possible.
- Use of certain portions of the band requires permission from authorities.

## Unidirectional Antenna for Micro Waves

Microwaves need unidirectional antennas that send out signals in one direction. Two types of antennas are used for microwave communications: Parabolic Dish and Horn.



a. Dish antenna



b. Horn antenna

A parabolic antenna works as a funnel, catching a wide range of waves and directing them to a common point. In this way, more of the signal is recovered than would be possible with a single-point receiver.

A horn antenna looks like a gigantic scoop. Outgoing transmissions are broadcast up a stem and deflected outward in a series of narrow parallel beams by the curved head. Received transmissions are collected by the scooped shape of the horn, in a manner similar to the parabolic dish, and are deflected down into the stem.

### **Applications of Micro Waves**

Microwaves, due to their unidirectional properties, are very useful when unicast(one-to-one) communication is needed between the sender and the receiver. They are used in cellular phones, satellite networks and wireless LANs.

There are 2 types of Microwave Transmission:

- Terrestrial Microwave
- Satellite Microwave

Advantages of Microwave Transmission

- Used for long distance telephone communication
- Carries 1000's of voice channels at the same time

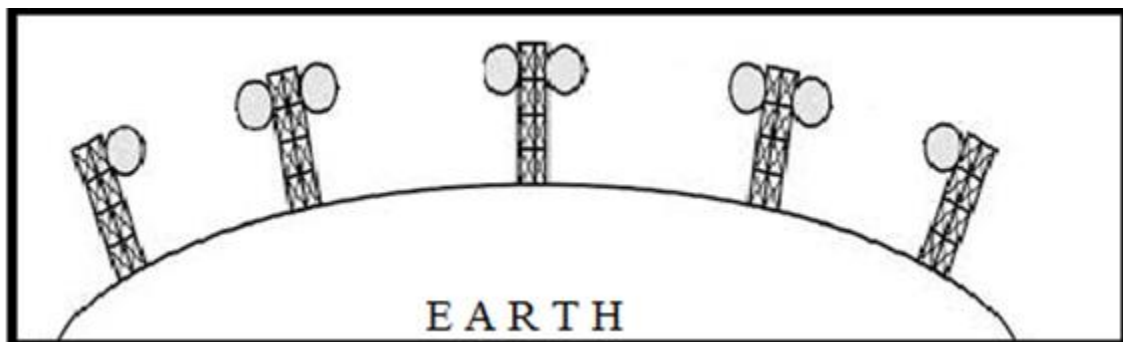
Disadvantages of Microwave Transmission

- It is very costly

### **Terrestrial Microwave**

For increasing the distance served by terrestrial microwave, repeaters can be installed with each antenna .The signal received by an antenna can be converted into transmittable form and relayed to next antenna as shown in below figure. It is an example of telephone systems all over the world

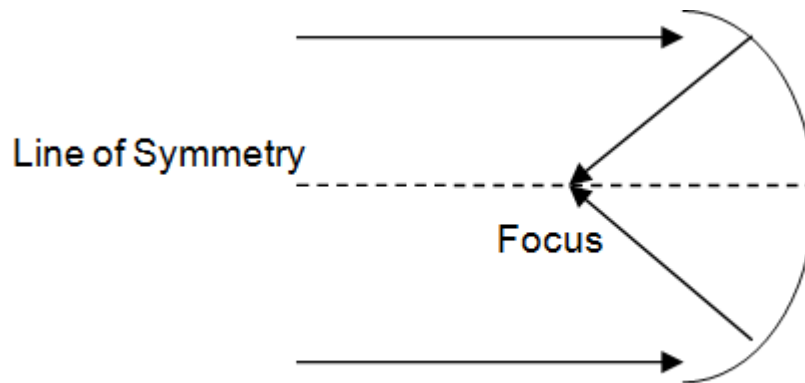
There are two types of antennas used for terrestrial microwave communication.



*Figure3. 16 Terrestrial Microwave*

#### **1. Parabolic Dish Antenna**

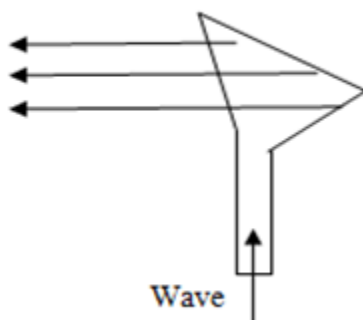
In this every line parallel to the line of symmetry reflects off the curve at angles in a way that they intersect at a common point called focus. This antenna is based on geometry of parabola.



*Figure 3.17 Parabolic Dish Antenna*

## **2. Horn Antenna**

It is a like gigantic scoop. The outgoing transmissions are broadcast up a stem and deflected outward in a series of narrow parallel beams by curved head.



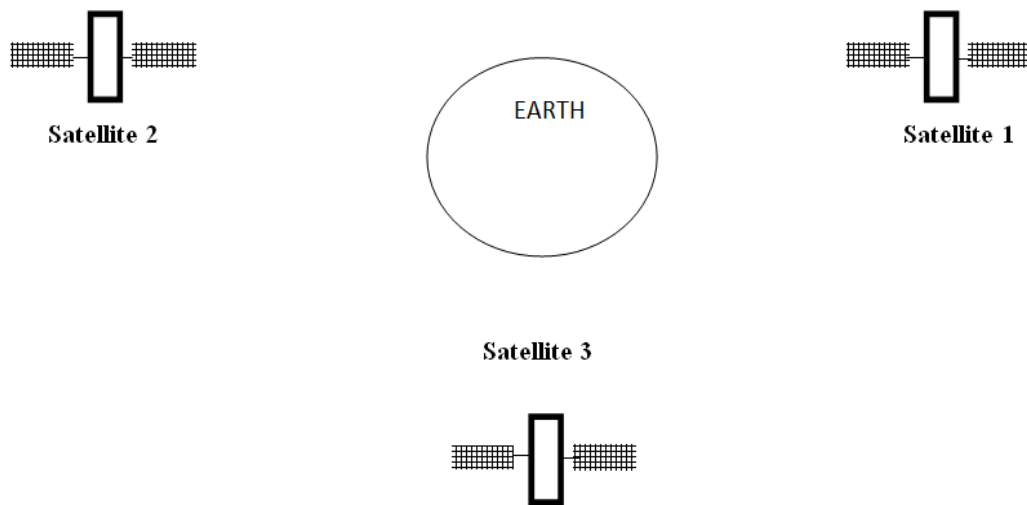
*Figure 3.18 Horn Antenna*

## **Satellite Microwave**

This is a microwave relay station which is placed in outer space. The satellites are launched either by rockets or space shuttles carry them.

These are positioned 36000 Km above the equator with an orbit speed that exactly matches the rotation speed of the earth. As the satellite is positioned in a geo-synchronous orbit, it is stationery relative to earth and always stays over the same point on the ground. This is usually

done to allow ground stations to aim antenna at a fixed point in the sky.



*Figure 3.19 Satellite Communication*

#### Features of Satellite Microwave

- Bandwidth capacity depends on the frequency used.
- Satellite microwave deployment for orbiting satellite is difficult.

#### Advantages of Satellite Microwave

- Transmitting station can receive back its own transmission and check whether the satellite has transmitted information correctly.
- A single microwave relay station which is visible from any point.

#### Disadvantages of Satellite Microwave

- Satellite manufacturing cost is very high
- Cost of launching satellite is very expensive
- Transmission highly depends on whether conditions, it can go down in bad weather

### **Infrared Waves**

Infrared waves, with frequencies from 300 GHz to 400 THz, can be used for short-range communication. Infrared waves, having high frequencies, cannot penetrate walls. This advantageous characteristic prevents interference between one system and another, a short-range communication system in one room cannot be affected by another system in the next room.

When we use infrared remote control, we do not interfere with the use of the remote by our neighbors. However, this same characteristic makes infrared signals useless for long-range communication. In addition, we cannot use infrared waves outside a building because the sun's rays contain infrared waves that can interfere with the communication.

## Applications of Infrared Waves

- The infrared band, almost 400 THz, has an excellent potential for data transmission. Such a wide bandwidth can be used to transmit digital data with a very high data rate.
- The Infrared Data Association (IrDA), an association for sponsoring the use of infrared waves, has established standards for using these signals for communication between devices such as keyboards, mouse, PCs and printers.
- Infrared signals can be used for short-range communication in a closed area using line-of-sight propagation.

## Bluetooth

Bluetooth wireless technology is a short range communications technology intended to replace the cables connecting portable unit and maintaining high levels of security. Bluetooth technology is based on **Ad-hoc technology** also known as **Ad-hoc Pico nets**, which is a local area network with a very limited coverage.

### History of Bluetooth

WLAN technology enables device connectivity to infrastructure based services through a wireless carrier provider. The need for personal devices to communicate wirelessly with one another without an established infrastructure has led to the emergence of **Personal Area Networks (PANs)**.

- Ericsson's Bluetooth project in 1994 defines the standard for PANs to enable communication between mobile phones using low power and low cost radio interfaces.
- In May 1988, Companies such as IBM, Intel, Nokia and Toshiba joined Ericsson to form the Bluetooth Special Interest Group (SIG) whose aim was to develop a defacto standard for PANs.
- IEEE has approved a Bluetooth based standard named IEEE 802.15.1 for Wireless Personal Area Networks (WPANs). IEEE standard covers MAC and Physical layer applications.

**Bluetooth** specification details the entire protocol stack. Bluetooth employs Radio Frequency (RF) for communication. It makes use of **frequency modulation** to generate radio waves in the **ISM** band.



**Symbol of Bluetooth**



**An example of a Bluetooth device**

The usage of Bluetooth has widely increased for its special features.

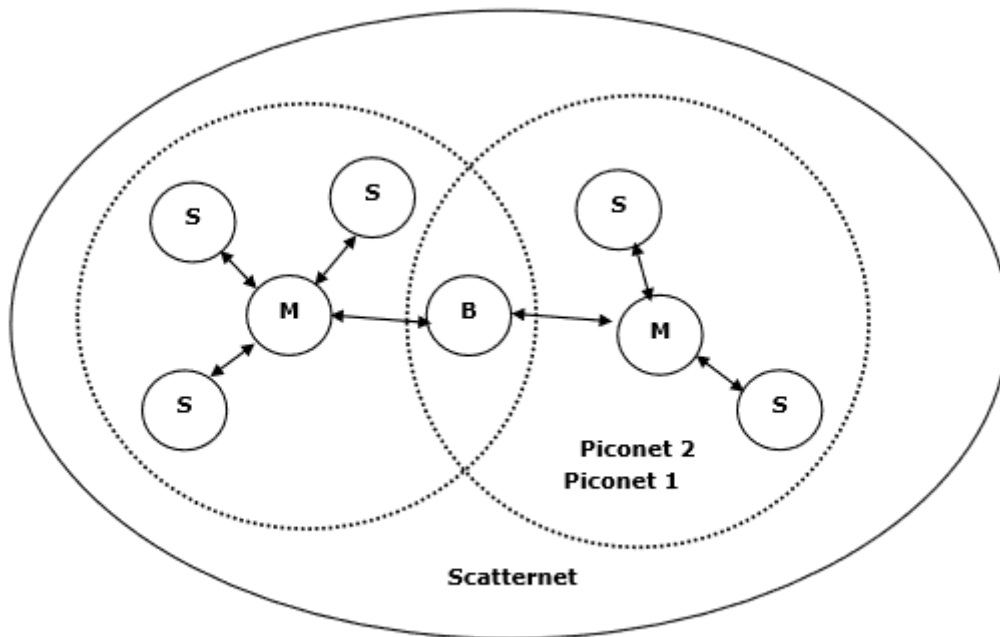
- Bluetooth offers a uniform structure for a wide range of devices to connect and communicate with each other.
- Bluetooth technology has achieved global acceptance such that any Bluetooth enabled device, almost everywhere in the world, can be connected with Bluetooth enabled devices.
- Low power consumption of Bluetooth technology and an offered range of up to ten meters has paved the way for several usage models.
- Bluetooth offers interactive conference by establishing an adhoc network of laptops.
- Bluetooth usage model includes cordless computer, intercom, cordless phone and mobile phones.

### **Piconets and Scatternet**

Bluetooth enabled electronic devices connect and communicate wirelessly through shortrange devices known as **Piconets**. Bluetooth devices exist in small ad-hoc configurations with the ability to act either as master or slave the specification allows a mechanism for **master** and **slave** to switch their roles. Point to point configuration with one master and one slave is the simplest configuration.

When more than two Bluetooth devices communicate with one another, this is called a **PICONET**. A Piconet can contain up to seven slaves clustered around a single master. The device that initializes establishment of the Piconet becomes the **master**.

The master is responsible for transmission control by dividing the network into a series of time slots amongst the network members, as a part of **time division multiplexing** scheme which is shown below.



**Figure: Piconets and Scatternets**

The features of Piconets are as follows –

- Within a Piconet, the timing of various devices and the frequency hopping sequence of individual devices is determined by the clock and unique **48-bit address** of master.
- Each device can communicate simultaneously with up to seven other devices within a single Piconet.
- Each device can communicate with several piconets simultaneously.
- Piconets are established dynamically and automatically as Bluetooth enabled devices enter and leave piconets.
- There is no direct connection between the slaves and all the connections are essentially master-to-slave or slave-to-master.
- Slaves are allowed to transmit once these have been polled by the master.
- Transmission starts in the slave-to-master time slot immediately following a polling packet from the master.
- A device can be a member of two or more piconets, jumping from one piconet to another by adjusting the transmission regime-timing and frequency hopping sequence dictated by the master device of the second piconet.
- It can be a slave in one piconet and master in another. It however cannot be a master in more than once piconet.

- Devices resident in adjacent piconets provide a bridge to support inner-piconet connections, allowing assemblies of linked piconets to form a physically extensible communication infrastructure known as **Scatternet**.

### **Spectrum**

Bluetooth technology operates in the unlicensed industrial, scientific and medical (ISM) band at 2.4 to 2.485 GHz, using a spread spectrum hopping, full-duplex signal at a nominal rate of 1600 hops/sec. the 2.4 GHz ISM band is available and unlicensed in most countries.

### **Range**

Bluetooth operating range depends on the device Class 3 radios have a range of up to 1 meter or 3 feet Class 2 radios are most commonly found in mobile devices have a range of 10 meters or 30 feet Class 1 radios are used primarily in industrial use cases have a range of 100 meters or 300 feet.

### **Data rate**

Bluetooth supports 1Mbps data rate for version 1.2 and 3Mbps data rate for Version 2.0 combined with Error Data Rate.

### **Switching**

Switching is process to forward packets coming in from one port to a port leading towards the destination. When data comes on a port it is called ingress, and when data leaves a port or goes out it is called egress. A communication system may include number of switches and nodes. At broad level, switching can be divided into two major categories:

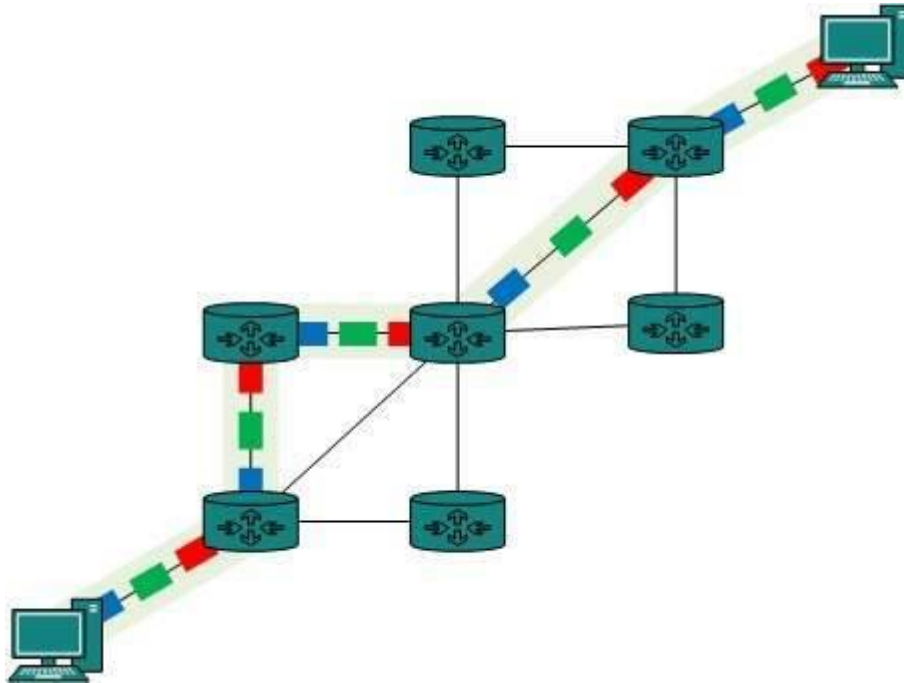
- **Connectionless:** The data is forwarded on behalf of forwarding tables. No previous handshaking is required and acknowledgements are optional.
- **Connection Oriented:** Before switching data to be forwarded to destination, there is a need to pre-establish circuit along the path between both endpoints. Data is then forwarded on that circuit. After the transfer is completed, circuits can be kept for future use or can be turned down immediately.

### **Circuit Switching**

When two nodes communicate with each other over a dedicated communication path, it is called circuit switching. There is a need of pre-specified route from which data will travels and no other data is permitted. In circuit switching, to transfer the data, circuit must be established so that the data transfer can take place.

Circuits can be permanent or temporary. Applications which use circuit switching may have to go through three phases:

- Establish a circuit
- Transfer the data
- Disconnect the circuit

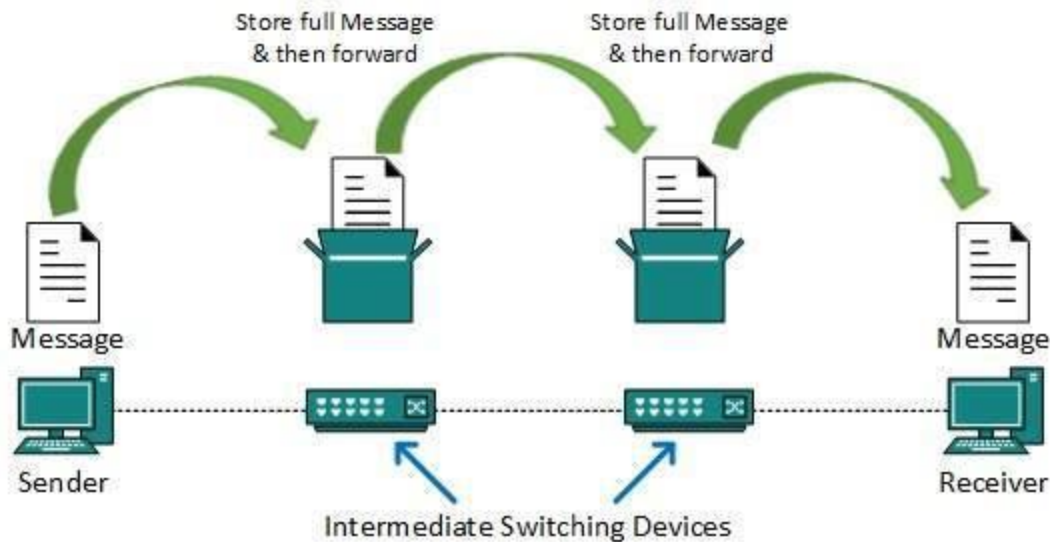


Circuit switching was designed for voice applications. Telephone is the best suitable example of circuit switching. Before a user can make a call, a virtual path between caller and callee is established over the network.

### Message Switching

This technique was somewhere in middle of circuit switching and packet switching. In message switching, the whole message is treated as a data unit and is switching / transferred in its entirety.

A switch working on message switching, first receives the whole message and buffers it until there are resources available to transfer it to the next hop. If the next hop is not having enough resource to accommodate large size message, the message is stored and switch waits.



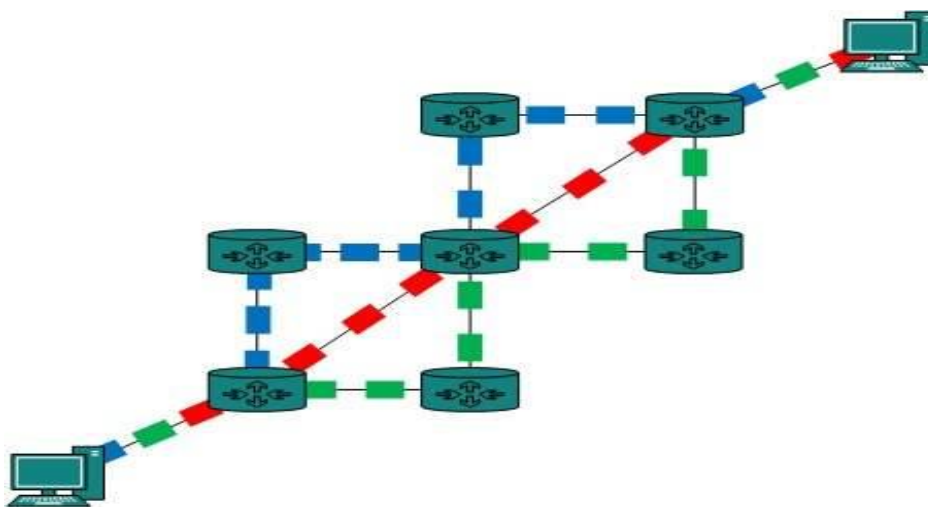
This technique was considered substitute to circuit switching. As in circuit switching the whole path is blocked for two entities only. Message switching is replaced by packet switching. Message switching has the following drawbacks:

- Every switch in transit path needs enough storage to accommodate entire message.
- Because of store-and-forward technique and waits included until resources are available, message switching is very slow.
- Message switching was not a solution for streaming media and real-time applications.

### Packet Switching

Shortcomings of message switching gave birth to an idea of packet switching. The entire message is broken down into smaller chunks called packets. The switching information is added in the header of each packet and transmitted independently.

It is easier for intermediate networking devices to store small size packets and they do not take much resources either on carrier path or in the internal memory of switches.

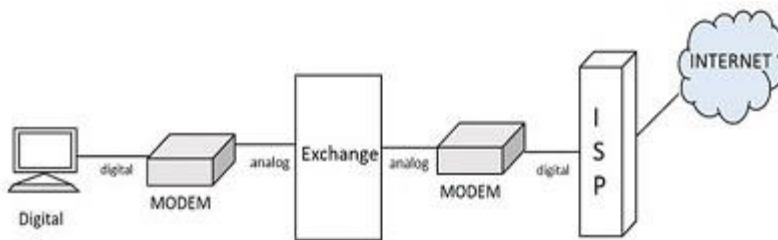


Packet switching enhances line efficiency as packets from multiple applications can be multiplexed over the carrier. The internet uses packet switching technique. Packet switching enables the user to differentiate data streams based on priorities. Packets are stored and forwarded according to their priority to provide quality of service.

### Why do we need ISDN?

**Integrated Services Digital Network.** Earlier, the transmission of data and voice both were possible through normal POTS, Plain Old Telephone Systems. With the introduction of Internet came the advancement in telecommunication too. Yet, the sending and receiving of data along with voice was not an easy task. One could use either the Internet or the Telephone. The invention of ISDN helped mitigate this problem.

The process of connecting a home computer to the Internet Service Provider used to take a lot of effort. The usage of the modulator-demodulator unit, simply called the MODEM was the essential thing to establish a connection. The following figure shows how the model worked in the past.



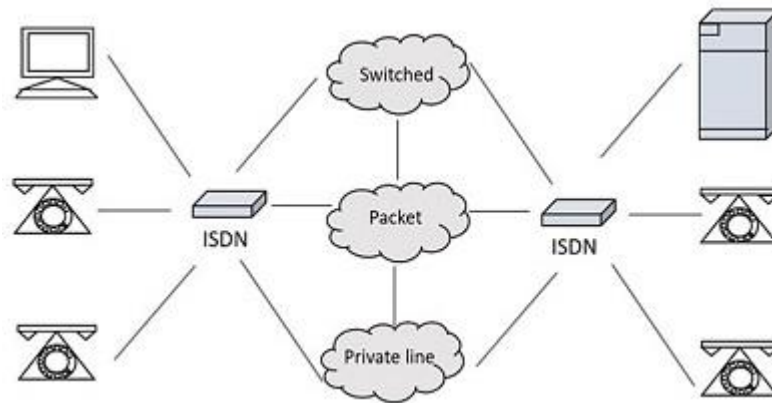
The above figure shows that the digital signals have to be converted into analog and analog signals to digital using modem during the whole path. What if the digital information at one end reaches to the other end in the same mode, without all these connections? It is this basic idea that lead to the development of **ISDN**.

As the system has to use the telephone cable through the telephone exchange for using the Internet, the usage of telephone for voice calls was not permitted. The introduction of ISDN has resolved this problem allowing the transmission of both voice and data simultaneously. This has many advanced features over the traditional PSTN, Public Switched Telephone Network.

### ISDN

ISDN was first defined in the CCITT red book in 1988. The **Integrated Services of Digital Networking**, in short **ISDN is a telephone network based infrastructure that allows the transmission of voice and data simultaneously at a high speed with greater efficiency.** This is a circuit switched telephone network system, which also provides access to Packet switched networks.

The model of a practical ISDN is as shown below.



ISDN supports a variety of services. A few of them are listed below

- Voice calls
- Facsimile
- Videotext
- Teletext
- Electronic Mail
- Database access
- Data transmission and voice
- Connection to internet
- Electronic Fund transfer
- Image and graphics exchange
- Document storage and transfer
- Audio and Video Conferencing
- Automatic alarm services to fire stations, police, medical etc.

### Types of ISDN

Among the types of several interfaces present, some of them contains channels such as the **B-Channels** or Bearer Channels that are used to transmit voice and data simultaneously; the **D-Channels** or Delta Channels that are used for signaling purpose to set up communication.

The ISDN has several kinds of access interfaces such as –

- Basic Rate Interface (BRI)
- Primary Rate Interface (PRI)
- Narrowband ISDN
- Broadband ISDN

### Basic Rate Interface (BRI)

The Basic Rate Interface or Basic Rate Access, simply called the **ISDN BRI Connection** uses the existing telephone infrastructure. The BRI configuration provides **two data** or bearer channels at **64 Kbits/sec** speed and one control or delta channel at **16 Kbits/sec**. This is a standard rate.

The ISDN BRI interface is commonly used by smaller organizations or home users or within a local group, limiting a smaller area.

### **Primary Rate Interface (PRI)**

The Primary Rate Interface or Primary Rate Access, simply called the ISDN PRI connection is used by enterprises and offices. The PRI configuration is based on T-carrier or T1 in the US, Canada and Japan countries consisting of **23 data** or bearer channels and one control or delta channel, with 64kbps speed for a bandwidth of 1.544 M bits/sec. The PRI configuration is based on E-carrier or E1 in Europe, Australia and few Asian countries consisting of **30 data** or bearer channels and **two-control** or delta channel with 64kbps speed for a bandwidth of 2.048 M bits/sec.

The ISDN BRI interface is used by larger organizations or enterprises and for Internet Service Providers.

### **Narrowband ISDN**

The Narrowband Integrated Services Digital Network is called the **N-ISDN**. This can be understood as a telecommunication that carries voice information in a narrow band of frequencies. This is actually an attempt to digitize the analog voice information. This uses 64kbps circuit switching.

The narrowband ISDN is implemented to carry voice data, which uses lesser bandwidth, on a limited number of frequencies.

### **Broadband ISDN**

The Broadband Integrated Services Digital Network is called the **B-ISDN**. This integrates the digital networking services and provides digital transmission over ordinary telephone wires, as well as over other media. The CCITT defined it as, “Qualifying a service or system requiring transmission channels capable of supporting rates greater than primary rates.”

The broadband ISDN speed is around 2 MBPS to 1 GBPS and the transmission is related to ATM, i.e., Asynchronous Transfer Mode. The broadband ISDN communication is usually made using the fiber optic cables.

As the speed is greater than 1.544 Mbps, the communications based on this are called **Broadband Communications**. The broadband services provide a continuous flow of information, which is distributed from a central source to an unlimited number of authorized

receivers connected to the network. Though a user can access this flow of information, he cannot control it.

### **Advantages of ISDN**

ISDN is a telephone network based infrastructure, which enables the transmission of both voice and data simultaneously. There are many advantages of ISDN such as –

- As the services are digital, there is less chance for errors.
- The connection is faster.
- The bandwidth is higher.
- Voice, data and video – all of these can be sent over a single ISDN line.

### **Disadvantages of ISDN**

The disadvantage of ISDN is that it requires specialized digital services and is costlier.

However, the advent of ISDN has brought great advancement in communications. Multiple transmissions with greater speed are being achieved with higher levels of accuracy.

## **Network Performance**

### **1. Latency**

Latency is the delay from input into a system to desired outcome; the term is understood slightly differently in various contexts and latency issues also vary from one system to another. Latency greatly affects how usable and enjoyable electronic and mechanical devices as well as communications are.

Latency in communication is demonstrated in live transmissions from various points on the earth as the communication hops between a ground transmitter and a satellite and from a satellite to a receiver each take time. People connecting from distances to these live events can be seen to have to wait for responses. This latency is the wait time introduced by the signal travelling the geographical distance as well as over the various pieces of communications equipment. Even fiber optics are limited by more than just the speed of light, as the refractive index of the cable and all repeaters or amplifiers along their length introduce delays.

### **2. Throughput**

Throughput is a measure of how many units of information a system can process in a given amount of time. It is applied broadly to systems ranging from various aspects of computer and network systems to organizations. Related measures of system productivity include, the speed with which some specific workload can be completed, and response time, the amount of time between a single interactive user request and receipt of the response.

In data transmission, network throughput is the amount of data moved successfully from one place to another in a given time period, and typically measured in bits per second (bps), as in megabits per second (Mbps) or gigabits per second (Gbps).

### 3. Jitter

Jitter is the undesired deviation from true periodicity of an assumed periodic signal in electronics and telecommunications, often in relation to a reference clock source. Jitter may be observed in characteristics such as the frequency of successive pulses, the signal amplitude, or phase of periodic signals. Jitter is a significant, and usually undesired, factor in the design of almost all communications links (e.g., USB, PCI-e, SATA, and OC-48). In clock recovery applications it is called timing jitter.

### 4. Bandwidth

Bandwidth is also the amount of data that can be transmitted in a fixed amount of time. For digital devices, the bandwidth is usually expressed in bits per second (bps) or bytes per second. For analog devices, the bandwidth is expressed in cycles per second, or Hertz (Hz).

### 5. Bandwidth- Delay Products

In data communications, bandwidth-delay product is the product of a data link's capacity (in bits per second) and its round-trip delay time (in seconds).[1][2] The result, an amount of data measured in bits (or bytes), is equivalent to the maximum amount of data on the network circuit at any given time, i.e., data that has been transmitted but not yet acknowledged.

A network with a large bandwidth-delay product is commonly known as a long fat network (shortened to LFN). As defined in RFC 1072, a network is considered an LFN if its bandwidth-delay product is significantly larger than 105 bits (12500 bytes).

Ultra-high speed LANs may fall into this category, where protocol tuning is critical for achieving peak throughput, on account of their extremely high bandwidth, even though their delay is not great. While a connection with 1 Gbit/s and a round-trip time below 100  $\mu$ s is no LFN, a connection with 100 Gbit/s would need to stay below 1  $\mu$ s RTT to not be considered an LFN.

An important example of a system where the bandwidth-delay product is large is that of geostationary satellite connections, where end-to-end delivery time is very high and link throughput may also be high. The high end-to-end delivery time makes life difficult for stop-and-wait protocols and applications that assume rapid end-to-end response.

A high bandwidth-delay product is an important problem case in the design of protocols such as Transmission Control Protocol (TCP) in respect of TCP tuning, because the protocol can only achieve optimum throughput if a sender sends a sufficiently large quantity of data before being required to stop and wait until a confirming message is received from the receiver, acknowledging successful receipt of that data. If the quantity of data sent is insufficient compared with the bandwidth-delay product, then the link is not being kept busy and the protocol is operating below peak efficiency for the link. Protocols that hope to succeed in this respect need carefully designed self-monitoring, self-tuning algorithms. The TCP window scale option may be used to solve this problem caused by insufficient window size, which is limited to 65535 bytes without scaling.

#### Examples

- Moderate speed satellite network: 512 kbit/s, 900 ms RTT

$B \times D = 512 \times 103 \text{ b/s} \times 900 \times 10^{-3} \text{ s} = 460,800 \text{ b.}, / 8 = 57,600 \text{ B (or / 1,000 = 57.6 kB, or / 1,024 = 56.25 KiB)}$

- Residential DSL: 2 Mbit/s, 50 ms RTT  
 $B \times D = 2 \times 10^6 \text{ b/s} \times 50 \times 10^{-3} \text{ s} = 100 \times 10^3 \text{ b, or 100 kb, or 12.5 kB.}$
- Mobile broadband (HSDPA): 6 Mbit/s, 100 ms RTT  
 $B \times D = 6 \times 10^6 \text{ b/s} \times 10^{-1} \text{ s} = 6 \times 10^5 \text{ b, or 600 kb, or 75 kB.}$
- Residential ADSL2+: 20 Mbit/s (from DSLAM to residential modem), 50 ms RTT  
 $B \times D = 20 \times 10^6 \text{ b/s} \times 50 \times 10^{-3} \text{ s} = 10^6 \text{ b, or 1 Mb, or 125 kB.}$
- High-speed terrestrial network: 1 Gbit/s, 1 ms RTT  
 $B \times D = 10^9 \text{ b/s} \times 10^{-3} \text{ s} = 10^6 \text{ b, or 1 Mb, or 125 kB.}$
- Ultra-high speed LAN: 100 Gbit/s, 30  $\mu\text{s}$  RTT  
 $B \times D = 100 \times 10^9 \text{ b/s} \times 30 \times 10^{-6} \text{ s} = 3 \times 10^6 \text{ b, or 3 Mb, or 375 kB.}$

## CHAPTER 4

### Data Link Layers

The data link layer is the protocol layer in a program that handles the moving of data into and out of a physical link in a network. The data link layer is Layer 2 in the Open Systems Interconnection (OSI) architecture model for a set of telecommunication protocols. Data bits are encoded, decoded and organized in the data link layer, before they are transported as frames between two adjacent nodes on the same LAN or WAN. The data link layer also determines how devices recover from collisions that may occur when nodes attempt to send frames at the same time.

The data link layer has two sublayers: the **Logical Link Control (LLC)** sublayer and the **Media Access Control (MAC)** sublayer.

#### LLC / MAC

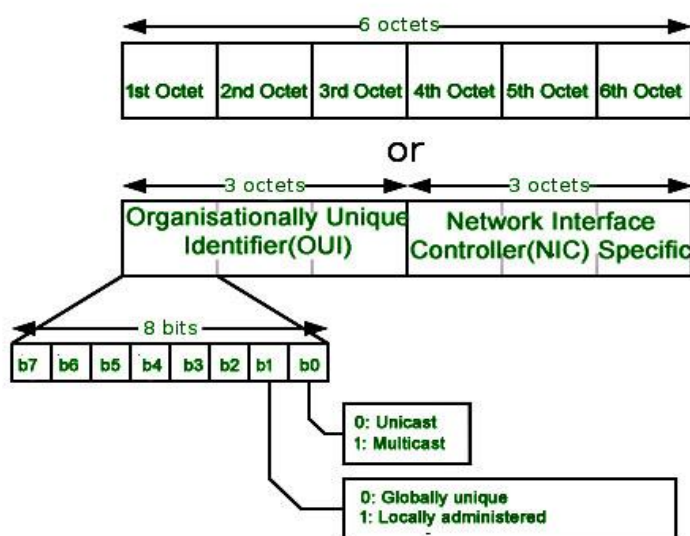
- Logical Link Control: It deals with protocols, flow-control, and error control
- Media Access Control: It deals with actual control of media
- The Logical Link Control (LLC) data communication protocol layer is the upper sub-layer of the Data Link Layer (which is itself layer 2, just above the Physical Layer) in the seven-layer OSI reference model.
- It provides multiplexing mechanisms that make it possible for several network protocols (IP, IPX) to coexist within a multipoint network and to be transported over the same network media, and can also provide flow control mechanisms.
- The LLC sub-layer acts as an interface between the Media Access Control (MAC) sublayer and the network layer.
- As the Ether type in an Ethernet II framing formatted frame is used to multiplex different protocols on top of the Ethernet MAC header it can be seen as LLC identifier.
- The LLC sub layer is primarily concerned with:
  - Multiplexing protocols transmitted over the MAC layer (when transmitting) and decoding them (when receiving).
  - Providing flow and error control
  - The Media Access Control (MAC) data communication protocol sub-layer, also known as the Medium Access Control, is a sub layer of the Data Link Layer specified in the seven-layer OSI model (layer 2).
  - It provides addressing and channel access control mechanisms that make it possible for several terminals or network nodes to communicate within a multi-point network, typically a local area network (LAN) or metropolitan area network (MAN). - The hardware that implements the MAC is referred to as a Medium Access Controller.
- The MAC sub-layer acts as an interface between the Logical Link Control (LLC) sublayer and the network's physical layer. - The MAC layer emulates a full-duplex logical communication channel in a multi-point network. This channel may provide unicast, multicast or broadcast communication service.

## Media Access Control (MAC) Address

MAC Addresses are **unique 48-bits hardware number of a computer**, which is **embedded into network card** (known as **Network Interface Card**) during the time of manufacturing. MAC Address is also known as **Physical Address** of a network device. In IEEE 802 standard, Data Link Layer is divided into two sublayers:

1. **Logical Link Control(LLC) Sublayer**
2. **Media Access Control(MAC) Sublayer**

MAC address is used by Media Access Control (MAC) sublayer of Data-Link Layer. MAC Address is word wide unique, since millions of network devices exists and we need to uniquely identify each.



## Format of MAC Address –

MAC Address is a 12-digit hexadecimal number (6-Byte binary number), which is mostly represented by Colon-Hexadecimal notation. First 6-digits (say 00:40:96) of MAC Address identifies the manufacturer, called as OUI (**Organizational Unique Identifier**). IEEE Registration Authority Committee assign these MAC prefixes to its registered vendors. Here are some OUI of well-known manufacturers:

CC: 46:D6 - Cisco

3C:5A:B4 - Google, Inc.

3C:D9:2B - Hewlett Packard

00:9A:CD - HUAWEI TECHNOLOGIES CO.,LTD

The rightmost six digits represents **Network Interface Controller**, which is assigned by manufacturer.

As discussed above, MAC address is represented by Colon-Hexadecimal notation. But this is just a conversion, not mandatory. MAC address can be represented using any of the following formats

Hyphen-Hexadecimal notation

00-0a-83-b1-c0-8e

Colon-Hexadecimal notation

00:0a:83:b1:c0:8e

Period-separated hexadecimal notation

000.a83.b1c.08e

**Note:** Colon-Hexadecimal notation is used by *Linux OS* and Period-separated Hexadecimal notation is used by *Cisco Systems*.

### How to find MAC address –

Command for UNIX/Linux - *ifconfig -a*

*ip link list*

*ip address show*

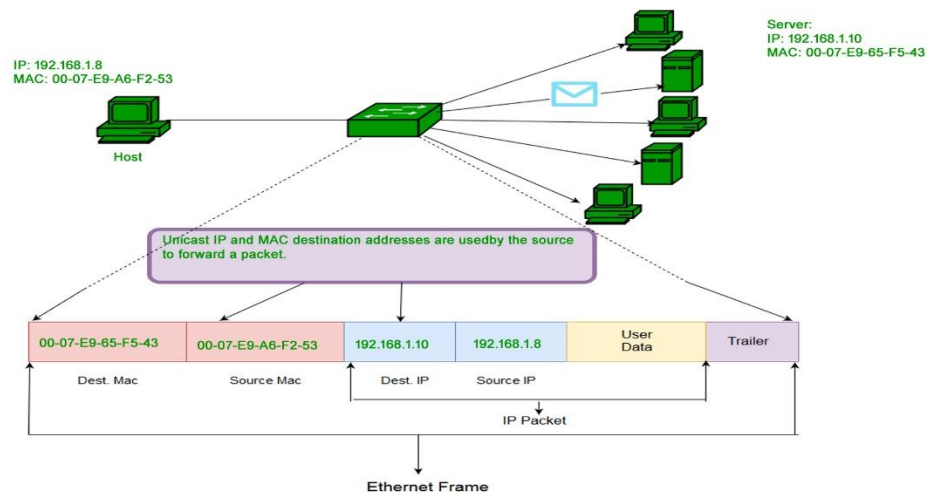
Command for Windows OS - *ipconfig /all*

MacOS - *TCP/IP Control Panel*

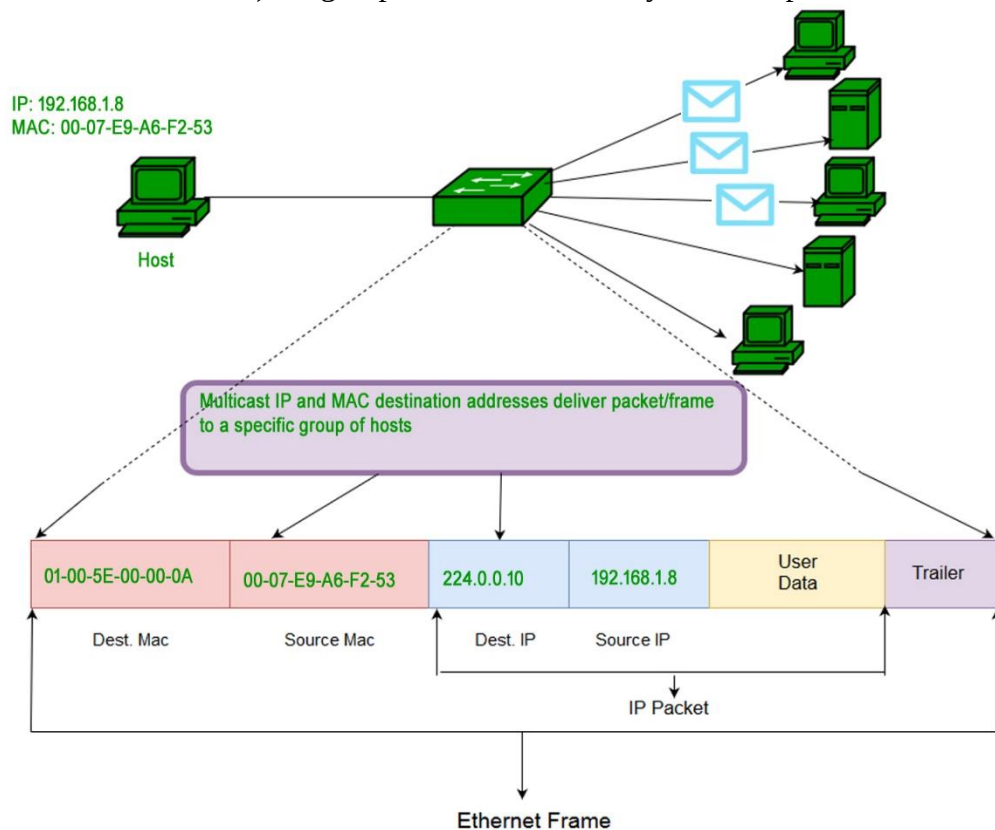
**Note** – LAN technologies like Token Ring, Ethernet use MAC Address as their Physical address but there are some networks (AppleTalk) which does not use MAC address.

### Types of MAC Address –

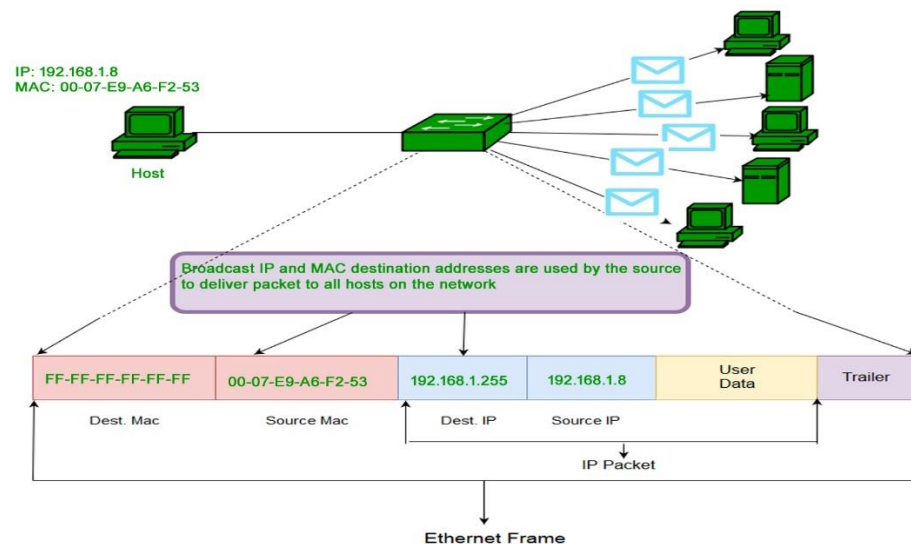
1. **Unicast** – A Unicast addressed frame is only sent out to the interface leading to specific NIC. If the LSB (least significant bit) of first octet of an address is set to zero, the frame is meant to reach only one receiving NIC. MAC Address of source machine is always Unicast.



2. **Multicast** – Multicast address allow the source to send a frame to group of devices. In Layer-2 (Ethernet) Multicast address, LSB (least significant bit) of first octet of an address is set to one. IEEE has allocated the address block 01-80-C2-xx-xx-xx (01-80-C2-00-00-00 to 01-80-C2-FF-FF-FF) for group addresses for use by standard protocols.

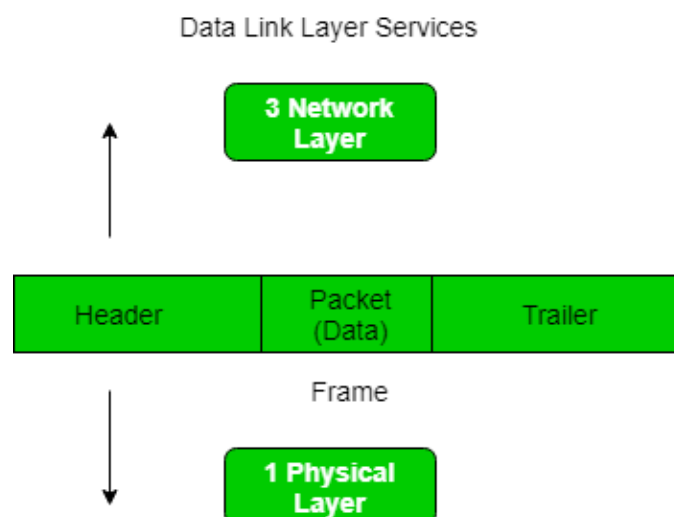


3. **Broadcast** – Similar to Network Layer, Broadcast is also possible on underlying layer (Data Link Layer). Ethernet frames with ones in all bits of the destination address (FF-FF-FF-FF-FF-FF) are referred as broadcast address. Frames which are destined with MAC address FF-FF-FF-FF-FF-FF will reach to every computer belong to that LAN segment.



### Framing In Data Link Layer

Framing is a point-to-point connection between two computers or devices consists of a wire in which data is transmitted as a stream of bits. However, these bits must be framed into discernible blocks of information. Framing is a function of the data link layer. It provides a way for a sender to transmit a set of bits that are meaningful to the receiver. Ethernet, token ring, frame relay, and other data link layer technologies have their own frame structures. Frames have headers that contain information such as error-checking codes.



At data link layer, it extracts message from sender and provide it to receiver by providing sender's and receiver's address. The advantage of using frames is that data is broken up into recoverable chunks that can easily be checked for corruption.

### Problems in Framing

**Detecting start of the frame:** When a frame is transmitted, every station must be able to detect it. Station detect frames by looking out for special sequence of bits that marks the beginning of the frame i.e. SFD (Starting Frame Delimiters).

**How do station detect a frame:** Every station listen to link for SFD pattern through a sequential circuit? If SFD is detected, sequential circuit alerts station. Station checks destination address to accept or reject frame.

**Detecting end of frame:** When to stop reading the frame.

**Types of framing** – There are two types of framing:

1. **Fixed size** - The frame is of fixed size and there is no need to provide boundaries to the frame, length of the frame itself acts as delimiter.

**Drawback:** It suffers from internal fragmentation if data size is less than frame size

**Solution:** Padding

2. **Variable size** – In this there is need to define end of frame as well as beginning of next frame to distinguish. This can be done in two ways:

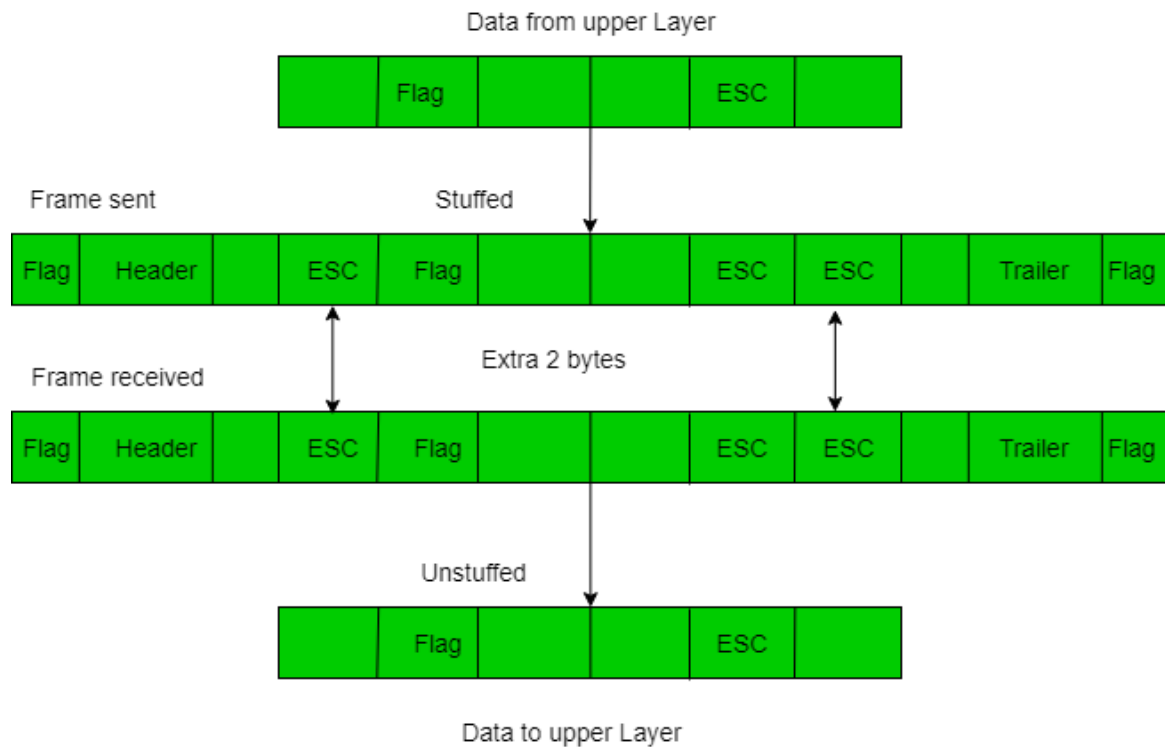
**Length field** – We can introduce a length field in the frame to indicate the length of the frame. Used in Ethernet (802.3). The problem with this is that sometimes the length field might get corrupted.

**End Delimiters (ED)** – We can introduce an ED (pattern) to indicate the end of the frame. Used in Token Ring. The problem with this is that ED can occur in the data. This can be solved by:

1. **Character/Byte Stuffing:** Used when frames consist of character. If data contains ED then, byte is stuffed into data to differentiate it from ED.

Let ED = “\$”

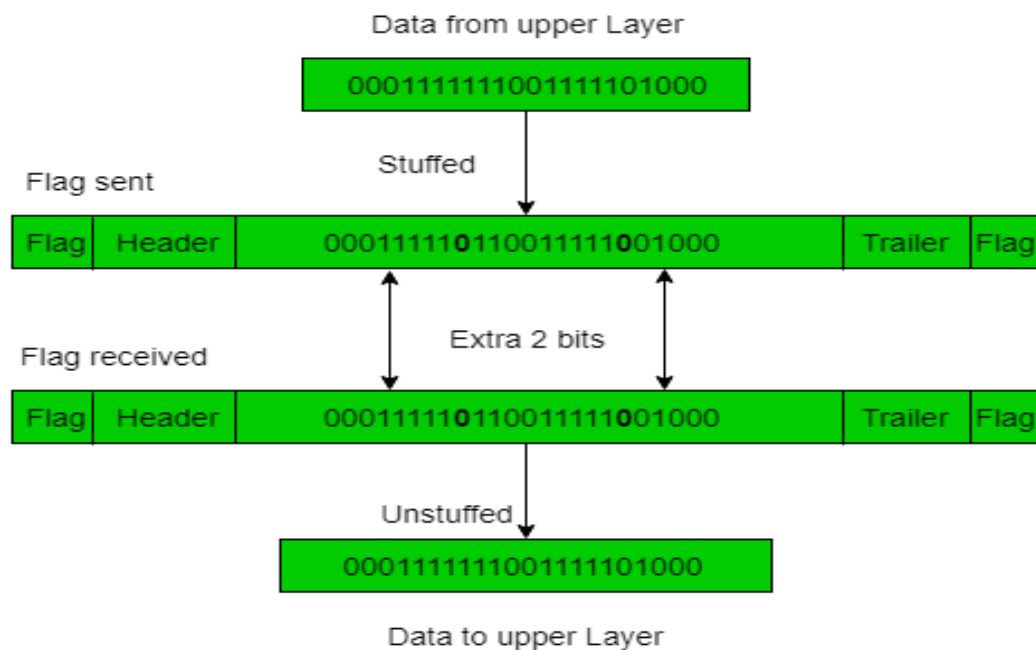
- ➔ if data contains ‘\$’ anywhere, it can be escaped using ‘\O’ character.
- ➔ if data contains ‘\O\$’ then, use ‘\O\O\O\$’(\$ is escaped using \O and \O is escaped using \O).



**Disadvantage** – It is very costly and obsolete method.

**2. Bit Stuffing:** Let ED = 01111 and if data = 01111

- Sender stuffs a bit to break the pattern i.e. here appends a 0 in data = 011101.
- Receiver receives the frame.
- If data contains 011101, receiver removes the 0 and reads the data.



Examples –

- If Data → 011100011110 and ED → 01111 then, find data after bit stuffing?  
→ 01110000111010
- If Data → 110001001 and ED → 1000 then, find data after bit stuffing?  
→ 11001010011

### Flow Control

Flow Control is a set of procedures that tells the sender how much data it can transmit before it must wait for an acknowledgment from the receiver. The flow of data should not be allowed to overwhelm the receiver. Receiver should also be able to inform the transmitter before its limits (this limit may be amount of memory used to store the incoming data or the processing power at the receiver end) are reached and the sender must send fewer frames. Hence, Flow control refers to the set of procedures used to restrict the amount of data the transmitter can send before waiting for acknowledgment.

There are two methods developed for flow control namely **Stop-and-wait** and **Sliding-window**. Stop-and-wait is also known as Request/reply sometimes. Request/reply (Stop-and-wait) flow control requires each data packet to be acknowledged by the remote host before the next packet is sent. This is discussed in detail in the following subsection. Sliding window algorithms, used by TCP, permit multiple data packets to be in simultaneous transit, making more efficient use of network bandwidth.

#### 1. Stop-and-Wait

This is the simplest form of flow control where a sender transmits a data frame. After receiving the frame, the receiver indicates its willingness to accept another frame by sending back an ACK frame acknowledging the frame just received. The sender must wait until it receives the ACK frame before sending the next data frame. This is sometimes referred to as ping-pong behavior, request/reply is simple to understand and easy to implement, but not very efficient. In LAN environment with fast links, this isn't much of a concern, but WAN links will spend most of their time idle, especially if several hops are required.

Figure below illustrates the operation of the stop-and-wait protocol. The blue arrows show the sequence of data frames being sent across the link from the sender (top to the receiver (bottom)). The protocol relies on two-way transmission (full duplex or half duplex) to allow the receiver at the remote node to return frames acknowledging the successful transmission. The acknowledgements are shown in green in the diagram, and flow back to the original sender. A small processing delay may be introduced between reception of the last byte of a Data PDU and generation of the corresponding ACK.

Major drawback of Stop-and-Wait Flow Control is that only one frame can be in transmission at a time, this leads to inefficiency if propagation delay is much longer than the transmission delay.

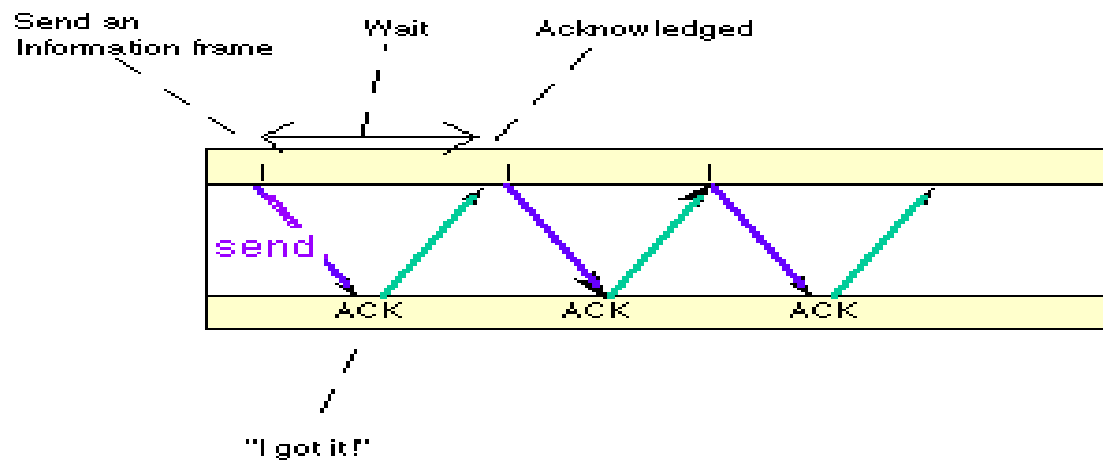


Figure 20 Stop and Wait Protocol

## Link Utilization in Stop-and-Wait

Let us assume the following:

- Transmission time: The time it takes for a station to transmit a frame (normalized to a value of 1).
- Propagation delay: The time it takes for a bit to travel from sender to receiver (expressed as  $a$ ).
  - $a < 1$ : The frame is sufficiently long such that the first bits of the frame arrive at the destination before the source has completed transmission of the frame.
  - $a > 1$ : Sender completes transmission of the entire frame before the leading bits of the frame arrive at the receiver.
- The link utilization  $U = 1/(1+2a)$ ,

$$a = \text{Propagation time} / \text{transmission time}$$

It is evident from the above equation that the link utilization is strongly dependent on the ratio of the propagation time to the transmission time. When the propagation time is small, as in case of LAN environment, the link utilization is good. But, in case of long propagation delays, as in case of satellite communication, the utilization can be very poor. To improve the link utilization, we can use the following (sliding-window) protocol instead of using stop-and-wait protocol.

## 2. Sliding Window

With the use of multiple frames for a single message, the stop-and-wait protocol does not perform well. Only one frame at a time can be in transit. In stop-and-wait flow control, if a  $> 1$ , serious inefficiencies result. Efficiency can be greatly improved by allowing multiple frames to be in transit at the same time. Efficiency can also be improved by making use of the full-duplex line. To keep track of the frames, sender station sends sequentially numbered frames. Since the sequence number to be used occupies a field in the frame, it should be of limited size. If the header of the frame allows  $k$  bits, the sequence numbers range from 0 to  $2^k - 1$ . Sender maintains a list of sequence numbers that it is allowed to send (sender window). The size of the sender's window is at most  $2^k - 1$ . The sender is provided with a buffer equal to the window size. Receiver also maintains a window of size  $2^k - 1$ . The receiver acknowledges a frame by sending an ACK frame that includes the sequence number of the next frame expected. This also explicitly announces that it is prepared to receive the next  $N$  frames, beginning with the number specified. This scheme can be used to acknowledge multiple frames. It could receive frames 2, 3, 4 but withhold ACK until frame 4 has arrived. By returning an ACK with sequence number 5, it acknowledges frames 2, 3, 4 in one go. The receiver needs a buffer of size 1.

Sliding window algorithm is a method of flow control for network data transfers. TCP, the Internet's stream transfer protocol, uses a sliding window algorithm.

A sliding window algorithm places a buffer between the application program and the network data flow. For TCP, the buffer is typically in the operating system kernel, but this is more of an implementation detail than a hard-and-fast requirement. Data received from the network is stored in the buffer, from where the application can read at its own pace. As the application reads data, buffer space is freed up to accept more input from the network. The window is the amount of data that can be "read ahead" - the size of the buffer, less the amount of valid data stored in it. Window announcements are used to inform the remote host of the current window size.

**Sender sliding Window:** At any instant, the sender is permitted to send frames with sequence numbers in a certain range (the sending window) as shown in Fig. 2.

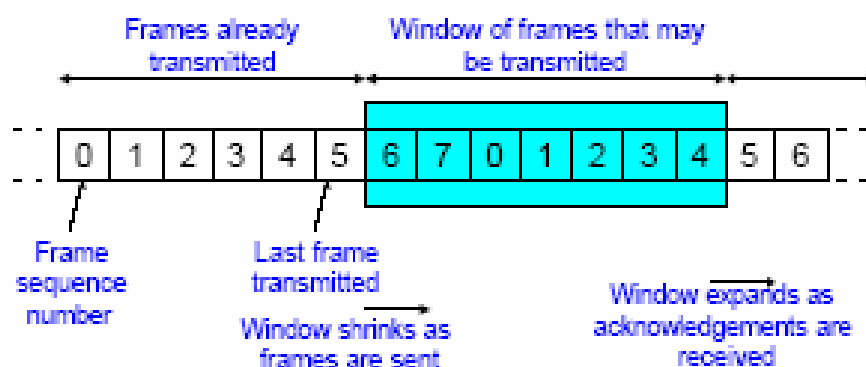


Figure 21 Sender's Window

**Receiver sliding Window:** The receiver always maintains a window of size 1 as shown in Fig. 3. It looks for a specific frame (frame 4 as shown in the figure) to arrive in a specific

order. If it receives any other frame (out of order), it is discarded and it needs to be resent. However, the receiver window also slides by one as the specific frame is received and accepted as shown in the figure. The receiver acknowledges a frame by sending an ACK frame that includes the sequence number of the next frame expected. This also explicitly announces that it is prepared to receive the next N frames, beginning with the number specified. This scheme can be used to acknowledge multiple frames. It could receive frames 2, 3, 4 but withhold ACK until frame 4 has arrived. By returning an ACK with sequence number 5, it acknowledges frames 2, 3, 4 at one time. The receiver needs a buffer of size 1.

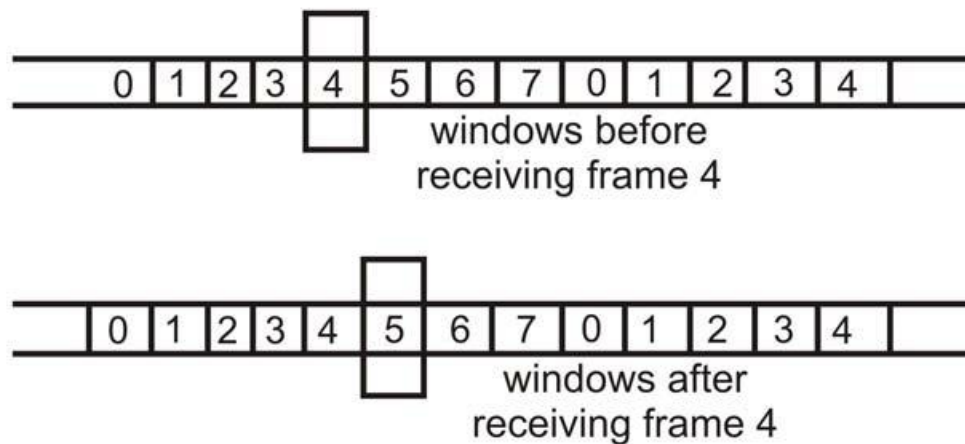


Figure 22 Receiver Sliding Window

On the other hand, if the local application can process data at the rate it's being transferred; sliding window still gives us an advantage. If the window size is larger than the packet size, then multiple packets can be outstanding in the network, since the sender knows that buffer space is available on the receiver to hold all of them. Ideally, a steady-state condition can be reached where a series of packets (in the forward direction) and window announcements (in the reverse direction) are constantly in transit. As each new window announcement is received by the sender, more data packets are transmitted. As the application reads data from the buffer (remember, we're assuming the application can keep up with the network), more window announcements are generated. Keeping a series of data packets in transit ensures the efficient use of network resources.

Hence, Sliding Window Flow Control

- Allows transmission of multiple frames
- Assigns each frame a k-bit sequence number
- Range of sequence number is  $[0 \dots 2k-1]$ , i.e., frames are counted modulo  $2k$ .

The link utilization in case of Sliding Window Protocol

$$U = 1, \quad \text{for } N > 2a + 1$$

$$N/(1+2a), \quad \text{for } N < 2a + 1$$

Where  $N$  = the window size,

And  $a$  = Propagation time / transmission time

### Error Control Techniques

When an error is detected in a message, the receiver sends a request to the transmitter to retransmit the ill-fated message or packet. The most popular retransmission scheme is known as Automatic-Repeat-Request (ARQ). Such schemes, where receiver asks transmitter to re-transmit if it detects an error, are known as reverse error correction techniques. There exist three popular ARQ techniques, as shown in Fig. 4

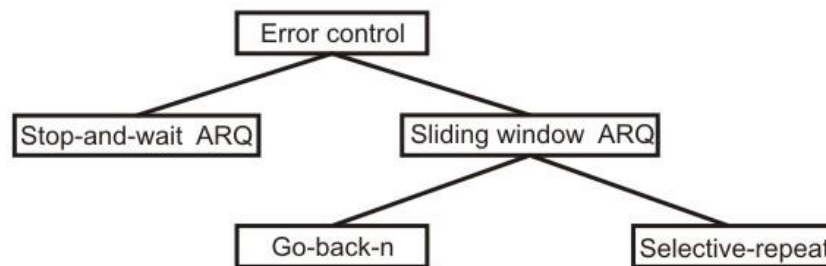


Figure 23 Error Control Techniques

### Stop-and-Wait ARQ

In Stop-and-Wait ARQ, which is simplest among all protocols, the sender (say station A) transmits a frame and then waits till it receives positive acknowledgement (ACK) or negative acknowledgement (NACK) from the receiver (say station B). Station B sends an ACK if the frame is received correctly, otherwise it sends NACK. Station A sends a new frame after receiving ACK; otherwise it retransmits the old frame, if it receives a NACK. This is illustrated in Fig 5

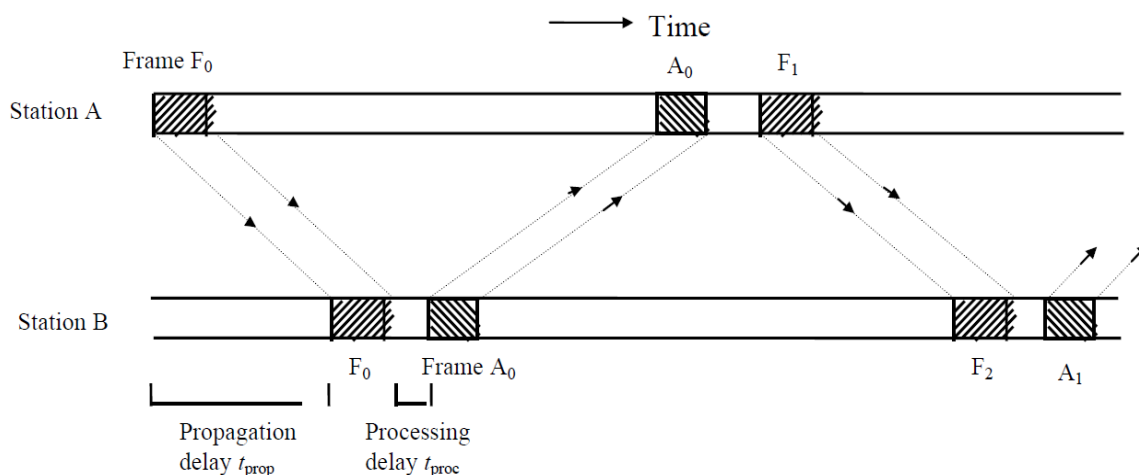


Figure 24 Stop and Wait ARQ Technique

To tackle the problem of a lost or damaged frame, the sender is equipped with a timer. In case of a lost ACK, the sender transmits the old frame. In the Fig. 3.3.7, the second PDU of Data is lost during transmission. The sender is unaware of this loss, but starts a timer after sending each

PDU. Normally an ACK PDU is received before the timer expires. In this case no ACK is received, and the timer counts down to zero and triggers retransmission of the same PDU by the sender. The sender always starts a timer following transmission, but in the second transmission receives an ACK PDU before the timer expires, finally indicating that the data has now been received by the remote node.

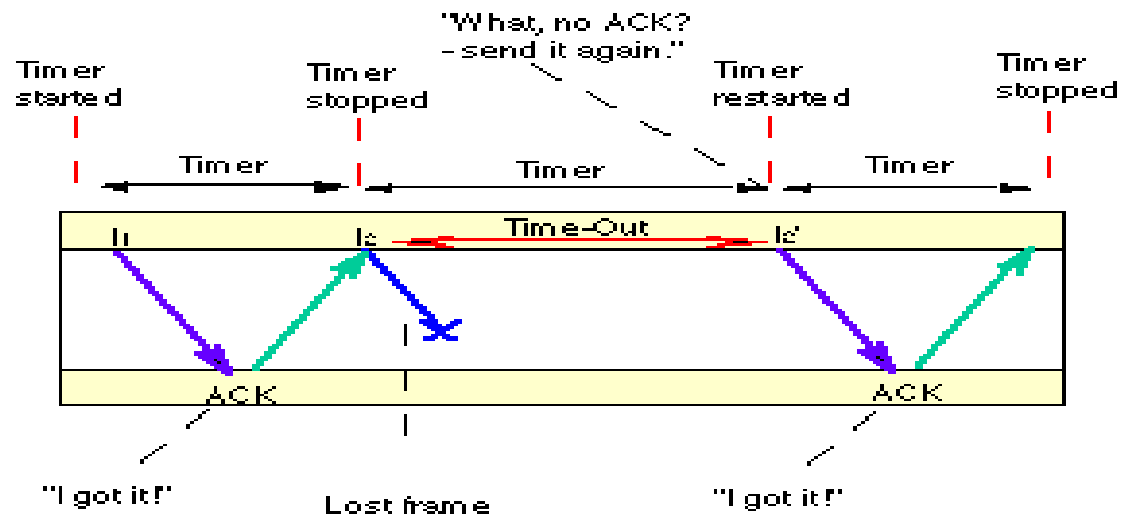


Figure 25 Retransmission due to lost of Frame

The receiver now can identify that it has received a duplicate frame from the label of the frame and it is discarded

To tackle the problem of damaged frames, say a frame that has been corrupted during the transmission due to noise, there is a concept of NACK frames, i.e. Negative Acknowledge frames. Receiver transmits a NACK frame to the sender if it finds the received frame to be corrupted. When a NACK is received by a transmitter before the time-out, the old frame is sent again as shown in Fig. 7

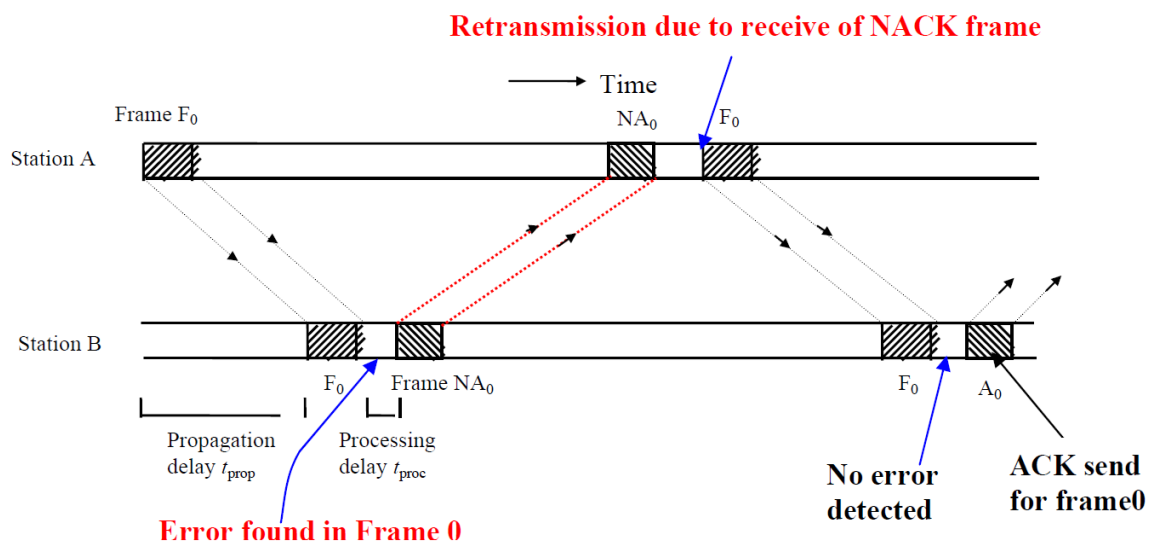


Figure 26 Retransmission due to damaged frame

### Go-back-N ARQ

The most popular ARQ protocol is the go-back-N ARQ, where the sender sends the frames continuously without waiting for acknowledgement. That is why it is also called as *continuous ARQ*. As the receiver receives the frames, it keeps on sending ACKs or a NACK, in case a frame is incorrectly received. When the sender receives a NACK, it retransmits the frame in error plus all the succeeding frames as shown in Fig.8 Hence, the name of the protocol is go-back-N ARQ. If a frame is lost, the receiver sends NAK after receiving the next frame as shown in Fig. 9. In case there is long delay before sending the NAK, the sender will resend the lost frame after its timer times out. If the ACK frame sent by the receiver is lost, the sender resends the frames after its timer times out as shown in Fig. 3.3.10.

Assuming full-duplex transmission, the receiving end sends piggybacked acknowledgement by using some number in the ACK field of its data frame. Let us assume that a 3-bit sequence number is used and suppose that a station sends frame 0 and gets back an RR1, and then sends frames 1, 2, 3, 4, 5, 6, 7, 0 and gets another RR1. This might either mean that RR1 is a cumulative ACK or all 8 frames were damaged. This ambiguity can be overcome if the maximum window size is limited to 7, i.e. for a k-bit sequence number field it is limited to  $2^k - 1$ . The number N ( $=2^k - 1$ ) specifies how many frames can be sent without receiving acknowledgement.

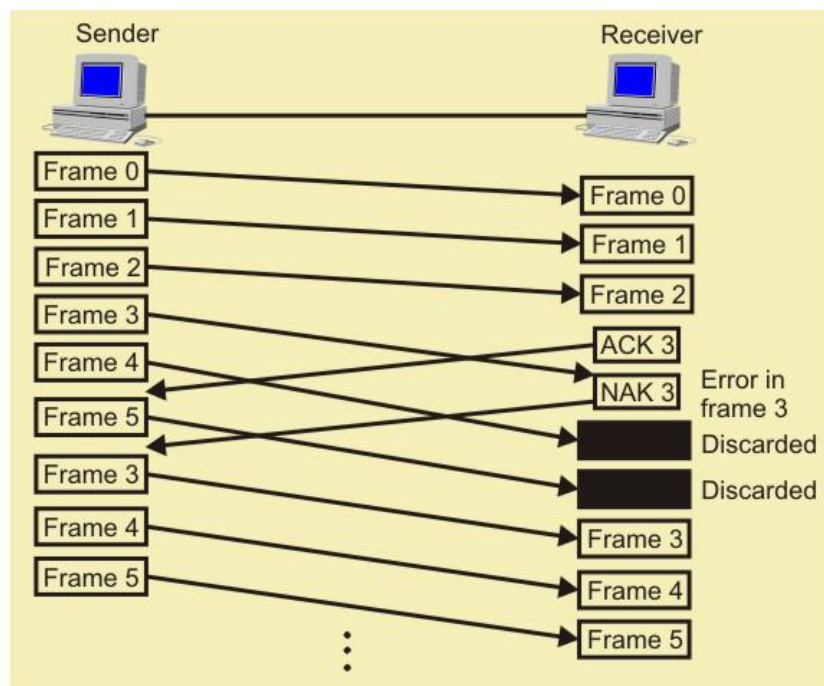


Figure 27 Frames in error in go-Back-N ARQ

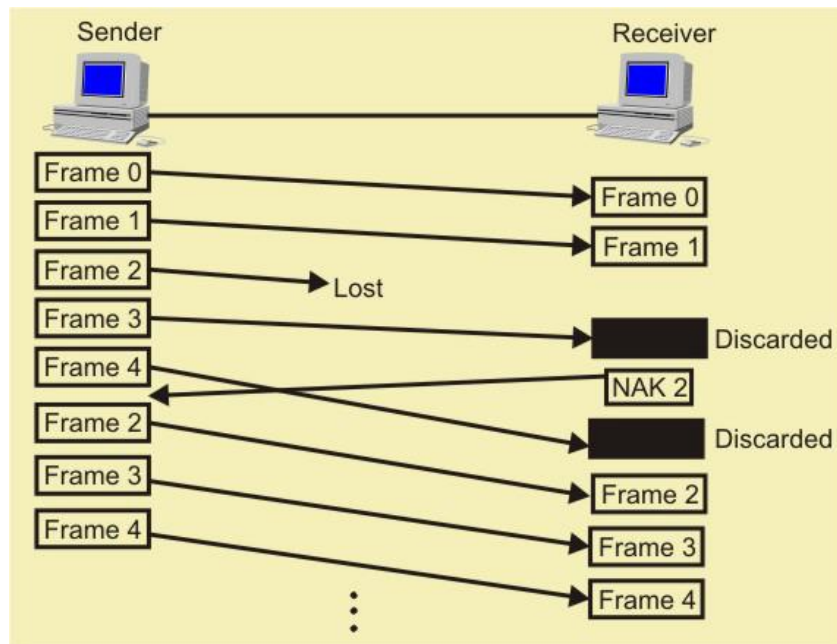


Figure 28 Lost Frames in Go-Back-N ARQ

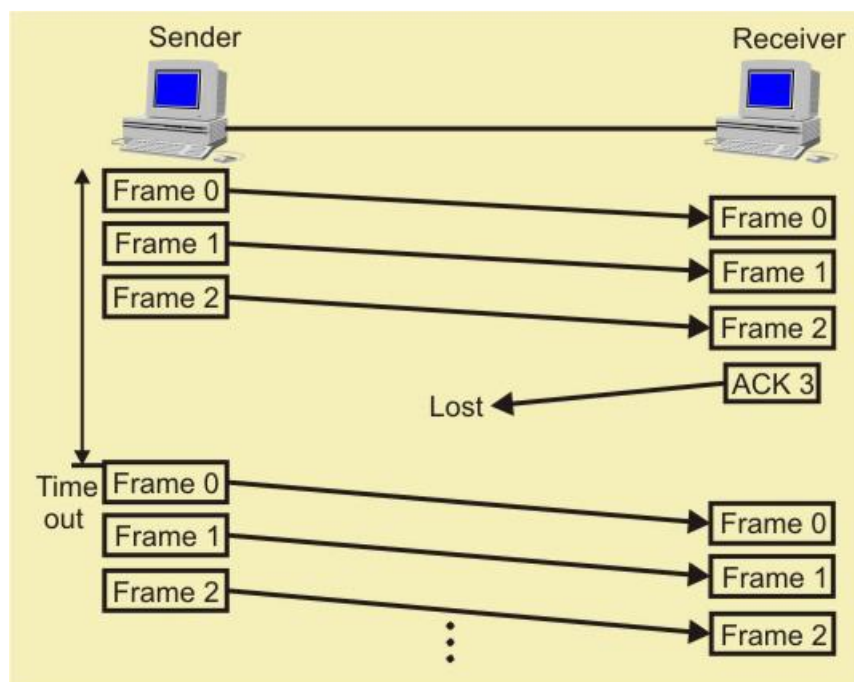


Figure 29 Lost ACK in Go-Back-N ARQ

If no acknowledgement is received after sending  $N$  frames, the sender takes the help of a timer. After the time-out, it resumes retransmission. The go-back- $N$  protocol also takes care of damaged frames and damaged ACKs. This scheme is little more complex than the previous one but gives much higher throughput.

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### Selective-Repeat ARQ

The selective-repetitive ARQ scheme retransmits only those for which NAKs are received or for which timer has expired, this is shown in the Fig.3.3.12. This is the most efficient among the ARQ schemes, but the sender must be more complex so that it can send out-of-order frames. The receiver also must have storage space to store the post-NAK frames and processing power to reinsert frames in proper sequence.

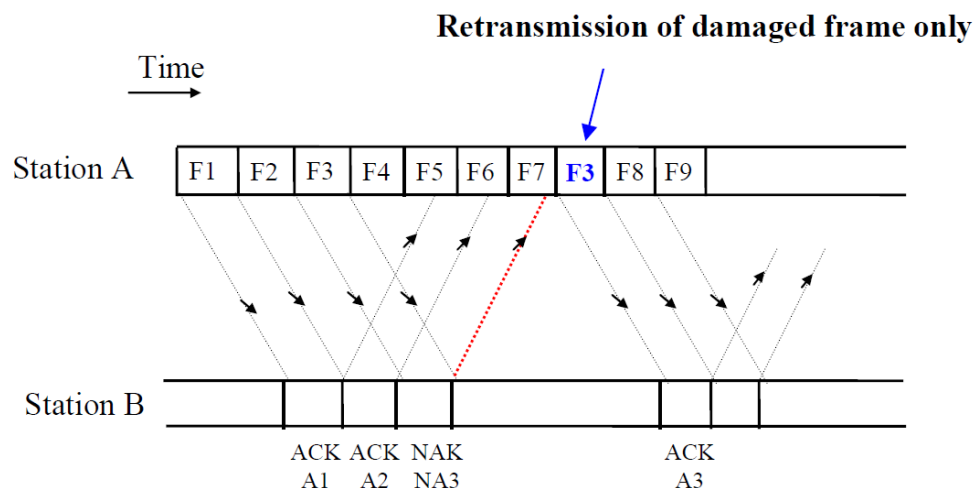


Figure 30 Selective-repeat Reject

### Error Detection and Correction

Environmental interference and physical defects in the communication medium can cause random bit errors during data transmission. Error coding is a method of detecting and correcting these errors to ensure information is transferred intact from its source to its destination. Error coding is used for fault tolerant computing in computer memory, magnetic and optical data storage media, satellite and deep space communications, network communications, cellular telephone networks, and almost any other form of digital data communication. Error coding uses mathematical formulas to encode data bits at the source into longer bit words for transmission. The "code word" can then be decoded at the destination to retrieve the information. The extra bits in the code word provide redundancy that, according to the coding scheme used,

will allow the destination to use the decoding process to determine if the communication medium introduced errors and in some cases correct them so that the data need not be retransmitted. Different error coding schemes are chosen depending on the types of errors expected, the communication medium's expected error rate, and whether or not data retransmission is possible. Faster processors and better communications technology make more complex coding schemes, with better error detecting and correcting capabilities, possible for smaller embedded systems, allowing for more robust communications. However, tradeoffs between bandwidth and coding overhead, coding complexity and allowable coding delay between transmissions, must be considered for each application.

Even if we know what type of errors can occur, we can't simply recognize them. We can do this simply by comparing this copy received with another copy of intended transmission. In this mechanism the source data block is sent twice. The receiver compares them with the help of a comparator and if those two blocks differ, a request for re-transmission is made. To achieve forward error correction, three sets of the same data block are sent and majority decision selects the correct block. These methods are very inefficient and increase the traffic two or three times. Fortunately there are more efficient error detection and correction codes. There are two basic strategies for dealing with errors. One way is to include enough redundant information (extra bits are introduced into the data stream at the transmitter on a regular and logical basis) along with each block of data sent to enable the receiver to deduce what the transmitted character must have been. The other way is to include only enough redundancy to allow the receiver to deduce that error has occurred, but not which error has occurred and the receiver asks for a retransmission. The former strategy uses Error-Correcting Codes and latter uses Error-detecting Codes.

To understand how errors can be handled, it is necessary to look closely at what error really is. Normally, a frame consists of  $m$ -data bits (i.e., message bits) and  $r$ -redundant bits (or check bits). Let the total number of bits be  $n$  ( $m + r$ ). An  $n$ -bit unit containing data and check-bits is often referred to as an  $n$ -bit code word.

Given any two code-words, say 10010101 and 11010100, it is possible to determine how many corresponding bits differ, just EXCLUSIVE OR the two code-words, and count the number of 1's in the result. The number of bits position in which code words differ is called the Hamming distance. If two code words are a Hamming distance  $d$ -apart, it will require  $d$  single-bit errors to convert one code word to other. The error detecting and correcting properties depends on its Hamming distance.

- To detect  $d$  errors, you need a distance  $(d+1)$  code because with such a code there is no way that  $d$ -single bit errors can change a valid code word into another valid code word. Whenever receiver sees an invalid code word, it can tell that a transmission error has occurred.
- Similarly, to correct  $d$  errors, you need a distance  $2d+1$  code because that way the legal code words are so far apart that even with  $d$  changes, the original code word is still closer than any other code-word, so it can be uniquely determined.

First, various types of errors have been introduced in Sec. 3.2.2 followed by different error detecting codes in Sec. 3.2.3. Finally, error correcting codes have been introduced in Sec. 3.2.4.

## Types of errors

These interferences can change the timing and shape of the signal. If the signal is carrying binary encoded data, such changes can alter the meaning of the data. These errors can be divided into two types: Single-bit error and Burst error.

### Single-bit Error

The term single-bit error means that only one bit of given data unit (such as a byte, character, or data unit) is changed from 1 to 0 or from 0 to 1 as shown in Fig.

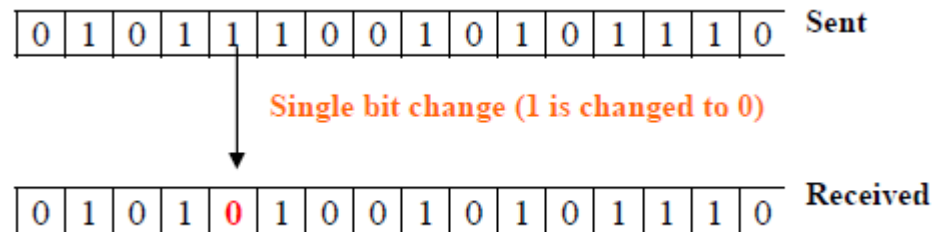


Figure 31 Single bit Error

Single bit errors are least likely type of errors in serial data transmission. To see why, imagine a sender sends data at 10 Mbps. This means that each bit lasts only for  $0.1 \mu\text{s}$  (micro-second). For a single bit error to occur noise must have duration of only  $0.1 \mu\text{s}$  (micro-second), which is very rare. However, a single-bit error can happen if we are having a parallel data transmission. For example, if 16 wires are used to send all 16 bits of a word at the same time and one of the wires is noisy, one bit is corrupted in each word.

### Burst Error

The term burst error means that two or more bits in the data unit have changed from 0 to 1 or vice-versa. Note that burst error doesn't necessarily means that error occurs in consecutive bits. The length of the burst error is measured from the first corrupted bit to the last corrupted bit. Some bits in between may not be corrupted.

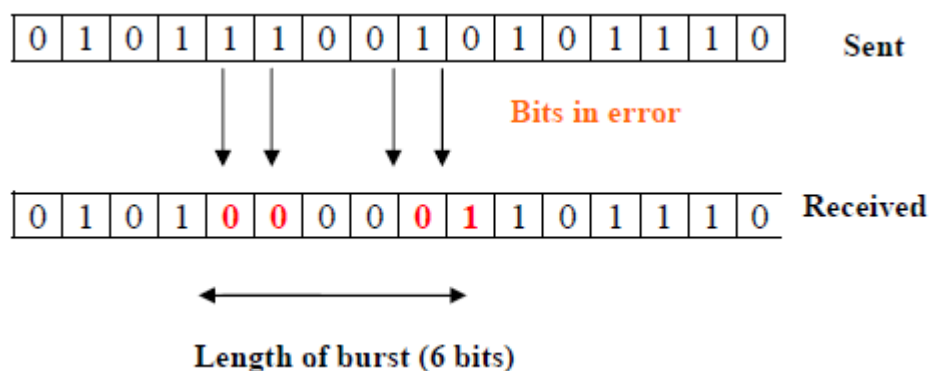


Figure 32 Burst Error

Burst errors are mostly likely to happen in serial transmission. The duration of the noise is normally longer than the duration of a single bit, which means that the noise affects data; it

affects a set of bits as shown in Fig. . The number of bits affected depends on the data rate and duration of noise.

### Error Detecting Codes

Basic approach used for error detection is the use of redundancy, where additional bits are added to facilitate detection and correction of errors. Popular techniques are:

- Simple Parity check
- Two-dimensional Parity check
- Checksum
- Cyclic redundancy check

### Simple Parity Checking or One-dimension Parity Check

The most common and least expensive mechanism for error- detection is the **simple parity check**. In this technique, a redundant bit called parity bit, is appended to every data unit so that the number of 1s in the unit (including the parity becomes even).

Blocks of data from the source are subjected to a check bit or Parity bit generator form, where a parity of 1 is added to the block if it contains an odd number of 1's (ON bits) and 0 is added if it contains an even number of 1's. At the receiving end the parity bit is computed from the received data bits and compared with the received parity bit, as shown in Fig. This scheme makes the total number of 1's even, that is why it is called even parity checking. Considering a 4-bit word, different combinations of the data words and the corresponding code words are given in Table.

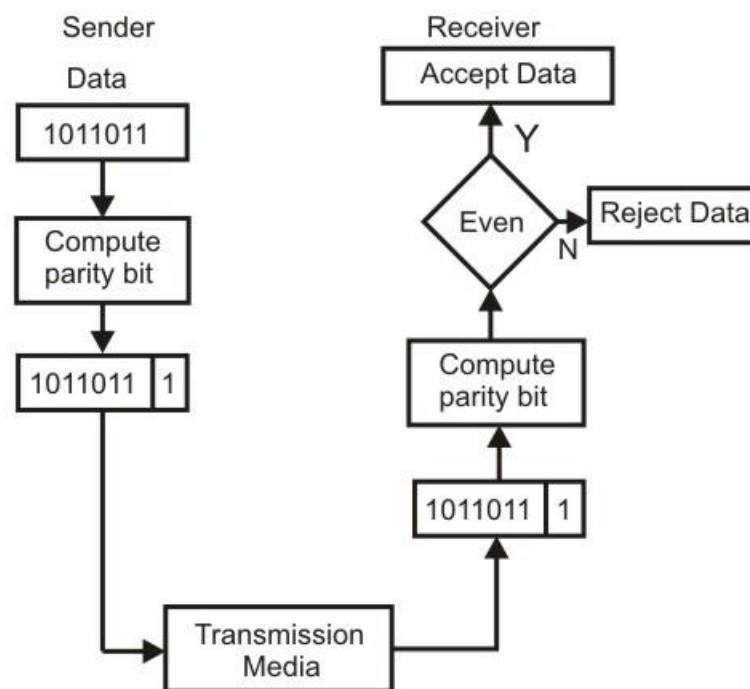


Figure 33 Even-parity checking scheme

Table 1: Possible 4-bit data words and corresponding code words

| Decimal value | Data Block | Parity bit | Code word |
|---------------|------------|------------|-----------|
| 0             | 0000       | 0          | 00000     |
| 1             | 0001       | 1          | 00011     |
| 2             | 0010       | 1          | 00101     |
| 3             | 0011       | 0          | 00110     |
| 4             | 0100       | 1          | 01001     |
| 5             | 0101       | 0          | 01010     |
| 6             | 0110       | 0          | 01100     |
| 7             | 0111       | 1          | 01111     |
| 8             | 1000       | 1          | 10001     |
| 9             | 1001       | 0          | 10010     |
| 10            | 1010       | 0          | 10100     |
| 11            | 1011       | 1          | 10111     |
| 12            | 1100       | 0          | 11000     |
| 13            | 1101       | 1          | 11011     |
| 14            | 1110       | 1          | 11101     |
| 15            | 1111       | 0          | 11110     |

Note that for the sake of simplicity, we are discussing here the even-parity checking, where the number of 1's should be an even number. It is also possible to use odd-parity checking, where the number of 1's should be odd.

### Two-dimension Parity Check

Performance can be improved by using two-dimensional parity check, which organizes the block of bits in the form of a table. Parity check bits are calculated for each row, which is equivalent to a simple parity check bit. Parity check bits are also calculated for all columns then both are sent along with the data. At the receiving end these are compared with the parity bits calculated on the received data.

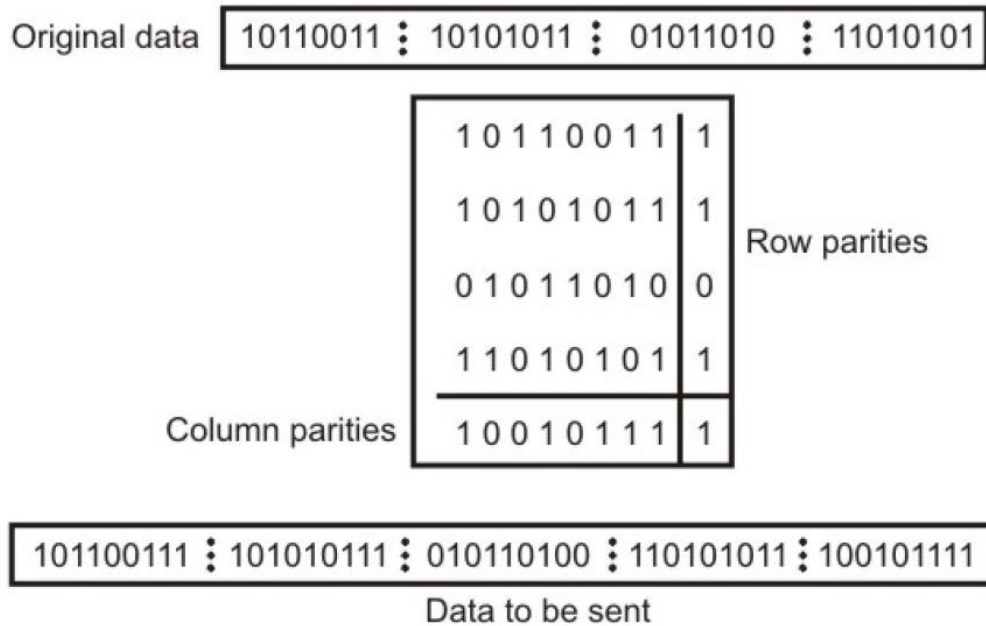


Figure 34 Two-dimension Parity Checking

## Checksum

In checksum error detection scheme, the data is divided into  $k$  segments each of  $m$  bits. In the sender's end the segments are added using 1's complement arithmetic to get the sum. The sum is complemented to get the checksum. The checksum segment is sent along with the data segments as shown in Fig. At the receiver's end, all received segments are added using 1's complement arithmetic to get the sum. The sum is complemented. If the result is zero, the received data is accepted; otherwise discarded, as shown in Fig.

The checksum detects all errors involving an odd number of bits. It also detects most errors involving even number of bits.

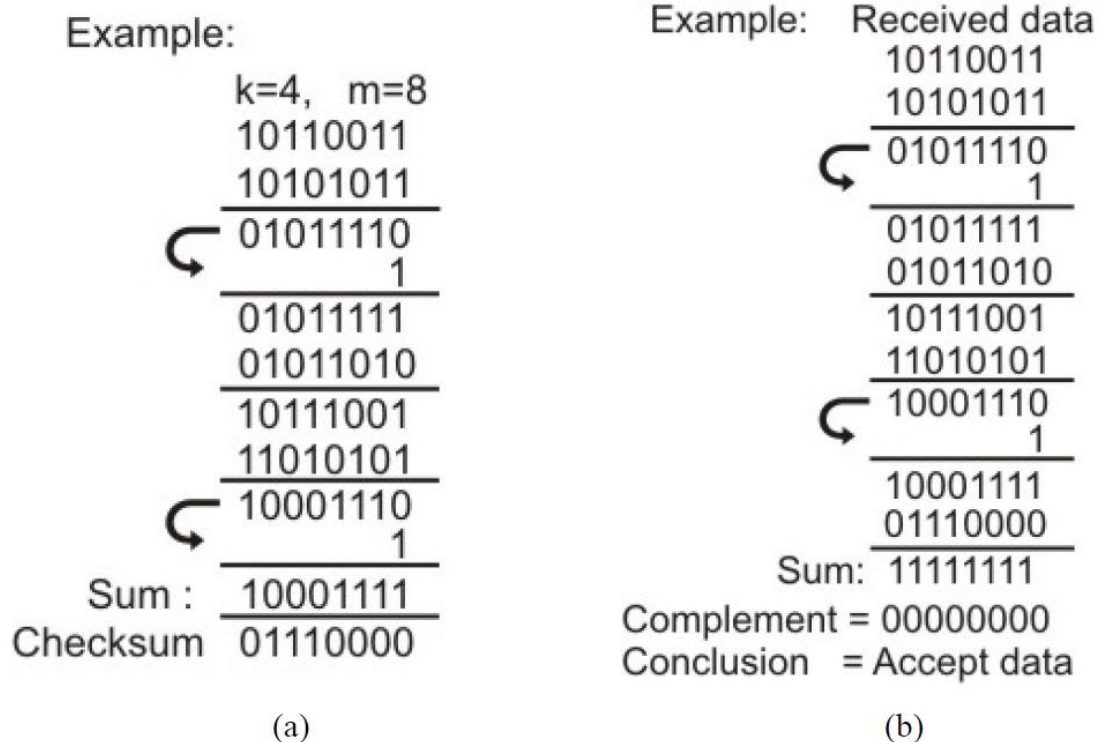


Figure 35 (a) Sender's end for the calculation of the checksum, (b) Receiving end for checking the checksum

### Cyclic Redundancy Checks (CRC)

This Cyclic Redundancy Check is the most powerful and easy to implement technique. Unlike checksum scheme, which is based on addition, CRC is based on binary division. In CRC, a sequence of redundant bits, called cyclic redundancy check bits, are appended to the end of data unit so that the resulting data unit becomes exactly divisible by a second, predetermined binary number. At the destination, the incoming data unit is divided by the same number. If at this step there is no remainder, the data unit is assumed to be correct and is therefore accepted. A remainder indicates that the data unit has been damaged in transit and therefore must be rejected. The generalized technique can be explained as follows.

If a  $k$  bit message is to be transmitted, the transmitter generates an  $r$ -bit sequence, known as Frame Check Sequence (FCS) so that the  $(k + r)$  bits are actually being transmitted. Now this  $r$ -bit FCS is generated by dividing the original number, appended by  $r$  zeros, by a predetermined number. This number, which is  $(r+1)$  bit in length, can also be considered as the coefficients of a polynomial, called Generator Polynomial. The remainder of this division process generates the  $r$ -bit FCS. On receiving the packet, the receiver divides the  $(k + r)$  bit frame by the same predetermined number and if it produces no remainder, it can be assumed that no error has occurred during the transmission. Operations at both the sender and receiver end

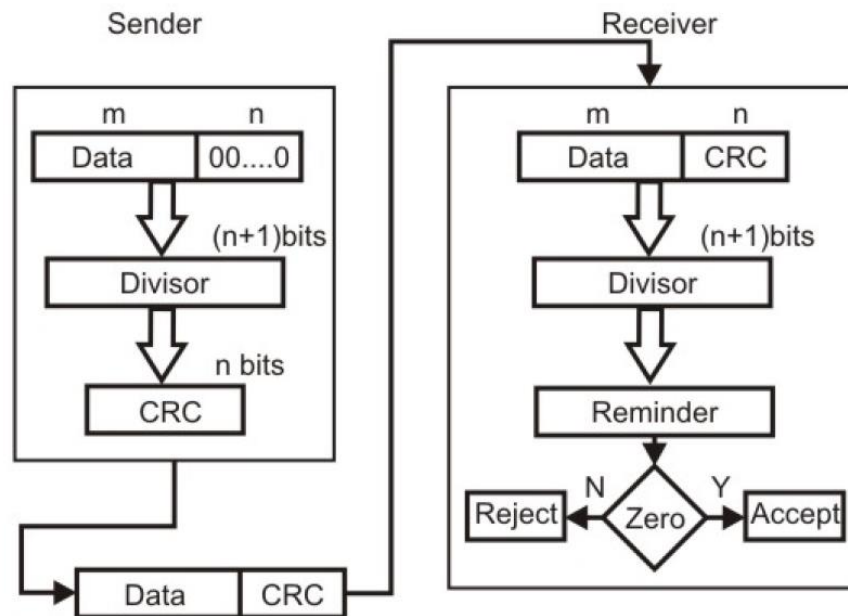


Figure 36 Basic scheme for Cyclic Redundancy Checking

This mathematical operation performed is illustrated in Fig. by dividing a sample 4-bit number by the coefficient of the generator polynomial  $x^3+x+1$ , which is 1011, using the modulo-2 arithmetic. Modulo-2 arithmetic is a binary addition process without any carry over, which is just the Exclusive-OR operation. Consider the case where  $k=1101$ . Hence we have to divide 1101000 (i.e.  $k$  appended by 3 zeros) by 1011, which produces the remainder  $r=001$ , so that the bit frame  $(k + r) = 1101001$  is actually being transmitted through the communication channel. At the receiving end, if the received number, i.e., 1101001 is divided by the same generator polynomial 1011 to get the remainder as 000, it can be assumed that the data is free of errors.

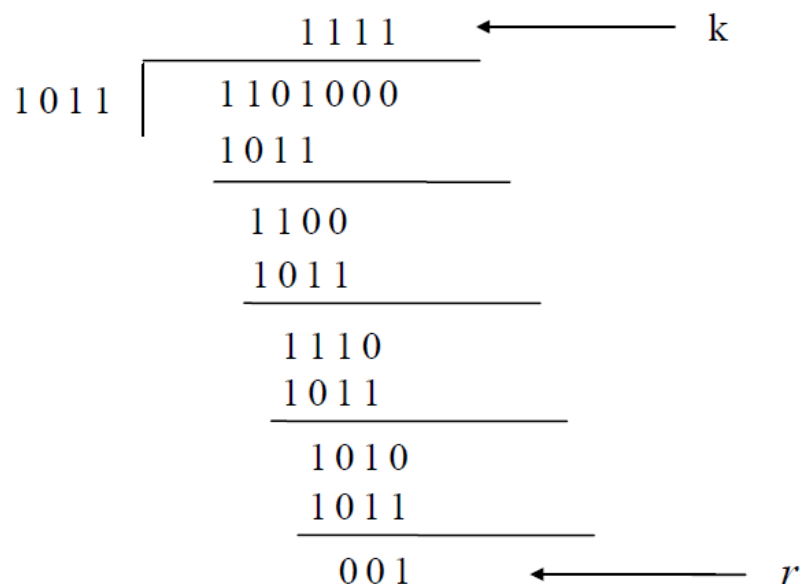


Figure 37 Cyclic Redundancy Checks (CRC)

## Error Correcting Codes

The techniques that we have discussed so far can detect errors, but do not correct them. Error Correction can be handled in two ways.

- One is when an error is discovered; the receiver can have the sender retransmit the entire data unit. This is known as backward error correction.
- In the other, receiver can use an error-correcting code, which automatically corrects certain errors. This is known as forward error correction.

In theory it is possible to correct any number of errors atomically. Error-correcting codes are more sophisticated than error detecting codes and require more redundant bits. The number of bits required to correct multiple-bit or burst error is so high that in most of the cases it is inefficient to do so. For this reason, most error correction is limited to one, two or at the most three-bit errors.

### Single-bit error correction

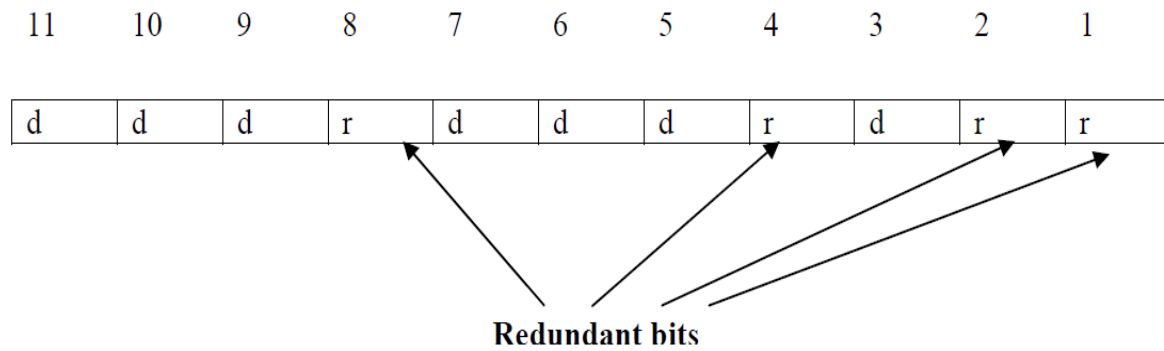
Concept of error-correction can be easily understood by examining the simplest case of single-bit errors. As we have already seen that a single-bit error can be detected by addition of a parity bit (VRC) with the data, which needed to be send. A single additional bit can detect error, but it's not sufficient enough to correct that error too. For correcting an error one has to know the exact position of error, i.e. exactly which bit is in error (to locate the invalid bits). For example, to correct a single-bit error in an ASCII character, the error correction must determine which one of the seven bits is in error. To this, we have to add some additional redundant bits.

To calculate the numbers of redundant bits ( $r$ ) required to correct  $d$  data bits, let us find out the relationship between the two. So we have  $(d+r)$  as the total number of bits, which are to be transmitted; then  $r$  must be able to indicate at least  $d+r+1$  different values. Of these, one value means no error, and remaining  $d+r$  values indicate error location of error in each of  $d+r$  locations. So,  $d+r+1$  states must be distinguishable by  $r$  bits, and  $r$  bits can indicates  $2^r$  states. Hence,  $2^r$  must be greater than  $d+r+1$ .

$$2^r \geq d+r+1$$

The value of  $r$  must be determined by putting in the value of  $d$  in the relation. For example, if  $d$  is 7, then the smallest value of  $r$  that satisfies the above relation is 4. So the total bits, which are to be transmitted is 11 bits ( $d+r = 7+4 = 11$ ).

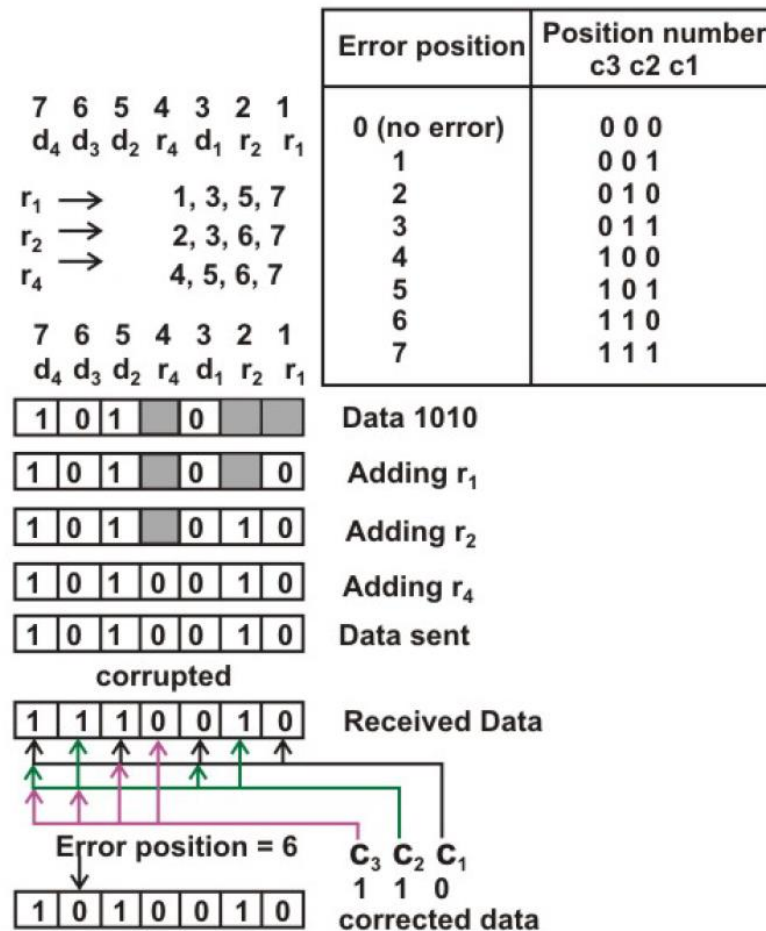
Now let us examine how we can manipulate these bits to discover which bit is in error. A technique developed by R.W. Hamming provides a practical solution. The solution or coding scheme he developed is commonly known as Hamming Code. Hamming code can be applied to data units of any length and uses the relationship between the data bits and redundant bits as discussed.



*Figure 38 Positions of redundancy bits in hamming code*

Basic approach for error detection by using Hamming code is as follows:

- To each group of  $m$  information bits  $k$  parity bits are added to form  $(m + k)$  bit code as shown in Fig.
- Location of each of the  $(m + k)$  digits is assigned a decimal value.
- The  $k$  parity bits are placed in positions 1, 2, ...,  $2^{k-1}$  positions.  $K$  parity checks are performed on selected digits of each code word.
- At the receiving end the parity bits are recalculated. The decimal value of the  $k$  parity bits provides the bit-position in error, if any.



**Figure 39 Use of Hamming code for error correction for a 4-bit data**

Figure 20 shows how hamming code is used for correction for 4-bit numbers (d4d3d2d1) with the help of three redundant bits (r3r2r1). For the example data 1010, first r1 (0) is calculated considering the parity of the bit positions, 1, 3, 5 and 7. Then the parity bits r2 is calculated considering bit positions 2, 3, 6 and 7. Finally, the parity bits r4 is calculated considering bit positions 4, 5, 6 and 7 as shown. If any corruption occurs in any of the transmitted code 1010010, the bit position in error can be found out by calculating r3r2r1 at the receiving end. For example, if the received code word is 1110010, the recalculated value of r3r2r1 is 110, which indicates that bit position in error is 6, the decimal value of 110.

**Example:**

Let us consider an example for 5-bit data. Here 4 parity bits are required. Assume that during transmission bit 5 has been changed from 1 to 0 as shown in Fig. 3.2.11. The receiver receives the code word and recalculates the four new parity bits using the same set of bits used by the sender plus the relevant parity (r) bit for each set (as shown in Fig. 3.2.11). Then it assembles the new parity values into a binary number in order of r positions (r8, r4, r2, r1).

|   |   |   |  |   |   |   |  |   |  |  |
|---|---|---|--|---|---|---|--|---|--|--|
| 1 | 1 | 0 |  | 1 | 0 | 1 |  | 1 |  |  |
|---|---|---|--|---|---|---|--|---|--|--|

**Data to be send**

|   |   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 0 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|

**Data to be send along with redundant bits**

|   |   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|

**Data Received**

|   |   |   |  |   |   |   |  |   |  |  |
|---|---|---|--|---|---|---|--|---|--|--|
| 1 | 1 | 0 |  | 1 | 0 | 0 |  | 1 |  |  |
|---|---|---|--|---|---|---|--|---|--|--|

**Data Received Minus Parity Bits**

|  |  |  |   |  |  |  |   |   |   |
|--|--|--|---|--|--|--|---|---|---|
|  |  |  | 0 |  |  |  | 1 | 0 | 1 |
|--|--|--|---|--|--|--|---|---|---|

**Parity bits recalculated****Calculations:**

Parity recalculated ( $r_8, r_4, r_2, r_1$ ) = 01012 = 510.

Hence, bit 5th is in error i.e. d5 is in error.

So, correct code-word which was transmitted is:

|   |   |   |  |   |   |   |  |   |  |  |
|---|---|---|--|---|---|---|--|---|--|--|
| 1 | 1 | 0 |  | 1 | 0 | 1 |  | 1 |  |  |
|---|---|---|--|---|---|---|--|---|--|--|

Figure 40 Hamming code for error correction for a 5-bit data

**Contention-based Approaches**

Round-Robin techniques work efficiently when majority of the stations have data to send most of the time. But, in situations where only a few nodes have data to send for brief periods of time, Round-Robin techniques are unsuitable. Contention techniques are suitable for busy nature of traffic. In contention techniques, there is no centralized control and when a node has data to send, it contends for gaining control of the medium. The principle advantage of contention techniques is their simplicity. They can be easily implemented in each node. The techniques work efficiently under light to moderate load, but performance rapidly falls under heavy load.

**ALOHA**

The ALOHA scheme was invented by Abramson in 1970 for a packet radio network connecting remote stations to a central computer and various data terminals at the campus of the university of Hawaii. A simplified situation is shown in Fig. Users are allowed random access of the central computer through a common radio frequency band  $f_1$  and the computer Centre broadcasts all received signals on a different frequency band  $f_2$ . This enables the users to monitor packet collisions, if any. The protocol followed by the users is simple; whenever a node has a packet to send, it simply does so. The scheme, known as Pure ALOHA, is truly a free-for-all scheme. Of course, frames will suffer collision and colliding frames will be destroyed. By monitoring the

signal sent by the central computer, after the maximum round-trip propagation time, and user comes to know whether the packet sent by him has suffered a collision or not.

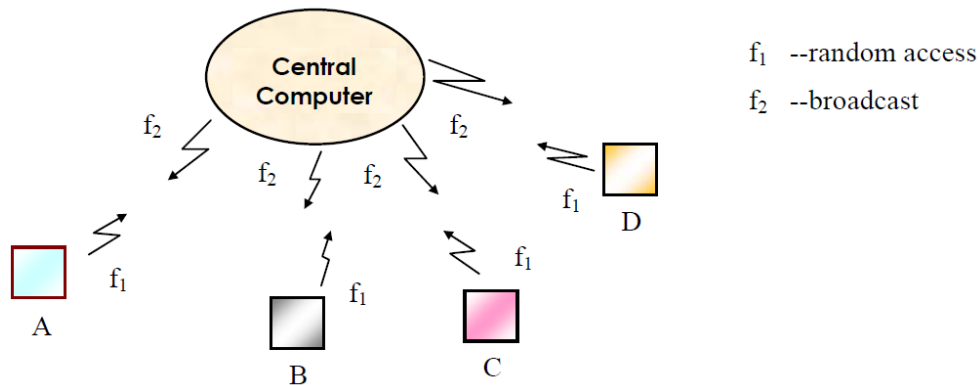


Figure 41 Simplified ALOHA scheme for a packet radio system

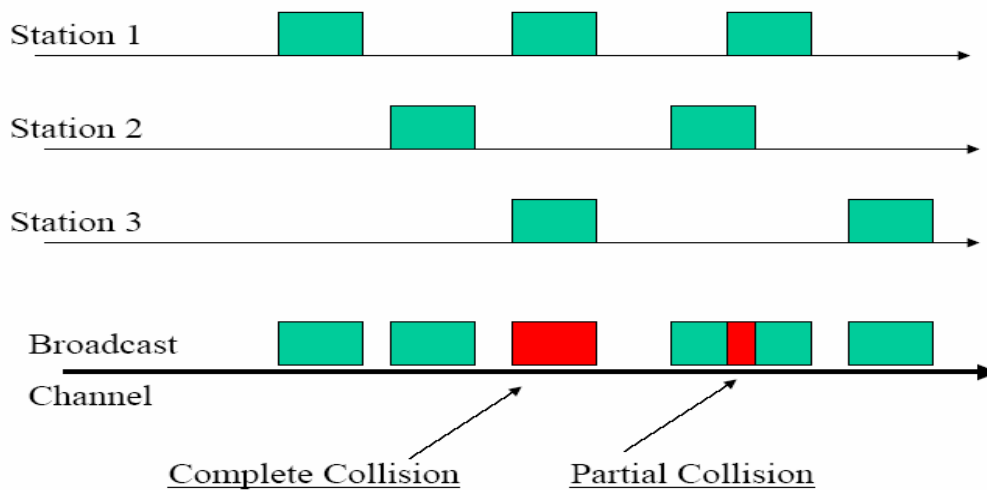


Figure 42 Collision in Pure ALOHA

It may be noted that if all packets have a fixed duration of  $\tau$  (shown as F in Figure), then a given packet A will suffer collision if another user starts to transmit at any time from  $\tau$  before to until  $\tau$  after the start of the packet A as shown in Fig. This gives a vulnerable period of  $2\tau$ . Based on this assumption, the channel utilization can be computed. The channel utilization, expressed as throughput S, in terms of the offered load G is given by  $S = Ge^{-2G}$ .

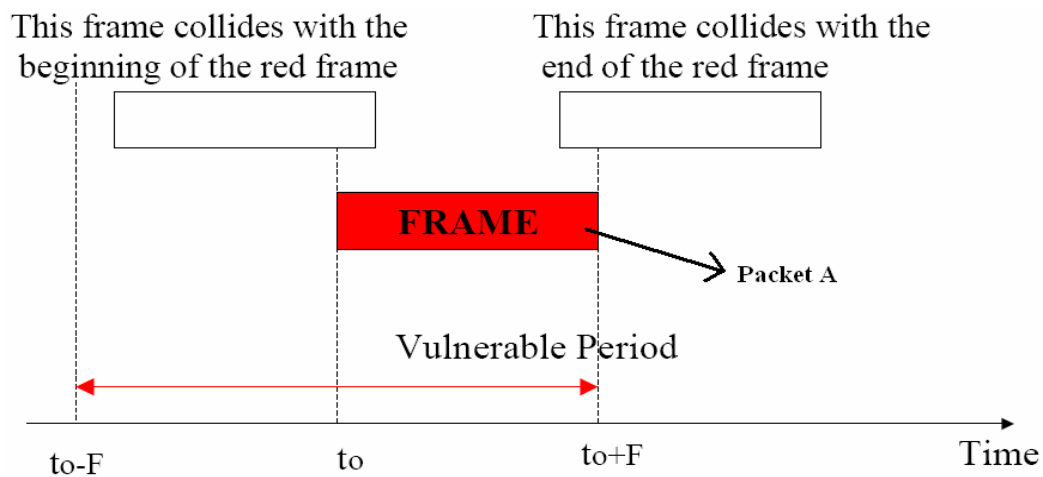


Figure 43 Vulnerable period in Pure ALOHA

Based on this, the best channel utilization of 18% can be obtained at 50 percent of the offered load as shown in Fig. At smaller offered load, channel capacity is underused and at higher offered load too many collisions occur reducing the throughput. The result is not encouraging, but for such a simple scheme high throughput was also not expected.

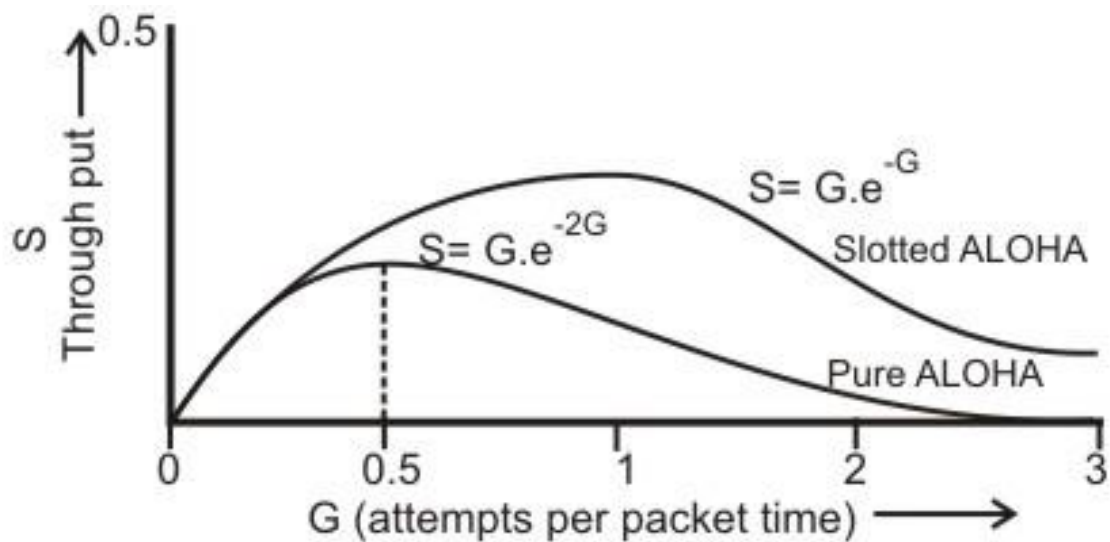


Figure 44 Throughput versus offered load for ALOHA protocol

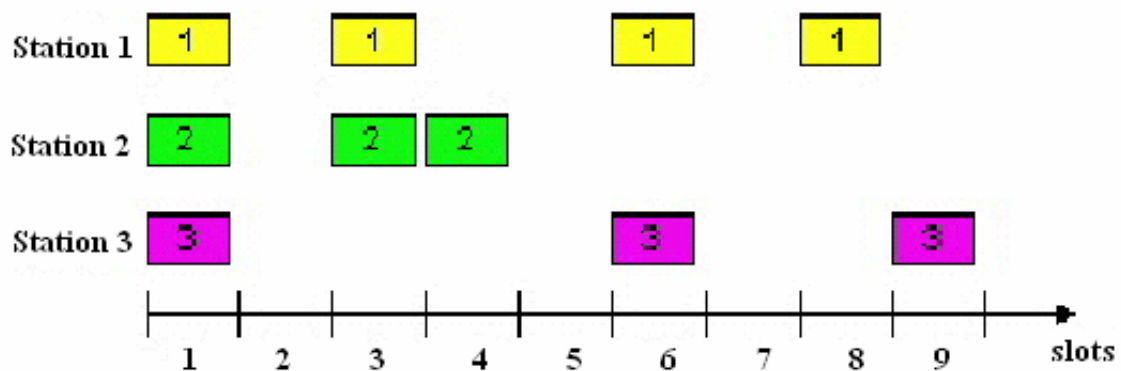


Figure 45 Slotted ALOHA: Single active node can continuously transmit at full rate of channel

Subsequently, in a new scheme, known as *Slotted ALOHA*, was suggested to improve upon the efficiency of pure ALOHA. In this scheme, the channel is divided into slots equal to  $\tau$  and packet transmission can start only at the beginning of a slot as shown in Fig. This reduces the vulnerable period from  $2\tau$  to  $\tau$  and improves efficiency by reducing the probability of collision as shown in Fig. This gives a maximum throughput of 37% at 100 percent of offered load, as shown in Figure

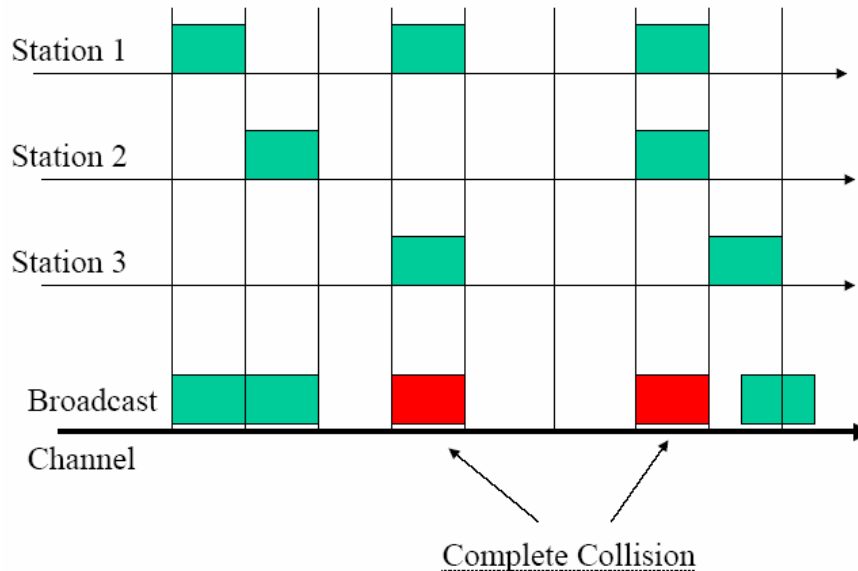


Figure 46 Collision in Slotted ALOHA

## CSMA

The poor efficiency of the ALOHA scheme can be attributed to the fact that a node starts transmission without paying any attention to what others are doing. In situations where propagation delay of the signal between two nodes is small compared to the transmission time of a packet, all other nodes will know very quickly when a node starts transmission. This observation is the basis of the carrier-sense multiple-access (CSMA) protocol. In this scheme, a node having data to transmit first listens to the medium to check whether another transmission is in progress or not. The node starts sending only when the channel is free, that is there is no carrier. That is why the scheme is also known as listen-before-talk. There are three variations of this basic scheme as outlined below.

- (i) 1-persistent CSMA: In this case, a node having data to send, starts sending, if the channel is sensed free. If the medium is busy, the node continues to monitor until the channel is idle. Then it starts sending data.
- (ii) Non-persistent CSMA: If the channel is sensed free, the node starts sending the packet. Otherwise, the node waits for a random amount of time and then monitors the channel.
- (iii) p-persistent CSMA: If the channel is free, a node starts sending the packet. Otherwise the node continues to monitor until the channel is free and then it sends with probability  $p$ .

The efficiency of CSMA scheme depends on the propagation delay, which is represented by a parameter  $a$ , as defined below:

$$a = \frac{\text{Propagation delay}}{\text{Packet transmission time.}}$$

The throughput of 1-persistent CSMA scheme is shown in Fig. 5.2.11 for different values of  $a$ . It may be noted that smaller the value of propagation delay, lower is the vulnerable period and higher is the efficiency.

### CSMA/CD

CSMA/CD protocol can be considered as a refinement over the CSMA scheme. It has evolved to overcome one glaring inefficiency of CSMA. In CSMA scheme, when two packets collide the channel remains unutilized for the entire duration of transmission time of both the packets. If the propagation time is small (which is usually the case) compared to the packet transmission time, wasted channel capacity can be considerable. This wastage of channel capacity can be reduced if the nodes continue to monitor the channel while transmitting a packet and immediately cease transmission when collision is detected. This refined scheme is known as Carrier Sensed Multiple Access with Collision Detection (CSMA/CD) or Listen-While-Talk.

On top of the CSMA, the following rules are added to convert it into CSMA/CD:

- (i) If a collision is detected during transmission of a packet, the node immediately ceases transmission and it transmits jamming signal for a brief duration to ensure that all stations know that collision has occurred.
- (ii) After transmitting the jamming signal, the node waits for a random amount of time and then transmission is resumed.

The random delay ensures that the nodes, which were involved in the collision are not likely to have a collision at the time of retransmissions. To achieve stability in the back off scheme, a technique known as binary exponential back off is used. A node will attempt to transmit repeatedly in the face of repeated collisions, but after each collision, the mean value of the random delay is doubled. After 15 retries (excluding the original try), the unlucky packet is discarded and the node reports an error. A flowchart representing the binary exponential back off algorithm is given in Fig

**Performance Comparisons:** The throughput of the three contention based schemes with respect to the offered load is given in Fig. The figure shows that pure ALHOA gives a maximum throughput of only 18 percent and is suitable only for very low offered load. The slotted ALHOA gives a modest improvement over pure ALHOA with a maximum throughput of 36 percent. Non persistent CSMA gives a better throughput than 1-persistent CSMA because of smaller probability of collision for the retransmitted packets. The non-persistent CSMA/CD provides a high throughput and can tolerate a very heavy offered load. Figure provides a plot of the offered load versus throughput for the value of  $a = 0.01$ .

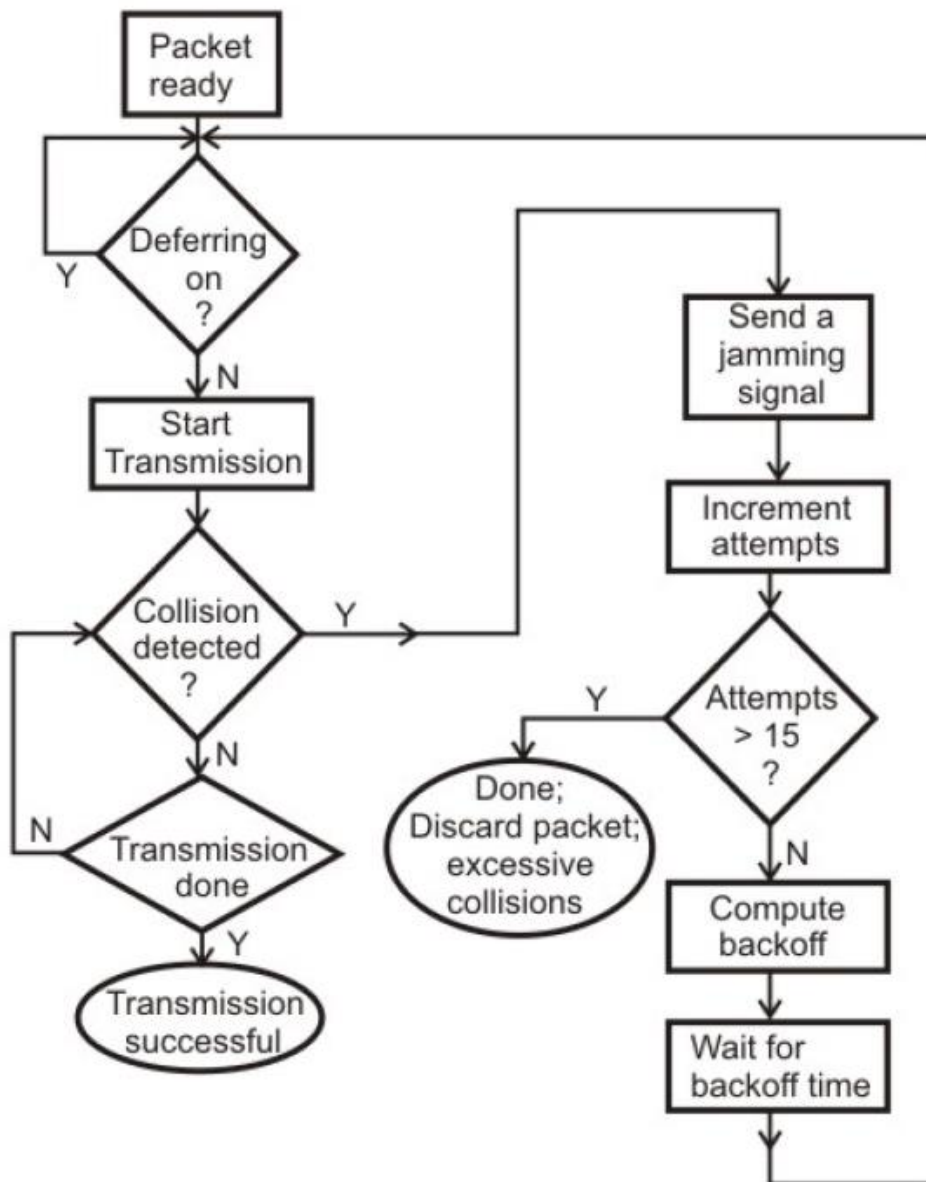


Figure 47 Binary exponential back off algorithm used in CSMA/CD

| Protocol              | Throughput                                                   |
|-----------------------|--------------------------------------------------------------|
| ALOHA                 | $S = Ge^{-2G}$                                               |
| Slotted ALOHA         | $S = Ge^{-G}$                                                |
| Nonpersistent CSMA    | $S = \frac{Ge^{-aG}}{[G(1+2a)+e^{-aG}]}$                     |
| Nonpersistent CSMA/CD | $S = \frac{Ge^{-aG}}{[Ge^{-aG}+3aG(1-e^{-aG})+(2-e^{-aG})]}$ |

Figure 48 Comparison of the throughputs for the contention-based MACs

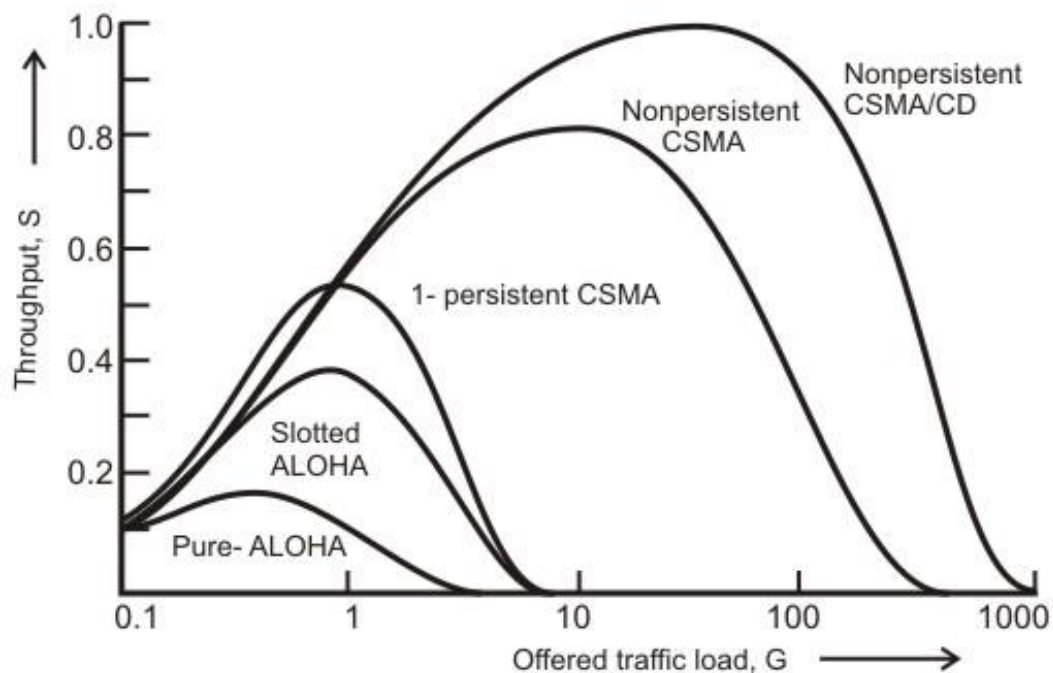


Figure 49 A plot of the offered load versus throughput for the value of  $a = 0.01$

Performance Comparison between CSMA/CD and Token ring: It has been observed that smaller the mean packet length, the higher the maximum mean throughput rate for token passing compared to that of CSMA/CD. The token ring is also least sensitive to workload and propagation effects compared to CSMA/CD protocol. The CSMA/CD has the shortest delay under light load conditions, but is most sensitive to variations to load, particularly when the load

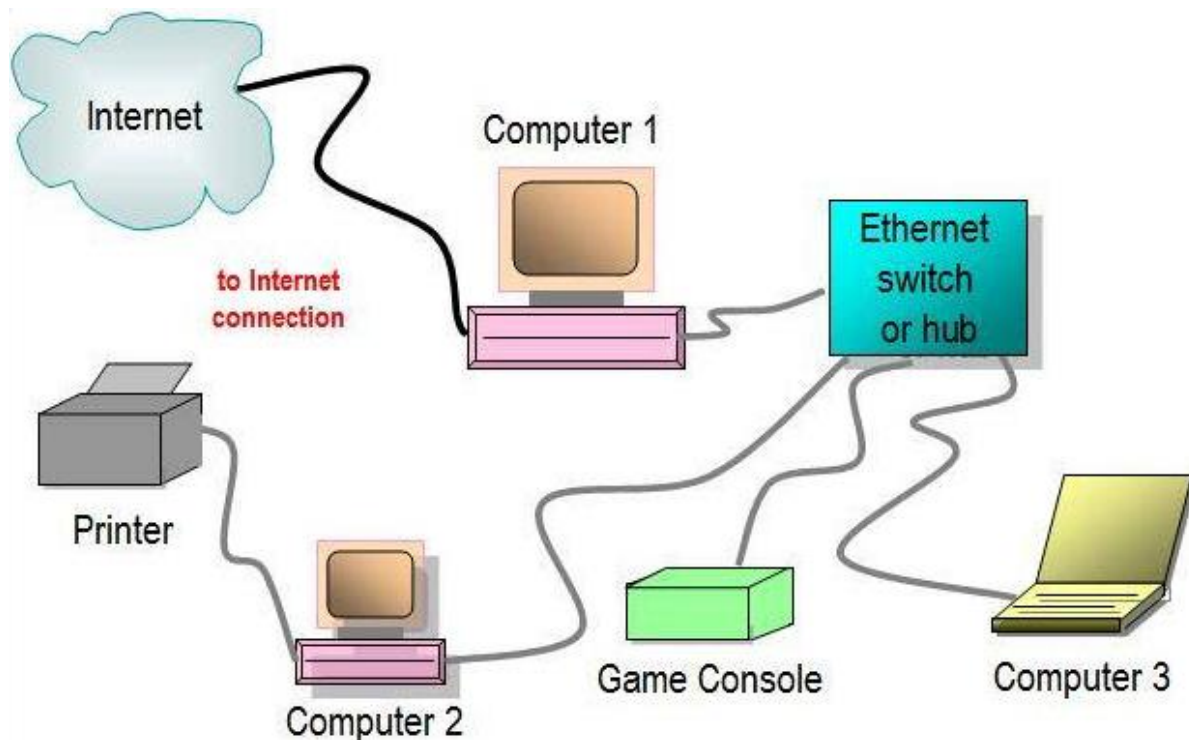
is heavy. In CSMA/CD, the delay is not deterministic and a packet may be dropped after fifteen collisions based on binary exponential back off algorithm. As a consequence, CSMA/CD is not suitable for real-time traffic.

## Ethernet communication

### 1. Wired Ethernet Network

The Ethernet technology mainly works with the fiber optic cables that connect devices within a distance of 10 km. The Ethernet supports 10 Mbps.

A computer network interface card (NIC) is installed in each computer, and is assigned to a unique address. An Ethernet cable runs from each NIC to the central switch or hub. The switch and hub act as a relay though they have significant differences in the manner in which they handle network traffic – receiving and directing packets of data across the LAN. Thus, Ethernet networking creates a communications system that allows sharing of data and resources including printers, fax machines and scanners.



*Figure 50 Ethernet Communication*

## 2. Wireless Ethernet

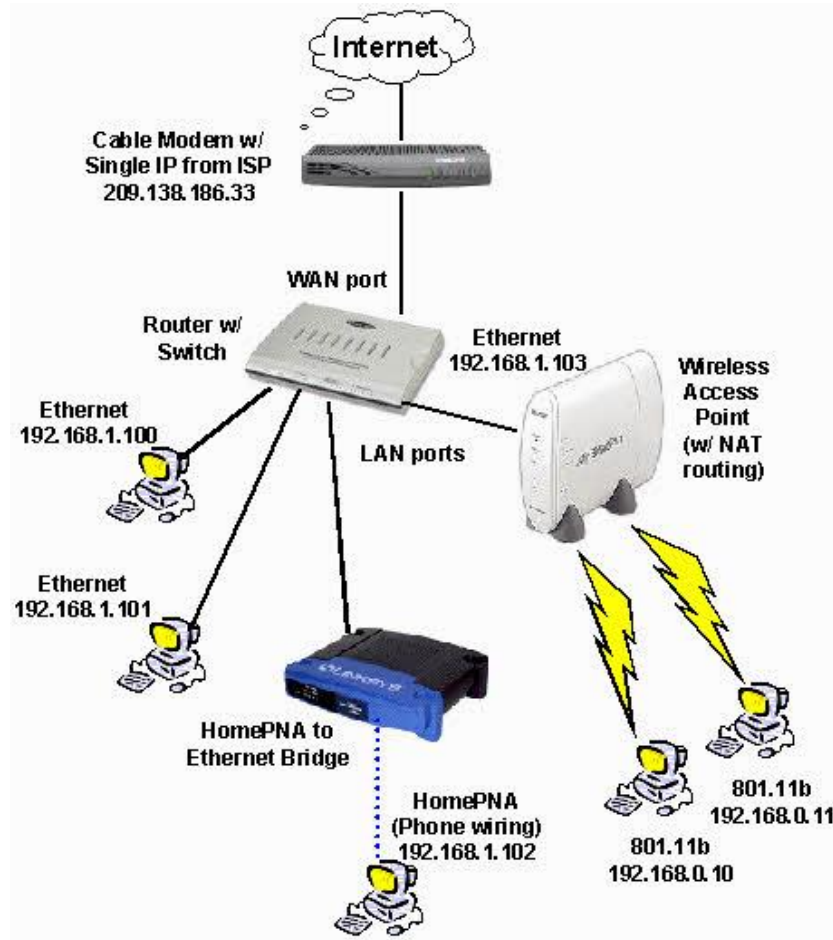


Figure 51 Wireless Network

Ethernet networks can also be wireless. Rather than using Ethernet cable to connect the computers, **wireless NICs use radio waves for two-way communication with a wireless switch or hub.** It consists of Ethernet ports, wireless NICs, switches and hubs. Wireless network technology can be more flexible to use, but also require extra care in configuring security.

### Types of Ethernet Networks

There are several types of Ethernet networks, such as Fast Ethernet, Gigabit Ethernet, and Switch Ethernet. A network is a group of two or more computer systems connected together.

#### 1. Fast Ethernet

The **fast Ethernet is a type of Ethernet network that can transfer data at a rate of 100 Mbps using a twisted-pair cable or a fiber-optic cable.** The older 10 Mbps Ethernet is still used, but such networks do not provide necessary bandwidth for some network-based video applications.

Fast Ethernet is based on the proven CSMA/CD Media Access Control (MAC) protocol, and **uses existing 10BaseT cabling.** Data can move from 10 Mbps to 100 Mbps without any protocol translation or changes to the application and networking software.

### What is Ethernet Port Speed?

When compare to a 10 Mb port, a 100 Mb port is theoretically 10 times faster than the standard port. Therefore, with a 100 Mb port more information can stream to and from your server. This will be of great help to you if you really need to explore very high speed, but not if you are under DDOS attack because you will find yourself running out of traffic allocation very fast.



*Figure 52 100Mbit/s Ethernet port*

If you are doing standard web hosting, the bigger 100 Mbps pipe will not offer true benefit to you because you may not even use more than 1 mbps at any given time. If you are hosting games or streaming media, then the bigger pipe of 100 Mbps would indeed be helpful to you.

With a 10 mbps pipe, you can transfer up to 1.25 Mbps, while a 100 mbps pipe, would allow you to transfer up to 12.5 Mbps.

However, if you leave your server unattended and running at full steam, a 10 Mbps pipe can consume about 3,240 GB a month and a 100 Mbps pipe can consume up to 32,400 GB a month. It would be really disgusting when you receive your bill.

## 2. Gigabit Ethernet

The Gigabit Ethernet is a type of Ethernet network capable of transferring data **at a rate of 1000 Mbps based on a twisted-pair or fiber optic cable**, and it is very popular. The type of twisted-pair cables that support Gigabit Ethernet is **Cat 5e cable**, where all the four pairs of twisted wires of the cable are used to achieve high data transfer rates. The 10 Gigabit Ethernet is a latest generation Ethernet capable of transferring data at a rate of 10 Gbps using twisted-pair or fiber optic cable.

### 3. Switch Ethernet

Multiple network devices in a LAN require network equipment's such as a network switch or hub. When using a network switch, a regular network cable is used instead of a crossover cable. The crossover cable consists of a transmission pair at one end and a receiving pair at the other end. The main function of a network switch is to forward data from one device to another device on the same network. Thus a network switch performs this task efficiently as the data is transferred from one device to another without affecting other devices on the same network.

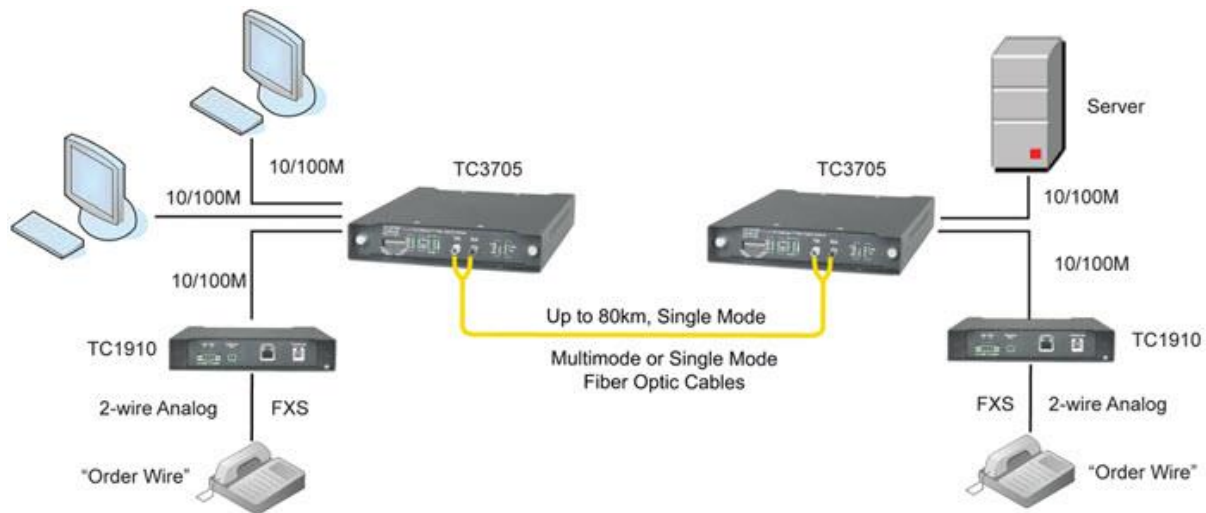


Figure 53 Switch Ethernet

The network switch normally supports different data transfer rates. The most common data transfer rates include 10 Mbps – 100 Mbps for fast Ethernet, and 1000 Mbps – 10 Gbps for the latest Ethernet.

Switch Ethernet uses star topology, which is organized around a switch. The switch in a network uses a filtering and switching mechanism similar to the one used by the gateways, in which these techniques have been in use for a long time.

### 802.4 Token Bus

Token Bus was a 4 Mbps Local Area Networking technology created by IBM to connect their terminals to IBM mainframes. Token bus utilized a copper coaxial cable to connect multiple end stations (terminals, workstations, shared printers etc.) to the mainframe. The coaxial cable served as a common communication bus and a token was created by the Token Bus protocol to manage or 'arbitrate' access to the bus. Any station that holds the token packet has permission to transmit data. The station releases the token when it is done communicating or when a higher priority device needs to transmit (such as the mainframe). This keeps two or more devices from transmitting information on the bus at the same time and accidentally destroying the transmitted data.

Token Bus suffered from two limitations. Any failure in the bus caused all the devices beyond the failure to be unable to communicate with the rest of the network. Second, adding more stations to the bus was somewhat difficult. Any new station that was improperly attached was unlikely to be able to communicate and all devices beyond it were also affected. Thus, token bus networks were seen as somewhat unreliable and difficult to expand and upgrade.

### **802.5 Token Ring**

Token Ring was created by IBM to compete with what became known as the DIX Standard of Ethernet (DEC/Intel/Xerox) and to improve upon their previous Token Bus technology. Up until that time, IBM had produced solutions that started from the mainframe and ran all the way to the desktop (or dumb terminal), allowing them to extend their SNA protocol from the AS400's all the way down to the end user. Mainframes were so expensive that many large corporations that purchased a mainframe as far back as 30-40 years ago are still using these mainframe devices, so Token Ring is still out there and you will encounter it. Token Ring is also still in use where high reliability and redundancy are important--such as in large military craft.

Token Ring comes in standard 4 and 16 Mbps and high-speed Token Ring at 100Mbps (IEEE 802.5t) and 1Gbps (IEEE 802.5v). Many mainframes (and until recently, ALL IBM mainframes) used a Front End Processor (FEP) with either a Line Interface Coupler (LIC) at 56kbps, or a Token-ring Interface Coupler (TIC) at 16 Mbps. Cisco still produces FEP cards for their routers (as of 2004).

Token Ring uses a ring based topology and passes a token around the network to control access to the network wiring. This token passing scheme makes conflicts in accessing the wire unlikely and therefore total throughput is as high as typical Ethernet and Fast Ethernet networks. The Token Ring protocol also provides features for allowing delay-sensitive traffic, to share the network with other data, which is key to a mainframe's operation. This feature is not available in any other LAN protocol, except Asynchronous Transfer Mode (ATM).

Token Ring does come with a higher price tag because token ring hardware is more complex and more expensive to manufacture. As a network technology, token ring is passing out of use because it has a maximum speed of 16 Mbps which is slow by today's gigabit Ethernet standards.

### **FDDI**

- Fiber Distributed Data Interface
- Similar to Token ring in the sense that it share some features such as topology(ring) and Media access technique (token-passing)
- High performance Fiber Optic token ring running at 100 mbps over distance 200 KM and permits up to 1000 stations
- FDDI deals with network reliable issues as mission-critical applications were implemented on high speed networks. It is frequently used as a backbone technology, and to connect high speed computer on LAN
- Based on two counter-rotating fiber rings, only one used at a time and next is for backup.
- So if there is any problem in one ring, next ring works automatically

- It allows 16 to 48 bits address and maximum frame size is 4500 bytes
- It prefers multimode fiber optic cable rather than single mode as multimode reduces cost for high data transmission
- It prefers LEDs instead of Laser for light source not only for cheaper but also to remove Accidental chances at user end connector (if user open connector and sees cable by naked eye, eye may damage on laser light)
- It operates at low error (1 bit error for  $2.5 \times 10^{10}$ )
- It uses 4B/5B encoding in place of Manchester encoding in Token Ring Computer Networks
- It capture token before transmitting and does not wait for acknowledgement to regenerate token as ring might be very long and may occurs much delay to wait for ACK.
- In normal operation, the token and frames travel only on the primary ring in a single direction. The second ring transmits idle signals in the opposite direction
- If a cable or device becomes disabled, the primary ring raps back around onto the secondary ring
- Stations may be directly connected to FDDI dual ring or attached to FDDI concentrator.
- There are three types of nodes:
  - DAS (Dual attachment station)
  - SAS (Single attachment station)
  - DAC (Dual attachment concentrator)
- FDDI deploys following timers:
  - Token holding time: upper limit on how long a station can hold token
  - Token Rotation time: how long it takes the token to traverse the ring or the interval between two successive arrivals of the token
- There are four specifications in FDDI.
  - Media Access control- deals with how medium is accessed, frame format, token handling, addressing, fair and equal access of the ring through the use of the timed token, guarantee bandwidth for special traffic etc.
  - Physical layer protocol-deals with data encoding/decoding procedures, establish clock synchronization, data recovery from incoming signal etc.
  - Physical layer medium- defines characteristics of transmission medium, fiber optic link type: single mode, multimode; power levels, bit error rates, optical components:
    - Connectors, switches, LEDs, Pin etc.
    - Station Management- defines FDDI station configuration, ring configuration, ring
    - Control features, station insertion and removal, initialization etc.

### X.25:

- The X.25 protocol, adopted as a standard by the Consultative Committee for International Telegraph and Telephone (CCITT), is a commonly-used network protocol.

- The X.25 protocol allows computers on different public networks (such as CompuServe, Tymnet, or a TCP/IP network) to communicate through an intermediary computer at the network layer level. - X.25's protocols correspond closely to the data-link and physical-layer protocols defined in the Open Systems - Interconnection (OSI) communication model.

- Three levels:

- ❖ **Physical Layer:**

Physical interface between and attached station (computer terminal and Packet Switching mode).

- ❖ **Link Level:**

- Provides reliable transfer of data across physical link.
- It is referred as Link Access protocol - Balanced (LAPB).

- ❖ **Packet Level:**

- Provides virtual circuit service
- Enables any subscriber to the network to setup logical conditions

**Following are the key features of X.25:**

- call control packets, used for setting up and cleaning virtual circuits are carried on same channel and same virtual circuit as data packets
- multiplexing of virtual circuits takes place at layer 3
- Both layer 2 and layer 3 include flow control and error control mechanisms

**Frame Relay:**

- More efficient transmission scheme than X.25
- Call control signaling is carried on separate logical connection from user data.
- Intermediate nodes need not to maintain state tables or process messages relating to call control
- Multiplexing and switching of logical connections takes place at layer 2 instead of layer 3, eliminating one entire layer of processing.
- There is no hop-by hop flow control and error control. End to end flow control and error control are the responsibility of a higher layer, if they are employed at all.
- Frame relay used access speed up to 2Mbps Frame relay service at even higher data rates are now available
- Frame Relay is designed to provide efficient transmission than X.25.
- The X.25 approach results in considerable overhead at each hop through the network.

- The data link control protocol involves the exchange of a data frame and acknowledgement frame.
- At each intermediate node, state tables must be maintained for each virtual circuit to deal with cost management and flow/error, control aspects of X.25 protocol.
- All these overheads may be justified when there is significant probability of error in any of the links in the network.
- Today's network employee reliable digital transmission technology over high quality reliable digital transmission technology over high quality reliable transmission links such as optical fiber.
- In this environment, the overhead of X.25 is not only unnecessary but degrades the effective utilization of the available high data rates.
- Frame Relay is designed to eliminate much of the overhead that X.25 imposes on end user systems.

### ATM:

- Asynchronous Transfer Mode
- It is a streamlined packet transfer interface.
- ATM makes use of a fixed size packets called cells.
- The use of fixed size and fixed formats results an efficient scheme for transmission over high speed networks.
- Data rate ranges from 25.6 Mbps to 622.08 Mbps
- Physical layer specifies transmission medium and signal encoding scheme
- ATM layer defines transmission of data in fixed size cells and defines the use of logical connection.
- ATM adaptation layer maps higher layer information into ATM cells to be transported over an ATM network.
- User plane provides user information into ATM cells to be transported over an ATM network
- User plane provides user information transfer (eg. flow control, error control)
- Control plane provides call control and connection control functions.
- Management plane performs coordination between all the planes and layers management.

## CHAPTER 5

### Network / Internet Layer Protocols and Addressing

The network layer is the third layer (from bottom) in the OSI Model. The network layer is concerned with the delivery of a packet across multiple networks. The network layer is considered the backbone of the OSI Model. It selects and manages the best logical path for data transfer between nodes. This layer contains hardware devices such as routers, bridges, firewalls, and switches, but it actually creates a logical image of the most efficient communication route and implements it with a physical medium. Network layer protocols exist in every host or router. The router examines the header fields of all the IP packets that pass through it. Internet Protocol and Netware IPX/SPX are the most common protocols associated with the network layer.

In the OSI model, the network layer responds to requests from the layer above it (transport layer) and issues requests to the layer below it (data link layer).

Responsibilities of Network Layer:

- **Packet forwarding/Routing of packets:** Relaying of data packets from one network segment to another by nodes in a computer network
- **Connectionless communication (IP):** A data transmission method used in packet-switched networks in which each data unit is separately addressed and routed based on information carried by it
- **Fragmentation of data packets:** Splitting of data packets that are too large to be transmitted on the network

### Logical Addressing

In computing, a logical address is the address at which an item (memory cell, storage element and network host) appears to reside from the perspective of an executing application program. A logical address may be different from the physical address due to the operation of an address translator or mapping function. Such mapping functions may be, in the case of a computer memory architecture, a memory management unit (MMU) between the CPU and the memory bus, or an address translation layer, e.g., the Data Link Layer, between the hardware and the internetworking protocols (Internet Protocol) in a computer networking system.

### IPv4

- IPv4 is a connectionless protocol used for packet switched networks. It operates on a best effort delivery model, in which neither delivery is guaranteed, nor is proper sequencing or avoidance of duplicate delivery assured. Internet Protocol Version 4 (IPv4) is the fourth revision of the Internet Protocol and a widely used protocol in data communication over different kinds of networks. IPv4 is a connectionless protocol used in packet-switched layer networks, such as Ethernet. It provides the logical connection between network devices by providing identification for each device. There are many ways to configure IPv4 with all kinds of devices – including manual and automatic configurations – depending on the network type.
- IPv4 is defined and specified in IETF publication RFC 791.

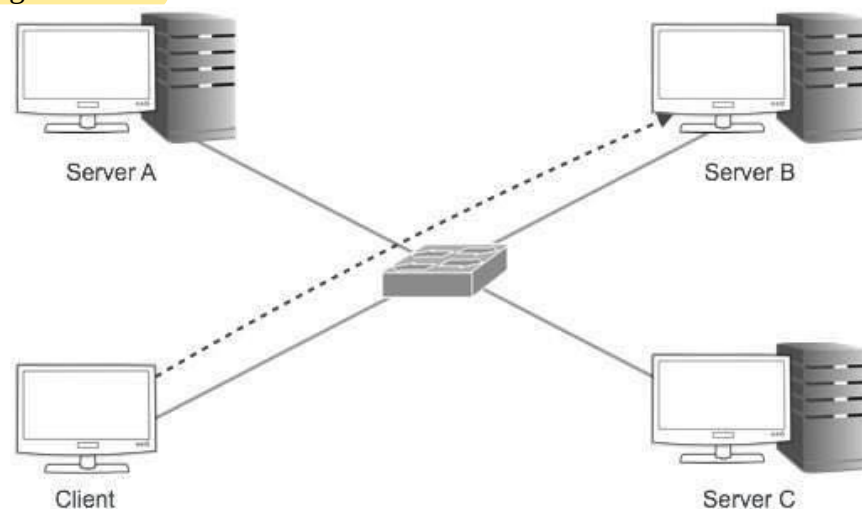
- IPv4 uses 32-bit addresses for Ethernet communication in five classes: A, B, C, D and E. Classes A, B and C have a different bit length for addressing the network host. Class D addresses are reserved for military purposes, while class E addresses are reserved for future use.
- IPv4 uses 32-bit (4 byte) addressing, which gives  $2^{32}$  (4,294,967,296 i.e. nearly 4 billion) IP address are possible in IPv4. IPv4 addresses are written in the dot-decimal notation, which comprises of four octets of the address expressed individually in decimal and separated by periods, for instance, e.g. 192.168.1.5.
- 32 bits of IP address is divided into network and host portion.



- IPv4 supported address types:
  - Unicast Address
  - Multicast Address
  - Broadcast Address

➤ **Unicast Addressing Mode:**

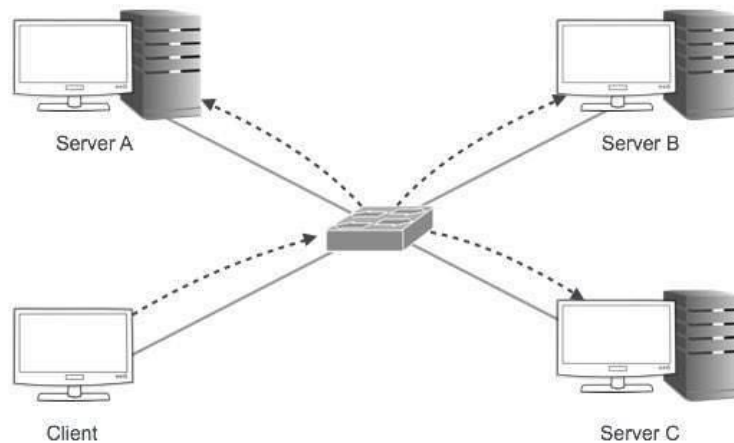
In this mode, data is sent only to one destined host. The Destination Address field contains 32-bit IP address of the destination host. Here the client sends data to the targeted server:



*Figure 54 Unicast Addressing*

➤ **Broadcast Addressing Mode:**

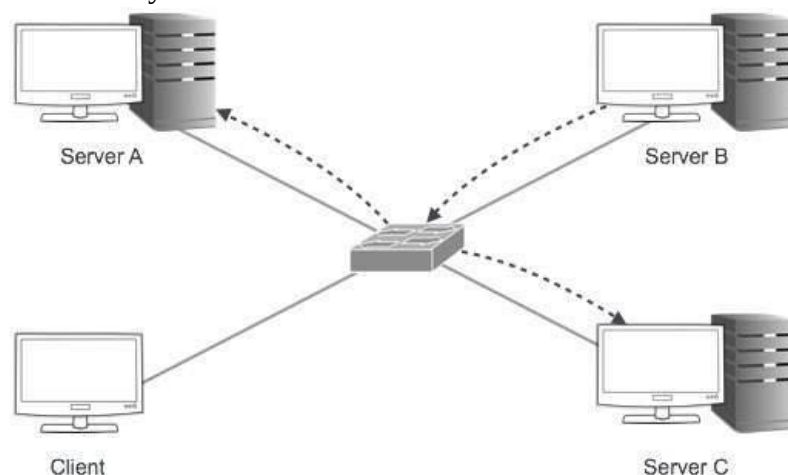
In this mode, the packet is addressed to all the hosts in a network segment. The Destination Address field contains a special broadcast address, i.e. 255.255.255.255. When a host sees this packet on the network, it is bound to process it. Here the client sends a packet, which is entertained by all the Servers:



*Figure 55 Broadcasting Addressing*

➤ **Multicast Addressing Mode:**

This mode is a mix of the previous two modes, i.e. the packet sent is **neither destined to a single host nor all the hosts on the segment**. In this packet, the **Destination Address contains a special address which starts with 224.x.x.x** and can be entertained by more than one host.



*Figure 56 Multicast Addressing*

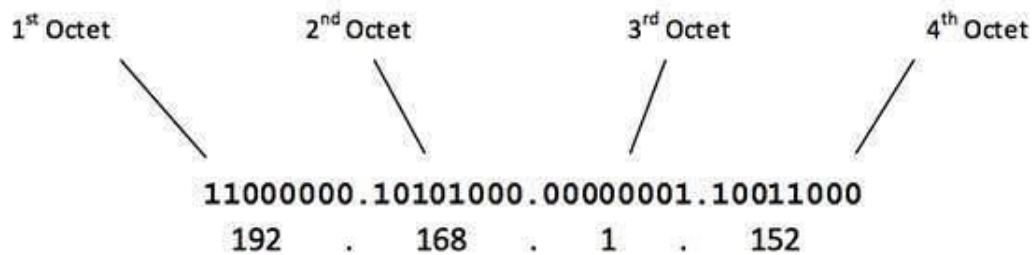
Here a server sends packets which are entertained by more than one servers. Every network has one IP address reserved for the Network Number which represents the network and one IP address reserved for the Broadcast Address, which represents all the hosts in that network.

➤ **IPv4 - Address Classes**

Internet Protocol hierarchy contains several classes of IP Addresses to be used efficiently in various situations as per the requirement of hosts per network. Broadly, the IPv4 Addressing system is divided into five classes of IP Addresses. **All the five classes are identified by the first octet of IP Address.**

**Internet Corporation for Assigned Names and Numbers** is responsible for assigning IP addresses.

The first octet referred here is the left most of all. The octets numbered as follows depicting dotted decimal notation of IP Address:



The number of networks and the number of hosts per class can be derived by this formula:

$$\text{Number of networks} = 2^{\text{networks\_bits}}$$

$$\text{Number of Hosts/Network} = 2^{\text{host\_bits}} - 2$$

When calculating hosts' IP addresses, 2 IP addresses are decreased because they cannot be assigned to hosts, i.e. the first IP of a network is network number and the last IP is reserved for Broadcast IP.

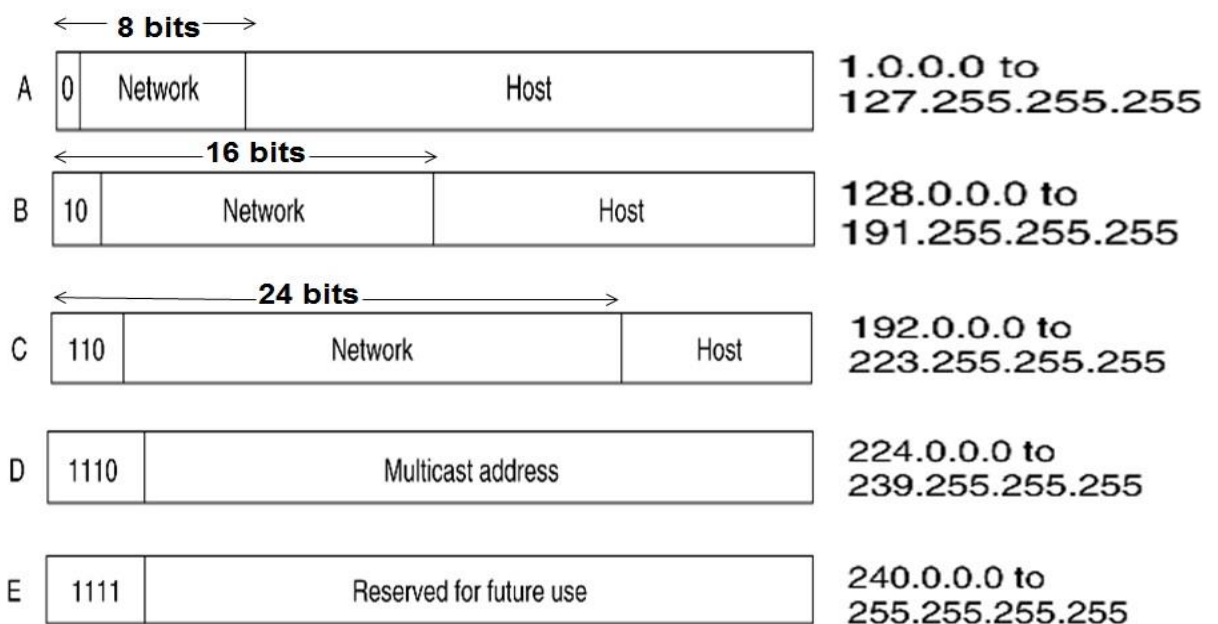


Figure 57 An Overview of IPv4 address structure

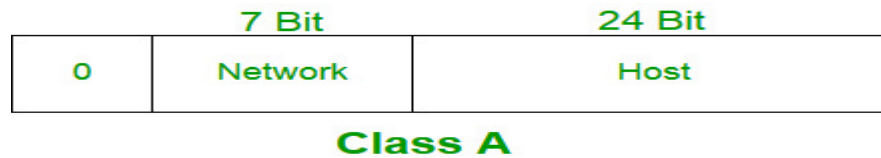
### Class A:

IP address belonging to class A are assigned to the networks that contain a large number of hosts.

- The network ID is 8 bits long.
- The host ID is 24 bits long.

The higher order bits of the first octet in class A is always set to 0. The remaining 7 bits in first octet are used to determine network ID. The 24 bits of host ID are used to determine the host in any network. The default sub-net mask for class A is 255.x.x.x. Therefore, class A has a total of:

- $2^7 = 128$  network ID
  - $2^{24} - 2 = 16,777,214$  host ID
- IP addresses belonging to class A ranges from 1.x.x.x – 126.x.x.x

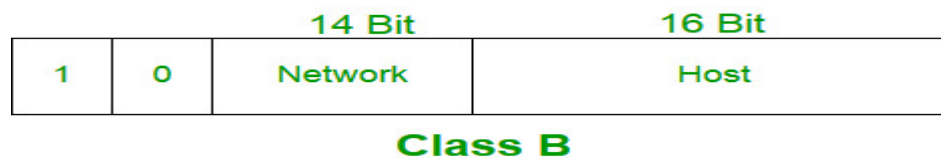
**Class B:**

IP address belonging to class B are assigned to the networks that ranges from medium-sized to large-sized networks.

- The network ID is 16 bits long.
- The host ID is 16 bits long.

The higher order bits of the first octet of IP addresses of class B are always set to 10. The remaining 14 bits are used to determine network ID. The 16 bits of host ID is used to determine the host in any network. The default sub-net mask for class B is 255.255.x.x. Class B has a total of:

- $2^{14} = 16384$  network address
  - $2^{16} - 2 = 65534$  host address
- IP addresses belonging to class B ranges from 128.0.x.x – 191.255.x.x.

**Class C:**

IP address belonging to class C are assigned to small-sized networks.

- The network ID is 24 bits long.
- The host ID is 8 bits long.

The higher order bits of the first octet of IP addresses of class C are always set to 110. The remaining 21 bits are used to determine network ID. The 8 bits of host ID is used to determine the host in any network. The default sub-net mask for class C is 255.255.255.x. Class C has a total of:

- $2^{21} = 2097152$  network address
  - $2^8 - 2 = 254$  host address
- IP addresses belonging to class C ranges from 192.0.0.x – 223.255.255.x.

**Class D:**

IP address belonging to class D are reserved for multi-casting. The higher order bits of the first octet of IP addresses belonging to class D are always set to 1110. The remaining bits are for the address that interested hosts recognize. Class D does not possess any sub-net mask. IP addresses belonging to class D ranges from 224.0.0.0 – 239.255.255.255.



#### Class E:

IP addresses belonging to class E are reserved for experimental and research purposes. IP addresses of class E ranges from 240.0.0.0 – 255.255.255.254. This class doesn't have any sub-net mask. The higher order bits of first octet of class E are always set to 1111.



#### Range of special IP addresses:

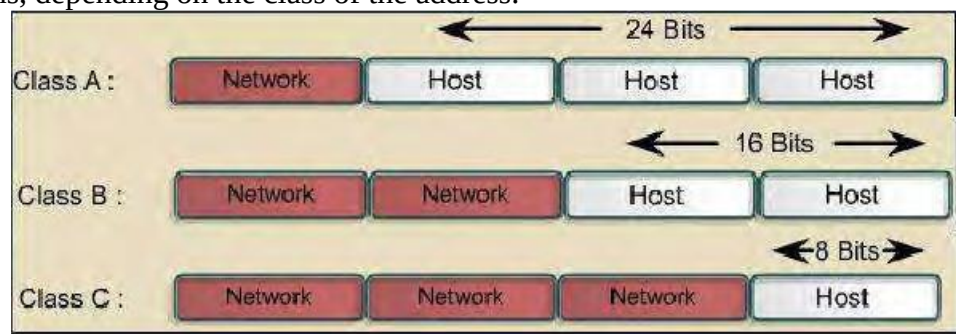
**169.254.0.0 – 169.254.0.16:** Link local addresses

**127.0.0.0 – 127.0.0.8:** Loop-back addresses

**0.0.0.0 – 0.0.0.8:** used to communicate within the current network.

#### ➤ NETID AND HOSTID

All IP addresses have a network and host portion. Classful addressing divides an IP address into the Network and Host portions along octet boundaries. These are represented as: net id or network id and host id or network number and host number. In classful addressing, an IP address in class A, B, or C is divided into netid and hostid are of varying lengths, depending on the class of the address.



*Figure shows some netid and hostid bytes.*

Note that the concept does not apply to classes D and E. In class A, one byte defines the netid and three bytes define the hostid. In class B, two bytes define the netid and two bytes define the hostid. In class C, three bytes define the netid and one byte defines the hostid. In class A, one byte defines the netid and three bytes define the hostid. In class B, two bytes define the netid and two bytes define the hostid. In class C, three bytes define the netid and one byte defines the hostid.

#### ➤ Mask

Although the length of the netid and hostid (in bits) is predetermined in classful addressing, we can also use a mask (also called the default mask), a 32-bit number made of contiguous 1s and followed by contiguous 0s. The masks for classes A, B, and C are shown in the below table and this concept does not apply to classes D and E.

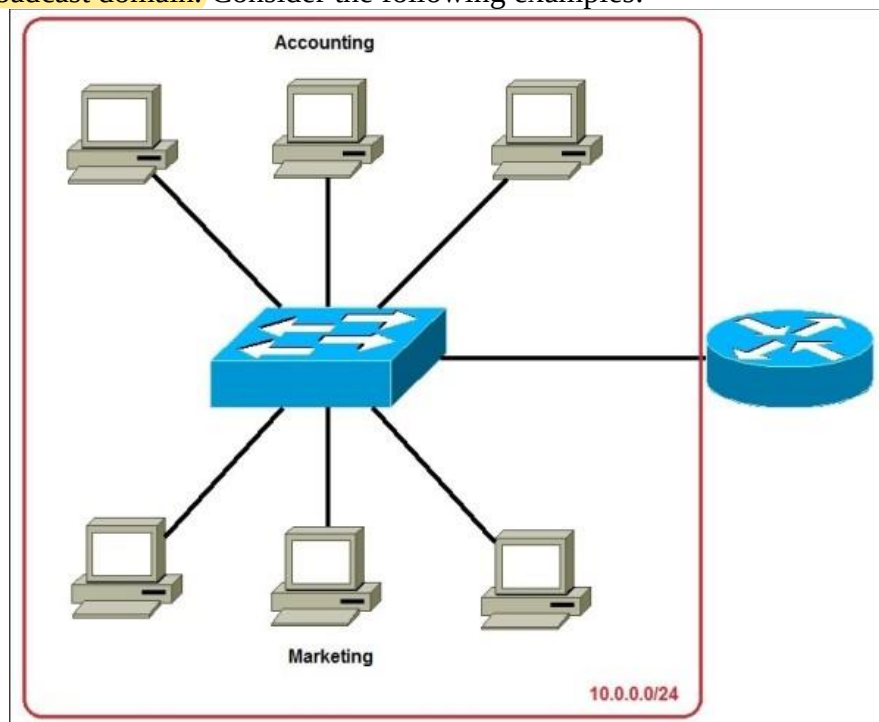
*Default masks for classful addressing*

| class | Binary                                 | Dotted-decimal | CIDR |
|-------|----------------------------------------|----------------|------|
| A     | 11111111 00000000 00000000<br>00000000 | 255.0.0.0      | /8   |
| B     | 11111111 11111111 00000000<br>00000000 | 255.255.0.0    | /16  |
| C     | 11111111 11111111 11111111<br>00000000 | 255.255.255.0  | /24  |

The mask can help us to find the netid and the hostid. For example, the mask for a class A address has eight 1s, which means the first 8 bits of any address in class A define the netid; the next 24 bits define the hostid. Similarly for class B and class C. The last column of Table above shows the mask in the form “/n” where n can be 8, 16, or 24 in classful addressing. This notation is also called **slash notation** or **Classless Interdomain Routing (CIDR)** notation. The notation is used in classless addressing can also be applied to classful addressing, Where classful addressing is a special case of classless addressing.

### ➤ Subnetting

Subnetting is the practice of dividing a network into two or more smaller networks. It increases routing efficiency, enhances the security of the network and reduces the size of the broadcast domain. Consider the following examples:



In the picture above we have one huge network: 10.0.0.0/24. All hosts on the network are in the same subnet, which has following disadvantages:

- **A Single Broadcast Domain** – all hosts are in the same broadcast domain. A broadcast sent by any device on the network will be processed by all hosts.
- **Network Security** – each device can reach any other device on the subnet, which can present security problems. For example, a server containing sensitive information would be in the same network as an ordinary end user workstation.
- **Organizational Problems** – in a large networks, different departments are usually grouped into different subnets. For example, you can group all devices from the Accounting department in the same subnet and then give access to sensitive financial data only to hosts from that subnet.

➤ **Subnetting**

Subnetting is the strategy used to partition a single physical network into more than one smaller logical sub-network (subnets). During the era of classful addressing, subnetting was introduced. Subnets are designed by accepting bits from the IP address's host part and using these bits to assign a number of smaller sub-networks inside the original network. Subnetting allows an organization to add sub-networks without the need to acquire a new network number via the Internet service provider (ISP). Subnets were initially designed for solving the shortage of IP addresses over the Internet. Each IP address consists of a subnet mask. All the class types, such as Class A, Class B and Class C include the subnet mask known as the default subnet mask. The subnet mask determines the type and number of IP addresses required for a given local network. The firewall or router is called the default gateway. The default subnet mask is as follows as in the above table:

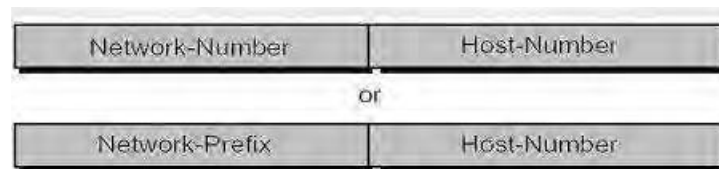
- Class A: 255.0.0.0
- Class B: 255.255.0.0
- Class C: 255.255.255.0

The subnetting process allows the administrator to divide a single Class A, Class B, or Class C network number into smaller portions. The subnets can be subnetted again into sub-subnets.

Dividing the network into a number of subnets provides the following benefits:

- Reduces the network traffic by reducing the volume of broadcasts
- Helps to surpass the constraints in a local area network (LAN), for example, the maximum number of permitted hosts.
- Enables users to access a work network from their homes; there is no need to open the complete network.

If an organization was granted a large block in class A or B, it could divide the addresses into several contiguous groups and assign each group to smaller networks (called subnets) or, in rare cases, share part of the addresses with neighbors. Subnetting increases the number of 1s in the mask.



This level is now known as Prefix and Host where all hosts (individual computers) in the network share the prefix (subnet mask).

### ➤ Subnet mask

- A subnet mask is a 32-bit number used to differentiate the network component of an IP address by dividing the IP address into a network address and host address. Like the IP address, a subnet mask is written using the "dotted-decimal" notation.
- Subnet masks are used to design subnetworks, or subnets, that connect local networks. It determines both the number and size of subnets where the size of a subnet is the number of hosts that can be addressed. we can create a subnet mask by taking the 32-bit value of an existing IP address, choosing how many subnets you want to create or alternatively, how many nodes you need on each subnet, and then setting all subsequent network bits to "1" and host bits to "0". The resulting 32-bit value is our subnet mask.
- In any given network, two host addresses are always reserved for special purposes. The "0" address becomes the network address or network identification and the "255" address is assigned as a broadcast address. Note that these cannot be assigned to a host.
- An IP address is divided into two parts: network and host parts. For example, an IP class A address consists of 8 bits identifying the network and 24 bits identifying the host. This is because the default subnet mask for a class A IP address is 8 bits long (or, written in dotted decimal notation, 255.0.0.0). What does it mean? Well, like an IP address, a subnet mask also consists of 32 bits. Computers use it to determine the network part and the host part of an address. The 1s in the subnet mask represent a network part, the 0s a host part.
- Computers works only with bits. The math used to determine a network range is binary AND.

| INPUT 1 | INPUT 2 | OUTPUT |
|---------|---------|--------|
| 0       | 0       | 0      |
| 0       | 1       | 0      |
| 1       | 0       | 0      |
| 1       | 1       | 1      |

- Let's say that we have the IP address of 10.0.0.1 with the default subnet mask of 8 bits (255.0.0.0). First, we need to convert the IP address to binary:
  - IP address: 10.0.0.1 = 00001010.00000000.00000000.00000001
  - Subnet mask 255.0.0.0 = 11111111.00000000.00000000.00000000
- Computers then use the AND operation to determine the network number:

00001010.00000000.00000000.00000001 = 10.0.0.1

11111111.00000000.00000000.00000000 = 255.0.0.0

-----  
00001010.00000000.00000000.00000000 = 10.0.0.0

- The computer can then determine the size of the network. Only IP addresses that begins with 10 will be in the same network. So, in this case, the range of addresses in this network is 10.0.0.0 – 10.255.255.255.

### Example

There are a couple of ways to create subnets. In this article we will subnet a class C address **192.168.0.0** that, by default, has **24** subnet bits and **8** host bits.

Before we start Subnetting, we have to ask ourselves these two questions:

#### **1. How many subnets do we need?**

$2^x$  = number of subnets.  $x$  is the number of 1s in the subnet mask. With 1 subnet bit, we can have  $2^1$  or 2 subnets. With 2 bits,  $2^2$  or 4 subnets, with 3 bits,  $2^3$  or 8 subnets, etc.

#### **2. How many hosts per subnet do we need?**

$2^y - 2$  = number of hosts per subnet.  $y$  is the number of 0s in the subnet mask.

### **Subnetting example**

An example will help you understand the Subnetting concept. Let's say that we need to subnet a class C address **192.168.0.0/24**. We need two subnets with **50** hosts per subnet. Here is our calculation:

1. Since we need only two subnets, we need  $2^1$  subnet bits. In our case, this means that we will take one bit from the host part. Here is the calculation:

First, we have a class C address 192.168.0.0 with the subnet mask of 24. Let's convert them to binary:

**192.168.0.0 = 11000000.10101000.00000000.00000000**

**255.255.255.0 = 11111111.11111111.11111111.00000000**

We need to take over a single zero from the host part of the subnet mask. Here is our new subnet mask:

**255.255.255.128 = 11111111.11111111.11111111.10000000**

Remember, the ones in the subnet mask represent the network.

2. We need 50 hosts per subnet. Since we took one bit from the host part, we are left with seven bits for the hosts. Is it enough for 50 hosts? The formula to

calculate the number of hosts is  $2^y - 2$ , with y representing the number of host bits. Since  $2^7 - 2$  is 126, we have more than enough bits for our hosts.

3. Our network will look like this:

**192.168.0.0/25** – the first subnet has the subnet number of **192.168.0.0**. The range of IP addresses in this subnet is **192.168.0.0 – 192.168.0.127**.

**192.168.0.128/25** – the second subnet has the subnet number of **192.168.0.128**. The range of IP addresses in this subnet is **192.168.0.128 – 192.168.0.255**.

**For example:** The subnet mask is the network address plus the bits reserved for identifying the subnetwork. Given

**Full address is:** 10010110.11010111.00010001.00001001

The Class B network part is: 10010110.11010111

The host address is: 00010001.00001001

If this network is divided into 14 subnets, then the first 4 bits of the host address (0001) are reserved for identifying the subnet.

The bits for the network address are all set to 1. In this case, therefore, the subnet mask would be 11111111.11111111.11110000.00000000. It's called a mask because it can be used to identify the subnet to which an IP address belongs. On performing bitwise AND operation on the mask and IP resulted sub network address i.e. 150.215.16.0

|                            |                                     |
|----------------------------|-------------------------------------|
| Subnet Mask: 255.255.240.0 | 11111111.11111111.11110000.00000000 |
|----------------------------|-------------------------------------|

|                          |                                     |
|--------------------------|-------------------------------------|
| IP address: 150.215.17.9 | 10010110.11010111.00010001.00001001 |
|--------------------------|-------------------------------------|

|                              |                                     |
|------------------------------|-------------------------------------|
| Subnet address: 150.215.16.0 | 10010110.11010111.00010000.00000000 |
|------------------------------|-------------------------------------|

➤ **CIDR (Classless inter-domain routing)** is a method of public IP address assignment. It was introduced in 1993 by Internet Engineering Task Force with the following goals:

- to deal with the IPv4 address exhaustion problem
- to slow down the growth of routing tables on Internet routers

Before CIDR, public IP addresses were assigned based on the class boundaries:

- Class A – the classful subnet mask is /8. The number of possible IP addresses is 16,777,216 ( $2$  to the power of 24).
- Class B – the classful subnet mask is /16. The number of addresses is 65,536
- Class C – the classful subnet mask is /24. Only 256 addresses available.

Some organizations were known to have gotten an entire Class A public IP address (for example, IBM got all the addresses in the 9.0.0.0/8 range). Since these addresses can't

be assigned to other companies, there was a shortage of available IPv4 addresses. Also, since IBM probably didn't need more than 16 million IP addresses, a lot of addresses were unused. To combat this, the classful network scheme of allocating the IP address was abandoned. The new system was classless – a classful network was split into multiple smaller networks. For example, if a company needs 12 public IP addresses, it would get something like this: 190.5.4.16/28.

The number of usable IP addresses can be calculated with the following formula:

### 2 to the power of host bits – 2

In the example above, the company got 14 usable IP addresses from the 190.5.4.16 – 190.5.4.32 range because there are 4 host bits and 2 to the power of 4 minus 2 is 14. The first and the last address are the network address and the broadcast address, respectively. All other addresses inside the range could be assigned to Internet hosts.

### ➤ VLSM - Variable Length Subnet Masking

This is a specialized form of CIDR, in which you take a classful network and then you subnet it such that each subnet would have different number of hosts in it resulting in different masks, but number of hosts in all subnets when added equal to the total number of hosts in the original classful network.

### ➤ Supernetting

Supernetting is combining several small networks (e.g. of class C) into a big one to create a large range of addresses. In Supernetting, the first address of the super net and the super net mask define the range of addresses.

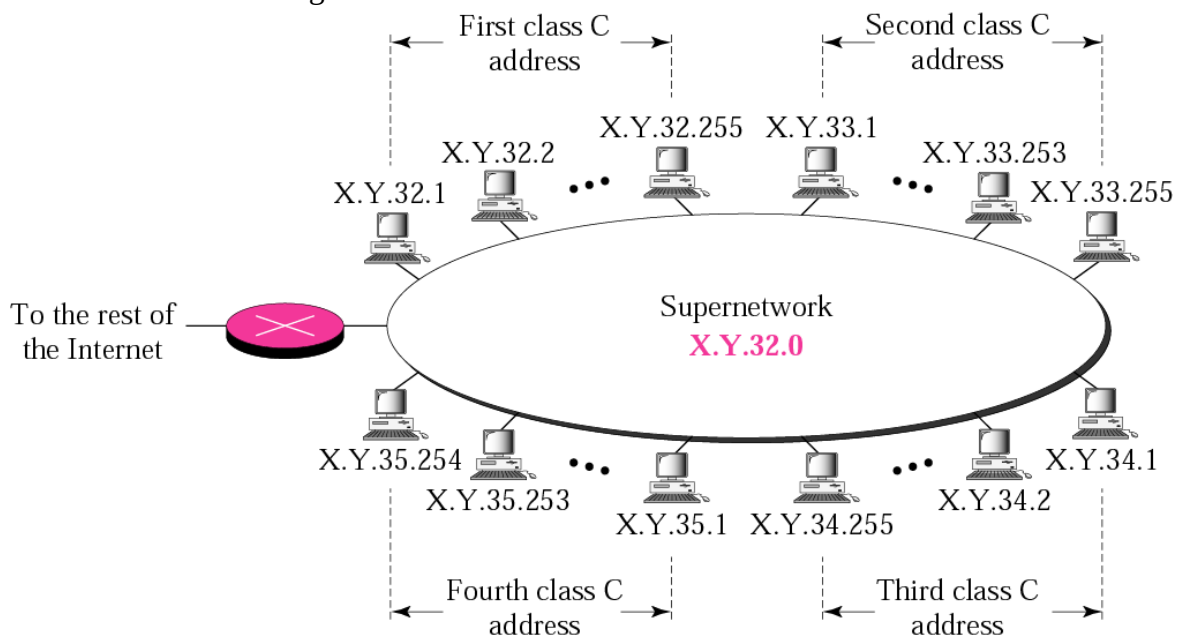


Figure 58 Supernetting

### For Example

A block of addresses is granted to a small organization. We know that one of the addresses is 205.16.37.39/28. What is the first address in the block?



a problem if you want to access the Internet. Each device in the network (in this case, this includes the Internet) must have a unique IP address. If two networks are using the same private addresses, it would run into reach ability issues. To access the Internet, our source IP addresses must have a unique Internet public address. This can be accomplished through address translation.

Here is a list of private addresses.

- Class A: 10.0.0.0 - 10.255.255.255 (1 Class A network)
- Class B: 172.16.0.0 - 172.31.255.255 (16 Class B networks)
- Class C: 192.168.0.0 - 192.168.255.255 (256 Class C networks)

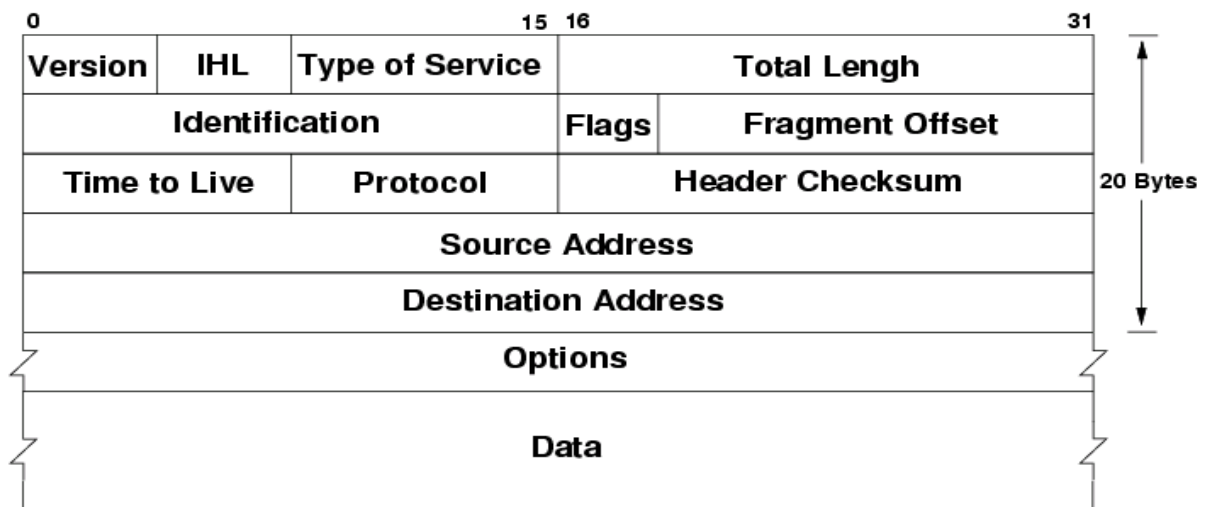
➤ IPv4 Datagram Header

Figure 60: IPv4 Datagram Header

- **VERSION:** Version of the IP protocol (4 bits), which is 4 for IPv4
- **HLEN:** IP header length (4 bits), which is the number of 32 bit words in the header. The minimum value for this field is 5 and the maximum is 15
- **Type of service:** Low Delay, High Throughput, Reliability (8 bits)
- **Total Length:** Length of header + Data (16 bits), which has a minimum value 20 bytes and the maximum is 65,535 bytes
- **Identification:** Unique Packet Id for identifying the group of fragments of a single IP datagram (16 bits)
- **Flags:** 3 flags of 1 bit each : reserved bit (must be zero), do not fragment flag, more fragments flag (same order)
- **Fragment Offset:** Specified in terms of number of 8 bytes, which has the maximum value of 65,528 bytes
- **Time to live:** Datagram's lifetime (8 bits), It prevents the datagram to loop through the network
- **Protocol:** Name of the protocol to which the data is to be passed (8 bits)
- **Header Checksum:** 16 bits header checksum for checking errors in the datagram header
- **Source IP address:** 32 bits IP address of the sender
- **Destination IP address:** 32 bits IP address of the receiver
- **Option:** Optional information such as source route, record route.

➤ IPv6

The wonder of IPv6 lies in its header. An IPv6 address is 4 times larger than IPv4, but surprisingly, the header of an IPv6 address is only 2 times larger than that of IPv4. IPv6 headers have one Fixed Header and zero or more Optional (Extension) Headers. All the necessary information that is essential for a router is kept in the Fixed Header. The

Extension Header contains optional information that helps routers to understand how to handle a packet/flow.

128-bit address is divided along 16-bit boundaries, and each 16-bit block is converted to a 4-digit hexadecimal number and separated by colons (Colon-Hex Notation)

FEDC:BA98:7654:3210: FEDC:BA98:7654:3

3FFE:085B:1F1F:0000:0000:0000:00A9:1234

8 groups of 16-bit hexadecimal numbers separated by ":" Leading zeros can be removed

3FFE:85B:1F1F::A9:1234

:: = all zeros in one or more group of 16-bit hexadecimal numbers

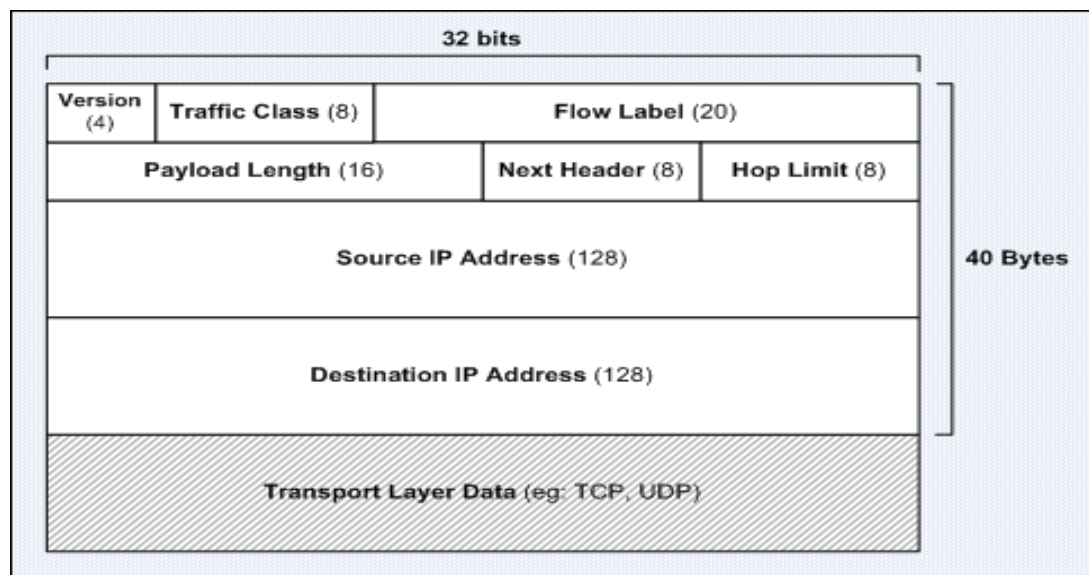


Figure 61: IPv6 Datagram Header

- **Version:** The size of the Version field is 4 bits. The Version field shows the version of IP and is set to 6.
- **Traffic Class:** The size of Traffic Class field is 8 bits. Traffic Class field is similar to the IPv4 Type of Service (ToS) field. The Traffic Class field indicates the IPv6 packet's class or priority.
- **Flow Label:** The size of Flow Label field is 20 bits. The Flow Label field provide additional support for real-time datagram delivery and quality of service features. The purpose of Flow Label field is to indicate that this packet belongs to a specific sequence of packets between a source and destination and can be used to prioritized delivery of packets for services like voice. The flow label is a 3-byte (24-bit) field that is designed to provide special handling for a particular flow of data.
- **Payload Length:** The size of the Payload Length field is 16 bits. The Payload Length field shows the length of the IPv6 payload, including the extension headers and the upper layer protocol data. **Next Header:** The size of the Next Header field is 8 bits. The Next Header field shows either the type of the first extension (if any extension header is available) or the protocol in the upper layer such as TCP, UDP, or ICMPv6.

- **Hop Limit:** The size of the Hop Limit field is 8 bits. The Hop Limit field shows the maximum number of routers the IPv6 packet can travel. This Hop Limit field is similar to IPv4 Time to Live (TTL) field. This field is typically used by distance vector routing protocols, like Routing Information Protocol (RIP) to prevent layer 3 loops (routing loops).
- **Source Address:** The size of the Source Address field is 128 bits. The Source Address field shows the IPv6 address of the source of the packet.
- **Destination Address:** The size of the Destination Address field is 128 bits. The Destination Address field shows the IPv6 address of the destination of the packet.

### ➤ The Benefits of IPv6

While increasing the pool of addresses is one of the most often-talked about benefit of IPv6, there are other important technological changes in IPv6 that will improve the IP protocol:

- No more NAT (Network Address Translation)
- Auto-configuration
- No more private address collisions
- Better multicast routing
- Simpler header format
- Simplified, more efficient routing
- True quality of service (QoS), also called "flow labeling"
- Built-in authentication and privacy support
- Flexible options and extensions
- Easier administration (say good-bye to DHCP)

### ➤ Differences Between IPv4 and IPv6

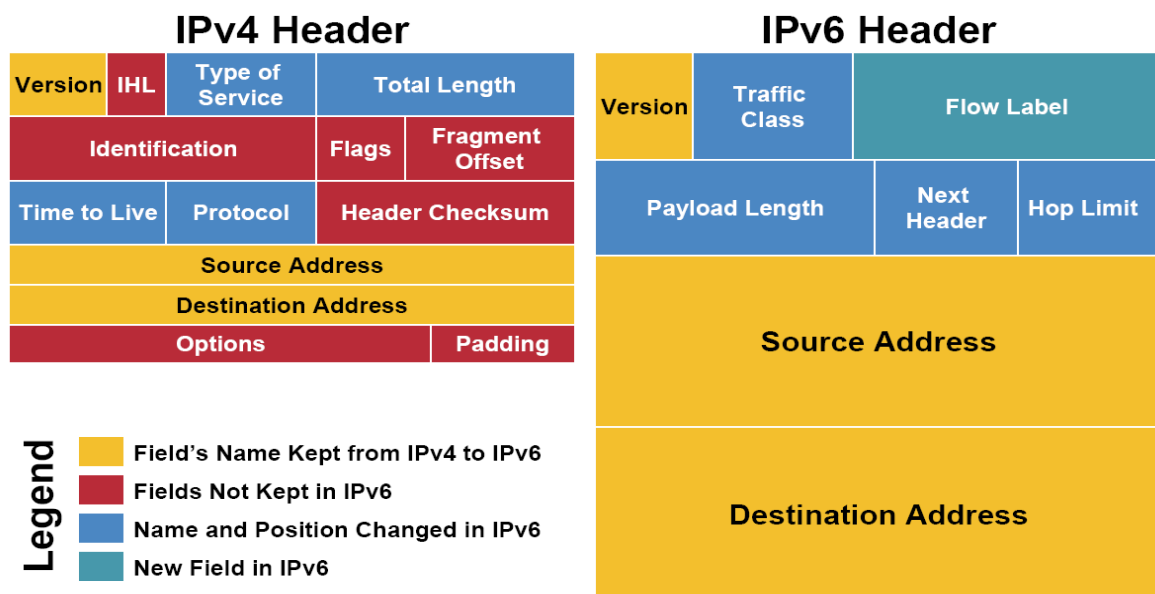


Figure 62 Header Differences

| IPv4                                                                                                                                                                      | IPv6                                                                                                                                                                                                         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>IPv4 addresses are 32 bit length.</li> </ul>                                                                                       | <ul style="list-style-type: none"> <li>IPv6 addresses are 128 bit length.</li> </ul>                                                                                                                         |
| <ul style="list-style-type: none"> <li>IP v4 addresses are binary numbers represented in decimals</li> </ul>                                                              | <ul style="list-style-type: none"> <li>IPv6 addresses are binary numbers represented in hexadecimals.</li> </ul>                                                                                             |
| <ul style="list-style-type: none"> <li>IPsec support is only optional</li> </ul>                                                                                          | <ul style="list-style-type: none"> <li>Inbuilt IPsec support</li> </ul>                                                                                                                                      |
| <ul style="list-style-type: none"> <li>Fragmentation is done by sender and forwarding routers.</li> </ul>                                                                 | <ul style="list-style-type: none"> <li>Fragmentation is done only by sender.</li> </ul>                                                                                                                      |
| <ul style="list-style-type: none"> <li>No packet flow identification.</li> </ul>                                                                                          | <ul style="list-style-type: none"> <li>Packet flow identification is available within the IPv6 header using the Flow Label field.</li> </ul>                                                                 |
| <ul style="list-style-type: none"> <li>Checksum field is available in IPv4 header</li> </ul>                                                                              | <ul style="list-style-type: none"> <li>No checksum field in IPv6 header.</li> </ul>                                                                                                                          |
| <ul style="list-style-type: none"> <li>Options fields are available in IPv4 header.</li> </ul>                                                                            | <ul style="list-style-type: none"> <li>No option fields, but IPv6 Extension headers are available.</li> </ul>                                                                                                |
| <ul style="list-style-type: none"> <li>Address Resolution Protocol (ARP) is available to map IPv4 addresses to MAC addresses.</li> </ul>                                  | <ul style="list-style-type: none"> <li>Address Resolution Protocol (ARP) is replaced with a function of Neighbor Discovery Protocol (NDP).</li> </ul>                                                        |
| <ul style="list-style-type: none"> <li>Internet Group Management Protocol (IGMP) is used to manage multicast group membership.</li> </ul>                                 | <ul style="list-style-type: none"> <li>IGMP is replaced with Multicast Listener Discovery (MLD) messages.</li> </ul>                                                                                         |
| <ul style="list-style-type: none"> <li>Broadcast messages are available.</li> </ul>                                                                                       | <ul style="list-style-type: none"> <li>Broadcast messages are not available. Instead a link-local scope "All nodes" multicast IPv6 address (FF02::1) is used for broadcast similar functionality.</li> </ul> |
| <ul style="list-style-type: none"> <li>Manual configuration (Static) of IPv4 addresses or DHCP (Dynamic configuration) is required to configure IPv4 addresses</li> </ul> | <ul style="list-style-type: none"> <li>Auto-configuration of addresses is available.</li> </ul>                                                                                                              |

- Transition mechanisms**

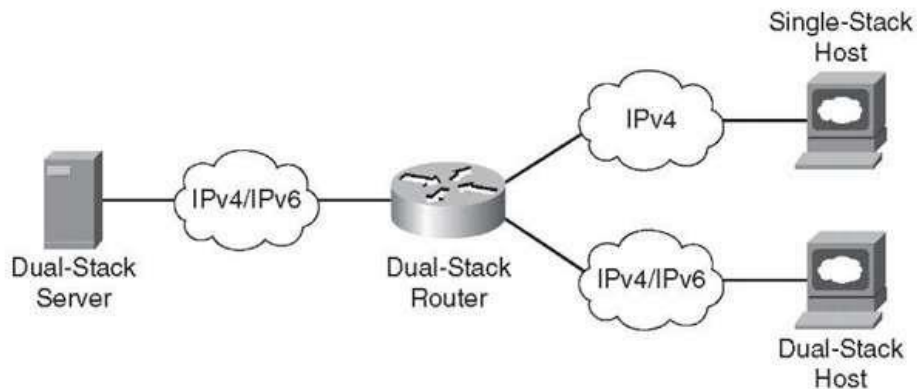
The transition from IPv4 to IPv6 is expected to take years, and in the meantime, both protocols will have to coexist and interoperate. For this to happen IETF has developed various tools that come to help the network administrator's transition to IPv6. There are three categories of migration techniques:

- Dual Stack:** Both IPv4 and IPv6 will run simultaneously on devices in the network, allowing them to coexist in the ISP network
  - Tunneling:** An IPv6 packet is encapsulated in IPv4 packet and send over an IPv4 network.
  - Translation:** A similar technique to NAT for IPv4 is used. Using NAT64 (Network Address Translation64), the IPv6 packet is translated to IPv4 packet.

### 1. End to end Dual Stack

Represents a major project for an ISP and it takes from 2 to 5 years to implement. The starting point of change is the core of the network, which is easy for most network operators, meaning a few months of work.

The advantage of dual-stack is that it makes available to use devices that support only one IP protocol or both, allowing older network services to still be used. On the other hand, the costs for implementation are very high and very few organizations can change from IPv4 to IPv6.



*Figure 63 End to end Dual Stack*

### 2. Tunneling (Generic routing Encapsulation)

Allows the use of IPv4 networks to carry IPv6 traffic and its basic principle as per the figure. This can be done either in a manual or in an automatic way. The manual configuration requires definite specification of the IPv4/IPv6 source and the tunnel IPv4/IPv6 destination. When the number of tunnels grows, administrating this technique becomes a major drawback.

The main advantage of using the tunneling technique is that it uses the existing infrastructure of the ISPs and it meets their standards in terms of administration and costs.

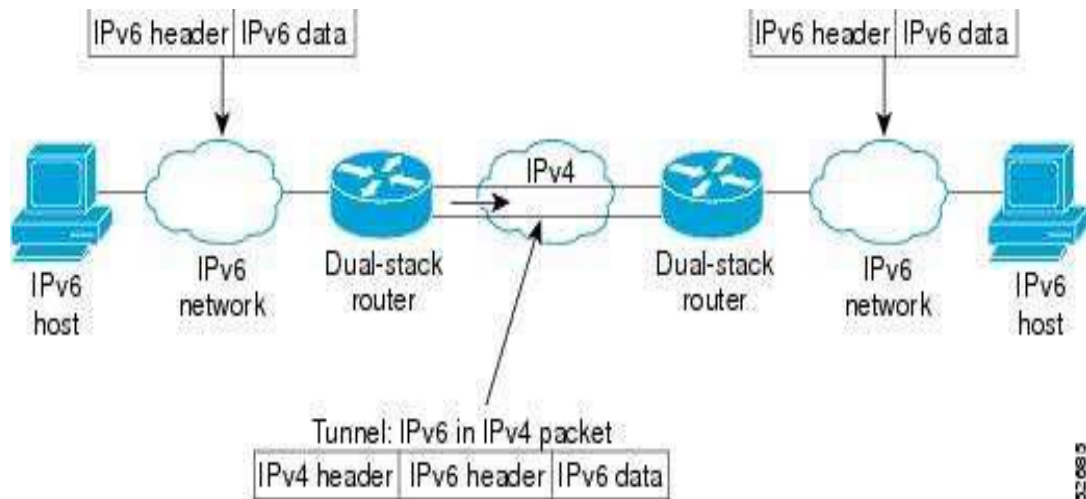


Figure 64 Tunneling

### 3. Translation

It is used to achieve direct communication between IPv4 and IPv6. The new protocol supports translation from IPv4 header to IPv6 format. When an IPv4 host tries to communicate with an IPv6 server, a NAT-PT (NAT –Protocol Translation) enabled device removes the IPv4 header of the packet, adds an IPv6 header and then sends it through to the server. When the reply comes it does the other way around.

The algorithm for all translation methods is known as Stateless IP/ICMP Translator (SIIT). For an ISP, translation is not seen as a viable solution because of NAT use with IPv4.

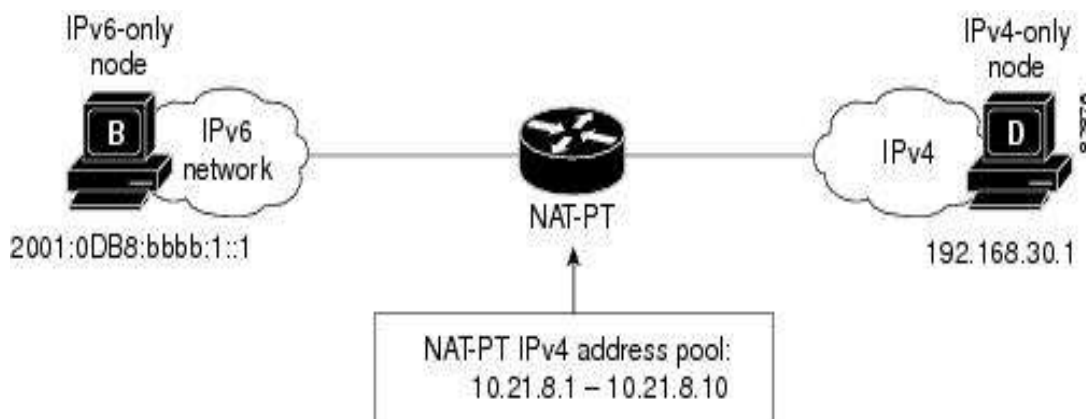


Figure 65 Translation

### Routing

Routing is the process of forwarding of a packet in a network so that it reaches its intended destination. The main goals of routing are

1. **Correctness:** The routing should be done properly and correctly so that the packets may reach their proper destination.

2. **Simplicity:** The routing should be done in a simple manner so that the overhead is as low as possible. With increasing complexity of the routing algorithms the overhead also increases.
3. **Robustness:** The algorithms designed for routing should be robust enough to handle hardware and software failures and should be able to cope with changes in the topology and traffic without requiring all jobs in all hosts to be aborted. And the network rebooted every time when some router goes down.
4. **Stability:** The routing algorithms should be stable under all possible circumstances.
5. **Fairness:** Every node connected to the network should get a fair chance of transmitting their packets. This is generally done on a first come first serve basis.
6. **Optimality:** The routing algorithms should be optimal in terms of throughput and minimizing mean packet delays.

**There are 3 types of routing:**

**1. Static routing –**

Static routing is a process in which we have to manually add routes in routing table.

**Advantages**

- No routing overhead for router CPU which means a cheaper router can be used to do routing.
- It adds security because only administrator can allow routing to particular networks only.
- No bandwidth usage between routers.

**Disadvantage**

- For a large network, it is a hectic task for administrator to manually add each route for the network in the routing table on each router.
- The administrator should have good knowledge of the topology. If a new administrator comes, then he has to manually add each route so he should have very good knowledge of the routes of the topology.

**2. Default Routing**

This is the method where the router is configured to send all packets towards a single router (next hop). It doesn't matter to which network the packet belongs, it is forwarded out to router which is configured for default routing. It is generally used with stub routers. A stub router is a router which has only one route to reach all other networks.

**3. Dynamic Routing**

Dynamic routing makes automatic adjustment of the routes according to the current state of the route in the routing table. Dynamic routing uses protocols to discover network

destinations and the routes to reach it. RIP and OSPF are the best examples of dynamic routing protocol. Automatic adjustment will be made to reach the network destination if one route goes down.

A dynamic protocol have following features:

1. The routers should have the same dynamic protocol running in order to exchange routes.
2. When a router finds a change in the topology then router advertises it to all other routers.

#### Advantages

- Easy to configure.
- More effective at selecting the best route to a destination remote network and also for discovering remote network.

#### Disadvantage

- Consumes more bandwidth for communicating with other neighbors.
- Less secure than static routing.

### Classification of Routing Algorithms

The routing algorithms may be classified as follows:

1. **Adaptive Routing Algorithm:** These algorithms change their routing decisions to reflect changes in the topology and in traffic as well. These get their routing information from adjacent routers or from all routers. The optimization parameters are the **distance, number of hops and estimated transit time**. This can be further classified as follows:
  - i. **Centralized:** In this type some central node in the network gets entire information about the network topology, about the traffic and about other nodes. This then transmits this information to the respective routers. The advantage of this is that only one node is required to keep the information. The disadvantage is that if the central node goes down the entire network is down, i.e. single point of failure.
  - ii. **Isolated:** In this method the node decides the routing without seeking information from other nodes. The sending node does not know about the status of a particular link. The disadvantage is that the packet may be send through a congested route resulting in a delay. Some examples of this type of algorithm for routing are:
    - **Hot Potato:** When a packet comes to a node, it tries to get rid of it as fast as it can, by putting it on the shortest output queue without regard to where that link leads.
    - **Backward Learning:** In this method the routing tables at each node gets modified by information from the incoming packets. One way to implement backward learning is to include the identity of the source node in each packet, together with a hop counter that is incremented on each hop. When a node receives a packet in a particular line, it notes down the number of hops it has taken to reach it from the source node. If the previous value of hop count stored in the node is better than the current one then nothing is done but if the current value is

better than the value is updated for future use. The problem with this is that when the best route goes down then it cannot recall the second best route to a particular node. Hence all the nodes have to forget the stored information's periodically and start all over again.

iii. **Distributed:** In this the node receives information from its neighboring nodes and then takes the decision about which way to send the packet. The disadvantage is that if in between the interval it receives information and sends the packet something changes then the packet may be delayed.

2. **Non-Adaptive Routing Algorithm:** These algorithms do not base their routing decisions on measurements and estimates of the current traffic and topology. Instead the route to be taken in going from one node to the other is computed in advance, off-line, and downloaded to the routers when the network is booted. This is also known as static routing. This can be further classified as:

i. **Flooding:** Flooding adapts the technique in which every incoming packet is sent on every outgoing line except the one on which it arrived. One problem with this method is that packets may go in a loop. As a result of this a node may receive several copies of a particular packet which is undesirable. Some techniques adapted to overcome these problems are as follows:

- **Sequence Numbers:** Every packet is given a sequence number. When a node receives the packet it sees its source address and sequence number. If the node finds that it has sent the same packet earlier then it will not transmit the packet and will just discard it.
- **Hop Count:** Every packet has a hop count associated with it. This is decremented (or incremented) by one by each node which sees it. When the hop count becomes zero (or a maximum possible value) the packet is dropped.
- **Spanning Tree:** The packet is sent only on those links that lead to the destination by constructing a spanning tree routed at the source. This avoids loops in transmission but is possible only when all the intermediate nodes have knowledge of the network topology.

Flooding is not practical for general kinds of applications. But in cases where high degree of robustness is desired such as in military applications, flooding is of great help.

ii. **Random Walk:** In this method a packet is sent by the node to one of its neighbors randomly. This algorithm is highly robust. When the network is highly interconnected, this algorithm has the property of making an alternative routes. It is usually implemented by sending the packet onto the least queued link.

## Distance Vector Routing

A **distance-vector routing protocol** in data networks determines the best route for data packets based on distance. Distance-vector routing protocols measure the distance by the number of routers a packet has to pass, one router counts as one hop.

Some distance-vector protocols also take into account network latency and other factors that influence traffic on a given route. To determine the best route across a network routers on which a distance-vector protocol is implemented exchange information with one another, usually routing tables plus hop counts for destination networks and possibly other traffic information. Distance-vector routing protocols also require that a router informs its neighbors of network topology changes periodically. Historically known as the old ARPANET routing algorithm (or known as Bellman-Ford algorithm).

**Bellman Ford Basics** – Each router maintains a Distance Vector table containing the distance between itself and ALL possible destination nodes. Distances, based on a chosen metric, are computed using information from the neighbors' distance vectors.

- Information kept by DV router -
  - Each router has an ID
  - Associated with each link connected to a router, there is a link cost (static or dynamic).
  - Intermediate hops
- Distance Vector Table Initialization -
  - Distance to itself = 0
  - Distance to ALL other routers = infinity number.

#### **Distance Vector Algorithm –**

- A router transmits its distance vector to each of its neighbors in a routing packet.
- Each router receives and saves the most recently received distance vector from each of its neighbors.
- A router recalculates its distance vector when:
  - **It receives a distance vector from a neighbor containing different information than before.**
  - **It discovers that a link to a neighbor has gone down.**
- From time-to-time, each node sends its own distance vector estimate to neighbors.

#### **Advantages**

- It is simpler to configure and maintain than link state routing.

#### **Disadvantages**

- It is slower to converge than link state.
- It is at risk from the count-to-infinity problem.
- It creates more traffic than link state since a hop count change must be propagated to all routers and processed on each router. Hop count updates take place on a periodic basis, even if there are no changes in the network topology, so bandwidth-wasting broadcasts still occur.

- For larger networks, distance vector routing results in larger routing tables than link state since each router must know about all other routers. This can also lead to congestion on WAN links.

### Link State Routing Protocols

Link state protocols are also called shortest-path-first protocols. Link state routing protocols have a complete picture of the network topology. Hence they know more about the whole network than any distance vector protocol.

Three separate tables are created on each link state routing enabled router. One table is used to hold details about directly connected neighbors, one is used to hold the topology of the entire internetwork and the last one is used to hold the actual routing table.

Link state protocols send information about directly connected links to all the routers in the network. Examples of Link state routing protocols include OSPF - Open Shortest Path First and IS-IS - Intermediate System to Intermediate System.

There are also routing protocols that are considered to be hybrid in the sense that they use aspects of both distance vector and link state protocols. EIGRP - Enhanced Interior Gateway Routing Protocol is one of those hybrid routing protocols.

### Advantages

- Link-state protocols use cost metrics to choose paths through the network. The cost metric reflects the capacity of the links on those paths.
- Link-state protocols use triggered updates and LSA floods to immediately report changes in the network topology to all routers in the network. This leads to fast convergence times.
- Each router has a complete and synchronized picture of the network. Therefore, it is very difficult for routing loops to occur.
- Routers use the latest information to make the best routing decisions.
- The link-state database sizes can be minimized with careful network design. This leads to smaller Dijkstra calculations and faster convergence.
- Every router, at the very least, maps the topology of its own area of the network. This attribute helps to troubleshoot problems that can occur.
- Link-state protocols support CIDR and VLSM.

### Disadvantages

- They require more memory and processor power than distance vector protocols. This makes it expensive to use for organizations with small budgets and legacy hardware.
- They require strict hierarchical network design, so that a network can be broken into smaller areas to reduce the size of the topology tables.
- They require an administrator who understands the protocols well.

- They flood the network with LSAs during the initial discovery process. This process can significantly decrease the capability of the network to transport data. It can noticeably degrade the network performance.

### Autonomous System

An Autonomous System (AS) is the unit of router policy, either a single network or a group of networks that is controlled by a common network administrator (or group of administrators) on behalf of a single administrative entity (such as a university, a business enterprise, or a business division). An autonomous system is also sometimes referred to as a routing domain. An autonomous system is assigned a globally unique number, sometimes called an Autonomous System Number (ASN).

Networks within an autonomous system communicate routing information to each other using an Interior Gateway Protocol (IGP). An autonomous system shares routing information with other autonomous systems using the Border Gateway Protocol (BGP). Previously, the Exterior Gateway Protocol (EGP) was used. In the future, the BGP is expected to be replaced with the OSI Inter-Domain Routing Protocol (IDRP).

### Interior Gateway Protocols

Interior Gateway Protocols (IGPs) handle routing within an Autonomous System (one routing domain). In plain English, IGP's figure out how to get from place to place between the routers you own. These dynamic routing protocols keep track of paths used to move data from one end system to another inside a network or set of networks that you administrate (all of the networks you manage combined are usually just one Autonomous System). IGP's are how you get all the networks communicating with each other.

IGP's fall into two categories:

- Distance Vector Protocols
  - Routing Information Protocol (RIP)
  - Interior Gateway Routing Protocol (IGRP)
- Link State Protocols
  - Open Shortest Path First (OSPF)
  - Intermediate System to Intermediate System (IS-IS)

### Exterior Gateway Protocols

To get from place to place outside your network(s), i.e. on the Internet, you must use an Exterior Gateway Protocol. Exterior Gateway Protocols handle routing outside an Autonomous System and get you from your network, through your Internet provider's network and onto any other network. BGP is used by companies with more than one

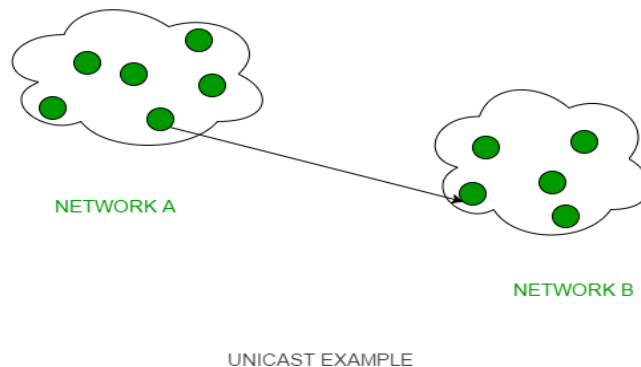
Internet provider to allow them to have redundancy and load balancing of their data transported to and from the Internet.

Examples of an EGP:

- Border Gateway Protocol (BGP)
- Exterior Gateway Protocol (Replaced by BGP)

## 1. Unicast

This type of information transfer is useful when there is a participation of single sender and single recipient. So, in short you can term it as a one-to-one transmission. For example, a device having IP address 10.1.2.0 in a network wants to send the traffic stream (data packets) to the device with IP address 20.12.4.2 in the other network, then unicast comes into picture. This is the most common form of data transfer over the networks.

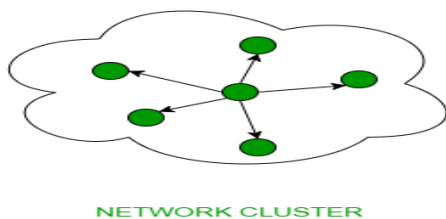


## 2. Broadcast

Broadcasting transfer (one-to-all) techniques can be classified into two types:

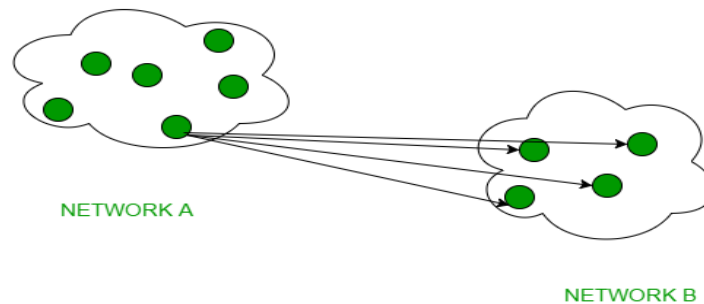
### ▪ Limited Broadcasting

Suppose you have to send stream of packets to all the devices over the network that you reside, this broadcasting comes handy. For this to achieve, it will append 255.255.255.255 (all the 32 bits of IP address set to 1) called as **Limited Broadcast Address** in the destination address of the datagram (packet) header which is reserved for information transfer to all the recipients from a single client (sender) over the network.



- **Direct Broadcasting**

This is useful when a device in one network wants to transfer packet stream to all the devices over the other network. This is achieved by translating all the Host ID part bits of the destination address to 1, referred as **Direct Broadcast Address** in the datagram header for information transfer.



This mode is mainly utilized by television networks for video and audio distribution. One important protocol of this class in Computer Networks is **Address Resolution Protocol (ARP)** that is used for resolving IP address into physical address which is necessary for underlying communication.

### 3. Multicast

In multicasting, one/more senders and one/more recipients participate in data transfer traffic. In this method traffic recline between the boundaries of unicast (one-to-one) and broadcast (one-to-all). Multicast lets server's direct single copies of data streams that are then simulated and routed to hosts that request it. IP multicast requires support of some other protocols like **IGMP (Internet Group Management Protocol)**, **Multicast routing** for its working. Also in Classful IP addressing **Class D** is reserved for multicast groups.

#### Routing Algorithm

##### 1. RIP

**Routing Information Protocol (RIP)** is a dynamic routing protocol which uses hop count as a routing metric to find the best path between the source and the destination network. It is a distance vector routing protocol which has AD value 120 and works on the application layer of OSI model. **RIP uses port number 520.**

##### **Hop Count:**

Hop count is the number of routers occurring in between the source and destination network. The path with the lowest hop count is considered as the best route to reach a network and therefore placed in the routing table. RIP prevents routing loops by limiting the number of hops allowed in a path from source and destination. The maximum hop count allowed for RIP is 15 and hop count of 16 is considered as network unreachable.

##### **Features of RIP:**

1. Updates of the network are exchanged periodically.
2. Updates (routing information) are always broadcast.
3. Full routing tables are sent in updates.

4. Routers always trust on routing information received from neighbor routers. This is also known as routing on rumors.

### Advantage

The biggest advantage of RIP is that it is simple to configure and implement. Stability in routing table. It is very easy to understand and configure. It is generally loop free. Conserve bandwidth, smaller routing updates sent & received. Minimized routing table and then faster lookup.

### Disadvantage

The main disadvantage of RIP is the inability to scale to large or very large networks. The maximum hop count used by RIP routers 15. Does not support discontinuous networks. Another disadvantage of RIP is its high recovery time. It is slow convergence in larger networks.

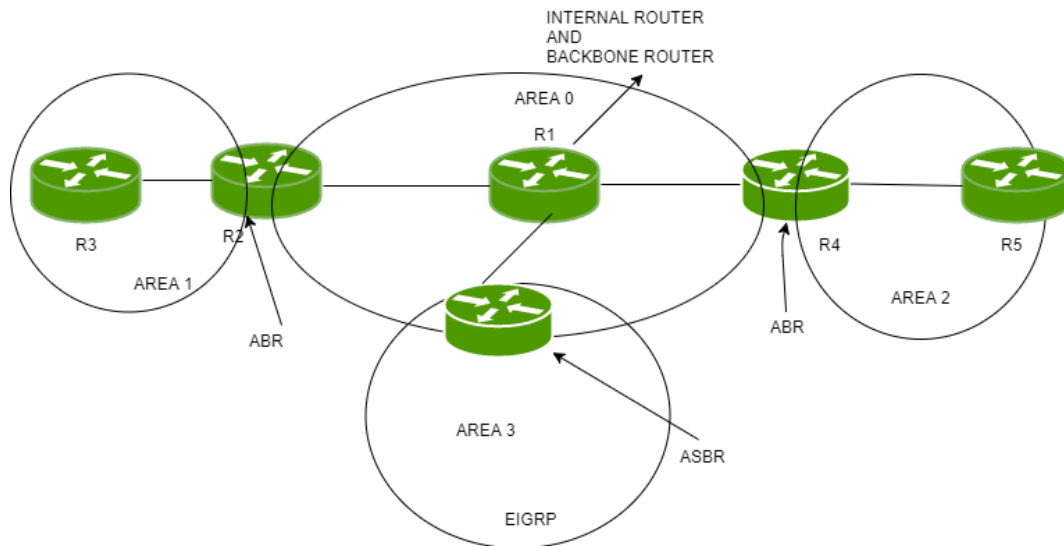
## 2. OSPF

**Open shortest path first (OSPF)** is a link-state routing protocol which is used to find the best path between the source and the destination router using its own SPF algorithm.

Open shortest path first (OSPF) router roles

An area is a group of contiguous network and routers. Routers belonging to same area shares a common topology table and area I'd. The area I'd is associated with router's interface as a router can belong to more than one area. There are some roles of router in OSPF:

1. **Backbone router** – The area 0 is known as backbone area and the routers in area 0 are known as backbone routers. If the routers exists partially in the area 0 then also it is a backbone router.
2. **Internal router** – An internal router is a router which have all of its interfaces in a single area.
3. **Area Boundary Router (ABR)** – The router which connects backbone area with another area is called Area Boundary Router. It belongs to more than one area. The ABRs therefore maintain multiple link-state databases that describe both the backbone topology and the topology of the other areas.
4. **Area Summary Border Router (ASBR)** – When an OSPF router is connected to a different protocol like EIGRP, or Border Gateway Protocol, or any other routing protocol then it is known as AS. The router which connects two different AS (in which one of the interface is operating OSPF) is known as Area Summary Border Router. These routers perform redistribution. ASBRs run both OSPF and another routing protocol, such as RIP or BGP. ASBRs advertise the exchanged external routing information throughout their AS.



### Features

- OSPF implements a two-layer hierarchy: the backbone (area 0) and areas off of the backbone (areas 1– 65,535)
- To provide scalability OSPF supports two important concepts: autonomous systems and areas.
- Synchronous serial links, no matter what the clock rate of the physical link is, the bandwidth always defaults to 1544 Kbps.
- OSPF uses cost as a metric, which is the inverse of the bandwidth of a link.

### Advantages

- It will run on most routers, since it is based on an open standard.
- It uses the SPF algorithm, developed by Dijkstra, to provide a loop-free topology.
- It provides fast convergence with triggered, incremental updates via Link State Advertisements (LSAs).
- It is a classless protocol and allows for a hierarchical design with VLSM and route summarization.

### Disadvantages

- It requires more memory to hold the adjacency (list of OSPF neighbors), topology and routing tables.
- It requires extra CPU processing to run the SPF algorithm
- It is complex to configure and more difficult to troubleshoot.

## 3. BGP

Border Gateway Protocol (BGP) is a routing protocol used to transfer data and information between different host gateways, the Internet or autonomous systems. BGP is a Path Vector Protocol (PVP), which maintains paths to different hosts, networks and gateway routers and determines the routing decision based on that. It does not use Interior Gateway Protocol (IGP) metrics for routing decisions, but only decides the route based on path, network policies and rule sets. Sometimes, BGP is described as a reachability protocol rather than a routing protocol.

BGP roles include:

- Because it is a PVP, BGP communicates the entire autonomous system/network path topology to other networks

- Maintains its routing table with topologies of all externally connected networks
- Supports classless inter domain routing (CIDR), which allocates Internet Protocol (IP) addresses to connected Internet devices

When used to facilitate communication between different autonomous systems, BGP is referred to as External BGP (EBGP). When used at host networks/autonomous systems, BGP is referred to as Internal BGP (IBGP). BGP was created to extend and replace Exterior Gateway Protocol (EGP).

BGP is a path vector protocol with the following properties:

- Reliable updates: BGP runs on top of TCP (port 179)
- Incremental, triggered updates only
- Periodic keep alive messages to verify TCP connectivity
- Rich metrics (called path vectors or attributes)
- Designed to scale to huge internetworks (for example, the Internet)

It has enhancements over distance vector protocols.

## Chapter 6

### Transport Layer and Protocols

Transport Layer is the fourth layer of TCP/IP model. It is an end-to-end layer used to deliver messages to a host. It is termed as end-to-end layer because it provides a point-to-point connection rather than hop-to-hop, between the source host and destination host to deliver the services reliably. The unit of data encapsulation in Transport Layer is a segment.

The standard protocols used by Transport Layer to enhance its functionalities are: TCP (Transmission Control Protocol), UDP (User Datagram Protocol), DCCP (Datagram Congestion Control Protocol) etc.

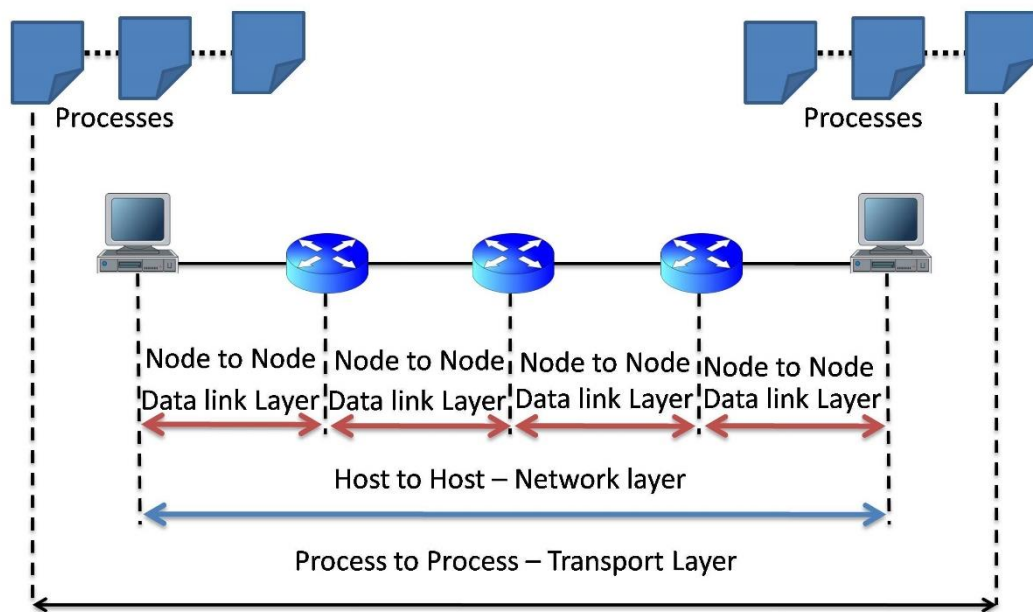
Various responsibilities of a Transport Layer

- **Process to process delivery** -While Data Link Layer requires the MAC address (48 bits address contained inside the Network Interface Card of every host machine) of source-destination hosts to correctly deliver a frame and Network layer requires the IP address for appropriate routing of packets, in a similar way Transport Layer requires a Port number to correctly deliver the segments of data to the correct process amongst the multiple processes running on a particular host. A port number is a 16 bit address used to identify any client-server program uniquely.
- **End-to-end Connection between hosts** –Transport layer is also responsible for creating the end-to-end Connection between hosts for which it mainly uses TCP and UDP. TCP is a secure, connection- orientated protocol which uses a handshake protocol to establish a robust connection between two end- hosts. TCP ensures reliable delivery of messages and is used in various applications. UDP on the other hand is a stateless and unreliable protocol which ensures best-effort delivery. It is suitable for the applications which have little concern with flow or error control and requires to send bulk of data like video conferencing. It is often used in multicasting protocols.
- **Multiplexing and Demultiplexing** –Multiplexing allows simultaneous use of different applications over a network which are running on a host. Transport layer provides this mechanism which enables us to send packet streams from various applications simultaneously over a network. Transport layer accepts these packets from different processes differentiated by their port numbers and passes them to network layer after adding proper headers. Similarly Demultiplexing is required at the receiver side to obtain the data coming from various processes. Transport receives the segments of data from network layer and delivers it to the appropriate process running on the receiver's machine.
- **Congestion Control** –Congestion is a situation in which too many sources over a network attempt to send data and the router buffers start overflowing due to which loss of packets occur. As a result retransmission of packets from the sources increase the congestion further. In this situation Transport layer provides Congestion Control in different ways. It uses open loop congestion control to prevent the congestion and closed loop congestion control to remove the congestion in a network once it occurred. TCP provides AIMD- additive increase multiplicative decrease, leaky bucket technique for congestion control.

- **Data integrity and Error correction** –Transport layer checks for errors in the messages coming from application layer by using error detection codes, computing checksums, it checks whether the received data is not corrupted and uses the ACK and NACK services to inform the sender if the data is arrived or not and checks for the integrity of data.
- **Flow control** –Transport layer provides a flow control mechanism between the adjacent layers of the TCP/IP model. TCP also prevents the data loss due to a fast sender and slow receiver by imposing some flow control techniques. It uses the method of sliding window protocol which is accomplished by receiver by sending a window back to the sender informing the size of data it can receive.

### Process to process delivery

The data link layer is responsible for delivery of frames between two neighboring nodes over a link is called node-to-node delivery. This layer responsible for delivery of datagram's between two hosts is called host-to-host delivery. Actual Communication on Internet is not defined by exchange of data but takes place between two processes (an application program), for that we need process to process delivery. Two processes communicate in a client/server relationship.



*Figure 66 Types of Data Delivery*

### Client/Server Paradigm

The most common one client/server paradigm is the ways to achieve process-to-process communication. A process on the local host, called a client, needs services from a process on the remote host, called a server.

Both processes (client and server) have the same name i.e Daytime. To get the day and time from a remote machine, we need a Daytime client process running on the local host and a Daytime server process running on a remote machine. A remote computer can run several server programs

at the same time, just as local computers can run one or more client programs at the same time. For communication, we must define the following:

- Local host
- Local process
- Remote host
- Remote process

### Addressing:

At a data link layer, whenever we need to deliver something to one specific destination among many, we need a destination MAC address to choose one node among several nodes if the connection is not point-to-point and source MAC address for reply. At network layer we need source and destination IP address similarly at the transport layer, we need a transport layer address, called a port number (destination port number), to choose among multiple processes running on the destination host and source port number to reply.

In the Internet model, the port numbers are 16-bit integers between 0 and 65,535. The client program defines itself with a port number, chosen randomly by the transport layer software running on the client host called ephemeral port number. The server process must also define itself with a port number, however, cannot be chosen randomly. If a server process assigns a random number as the port number, the process at the client site that wants to access that server will not know the port number then it can send a special packet and request the port number of a specific server, but this requires more overhead. Therefore Internet has decided to use universal port numbers for servers called well-known port numbers. To this rule exception are; there are clients that are assigned well-known port numbers. Every client process knows the well-known port number of the corresponding server process. For example, while the Daytime client process can use an ephemeral (temporary) port number 52,000 to identify itself, the Daytime server process must use the well-known (permanent) port number 13. The IP addresses and port numbers play different roles in selecting the final destination of data. The destination IP address defines the host among the different hosts in the world. After the host has been selected, the port number defines one of the processes on this particular host

### IANA Ranges

The IANA (Internet Assigned Number Authority) has divided the port numbers into three ranges: well known, registered, and dynamic or private.

- \* **Well-known ports:** The ports ranging from 0 to 1023 are assigned and controlled by IANA.
- \* **Registered ports:** The ports ranging from 1024 to 49,151 are not assigned or controlled by IANA. They can only be registered with IANA to prevent duplication.
- \* **Dynamic ports:** The ports ranging from 49,152 to 65,535 are neither controlled nor registered. They can be used by any process. These are the ephemeral ports.

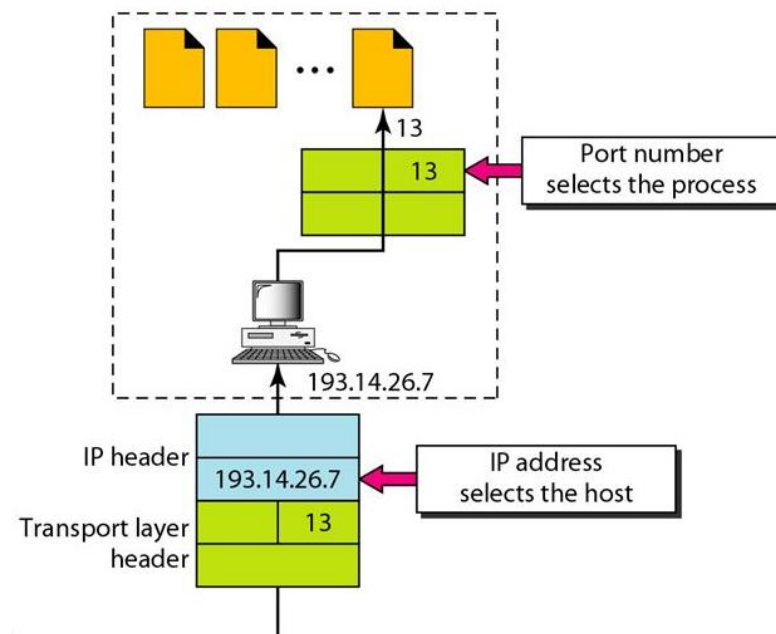
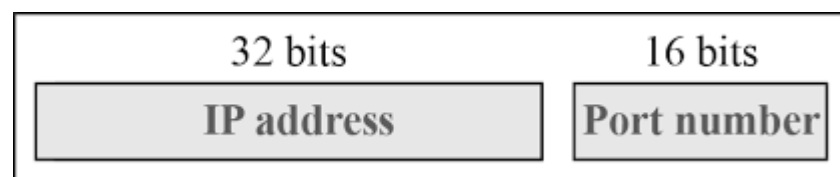


Figure 67: Socket Address

### Socket Addresses

Process-to-process delivery needs two identifiers, IP address and the port number, at each end to make a connection. The combination of an **IP address** and a **port number** is called a socket address. The client socket address defines the client process uniquely just as the server socket address defines the server process uniquely. A transport layer protocol needs a pair of socket addresses: the client socket address (IP address and port number) and the server socket address (IP address and port number). These four pieces of information are part of the IP header and the transport layer protocol header. The below figure showing socket address:



| value          | In X client        | In X server       |
|----------------|--------------------|-------------------|
| IP address     | 10.10.12.166       | 10.10.11.66       |
| Port           | 32825              | 55555             |
| Socket address | 10.10.12.166:32825 | 10.10.11.66:55555 |

Figure 68 Example of Socket Address

## Multiplexing and Demultiplexing:

The addressing mechanism allows multiplexing and Demultiplexing by the transport layer.

### Multiplexing:

At the sender site, there may be several processes that need to send packets. However, there is **only one transport layer protocol at any time**. This is a **many-to-one relationship and requires multiplexing**. The protocol accepts messages from different processes, differentiated by their assigned port numbers. After adding the header, the transport layer passes the packet to the network layer.

### Demultiplexing:

At the receiver site, the relationship is **one-to-many and requires demultiplexing**. The **transport layer receives datagram's from the network layer**. After error checking and dropping of the header, the transport layer delivers each message to the appropriate process based on the port number.

## TCP

The Transmission Control Protocol is the most common transport layer protocol. It works together with IP and provides a reliable transport service between processes using the network layer service provided by the IP protocol.

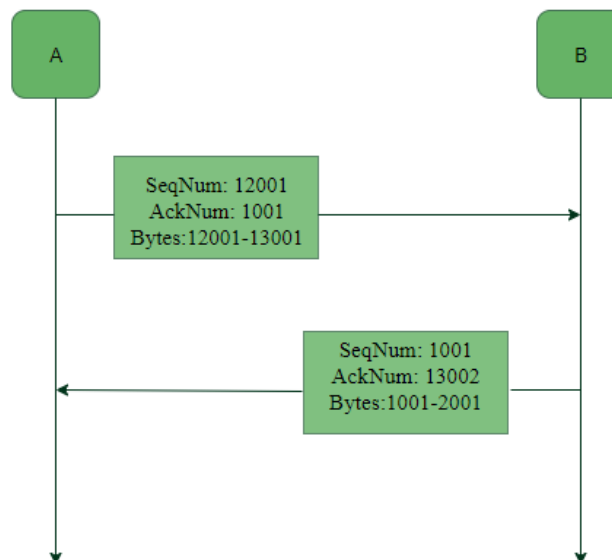
The various services provided by the TCP to the application layer are as follows:

- **Process-to-Process Communication**-TCP provides process to process communication, i.e., **the transfer of data takes place between individual processes executing on end systems**. This is done using port numbers or port addresses. Port numbers are 16 bit long that help identify which process is sending or receiving data on a host.
- **Stream oriented** -This means that the **data is sent and received as a stream of bytes (unlike UDP or IP that divides the bits into datagrams or packets)**. However, the network layer, that provides service for the TCP, sends packets of information not streams of bytes. Hence, **TCP groups a number of bytes together into a segment and adds a header to each of these segments and then delivers these segments to the network layer**. At the network layer, each of these segments are encapsulated in an IP packet for transmission. The TCP header has information that is required for control purpose which will be discussed along with the segment structure.
- **Full duplex service** -This means that the communication can take place in both directions at the same time.
- **Connection oriented service** -Unlike UDP, TCP provides connection oriented service. It defines 3 different phases:
  - Connection establishment
  - Data transfer
  - Connection termination

- **Reliability** -TCP is reliable as it uses checksum for error detection, attempts to recover lost or corrupted packets by re-transmission, acknowledgement policy and timers. It uses features like byte number and sequence number and acknowledgement number so as to ensure reliability. Also, it uses congestion control mechanisms.
- **Multiplexing** –TCP does multiplexing and de-multiplexing at the sender and receiver ends respectively as a number of logical connections can be established between port numbers over a physical connection.

### Byte number, Sequence number and Acknowledgement number:

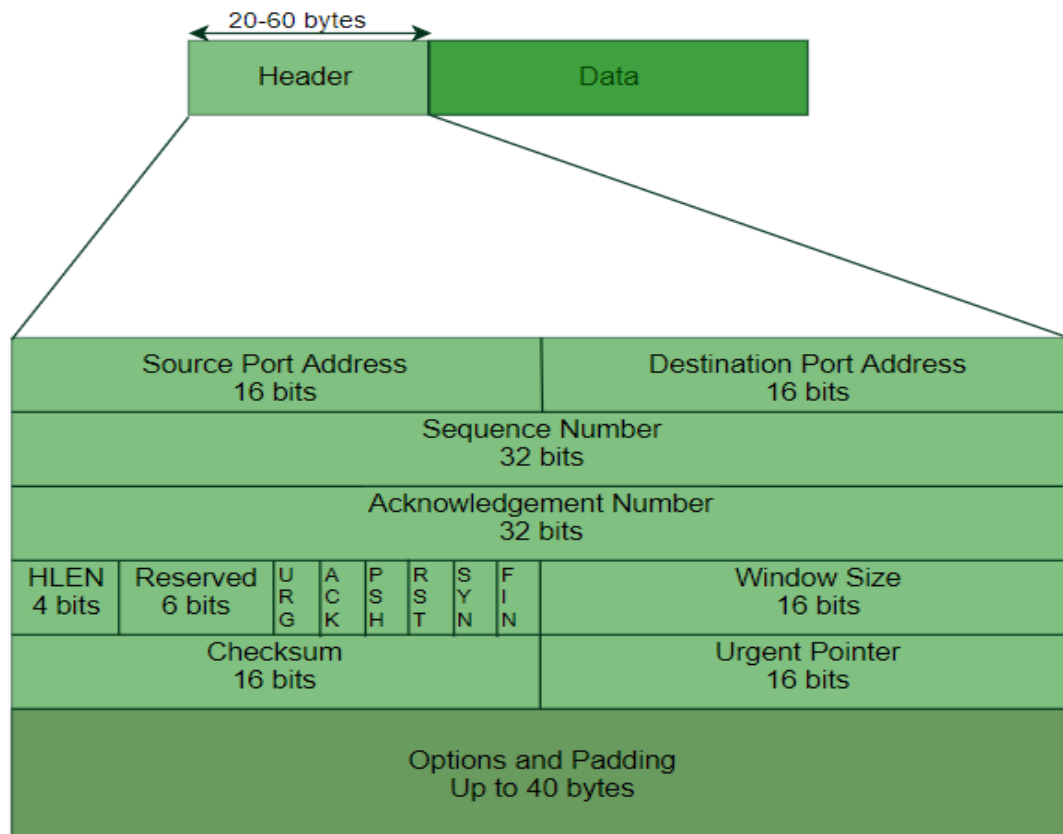
All the data bytes that are to be transmitted are numbered and the beginning of this numbering is arbitrary. Sequence numbers are given to the segments so as to reassemble the bytes at the receiver end even if they arrive in a different order. Sequence number of a segment is the byte number of the first byte that is being sent. Acknowledgement number is required since TCP provides full duplex service. Acknowledgement number is the next byte number that the receiver expects to receive which also provides acknowledgement for receiving the previous bytes.



In this example we see that, A sends acknowledgement number 1001, which means that it has received data bytes till byte number 1000 and expects to receive 1001 next, hence B next sends data bytes starting from 1001. Similarly, since B has received data bytes till byte number 13001 after the first data transfer from A to B, therefore B sends acknowledgement number 13002, the byte number that it expects to receive from A next.

### TCP Segment structure

TCP segment consists of data bytes to be sent and a header that is added to the data by TCP as shown:



The header of a TCP segment can range from 20-60 bytes. 40 bytes are for options. If there are no options, header is of 20 bytes else it can be of upmost 60 bytes.

Header fields:

- **Source Port Address** -16 bit field that holds the port address of the application that is sending the data segment.
- **Destination Port Address** -16 bit field that holds the port address of the application in the host that is receiving the data segment.
- **Sequence Number** -32 bit field that holds the sequence number, i.e, the byte number of the first byte that is sent in that particular segment. It is used to reassemble the message at the receiving end if the segments are received out of order.
- **Acknowledgement Number** -32 bit field that holds the acknowledgement number, i.e, the byte number that the receiver expects to receive next. It is an acknowledgment for the previous bytes being received successfully.
- **Header Length (HLEN)** -This is a 4 bit field that indicates the length of the TCP header by number of 4-byte words in the header, i.e, if the header is of 20 bytes(min length of TCP header), then this field will hold 5 (because  $5 \times 4 = 20$ ) and the maximum length: 60 bytes, then it'll hold the value 15(because  $15 \times 4 = 60$ ). Hence, the value of this field is always between 5 and 15.
- **Control flags** -These are 6 1-bit control bits that control connection establishment, connection termination, connection abortion, flow control, mode of transfer etc. Their function is:
  - URG: Urgent pointer is valid

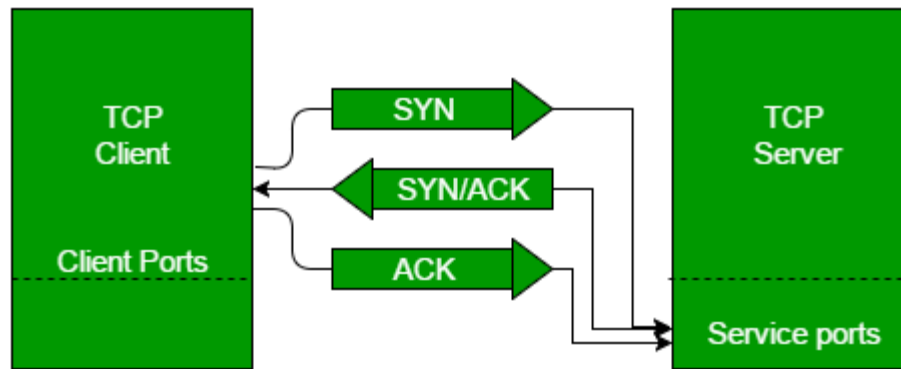
- ACK: Acknowledgement number is valid( used in case of cumulative acknowledgement)
- PSH: Request for push
- RST: Reset the connection
- SYN: Synchronize sequence numbers
- FIN: Terminate the connection
- **Window size** -This field tells the window size of the sending TCP in bytes.
- **Checksum** -This field holds the checksum for error control. It is mandatory in TCP as opposed to UDP.
- **Urgent pointer** -This field (valid only if the URG control flag is set) is used to point to data that is urgently required that needs to reach the receiving process at the earliest. The value of this field is added to the sequence number to get the byte number of the last urgent byte.

Some well-known ports of TCP

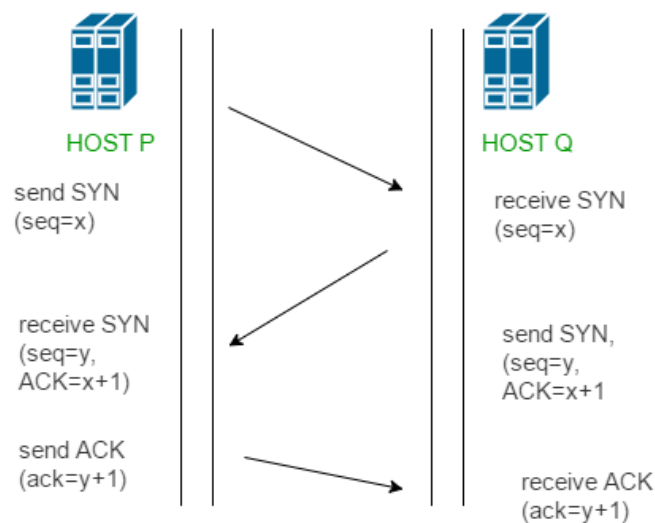
| Port | Protocol     | Description                                   |
|------|--------------|-----------------------------------------------|
| 7    | Echo         | Echoes a received datagram back to the sender |
| 9    | Discard      | Discards any datagram that is received        |
| 11   | Users        | Active users                                  |
| 13   | Daytime      | Returns the date and the time                 |
| 17   | Quote        | Returns a quote of the day                    |
| 19   | Chargen      | Returns a string of characters                |
| 20   | FIP, Data    | File Transfer Protocol (data connection)      |
| 21   | FIP, Control | File Transfer Protocol (control connection)   |
| 23   | TELNET       | Tenninal Network                              |
| 25   | SMTP         | Simple Mail Transfer Protocol                 |
| 53   | DNS          | Domain Name Server                            |
| 67   | BOOTP        | Bootstrap Protocol                            |
| 79   | Finger       | Finger                                        |
| 80   | HTTP         | Hypertext Transfer Protocol                   |
| 111  | RPC          | Remote Procedure Call                         |

### TCP 3-Way Handshake Process

The process of communication between devices over the internet happens according to the current **TCP/IP** suite model (stripped out version of OSI reference model). The Application layer is a top pile of stack of TCP/IP model from where network referenced application like web browser on the client side establish connection with the server. From the application layer, the information is transferred to the transport layer where our topic comes into picture. The two important protocols of this layer are – TCP, **UDP (User Datagram Protocol)** out of which TCP is prevalent (since it provides reliability for the connection established). However you can find application of UDP in querying the DNS server to get the binary equivalent of the Domain Name used for the website.



TCP provides reliable communication with something called **Positive Acknowledgement with Re-transmission (PAR)**. The Protocol Data Unit (PDU) of the transport layer is called segment. Now a device using PAR resend the data unit until it receives an acknowledgement. If the data unit received at the receiver's end is damaged (It checks the data with checksum functionality of the transport layer that is used for Error Detection), then receiver discards the segment. So the sender has to resend the data unit for which positive acknowledgement is not received. You can realize from above mechanism that three segments are exchanged between sender (client) and receiver (server) for a reliable TCP connection to get established. Let us delve how this mechanism works:



*Figure 69: 3-way handshake*

- **Step 1 (SYN)** : In the first step, client wants to establish a connection with server, so it sends a segment with SYN(Synchronize Sequence Number) which informs server that client is likely to start communication and with what sequence number it starts segments with
- **Step 2 (SYN + ACK)**: Server responds to the client request with SYN-ACK signal bits set. Acknowledgement(ACK) signifies the response of segment it received and SYN signifies with what sequence number it is likely to start the segments with
- **Step 3 (ACK)** : In the final part client acknowledges the response of server and they both establish a reliable connection with which they will start the actual data transfer

The steps 1, 2 establish the connection parameter (sequence number) for one direction and it is acknowledged. The steps 2, 3 establish the connection parameter (sequence number) for the other direction and it is acknowledged. With these, a full-duplex communication is established.

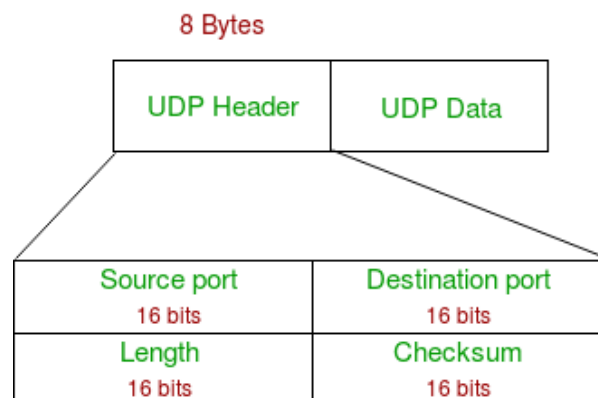
## UDP

**User Datagram Protocol (UDP)** is a Transport Layer protocol. UDP is a part of Internet Protocol suite, referred as UDP/IP suite. Unlike TCP, it is **unreliable and connectionless protocol**. So, there is no need to establish connection prior to data transfer.

Though Transmission Control Protocol (TCP) is the dominant transport layer protocol used with most of Internet services; provides assured delivery, reliability and much more but all these services cost us with additional overhead and latency. Here, UDP comes into picture. For the real-time services like computer gaming, voice or video communication, live conferences; we need UDP. Since high performance is needed, UDP permits packets to be dropped instead of processing delayed packets. There is no error checking in UDP, so it also save bandwidth. User Datagram Protocol (UDP) is more efficient in terms of both latency and bandwidth.

### UDP Header –

UDP header is **8-bytes** fixed and simple header, while for TCP it may vary from 20 bytes to 60 bytes. First 8 Bytes contains all necessary header information and remaining part consist of data. UDP port number fields are each 16 bits long, therefore range for port numbers defined from 0 to 65535; port number 0 is reserved. Port numbers help to distinguish different user requests or process.



- **Source Port:** Source Port is 2 Byte long field used to identify port number of source.
- **Destination Port:** It is 2 Byte long field, used to identify the port of destined packet.
- **Length:** Length is the length of UDP including header and the data. It is 16-bits field.
- **Checksum:** Checksum is 2 Bytes long field. It is the 16-bit one's complement of the one's complement sum of the UDP header, pseudo header of information from the IP header and the data, padded with zero octets at the end (if necessary) to make a multiple of two octets.

**Notes –** Unlike TCP, Checksum calculation is not mandatory in UDP. No Error control or flow control is provided by UDP. Hence UDP depends on IP and ICMP for error reporting.

### Applications of UDP:

- Used for simple request response communication when size of data is less and hence there is lesser concern about flow and error control.
- It is suitable protocol for multicasting as UDP supports packet switching.
- UDP is used for some routing update protocols like RIP (Routing Information Protocol).

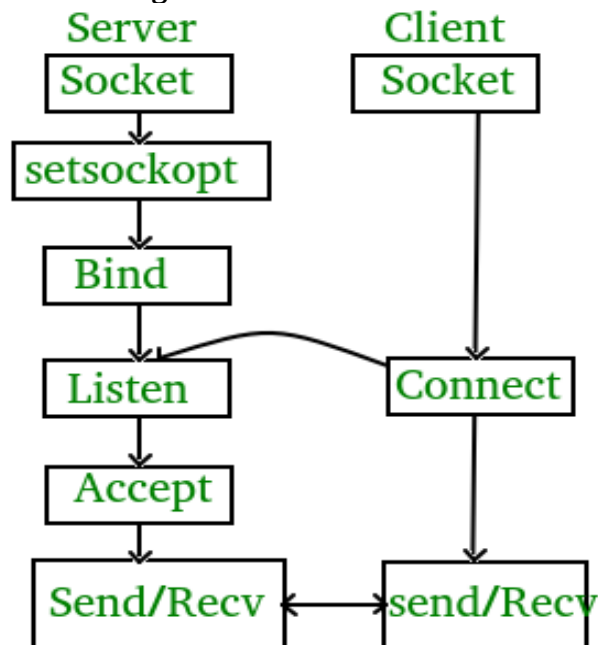
- Normally used for real time applications which cannot tolerate uneven delays between sections of a received message.
- Following implementations uses UDP as a transport layer protocol:
  - NTP (Network Time Protocol)
  - DNS (Domain Name Service)
  - BOOTP, DHCP.
  - NNP (Network News Protocol)
  - Quote of the day protocol
  - TFTP, RTSP, RIP, OSPF.
- Application layer can do some of the tasks through UDP-
  - Trace Route
  - Record Route
  - Time stamp
- UDP takes datagram from Network Layer, attach its header and send it to the user. So, it works fast.
- Actually UDP is null protocol if you remove checksum field.

## Socket Programming

### What is socket programming?

Socket programming is a way of connecting two nodes on a network to communicate with each other. One socket (node) listens on a particular port at an IP, while other socket reaches out to the other to form a connection. Server forms the listener socket while client reaches out to the server.

State diagram for server and client model



### Stages for server

- **Socket creation:**

```
int sockfd = socket(domain, type, protocol)
```

**sockfd:** socket descriptor, an integer (like a file-handle)

**domain:** integer, communication domain e.g., AF\_INET (IPv4 protocol) , AF\_INET6 (IPv6 protocol)

**type:** communication type

SOCK\_STREAM: TCP(reliable, connection oriented)

SOCK\_DGRAM: UDP(unreliable, connectionless)

**protocol:** Protocol value for Internet Protocol(IP), which is 0. This is the same number which appears on protocol field in the IP header of a packet.(man protocols for more details)

- **Setsockopt:**

```
int setsockopt(int sockfd, int level, int optname, const void *optval, socklen_t optlen);
```

This helps in manipulating options for the socket referred by the file descriptor sockfd. This is completely optional, but it helps in reuse of address and port. Prevents error such as: “address already in use”.

- **Bind:**

```
int bind(int sockfd, const struct sockaddr *addr, socklen_t addrlen);
```

After creation of the socket, bind function binds the socket to the address and port number specified in addr(custom data structure). In the example code, we bind the server to the localhost, hence we use INADDR\_ANY to specify the IP address.

- **Listen:**

```
int listen(int sockfd, int backlog);
```

It puts the server socket in a passive mode, where it waits for the client to approach the server to make a connection. The backlog, defines the maximum length to which the queue of pending connections for sockfd may grow. If a connection request arrives when the queue is full, the client may receive an error with an indication of ECONNREFUSED.

- **Accept:**

```
int new_socket= accept(int sockfd, struct sockaddr *addr, socklen_t *addrlen);
```

It extracts the first connection request on the queue of pending connections for the listening socket, sockfd, creates a new connected socket, and returns a new file descriptor referring to that socket. At this point, connection is established between client and server, and they are ready to transfer data.

### Stages for Client

- **Socket connection:** Exactly same as that of server’s socket creation
- **Connect:**

```
int connect(int sockfd, const struct sockaddr *addr, socklen_t addrlen);
```

The connect() system call connects the socket referred to by the file descriptor sockfd to the address specified by addr. Server's address and port is specified in addr.

## Chapter 7

### Congestion Control & Quality of Services

Congestion control and quality of service are two issues related to the three layers: the data link layer, the network layer, and the transport layer.

**Congestion:** it is an important issue in a packet-switched network. Congestion in a network may occur if the load on the network-the number of packets sent to the network-is greater than the capacity of the network (the number of packets a network can handle).

**Congestion control** refers to the mechanisms and techniques to control the congestion and keep the load below the capacity. Congestion happens in any system that involves waiting. For example, congestion happens on a freeway because any abnormality in the flow, such as an accident during rush hour, creates blockage. Congestion in a network (internetwork) occurs because routers and switches have queues-buffers that hold the packets before and after processing.

#### Data Traffic

The main focus of congestion control and quality of service is data traffic. In congestion control we try to avoid traffic congestion. In quality of service, we try to create an appropriate environment for the traffic. So, before discussing about congestion control and quality of service, we describe data traffic as given below:

#### Traffic Descriptor

Traffic descriptors are qualitative values that represent a data flow. Below Figure shows a traffic flow with some of these values.

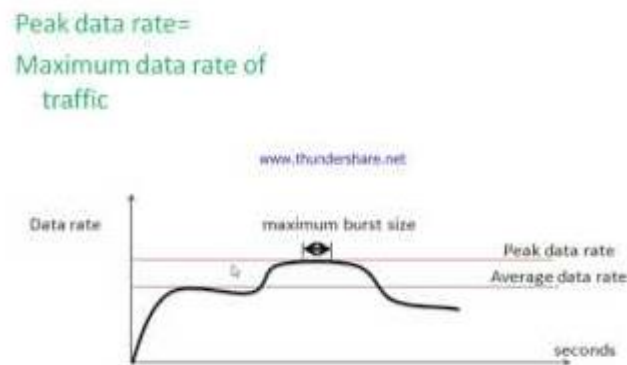


Figure 70 Traffic Descriptor

#### Average Data Rate

The average data rate is the number of bits sent during a period of time, divided by the number of seconds in that period. We use the following equation:

$$\text{Average data rate} = \text{amount of data} / \text{Time}$$

The average data rate is a very useful characteristic of traffic because, it indicates the average bandwidth needed by the traffic.

### Peak Data Rate

The peak data rate defines the maximum data rate of the traffic. In the figure of traffic descriptor (above), it is the maximum y axis value. The peak data rate is a very important measurement because it indicates the peak bandwidth that the network needs for traffic to pass through without changing its data flow.

### Maximum Burst Size

The maximum burst size normally refers to the maximum length of time the traffic is generated at the peak rate. Although the peak data rate is a critical value for the network, it can be ignored if the duration of the peak value is very short. For example, if data are flowing steadily at the rate of 1 Mbps with a sudden peak data rate of 2 Mbps for just 1 ms, the network probably can handle the situation. However, if the peak data rate lasts 60 ms, there may be a problem for the network.

### Effective Bandwidth

The effective bandwidth is the bandwidth that the network needs to allocate for the flow of traffic. The effective bandwidth is a function of three values: average data rate, peak data rate, and maximum burst size. The calculation of this value is very complex.

### Traffic Profiles

The data flow can have one of the following traffic profiles: constant bitrate, variable bit rate, or bursty as shown below:

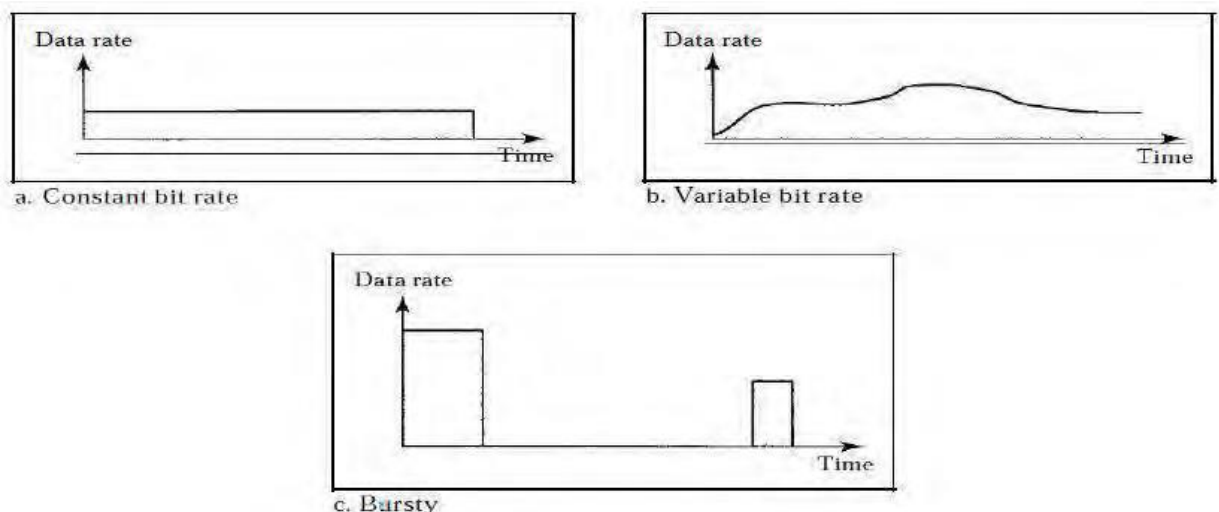


Figure 71: Three Traffic Profiles

#### a) Constant Bit Rate

A constant-bit-rate (CBR), or a fixed-rate, traffic model has a data rate that does not change. In this type of flow, the average data rate and the peak data rate are the same. The maximum burst size is not applicable. This type of traffic is very easy for a network to handle since

it is predictable. The network knows in advance how much bandwidth to allocate for this type of flow.

### b) Variable Bit Rate

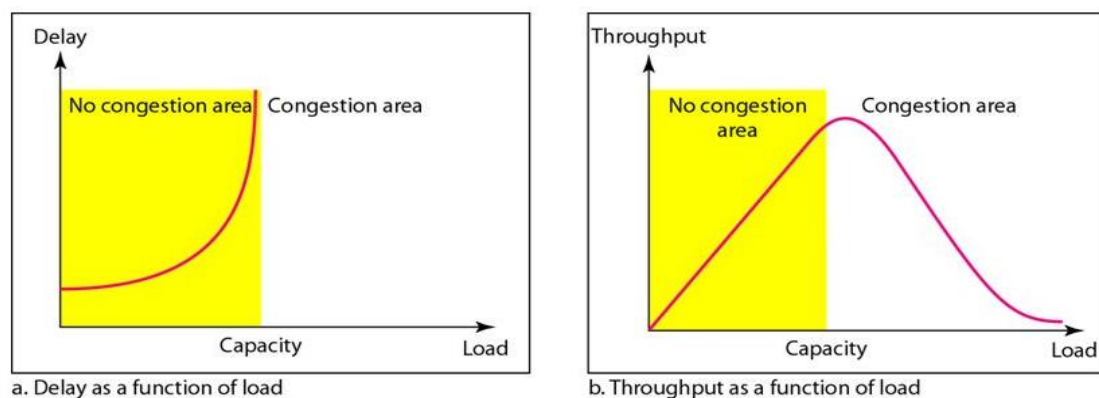
In the variable-bit-rate (VBR) category, the rate of the data flow changes in time, with the smooth changes instead of sudden and sharp. In this type of flow, the average data rate and the peak data rate are different. The maximum burst size is usually a small value. This type of traffic is more difficult to handle than constant bit-rate traffic, but it normally does not need to be reshaped.

#### Bursty

In this category, the data rate changes suddenly in a very short time. It may jump from zero, for example, to 1 Mbps in a few microseconds and vice versa. The average bit rate and the peak bit rate are very different values in this type of flow. The maximum burst size is significant. This is the most difficult type of traffic for a network to handle because the profile is very unpredictable. To handle bursty traffic, the network normally needs to reshape it, using reshaping techniques. Bursty traffic is one of the main causes of congestion in a network.

#### Network Performance

Congestion control involves two factors that measure the performance of a network: delay and throughput.



*Figure 72 Packet delay and throughput as functions of load*

#### Delay Versus Load

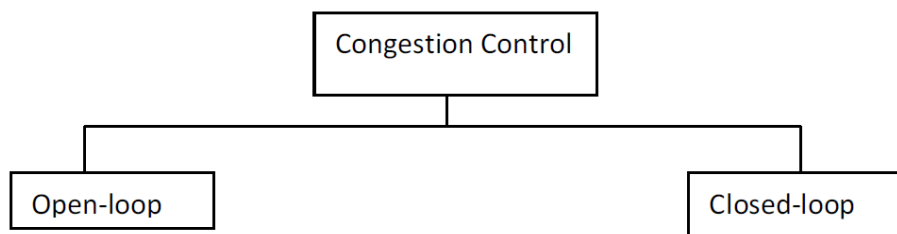
Note that when the load is much less than the capacity of the network, the delay is at a minimum. This minimum delay is composed of propagation delay and processing delay, both of which are negligible. When the load reaches network capacity, the delay increases sharply because, waiting time in the queues (for all routers in the path) is added to the total delay. Note that the delay becomes infinite when the load is greater than the capacity.

#### Throughput Versus Load

Throughput is the number of bits passing through a point in a second. We can define throughput in a network as the number of packets passing through the network in a unit of time. When the load is below the capacity of the network, the throughput increases proportionally with the load. When the load exceeds the capacity, the queues become full and the routers have to discard some packets. Discarding packets does not reduce the number of packets in the network because the sources retransmit the packets, using timeout mechanisms, when the packets have not reached the destinations.

## Congestion Control

Congestion control refers to techniques and mechanisms that can either prevent congestion, before it happens, or remove congestion, after it has happened. In general, we can divide congestion control mechanisms into two broad categories: open-loop congestion control (prevention) and closed-loop congestion control (removal).



1. *Retransmission Policy*
2. *Window Policy*
3. *Acknowledgment Policy*
4. *Discarding Policy*
5. *Admission Policy*

1. *Backpressure*
2. *Choke Packet*
3. *Implicit Signaling*
4. *Explicit Signaling*

## Open-Loop Congestion Control

In open-loop congestion control, policies are applied to prevent congestion before it happens. In these mechanisms, congestion control is handled by either the source or the destination. List of policies that can prevent congestion are as given below:

### 1. Retransmission Policy

Retransmission is sometimes unavoidable. If the sender feels that a sent packet is lost or corrupted, the packet is retransmitted. Retransmission Open-loop Closed-loop Congestion Control may increase congestion in the network. However, a good retransmission policy can prevent congestion. The retransmission policy and the retransmission timers must be designed to optimize efficiency and at the same time prevent congestion. For example, the retransmission policy used by TCP is designed to prevent or alleviate congestion.

### 2. Window Policy

The type of window at the sender may also affect congestion. The Selective Repeat window is better than the Go-Back-N window for congestion control. In the Go-Back-N window, when the timer for a packet times out, several packets may be resent, although some may have arrived safe and sound at the receiver. This duplication may make the congestion worse. The Selective Repeat window, tries to send only the specific packets that have been lost or corrupted.

### **3. Acknowledgment Policy**

The acknowledgment policy imposed by the receiver may also affect congestion. If the receiver does not acknowledge every packet it receives, it may slow down the sender and help prevent congestion. Several approaches are used in this case are: A receiver may send an acknowledgment only if it has a packet to be sent or a special timer expires. A receiver may decide to acknowledge only N packets at a time. Sending fewer acknowledgments means imposing fewer loads on the network.

### **4. Discarding Policy**

A good discarding policy by the routers may prevent congestion and at the same time may not harm the integrity of the transmission. For example, in audio transmission, if the policy is to discard less sensitive packets when congestion is likely to happen, the quality of sound is still preserved and congestion is prevented or alleviated.

### **5. Admission Policy**

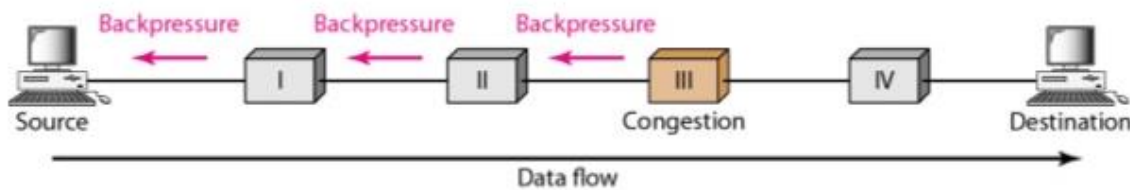
An admission policy is a quality-of-service mechanism. It can also prevent congestion in virtual-circuit networks. Switches in a flow, first check the resource requirement of a flow before admitting it to the network. A router can deny establishing a virtual circuit connection, if there is congestion in the network or if there is a possibility of future congestion.

## **Close Loop Congestion Control**

Closed-Loop Congestion Control: Closed-loop congestion control mechanisms try to alleviate congestion after it happens. Several mechanisms have been used by different protocols. We describe a few of them here.

### **1. Backpressure**

The technique of backpressure refers to a congestion control mechanism in which a congested node stops receiving data from the immediate upstream node or nodes. This may cause the upstream node or nodes to become congested, and they, in turn, reject data from their upstream nodes or nodes. And so on. Backpressure is a node-to-node congestion control that starts with a node and propagates, in the opposite direction of data flow, to the source. The backpressure technique can be applied only to virtual circuit networks, in which each node knows the upstream node from which a flow of data is coming.

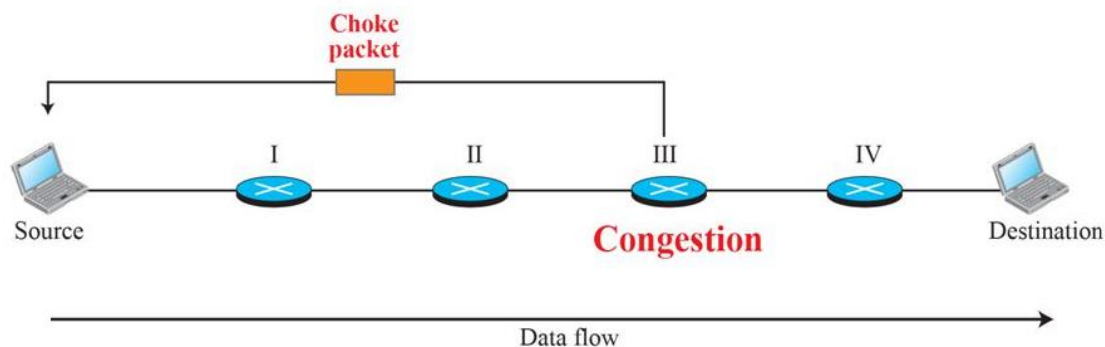


*Figure 73 Backpressure method for alleviating congestion*

Node III in the figure has more input data than it can handle. It drops some packets in its input buffer and informs node II to slow down the forwarding. Node II, in turn, may be congested because it is slowing down the output flow of data. If node II is congested, it informs node I to slow down, which in turn may create congestion. If so, node I informs the source of data to slow down. In this way alleviates the congestion. Note that the pressure on node III is moved backward to the source to remove the congestion. None of the virtual-circuit networks use backpressure. It is only implemented in the first virtual-circuit network, X.25. This technique cannot be implemented in a datagram network because in this type of network, a node (router) does not have the slightest knowledge of the upstream router.

## 2. Choke Packet:

A choke packet is a packet sent by a node to the source to inform it of congestion. Note the difference between the backpressure and choke packet methods. In backpressure, the warning is from one node to its upstream node, the warning may eventually reach the source station. In the choke packet method, the warning is from the router, which has encountered congestion, to the source station directly. The intermediate nodes through which the packet has traveled are not warned. We have seen example of this type of control in ICMP. It informs the source host, using a source quench ICMP message. The warning message goes directly to the source station, the intermediate routers, and does not take any action.



*Figure 74 Idea of a Choke packet*

## 3. Implicit Signaling

In implicit signaling, there is no communication between the congested node or nodes and the source. The source guesses that there is congestion somewhere in the network from other symptoms. For example, when a source sends several packets and there is no

acknowledgment for a while, one assumption is that the network is congested. The delay in receiving an acknowledgment is interpreted as congestion in the network, the source should slow down.

#### 4. Explicit Signaling

The node that experiences congestion can explicitly send a signal to the source or destination. The explicit signaling method, however, is different from the choke packet method. In the choke packet method, a separate packet is used for this purpose. In the explicit signaling method, the signal is included in the packets that carry data. Explicit signaling, as in Frame Relay congestion control, can occur in either the forward or the backward direction.

- **Backward Signaling:** A bit can be set in a packet moving in the direction opposite to the congestion. This bit can warn the source that there is congestion and that it needs to slow down, to avoid the discarding of packets.
- **Forward Signaling:** A bit can be set in a packet moving in the direction of the congestion. This bit can warn the destination that there is congestion. The receiver in this case can use policies, such as slowing down the acknowledgments, to alleviate the congestion.

#### Quality of Service

We can informally define quality of service as something a flow seeks to attain. Flow Characteristics

Traditionally, four types of characteristics are attributed to a flow:

1. Reliability
2. Delay
3. Jitter
4. Bandwidth

##### 1. Reliability

Reliability means safe and sound packet. Reliability is a characteristic that a flow needs. The Lack of reliability means, losing a packet or acknowledgment, which entails retransmission. However, the sensitivity of application programs to reliability is not the same. For example, it is more important that electronic mail, file transfer, and Internet access have reliable transmissions than telephony or audio conferencing.

##### 2. Delay

When packet is not transmitted with real time then delay factor is arises. Source-to-destination delay is another flow characteristic. Again applications can tolerate delay in different degrees. In this case, telephony, audio conferencing, video conferencing, and remote log-in need minimum delay, while delay in file transfer or e-mail is less important.

##### 3. Jitter

Jitter is the variation in delay for packets belonging to the same flow. Jitter is defined as the variation in the packet delay. High jitter means the difference between delays is large,

low jitter means the variation is small. For example, if four packets depart at times 0, 1, 2, 3 and arrive at 20, 21, 22, 23, all have the same delay, 20 units of time. On the other hand, if the above four packets arrive at 21, 23, 21, and 28, they will have different delays: 21, 22, 19, and 24. For audio and video applications, the first case is completely acceptable, the second case is not. For these applications, it does not matter if the packets arrive with a short or long delay as long as the delay is the same for all packets. The multimedia communication deals with jitter. If the jitter is high, some action is needed in order to use the received data.

#### 4. Bandwidth

Different applications need different bandwidths. In video conferencing we need to send millions of bits per second to refresh a color screen while the total number of bits in an e-mail may not reach even a million.

#### 5. Flow Classes

Based on the flow characteristics, we can classify flows into groups, with each group having similar levels of characteristics. This categorization is not formal or universal; some protocols such as ATM have defined classes.

#### Techniques to Improve QOS

Some techniques that can be used to improve the quality of service and common four Methods are:

- scheduling
- traffic shaping
- admission control
- Resource reservation.

##### ➤ Scheduling:

Packets from different flows arrive at a switch or router for processing. A good scheduling technique treats the different flows in a fair and appropriate manner. Several scheduling techniques are designed to improve the quality of service among that common three are: **FIFO queuing**, **priority queuing**, and **weighted fair queuing**.

##### i. FIFO Queuing

In first-in, first-out (FIFO) queuing, packets wait in a buffer (queue) until the node (router or switch) is ready to process them. If the average arrival rate is higher than the average processing rate, the queue will fill up and new packets will be discarded. A FIFO queue is familiar to those who have had to wait for a bus at a bus stop.

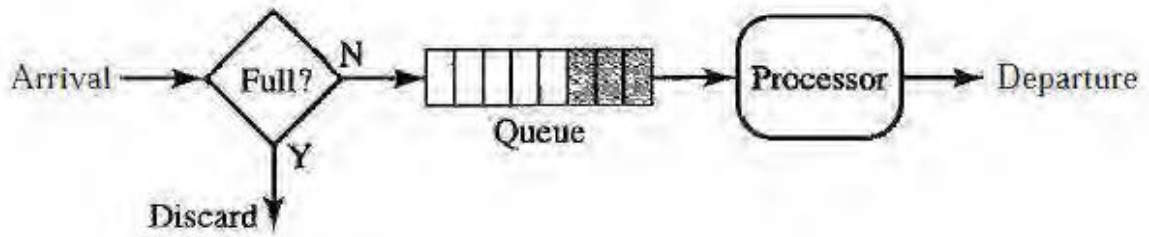


Figure 75 Conceptual view of FIFO

## ii. Priority Queuing

In priority queuing, packets are first assigned to a priority class. Each priority class has its own queue. The packets in the highest-priority queue are processed first. Packets in the lowest-priority queue are processed last. Note that the system does not stop serving a queue until it is empty.

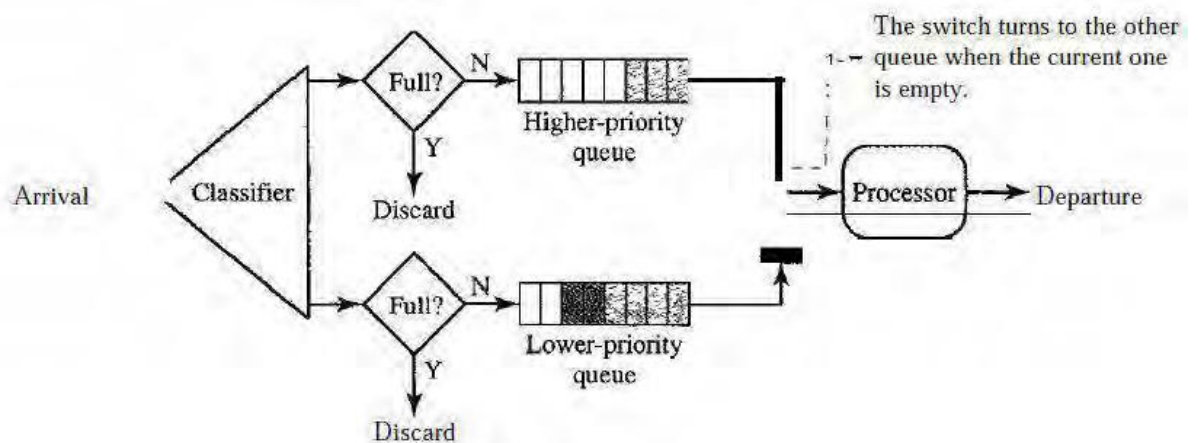


Figure 76 Priority queuing with two priority levels

A priority queue can provide better QoS than the FIFO queue because higher priority traffic, such as multimedia, can reach the destination with less delay. However, there is a potential drawback. If there is a continuous flow in a high-priority queue, the packets in the Lower priority queues will never have a chance to be processed. This condition is called starvation.

## iii. Weighted Fair Queuing

It is a better scheduling method. In this technique, the packets are assigned to different classes and admitted to different queues. The queues, however, are weighted based on the priority of the queues. Higher priority means a higher weight. The system processes packets in each queue in a round-robin fashion with the number of packets selected from

each queue based on the corresponding weight. For example, if the weights are 3, 2, and 1, three packets are processed from the first queue, two from the second queue, and one from the third queue. If the system does not impose priority on the classes, all weights can be equal. In this way, we have fair queuing with priority.

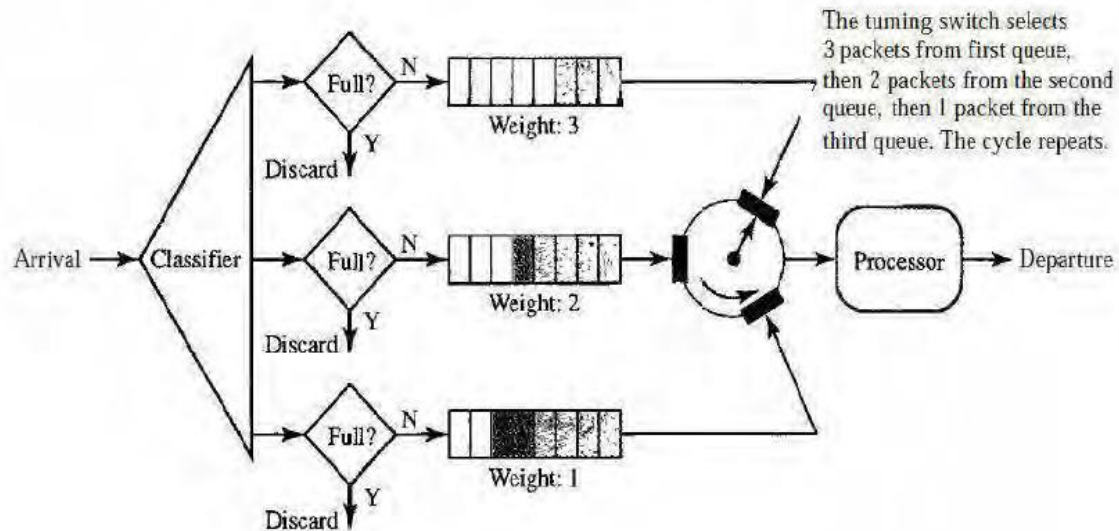


Figure 77 The technique with three classes

### ➤ Traffic Shaping

Traffic shaping is a mechanism to control the amount and the rate of the traffic sent to the network. Two techniques can shape traffic: leaky bucket and token bucket.

#### 1) Leaky Bucket

If a bucket has a small hole at the bottom, the water leaks from the bucket at a constant Rate as long as there is water in the bucket. The rate at which the water leaks does not depend on the rate at which the water is input to the bucket unless the bucket is empty.

The input rate can vary, but the output rate remains constant. Similarly, in networking, a technique called leaky bucket can smooth out bursty traffic. Bursty chunks are stored in the bucket and sent out at an average rate.

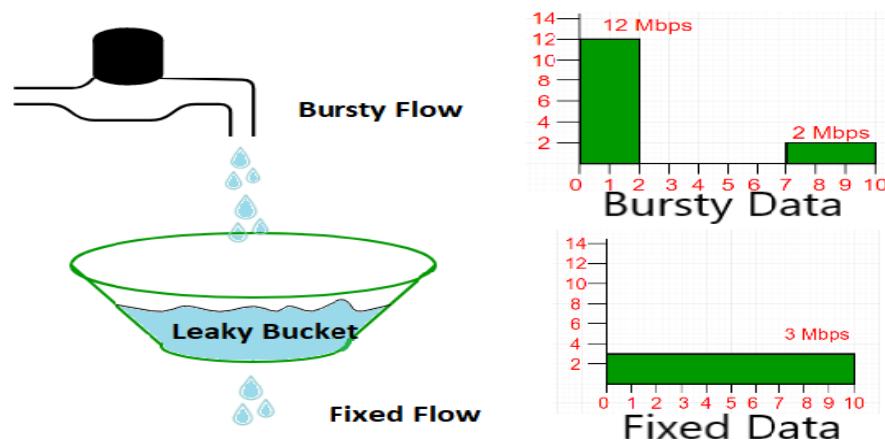
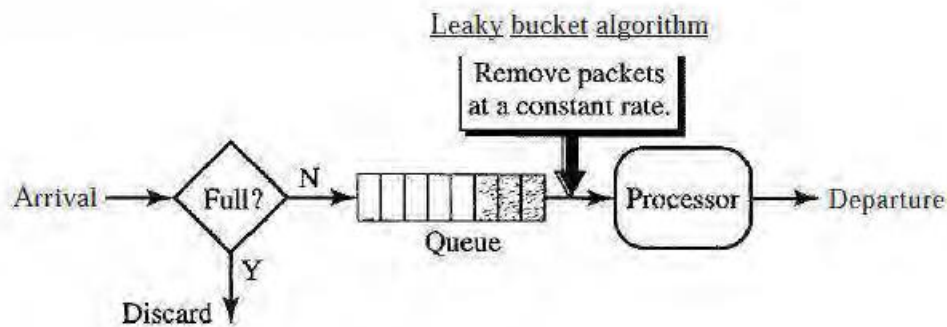


Figure 78 Leaky Bucket

### Leaky bucket implementation



If the traffic consists of variable-length packets, the fixed output rate must be based on the number of bytes or bits.

The following is an algorithm for variable-length packets:

1. Initialize a counter to  $n$  at the tick of the clock.
2. If  $n$  is greater than the size of the packet, send the packet and decrement the counter by the packet size. Repeat this step until  $n$  is smaller than the packet size.
3. Reset the counter and go to step 1.

## 2. Token Bucket

The **leaky bucket** is very restrictive. It does not credit an idle host. For example, if a host is not sending for a while, its bucket becomes empty. Now if the host has bursty data, the leaky bucket allows only an average rate. The time when the host was idle is not taken into account. On the other hand, the **token bucket algorithm** allows idle hosts to accumulate credit for the future in the form of tokens. For each tick of the clock, the system sends  $n$  tokens to the bucket. The system removes one token for every cell (or byte) of data sent. For example, if  $n$  is 100 and the host is idle for 100 ticks, the bucket collects 10,000 tokens. Now the host can consume all these tokens in one tick with 10,000

cells, or the host takes 1000 ticks with 10 cells per tick. In other words, the host can send bursty data as long as the bucket is not empty.

The token bucket can easily be implemented with a counter. The token is initialized to zero. Each time a token is added, the counter is incremented by 1. Each time a unit of data is sent, the counter is decremented by 1. When the counter is zero, the host cannot send data. Combining Token Bucket and Leaky Bucket the two techniques can be combined to credit an idle host and at the same time regulate the traffic. The leaky bucket is applied after the token bucket: the rate of the leaky bucket needs to be higher than the rate of tokens dropped in the bucket.

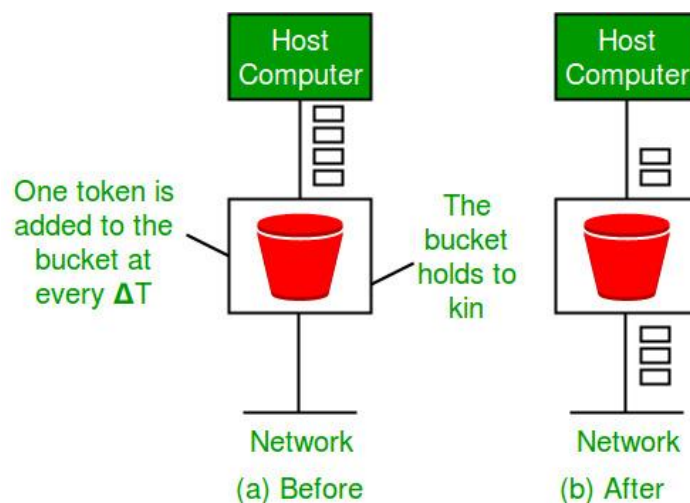


Figure 79 Concept of token bucket

### Resource Reservation

A flow of data needs resources such as a buffer, bandwidth, CPU time, and so on. The quality of service is improved if these resources are reserved beforehand. One of the QoS model called Integrated Services, which depends heavily on resource reservation to improve the quality of service.

### Admission Control

Admission control refers to the mechanism used by a router, or a switch, to accept or reject a flow based on predefined parameters called flow specifications. Before a router accepts a flow for processing, it checks the flow specifications to see if its capacity (in terms of bandwidth, buffer size, CPU speed, etc.) and its previous commitments to other flows can handle the new flow.

### TCP Congestion Control

TCP uses a congestion window and a congestion policy that avoid congestion. Previously, we assumed that only receiver can dictate the sender's window size. We ignored another entity here, the network. If the network cannot deliver the data as fast as it is created by the sender, it must tell the sender to slow down. In other words, in addition to the receiver, the network is a second entity that determines the size of the sender's window.

**Congestion policy in TCP –**

1. **Slow Start Phase:** starts slowly increment is exponential to threshold
2. **Congestion Avoidance Phase:** After reaching the threshold increment is by 1
3. **Congestion Detection Phase:** Sender goes back to slow start phase or Congestion avoidance phase.

**Slow Start Phase: exponential increment** – In this phase after every RTT the congestion window size increments exponentially.

Initially  $cwnd = 1$

After 1 RTT,  $cwnd = 2^{(1)} = 2$

2 RTT,  $cwnd = 2^{(2)} = 4$

3 RTT,  $cwnd = 2^{(3)} = 8$

**Congestion Avoidance Phase: additive increment** – This phase starts after the threshold value also denoted as *ssthresh*. The size of *cwnd*(congestion window) increases additive. After each RTT  $cwnd = cwnd + 1$ .

Initially  $cwnd = i$

After 1 RTT,  $cwnd = i+1$

2 RTT,  $cwnd = i+2$

3 RTT,  $cwnd = i+3$

**Congestion Detection Phase : multiplicative decrement** – If congestion occurs, the congestion window size is decreased. The only way a sender can guess that congestion has occurred is the need to retransmit a segment. Retransmission is needed to recover a missing packet which is assumed to have been dropped by a router due to congestion. Retransmission can occur in one of two cases: when the RTO timer times out or when three duplicate ACKs are received.

- **Case 1 : Retransmission due to Timeout** – In this case congestion possibility is high.
  - (a) *ssthresh* is reduced to half of the current window size.
  - (b) set  $cwnd = 1$
  - (c) start with slow start phase again.
- **Case 2 : Retransmission due to 3 Acknowledgement Duplicates** – In this case congestion possibility is less.
  - (a) *ssthresh* value reduces to half of the current window size.
  - (b) set  $cwnd = ssthresh$
  - (c) start with congestion avoidance phase

**Example** – Assume a TCP protocol experiencing the behavior of slow start. At 5th transmission round with a threshold (thresh) value of 32 goes into congestion avoidance phase and continues till 10th transmission. At 10th transmission round, 3 duplicate ACKs are received by the receiver and enter into additive increase mode. Timeout occurs at 16th transmission round. Plot the transmission round (time) vs congestion window size of TCP segments.

## Chapter 8

### Application Layer, Servers & Protocols

It is the top most layer of OSI Model. Manipulation of data (information) in various ways is done in this layer which enables user or software to get access to the network. Some services provided by this layer includes: E-Mail, transferring files, distributing the results to user, directory services, network resources, etc.

The Application Layer contains a variety of protocols that are commonly needed by users. One widely-used application protocol is HTTP (Hyper Text Transfer Protocol), which is the basis for the World Wide Web. When a browser wants a web page, it sends the name of the page it wants to the server using HTTP. The server then sends the page back.

Other Application protocols that are used are: **File Transfer Protocol (FTP)**, **Trivial File Transfer Protocol (TFTP)**, **Simple Mail Transfer Protocol (SMTP)**, **TELNET**, **Domain Name System (DNS)** etc.

#### Functions of application layer

1. **Mail Services:** This layer provides the basis for E-mail forwarding and storage.
2. **Network Virtual Terminal:** It allows a user to log on to a remote host. The application creates software emulation of a terminal at the remote host. User's computer talks to the software terminal which in turn talks to the host and vice versa. Then the remote host believes it is communicating with one of its own terminals and allows user to log on.
3. **Directory Services:** This layer provides access for global information about various services.
4. **File Transfer, Access and Management (FTAM):** It is a standard mechanism to access files and manages it. Users can access files in a remote computer and manage it. They can also retrieve files from a remote computer.

#### Application Layer protocol

##### 1. TELNET:

Telnet stands for telephone network. It helps in terminal emulation. It allows Telnet client to access the resources of Telnet server. It is used for managing the files on the internet. It is used for initial set up of devices like switches. Telnet command is a command that uses the Telnet protocol to communicate with a remote device or system.

Command

```
telnet [\\RemoteServer]
```

\\RemoteServer : Specifies the name of the server to which you want to connect

##### 2. FTP:

FTP stands for file transfer protocol. It is the protocol that actually lets us transfer files. It can facilitate this between any two machines using it. But FTP is not just a protocol but

it is also a program.FTP promotes sharing of files via remote computers with reliable and efficient data transfer

Command

```
ftp machinename
```

### 3. TFTP:

The Trivial File Transfer Protocol (TFTP) is the stripped-down, stock version of FTP, but it's the protocol of choice if you know exactly what you want and where to find it. It's a technology for transferring files between network devices, and is a simplified version of FTP

Command

```
tftp [ options... ] [host [port]] [-c command]
```

### 4. NFS:

It stands for network file system.It allows remote hosts to mount file systems over a network and interact with those file systems as though they are mounted locally. This enables system administrators to consolidate resources onto centralized servers on the network.

Command

```
service nfs start
```

### 5. SMTP:

It stands for Simple Mail Transfer Protocol.It is a part of TCP/IP protocol.Using a process called “store and forward,” SMTP moves your email on and across networks. It works closely with something called the Mail Transfer Agent (MTA) to send your communication to the right computer and email inbox.

Command

```
MAIL FROM:<mail@abc.com?
```

### 6. LPD:

It stands for Line Printer Daemon.It is designed for printer sharing.It is the part that receives and processes the request. A “daemon” is a server or agent.

Command

```
lpd [ -d ] [ -l ] [ -D DebugOutputFile]
```

### 7. X window:

It defines a protocol for the writing of graphical user interface–based client/server applications. The idea is to allow a program, called a client, to run on one computer. It is primarily used in networks of interconnected mainframes.

Command

```
Run xdm in runlevel 5
```

## 8. SNMP:

It stands for Simple Network Management Protocol. It gathers data by polling the devices on the network from a management station at fixed or random intervals, requiring them to disclose certain information. It is a way that servers can share information about their current state, and also a channel through which an administrator can modify pre-defined values.

Command

```
snmpget -mALL -v1 -cpublic snmp_agent_Ip_address sysName.0
```

## 9. DNS:

It stands for Domain Name Service. Every time you use a domain name, therefore, a DNS service must translate the name into the corresponding IP address. For example, the domain name www.abc.com might translate to 198.105.232.4.

Command

```
ipconfig /flushdns
```

## 10. DHCP:

It stands for Dynamic Host Configuration Protocol (DHCP). It gives IP addresses to hosts. There is a lot of information a DHCP server can provide to a host when the host is registering for an IP address with the DHCP server.

Command

```
clear ip dhcp binding {address | * }
```

## Domain Addressing

DNS is a host name to IP address translation service. DNS is a distributed database implemented in a hierarchy of name servers. It is an application layer protocol for message exchange between clients and servers.

### Requirement

Every host is identified by the IP address but remembering numbers is very difficult for the people and also the IP addresses are not static therefore a mapping is required to change the domain name to IP address. So DNS is used to convert the domain name of the websites to their numerical IP address.

### Domain:

There are various kinds of Domain:

1. **Generic domain:** .com (commercial) .edu (educational) .mil (military) .org (non-profit organization) .net (similar to commercial) all these are generic domain.
2. **Country domain** .in (India) .us .uk
3. Inverse domain if we want to know what is the domain name of the website. Ip to domain name mapping. So DNS can provide both the mapping for example to find the ip addresses of geeksforgeeks.org then we have to type nslookup www.geeksforgeeks.org.

### Organization of Domain

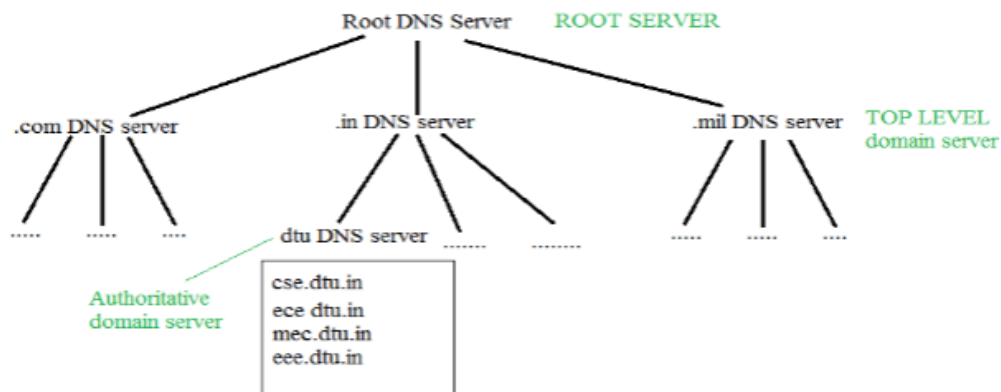


Figure 80 Organization of Domain

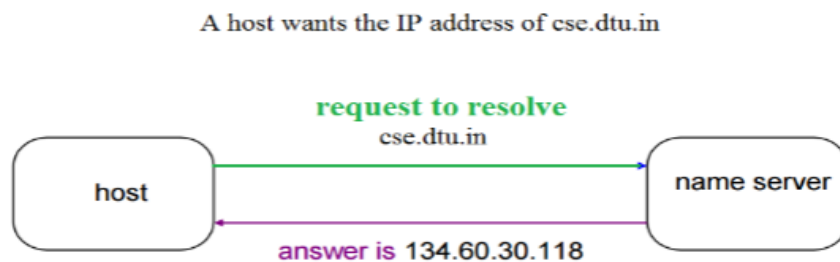
It is Very difficult to find out the ip address associated to a website because there are millions of websites and with all those websites we should be able to generate the ip address immediately, there should not be a lot of delay for that to happen organization of database is very important.

**DNS record** – Domain name, ip address what is the validity?? What is the time to live?? And all the information related to that domain name. These records are stored in tree like structure.

**Namespace** – Set of possible names, flat or hierarchical. Naming system maintains a collection of bindings of names to values – given a name, a resolution mechanism returns the corresponding value

**Name server** – It is an implementation of the resolution mechanism. DNS (Domain Name System) = Name service in Internet – Zone is an administrative unit, domain is a subtree.

### Name to Address Resolution



*Figure 81 Name to Address Resolution*

The host request the DNS name server to resolve the domain name. And the name server returns the IP address corresponding to that domain name to the host so that the host can future connect to that IP address.

### Hierarchy of Name Servers

- **Root name servers** – It is contacted by name servers that cannot resolve the name. It contacts authoritative name server if name mapping is not known. It then gets the mapping and return the IP address to the host.
- **Top level server** – It is responsible for com, org, edu etc. and all top level country domains like uk, fr, ca, in etc. They have info about authoritative domain servers and know names and IP addresses of each authoritative name server for the second level domains.
- **Authoritative name servers** This is organization's DNS server, providing authoritative hostname to IP mapping for organization servers. It can be maintained by organization or service provider. In order to reach cse.dtu.in we have to ask the root DNS server, then it will point out to the top level domain server and then to authoritative domain name server which actually contains the IP address. So the authoritative domain server will return the associative ip address.

### Domain Name Server

The client machine sends a request to the local name server, which, if root does not find the address in its database, sends a request to the root name server, which in turn, will route the query to an intermediate or authoritative name server. The root name server can also contain some hostname to IP address mappings. The intermediate name server always knows who the authoritative name server is. So finally the IP address is returned to the local name server which in turn returns the IP address to the host.

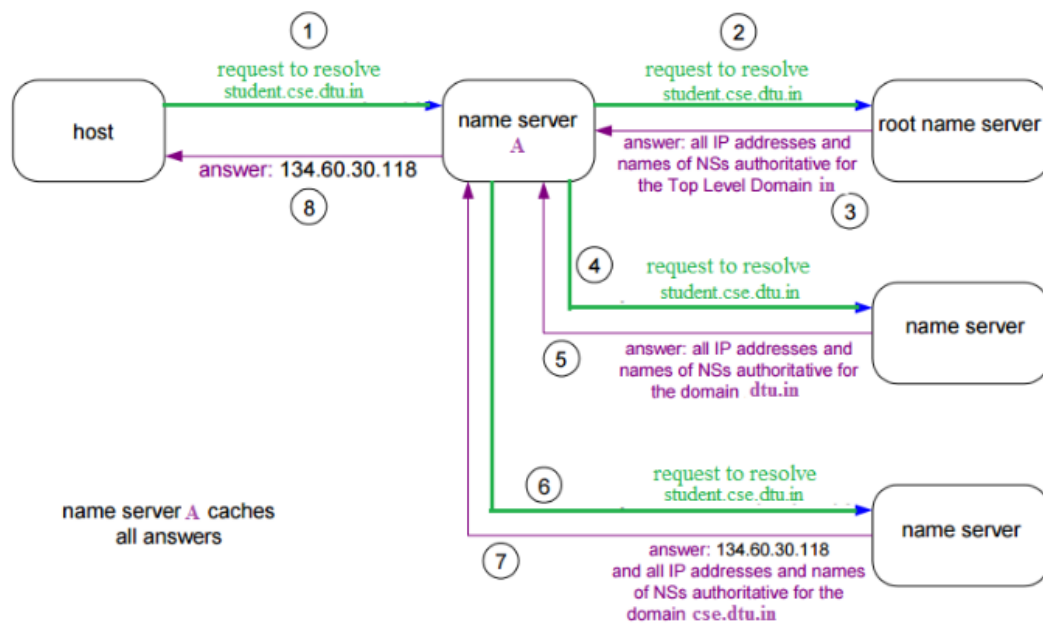


Figure 82 DNS

### How DNS Works

DNS is a global system for translating IP addresses to human-readable domain names. When a user tries to access a web address like “example.com”, their web browser or application performs a **DNS Query** against a DNS server, supplying the **hostname**. The DNS server takes the hostname and resolves it into a numeric IP address, which the web browser can connect to.

A component called a **DNS Resolver** is responsible for checking if the **hostname is available in local cache**, and if not, contacts a series of **DNS Name Servers**, until eventually it receives the IP of the service the user is trying to reach, and returns it to the browser or application. This usually takes less than a second.

### DNS Types: 3 DNS Query Types

There are three types of queries in the DNS system:

#### 1. Recursive Query

In a recursive query, a DNS client provides a hostname, and the **DNS Resolver “must” provide an answer**—it responds with either a relevant resource record, or an error message if it can't be found. The resolver starts a recursive query process, starting from the **DNS Root Server**, until it finds the **Authoritative Name Server** (for more on Authoritative Name Servers see DNS Server Types below) that holds the IP address and other information for the requested hostname.

#### 2. Iterative Query

In an iterative query, a DNS client provides a hostname, and the DNS Resolver returns the best answer it can. If the DNS resolver has the relevant DNS records in its cache, it returns them. If not, it refers the DNS client to the Root Server, or another Authoritative Name Server which is nearest to the required DNS zone. The DNS client must then repeat the query directly against the DNS server it was referred to.

### 3. Non-Recursive Query

A non-recursive query is a query in which the DNS Resolver already knows the answer. It either immediately returns a DNS record because it already stores it in local cache, or queries a DNS Name Server which is authoritative for the record, meaning it definitely holds the correct IP for that hostname. In both cases, there is no need for additional rounds of queries (like in recursive or iterative queries). Rather, a response is immediately returned to the client.

## DNS Types: 3 Types of DNS Servers

The following are the most common DNS server types that are used to resolve hostnames into IP addresses.

### 1. DNS Resolver

A DNS resolver (recursive resolver), is designed to receive DNS queries, which include a human-readable hostname such as “www.example.com”, and is responsible for tracking the IP address for that hostname.

### 2. DNS Root Server

The root server is the first step in the journey from hostname to IP address. The DNS Root Server extracts the Top Level Domain (TLD) from the user’s query—for example, www.example.com—and provides details for the .com TLD Name Server. In turn, that server will provide details for domains with the .com DNS zone, including “example.com”.

There are 13 root servers worldwide, indicated by the letters A through M, operated by organizations like the Internet Systems Consortium, Verisign, ICANN, the University of Maryland, and the U.S. Army Research Lab.

### 3. Authoritative DNS Server

Higher level servers in the DNS hierarchy define which DNS server is the “authoritative” name server for a specific hostname, meaning that it holds the up-to-date information for that hostname.

The Authoritative Name Server is the last stop in the name server query—it takes the hostname and returns the correct IP address to the DNS Resolver (or if it cannot find the domain, returns the message NXDOMAIN).

## DNS Types: 10 Top DNS Record Types

DNS servers create a DNS record to provide important information about a domain or hostname, particularly its current IP address. The most common DNS record types are:

- **Address Mapping record (A Record)**—also known as a DNS host record, stores a hostname and its corresponding IPv4 address.
- **IP Version 6 Address record (AAAA Record)**—stores a hostname and its corresponding IPv6 address.
- **Canonical Name record (CNAME Record)**—can be used to **alias a hostname to another hostname**. When a DNS client requests a record that contains a CNAME, which points to another hostname, the DNS resolution process is repeated with the new hostname.
- **Mail exchanger record (MX Record)**—specifies an SMTP email server for the domain, used to route outgoing emails to an email server.
- **Name Server records (NS Record)**—specifies that a DNS Zone, such as “example.com” is delegated to a specific Authoritative Name Server, and provides the address of the name server.
- **Reverse-lookup Pointer records (PTR Record)**—allows a DNS resolver to provide an IP address and receive a hostname (reverse DNS lookup).
- **Certificate record (CERT Record)**—stores encryption certificates—PKIX, SPKI, PGP, and so on.
- **Service Location (SRV Record)**—a service location record, like MX but for other communication protocols.
- **Text Record (TXT Record)**—typically carries machine-readable data such as opportunistic encryption, sender policy framework, DKIM, DMARC, etc.
- **Start of Authority (SOA Record)**—this record appears at the beginning of a DNS zone file, and indicates the Authoritative Name Server for the current DNS zone, contact details for the domain administrator, domain serial number, and information on how frequently DNS information for this zone should be refreshed.

### DNS Can Do Much More

Now that we’ve covered the major types of traditional DNS infrastructure, you should know that DNS can be more than just the “plumbing” of the Internet. Advanced DNS solutions can help do some amazing things, including:

- **Global server load balancing (GSLB)**—fast routing of connections between globally distributed data centers
- **Multi CDN**—routing users to the CDN that will provide the best experience
- **Geographical routing**—identifying the physical location of each user and ensuring they are routed to the nearest possible resource
- **Data center and cloud migration**—moving traffic in a controlled manner from on premise resources to cloud resources
- **Internet traffic management**—reducing network congestion and ensuring traffic flows to the appropriate resource in an optimal manner

These capabilities are made possible by next-generation DNS servers that are able to intelligently route and filter traffic. Learn more about NS1’s intelligent DNS platform and take DNS to the next level.

### HTTP

**The Hypertext Transfer Protocol (HTTP)** is an application-level protocol for distributed, collaborative, hypermedia information systems. This is the foundation for data

communication for the World Wide Web (i.e. internet) since 1990. HTTP is a generic and stateless **The Hypertext Transfer Protocol (HTTP)** protocol which can be used for other purposes as well using extensions of its request methods, error codes, and headers.

Basically, HTTP is a TCP/IP based communication protocol, that is used to deliver data (HTML files, image files, query results, etc.) on the World Wide Web. The default port is TCP 80, but other ports can be used as well. It provides a standardized way for computers to communicate with each other. HTTP specification specifies how clients' request data will be constructed and sent to the server, and how the servers respond to these requests.

### Basic Features of HTTP

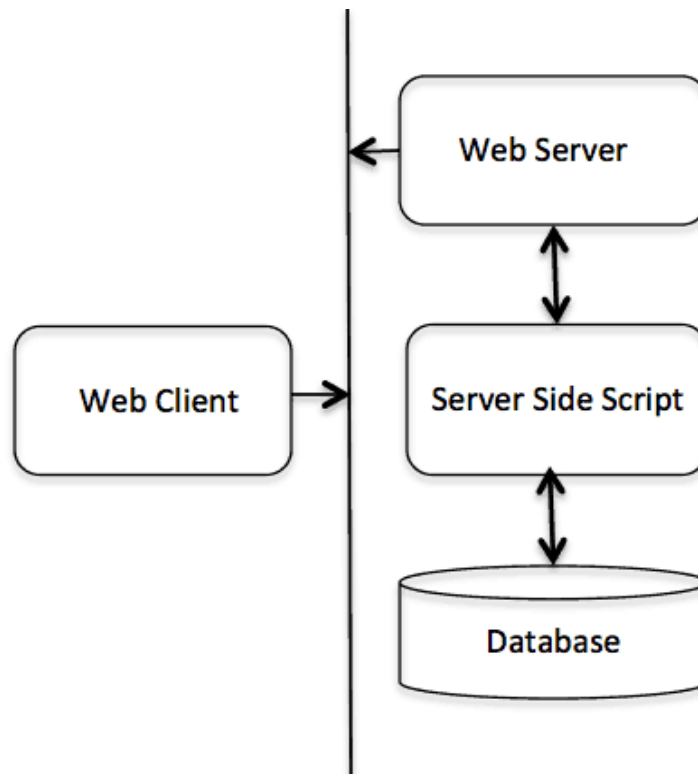
There are three basic features that make HTTP a simple but powerful protocol:

- **HTTP is connectionless:** The HTTP client, i.e., a browser initiates an HTTP request and after a request is made, the client disconnects from the server and waits for a response. The server processes the request and re-establishes the connection with the client to send a response back.
- **HTTP is media independent:** It means, any type of data can be sent by HTTP as long as both the client and the server know how to handle the data content. It is required for the client as well as the server to specify the content type using appropriate MIME-type.
- **HTTP is stateless:** As mentioned above, HTTP is connectionless and it is a direct result of HTTP being a stateless protocol. The server and client are aware of each other only during a current request. Afterwards, both of them forget about each other. Due to this nature of the protocol, neither the client nor the browser can retain information between different requests across the web pages.

HTTP/1.0 uses a new connection for each request/response exchange, where as HTTP/1.1 connection may be used for one or more request/response exchanges.

### Basic Architecture of HTTP

The following diagram shows a very basic architecture of a web application and depicts where HTTP sits:



*Figure 83 HTTP Protocol*

### HTTP Architecture

The HTTP protocol is a request/response protocol based on the client/server based architecture where web browsers, robots and search engines, etc. act like HTTP clients, and the Web server acts as a server.

#### Client

The HTTP client sends a request to the server in the form of a request method, URI, and protocol version, followed by a MIME-like message containing request modifiers, client information, and possible body content over a TCP/IP connection.

#### Server

The HTTP server responds with a status line, including the message's protocol version and a success or error code, followed by a MIME-like message containing server information, entity Meta information, and possible entity-body content.

### FTP

**File Transfer Protocol (FTP)** is an application layer protocol which moves files between local and remote file systems. It runs on the top of TCP, like HTTP. To transfer a file, 2 TCP connections are used by FTP in parallel: control connection and data connection.

**What is control connection?**

For sending control information like user identification, password, commands to change the remote directory, commands to retrieve and store files etc., FTP makes use of control connection. Control connection is initiated on port number 21.

### What is data connection?

For sending the actual file, FTP makes use of data connection. Data connection is initiated on port number 20.

FTP sends the control information out-of-band as it uses a separate control connection. Some protocols send their request and response header lines and the data in the same TCP connection. For this reason, they are said to send their control information in-band. HTTP and SMTP are such examples.

### FTP Session:

When a FTP session is started between a client and a server, the client initiates a control TCP connection with the server side. The client sends the control information over this. When the server receives this, it initiates a data connection to the client side. Only one file can be sent over one data connection. But the control connection remains active throughout the user session. As we know HTTP is stateless i.e. it does not have to keep track of any user state. But FTP needs to maintain a state about its user throughout the session.

**Data Structures:** FTP allows three types of data structures:

- **File Structure** – In file-structure there is no internal structure and the file is considered to be a continuous sequence of data bytes.
- **Record Structure** – In record-structure the file is made up of sequential records.
- **Page Structure** – In page-structure the file is made up of independent indexed pages.

**FTP Commands** – Some of the FTP commands are:

- **USER** – This command sends the user identification to the server.
- **PASS** – This command sends the user password to the server.
- **CWD** – This command allows the user to work with a different directory or dataset for file storage or retrieval without altering his login or accounting information.
- **RMD** – This command causes the directory specified in the path-name to be removed as a directory.
- **MKD** – This command causes the directory specified in the path name to be created as a directory.
- **PWD** – This command causes the name of the current working directory to be returned in the reply.
- **RETR** – This command causes the remote host to initiate a data connection and to send the requested file over the data connection.
- **STOR** – This command causes to store a file into the current directory of the remote host.
- **LIST** – Sends a request to display the list of all the files present in the directory.
- **ABOR** – This command tells the server to abort the previous FTP service command and any associated transfer of data.
- **QUIT** – This command terminates a USER and if file transfer is not in progress, the server closes the control connection.

**FTP Replies** – Some of the FTP replies are:

- 200 Command okay.
- 530 Not logged in.
- 331 User name okay, need password.
- 225 Data connection open; no transfer in progress.
- 221 Service closing control connection.
- 551 Requested action aborted: page type unknown.
- 502 Command not implemented.
- 503 Bad sequence of commands.
- 504 Command not implemented for that parameter.

**Anonymous FTP:**

Anonymous FTP is enabled on some sites whose files are available for public access. A user can access these files without having any username or password. Instead, username is set to anonymous and password to guest by default. Here, the user access is very limited. For example, the user can be allowed to copy the files but not to navigate through directories.

**Proxy Server**

A proxy server is a dedicated computer or a software system running on a computer that acts as an intermediary between an endpoint device, such as a computer, and another server from which a user or client is requesting a service. The proxy server may exist in the same machine as a firewall server or it may be on a separate server, which forwards requests through the firewall.

An advantage of a proxy server is that its cache can serve all users. If one or more Internet sites are frequently requested, these are likely to be in the proxy's cache, which will improve user response time. A proxy can also log its interactions, which can be helpful for troubleshooting.

**Here's a simple example of how proxy servers work:**

When a proxy server receives a request for an Internet resource (such as a Web page), it looks in its local cache of previously pages. If it finds the page, it returns it to the user without needing to forward the request to the Internet. If the page is not in the cache, the proxy server, acting as a client on behalf of the user, uses one of its own IP addresses to request the page from the server out on the Internet. When the page is returned, the proxy server relates it to the original request and forwards it on to the user.

Proxy servers are used for both legal and illegal purposes. In the enterprise, a proxy server is used to facilitate security, administrative control or caching services, among other purposes. In a personal computing context, proxy servers are used to enable user privacy and anonymous surfing. Proxy servers can also be used for the opposite purpose: To monitor traffic and undermine user privacy.

To the user, the proxy server is invisible; all Internet requests and returned responses appear to be directly with the addressed Internet server. (The proxy is not actually invisible; its IP address has to be specified as a configuration option to the browser or other protocol program.)

Users can access web proxies online or configure web browsers to constantly use a proxy server. Browser settings include automatically detected and manual options for HTTP, SSL, FTP, and SOCKS proxies. Proxy servers may serve many users or just one per server. These options are called shared and dedicated proxies, respectively. There are a number of reasons for proxies and thus a number of types of proxy servers, often in overlapping categories.

### **Forward and reverse proxy servers**

Forward proxies send the requests of a client onward to a web server. Users access forward proxies by directly surfing to a web proxy address or by configuring their Internet settings. Forward proxies allow circumvention of firewalls and increase the privacy and security for a user but may sometimes be used to download illegal materials such as copyrighted materials or child pornography.

### **Reverse proxies**

Reverse proxies transparently handle all requests for resources on destination servers without requiring any action on the part of the requester.

Reverse proxies are used:

- To enable indirect access when a website disallows direct connections as a security measure.
- To allow for load balancing between servers.
- To stream internal content to Internet users.
- To disable access to a site, for example when an ISP or government wishes to block a website.

Sites might be blocked for more or less legitimate reasons. Reverse proxies may be used to prevent access to immoral, illegal or copyrighted content. Sometimes these reasons are justifiable but sometimes justification is dubious. Reverse proxies sometimes prevent access news sites where users could view leaked information. They can also prevent users from accessing sites where they can disclose information about government or industry actions. Blocking access to such websites may violate free speech rights.

### **More types of proxies**

- **Transparent proxies** are typically found near the exit of a corporate network. These proxies centralize network traffic. On corporate networks, a proxy server is associated with -- or is part of -- a gateway server that separates the network from external networks (typically the Internet) and a firewall that protects the network from outside intrusion and allows data to be scanned for security purposes before delivery to a client on the network. These proxies help with monitoring and administering network traffic as the computers in a corporate network are usually safe devices that do not need anonymity for typically mundane tasks.
- **Anonymous proxies** hide the IP address of the client using them allow to access materials that are blocked by firewalls or to circumvent IP address bans. They may be used for enhanced privacy and / or protection from attack.
- **Highly anonymous proxies** hide even the fact that they are being used by clients and present a non-proxy public IP address. So not only do they hide the IP address of the

client using them, they also allow access to sites that might block proxy servers. Examples of highly anonymous proxies include I2P and TOR.

- **Socks 4 and 5 proxies** provide proxy service for UDP data and DNS look up operations in addition to Web traffic. Some proxy servers offer both Socks protocols.
- **DNS proxies** forward domain name service (DNS) requests from LANs to Internet DNS servers while caching for enhanced speed.

## DHCP

DHCP is an abbreviation for Dynamic Host Configuration Protocol. It is an application layer protocol used by hosts for obtaining network setup information. The DHCP is controlled by DHCP server that dynamically distributes network configuration parameters such as IP addresses, subnet mask and gateway address.

### What is Dynamic host configuration protocol?

- **Dynamic** – Automatically
- **Host** – Any computer that is connected to the network
- **Configuration** – To configure a host means to provide network information(ip address, subnet mask, Gateway address) to a host
- **Protocol** – Set of rules

Summing up, a DHCP server dynamically configures a host in a network.

Disadvantage of manually Configuring the host: Configuring a host when it is connected to the network can be done either manually i.e., by the network administrator or by the DHCP server. In case of home networks, manual configuration is quite easy. Whereas in the large networks, the network administrator might face many problems.

Also, the manual configuration is prone to mistakes. Say a Network administrator might assign an IP address which was already assigned. Thus, causing difficulty for both administrator as well as neighbors on network.

So, here comes the use of DHCP server. Before discussing about how DHCP server works, let's go through the DHCP entities.

### Configuring a host using DHCP

To configure a host, we require the following things:

- **Leased IP address** – IP address to a host which lasts for a particular duration which goes for a few hours, few days or few weeks.
- **Subnet Mask** – The host can know on which network it is on.
- **Gateway address** – The Gateway is the Internet Service Provider that connects user to the internet. The Gateway address lets the host know where the gateway is to connect to the internet.

### DHCP Entities

- **DHCP server:** It automatically provides network information (IP address, subnet mask, gateway address) on lease. Once the duration is expired, that network information can be assigned to other machine. It also maintains the data storage which stores the available IP addresses.
- **DHCP client:** Any node which request an IP address allocation to a network is considered as DHCP client.
- **DHCP Relay Agent:** In case, we have only one DHCP server for multiple LAN's then this Agent which presents in every network forwards the DHCP request to DHCP server. So, using DHCP Relay Agent we can configure multiple LAN's with single server.

#### How DHCP server assigns IP address to a host?

- **DHCPDISCOVER:** When a new node is connected to the network, it broadcasts the DHCPDISCOVER message which contains the source address as 0.0.0.0 to every node on the network including server. DHCP server on receiving the message, returns the DHCPOFFER message to the requested host which contains the server address and new IP address to the node.
- **DHCPOFFER:** If there are multiple servers on the network, host receives multiple DHCPOFFER messages. It is up to the host to select a particular message.
- **DHCPREQUEST:** The requested host on receiving the offer message, it again broadcasts the DHCPREQUEST message on the network with the address of the server whose offer message is accepted by the host. The server which pertains to that server address sent by the host checks whether the address to be assigned to the node is available in the data storage.
- **DHCPACK:** If the address is assigned, it marks the IP address in the storage as unavailable to ensure consistency. Now, the server sends DHCPACK packet to the requested host which contains network information (IP address, subnet mask, gateway address). In case, if the address is assigned to other machine meanwhile, then the server sends the packet DHCPNAK to the requested host indicating that the IP address is assigned to some other machine.
- **DHCPRELEASE:** And finally, if the host wants to move to other network or if it has finished its work, it sends the DHCPRELEASE packet to the server indicating that it wants to disconnect. Then the server marks the IP address as available in the storage so that it can be assigned to other machine.

## E-mail Protocols

E-mail Protocols are set of rules that help the client to properly transmit the information to or from the mail server. Here in this tutorial, we will discuss various protocols such as **SMTP**, **POP**, and **IMAP**.

### SMTP

**SMTP** stands for **Simple Mail Transfer Protocol**. It was first proposed in 1982. It is a standard protocol used for sending e-mail efficiently and reliably over the internet.

**Key Points:**

- SMTP is application level protocol.
- SMTP is **connection oriented** protocol.
- SMTP is **text based** protocol.
- It handles exchange of messages between e-mail servers over TCP/IP network.
- Apart from transferring e-mail, SMTP also **provides notification** regarding **incoming** mail.
- When you send e-mail, your e-mail client sends it to your e-mail server which further **contacts the recipient mail server using SMTP client**.
- These SMTP commands specify the sender's and receiver's e-mail address, along with the message to be send.
- The exchange of commands between servers is carried out without intervention of any user.
- In case, message cannot be delivered, **an error report is sent to the sender** which makes SMTP a reliable protocol.

**SMTP Commands**

The following table describes some of the SMTP commands:

| S.N. | Command Description                                                                                                                                           |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | <b>HELLO</b><br>This command initiates the SMTP conversation.                                                                                                 |
| 2    | <b>EHELLO</b><br>This is an alternative command to initiate the conversation. ESMTP indicates that the sender server wants to use extended SMTP protocol.     |
| 3    | <b>MAIL FROM</b><br>This indicates the sender's address.                                                                                                      |
| 4    | <b>RCPT TO</b><br>It identifies the recipient of the mail. In order to deliver similar message to multiple users this command can be repeated multiple times. |
| 5    | <b>SIZE</b><br>This command let the server know the size of attached message in bytes.                                                                        |
| 6    | <b>DATA</b><br>The <b>DATA</b> command signifies that a stream of data will follow. Here stream of data refers to the body of the message.                    |

|   |                                                                                                                             |
|---|-----------------------------------------------------------------------------------------------------------------------------|
| 7 | <b>QUIT</b><br>This commands is used to terminate the SMTP connection.                                                      |
| 8 | <b>VERFY</b><br>This command is used by the receiving server in order to verify whether the given username is valid or not. |
| 9 | <b>EXPN</b><br>It is same as VRFY, except it will list all the users name when it used with a distribution list.            |

## IMAP

**IMAP** stands for **Internet Mail Access Protocol**. It was first proposed in 1986. There exist five versions of IMAP as follows:

1. Original IMAP
2. IMAP2
3. IMAP3
4. IMAP2bis
5. IMAP4

### Key Points:

- IMAP allows the client program to **manipulate the e-mail message on the server without downloading** them on the local computer.
- The e-mail is **hold and maintained by the remote server**.
- It enables us to **take any action** such as downloading, delete the mail without reading the mail. It enables us to create, manipulate and delete remote **message folders called mail boxes**.
- IMAP enables the users to **search the e-mails**.
- It allows **concurrent access to multiple mailboxes on multiple mail servers**.

## IMAP Commands

The following table describes some of the IMAP commands:

| S.N. | Command Description                                                                               |
|------|---------------------------------------------------------------------------------------------------|
| 1    | <b>IMAP_LOGIN</b><br>This command opens the connection.                                           |
| 2    | <b>CAPABILITY</b><br>This command requests for listing the capabilities that the server supports. |

|   |                                                                                                                                                                                                 |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3 | <b>NOOP</b><br>This command is used as a periodic poll for new messages or message status updates during a period of inactivity.                                                                |
| 4 | <b>SELECT</b><br>This command helps to select a mailbox to access the messages.                                                                                                                 |
| 5 | <b>EXAMINE</b><br>It is same as SELECT command except no change to the mailbox is permitted.                                                                                                    |
| 6 | <b>CREATE</b><br>It is used to create mailbox with a specified name.                                                                                                                            |
| 7 | <b>DELETE</b><br>It is used to permanently delete a mailbox with a given name.                                                                                                                  |
| 8 | <b>RENAME</b><br>It is used to change the name of a mailbox.                                                                                                                                    |
| 9 | <b>LOGOUT</b><br>This command informs the server that client is done with the session. The server must send BYE untagged response before the OK response and then close the network connection. |

## POP

**POP** stands for **Post Office Protocol**. It is generally used to support a single client. There are several versions of POP but the POP 3 is the current standard.

### Key Points

- POP is an application layer internet standard protocol.
- Since POP supports offline access to the messages, thus requires less internet usage time.
- POP does not allow search facility.
- In order to access the messaged, it is necessary to download them.
- It allows only one mailbox to be created on server.
- It is not suitable for accessing non mail data.
- POP commands are generally abbreviated into codes of three or four letters. Eg. STAT.

### POP Commands

The following table describes some of the POP commands:

| S.N. | Command Description                                                                           |
|------|-----------------------------------------------------------------------------------------------|
| 1    | <b>LOGIN</b><br>This command opens the connection.                                            |
| 2    | <b>STAT</b><br>It is used to display number of messages currently in the mailbox.             |
| 3    | <b>LIST</b><br>It is used to get the summary of messages where each message summary is shown. |
| 4    | <b>RETR</b><br>This command helps to select a mailbox to access the messages.                 |
| 5    | <b>DELE</b><br>It is used to delete a message.                                                |
| 6    | <b>RSET</b><br>It is used to reset the session to its initial state.                          |
| 7    | <b>QUIT</b><br>It is used to log off the session.                                             |

### Comparison between POP and IMAP

| S.N. | POP                                      | IMAP                                                                 |
|------|------------------------------------------|----------------------------------------------------------------------|
| 1    | Generally used to support single client. | Designed to handle multiple clients.                                 |
| 2    | Messages are accessed offline.           | Messages are accessed online although it also supports offline mode. |
| 3    | POP does not allow search facility.      | It offers ability to search emails.                                  |
| 4    | All the messages have to be downloaded.  | It allows selective transfer of messages to the client.              |

|    |                                                                                       |                                                                                |
|----|---------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| 5  | Only one mailbox can be created on the server.                                        | Multiple mailboxes can be created on the server.                               |
| 6  | Not suitable for accessing non-mail data.                                             | Suitable for accessing non-mail data i.e. attachment.                          |
| 7  | POP commands are generally abbreviated into codes of three or four letters. Eg. STAT. | IMAP commands are not abbreviated, they are full. Eg. STATUS.                  |
| 8  | It requires minimum use of server resources.                                          | Clients are totally dependent on server.                                       |
| 9  | Mails once downloaded cannot be accessed from some other location.                    | Allows mails to be accessed from multiple locations.                           |
| 10 | The e-mails are not downloaded automatically.                                         | Users can view the headings and sender of e-mails and then decide to download. |
| 10 | POP requires less internet usage time.                                                | IMAP requires more internet usage time.                                        |

## Chapter 9

### Network Management and Security

#### Network Management

**Network management** is the process of administering and managing computer networks. Services provided by this discipline include **fault analysis, performance management, provisioning of networks and maintaining the quality of service.**

If an organization has 1000 of devices then to check all devices, one by one every day, are working properly or not is a hectic task. To ease these up, Simple Network Management Protocol (SNMP) is used.

#### Simple Network Management Protocol (SNMP)

SNMP is an **application layer protocol which uses UDP port number 161/162.** SNMP is used to **monitor network, detect network faults** and sometimes even used to **configure remote devices.**

#### SNMP components

There are 3 components of SNMP:

- **SNMP Manager** –It is a **centralized system used to monitor network.** It is also known as Network Management Station (NMS)
- **SNMP agent** –It is a **software management software module installed on a managed device.** Managed devices can be network devices like PC, router, switches, servers etc.
- **Management Information Base** –MIB consists of **information of resources that are to be managed.** These information is organized hierarchically. It consists of objects instances which are essentially variables.

#### SNMP messages

Different variables are:

- **GetRequest** –SNMP manager sends this message to request data from SNMP agent. It is simply used to retrieve data from SNMP agent. In response to this, SNMP agent responds with requested value through response message.
- **GetNextRequest** –This message can be sent to discover what data is available on a SNMP agent. The SNMP manager can request for data continuously until no more data is left. In this way, SNMP manager can take knowledge of all the available data on SNMP agent.
- **GetBulkRequest** –This message is used to retrieve large data at once by the SNMP manager from SNMP agent. It is introduced in SNMPv2c.
- **SetRequest** –It is used by SNMP manager to set the value of an object instance on the SNMP agent.
- **Response** – It is a message send from agent upon a request from manager. When sent in response to get messages, it will contain the data requested. When sent in response to set message, it will contain the newly set value as confirmation that the value has been set.

- **Trap** – These are the message send by the agent without being requested by the manager. It is sent when a fault has occurred.
- **InformRequest** – It was introduced in SNMPv2c, used to identify if the trap message has been received by the manager or not. The agents can be configured to set trap continuously until it receives an Inform message. It is same as trap but adds an acknowledgement that trap doesn't provide.
- **SNMP security levels** – It defines the type of security algorithm performed on SNMP packets. These are used in only SNMPv3. There are 3 security levels namely:
- **noAuthNoPriv** – This (no authentication, no privacy) security level uses community string for authentication and no encryption for privacy.
- **authNoPriv** – This security level (authentication, no privacy) uses HMAC with Md5 for authentication and no encryption is used for privacy.
- **authPriv** – This security level (authentication, privacy) uses HMAC with Md5 or SHA for authentication and encryption uses DES-56 algorithm.

### SNMP versions

There are 3 versions of SNMP:

- **SNMPv1** – It uses community strings for authentication and use UDP only.
- **SNMPv2c** – It uses community strings for authentication. It uses UDP but can be configured to use TCP.
- **SNMPv3** – It uses Hash based MAC with MD5 or SHA for authentication and DES-56 for privacy. This version uses TCP. Therefore, conclusion is the higher the version of SNMP, more secure it will be.

### Network Security

Security of a computer system is a crucial task. It is a process of ensuring confidentiality and integrity of the OS.

A system is said to be secure if its resources are used and accessed as intended under all the circumstances, but no system can guarantee absolute security from several of the various malicious threats and unauthorized access.

Security of a system can be threatened via two violations:

- **Threat**: A program which has the potential to cause serious damage to the system.
- **Attack**: An attempt to break security and make unauthorized use of an asset.

Security violations affecting the system can be categorized as malicious and accidental. Malicious threats, as the name suggests are a kind of harmful computer code or web script designed to create system vulnerabilities leading to back doors and security breaches. Accidental Threats, on the other hand, are comparatively easier to be protected against. Example: Denial of Service DDoS attack.

Security can be compromised via any of the breaches mentioned:

- **Breach of confidentiality:** This type of violation involves the unauthorized reading of data.
- **Breach of integrity:** This violation involves unauthorized modification of data.
- **Breach of availability:** It involves an unauthorized destruction of data.
- **Theft of service:** It involves an unauthorized use of resources.
- **Denial of service:** It involves preventing legitimate use of the system. As mentioned before, such attacks can be accidental in nature.

### Properties of Secure Communication

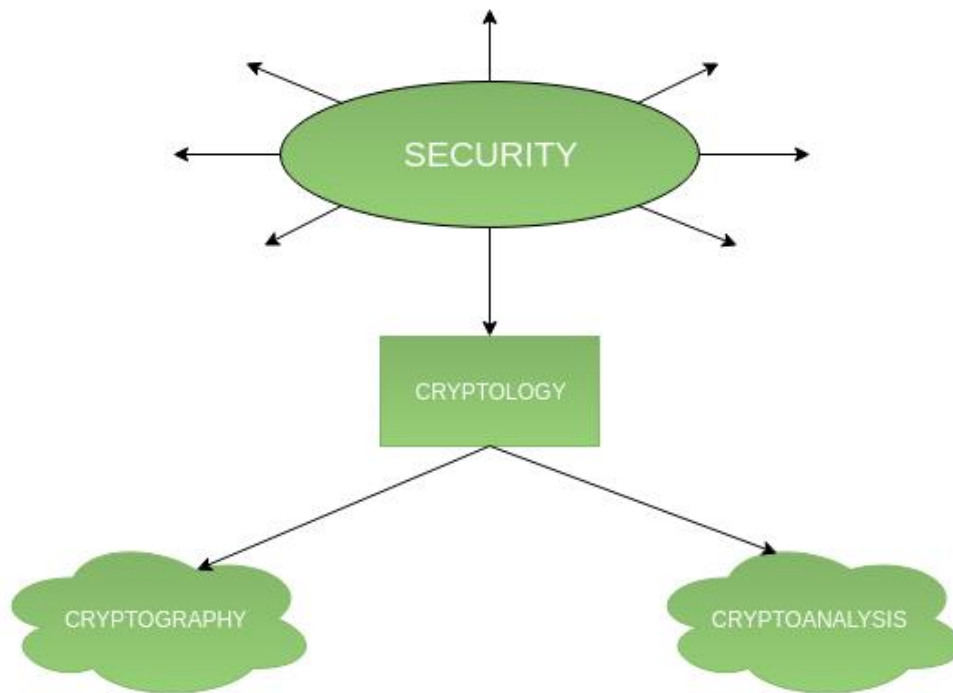
- **Confidentiality / Secrecy** - Confidentiality involves a set of rules or a promise usually executed through confidentiality agreements that limits access or places restrictions on certain types of information.
- **Authentication** - Authentication is the act of confirming the truth of an attribute of a single piece of data claimed true by an entity. In contrast with identification, which refers to the act of stating or otherwise indicating a claim purportedly attesting to a person or thing's identity, authentication is the process of actually confirming that identity.
- **Integrity**- Even if sender and receiver are able to authenticate each other, they must ensure that the data received is not altered either maliciously or by accident.
- **Non-Repudiation**- Non-repudiation refers to a situation where a statement's author cannot successfully dispute its authorship or the validity of an associated contract.
- **Authorization**- Authorization is the function of specifying access rights/privileges to resources related to information security and computer security in general and to access control in particular.

### Cryptography

Cryptography is an important aspect when we deal with network security. 'Crypto' means secret or hidden. **Cryptography is the science of secret writing with the intention of keeping the data secret.** Cryptanalysis, on the other hand, is the science or sometimes the art of breaking cryptosystems. These both terms are a subset of what is called as Cryptology.

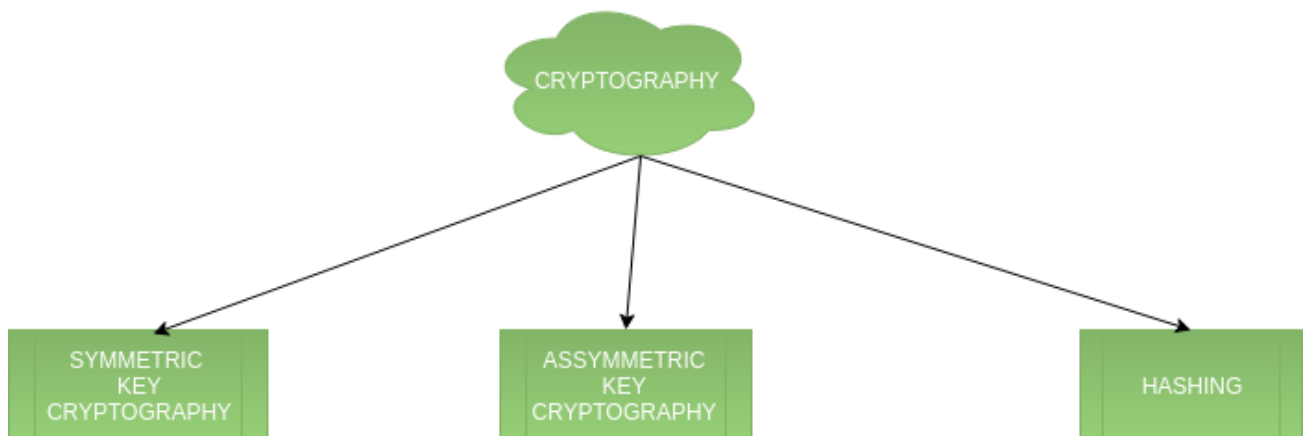
#### Classification

The flowchart depicts that cryptology is only one of the factors involved in securing networks. **Cryptology refers to study of codes, which involves both writing (cryptography) and solving (cryptanalysis) them.** Below is a classification of the crypto-terminologies and their various types.



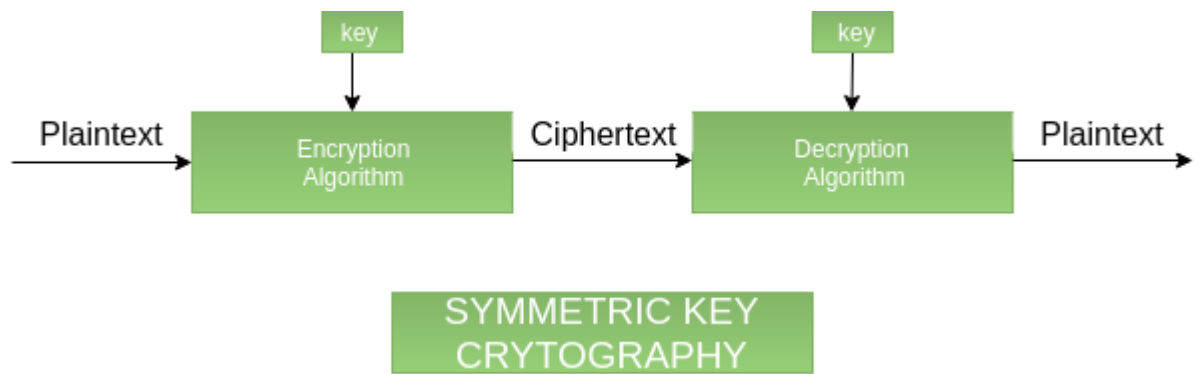
### 1. Cryptography –

Cryptography is classified into symmetric cryptography, asymmetric cryptography and hashing. Below are the description of these types.



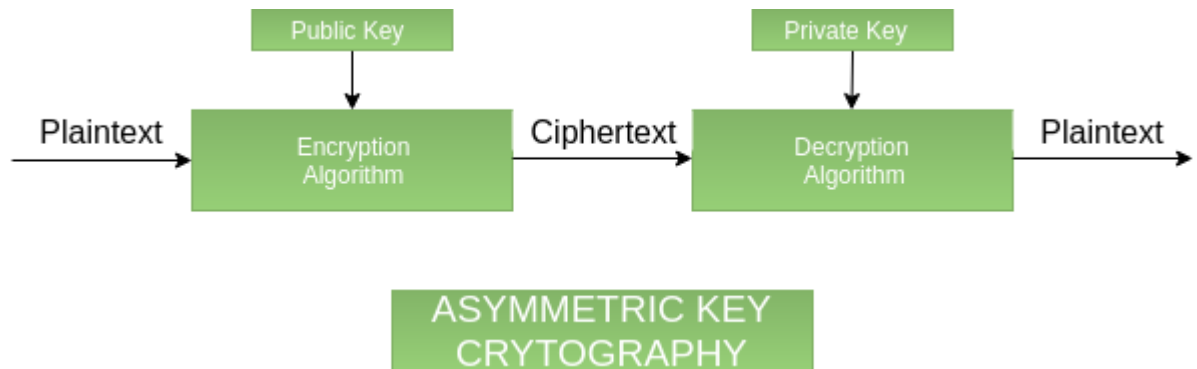
#### i. Symmetric key cryptography –

It involves usage of one secret key along with encryption and decryption algorithms which help in securing the contents of the message. The strength of symmetric key cryptography depends upon the number of key bits. It is relatively faster than asymmetric key cryptography. There arises a key distribution problem as the key has to be transferred from the sender to receiver through a secure channel.



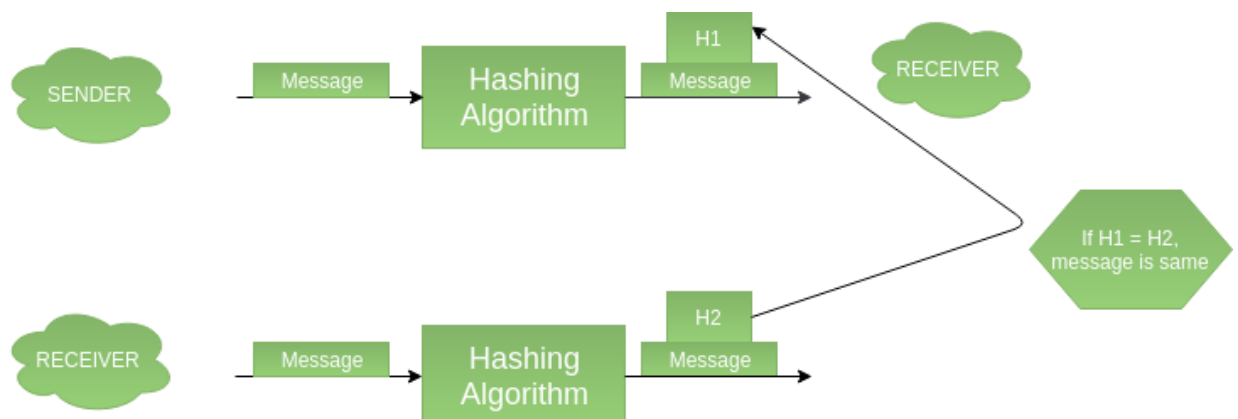
### ii. Asymmetric key cryptography –

It is also known as public key cryptography because it involves usage of a public key along with secret key. It solves the problem of key distribution as both parties' uses different keys for encryption/decryption. It is not feasible to use for decrypting bulk messages as it is very slow compared to symmetric key cryptography.

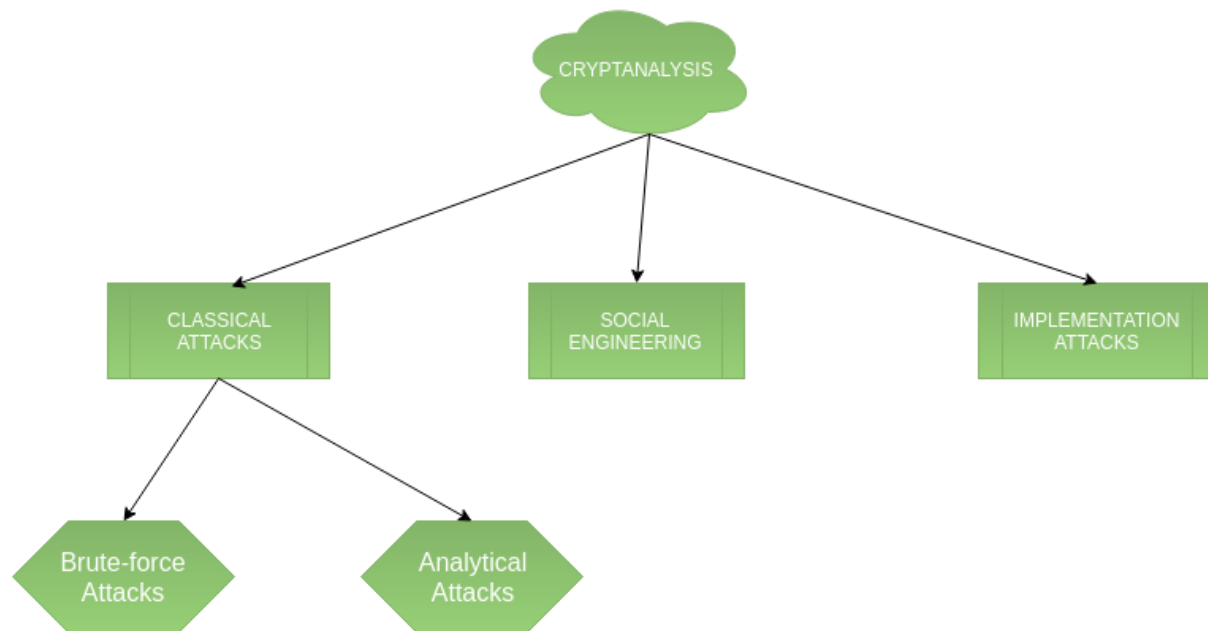


### iii. Hashing –

It involves taking the plain-text and converting it to a hash value of fixed size by a hash function. This process ensures integrity of the message as the hash value on both, sender's and receiver's side should match if the message is unaltered.



## 2. Cryptanalysis –



i. **Classical attacks –**

It can be divided into a) Mathematical analysis and b) Brute-force attacks. Brute-force attacks runs the encryption algorithm for all possible cases of the keys until a match is found. Encryption algorithm is treated as a black box. Analytical attacks are those attacks which focuses on breaking the cryptosystem by analyzing the internal structure of the encryption algorithm.

ii. **Social Engineering attack –**

It is something which is dependent on the human factor. Tricking someone to reveal their passwords to the attacker or allowing access to the restricted area comes under this attack. People should be cautious when revealing their passwords to any third party which is not trusted.

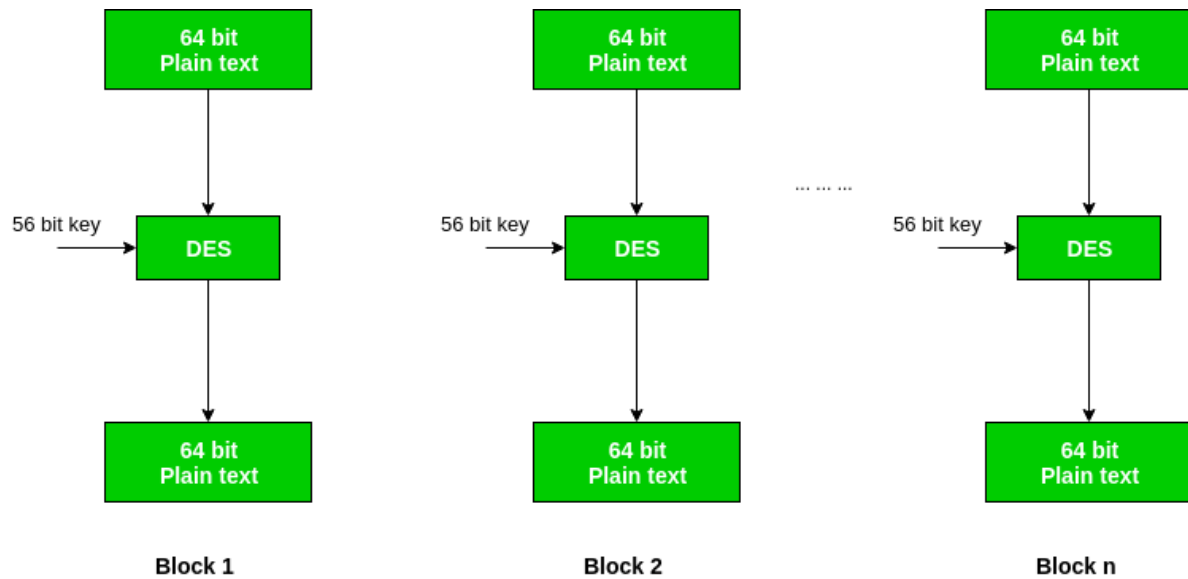
iii. **Implementation attacks –**

Implementation attacks such as side-channel analysis can be used to obtain a secret key. They are relevant in cases where the attacker can obtain physical access to the cryptosystem.

## Symmetric Key-DES

**Data encryption standard (DES)** has been found vulnerable against very powerful attacks and therefore, the popularity of DES has been found slightly on decline.

DES is a block cipher, and encrypts data in blocks of size of 64 bit each, means 64 bits of plain text goes as the input to DES, which produces 64 bits of cipher text. The same algorithm and key are used for encryption and decryption, with minor differences. The key length is 56 bits. The basic idea is show in figure.



We have mention that DES uses a 56 bit key. Actually, the initial key consists of 64 bits. However, before the DES process even starts, every 8th bit of the key is discarded to produce a 56 bit key. That is bit position 8, 16, 24, 32, 40, 48, 56 and 64 are discarded.

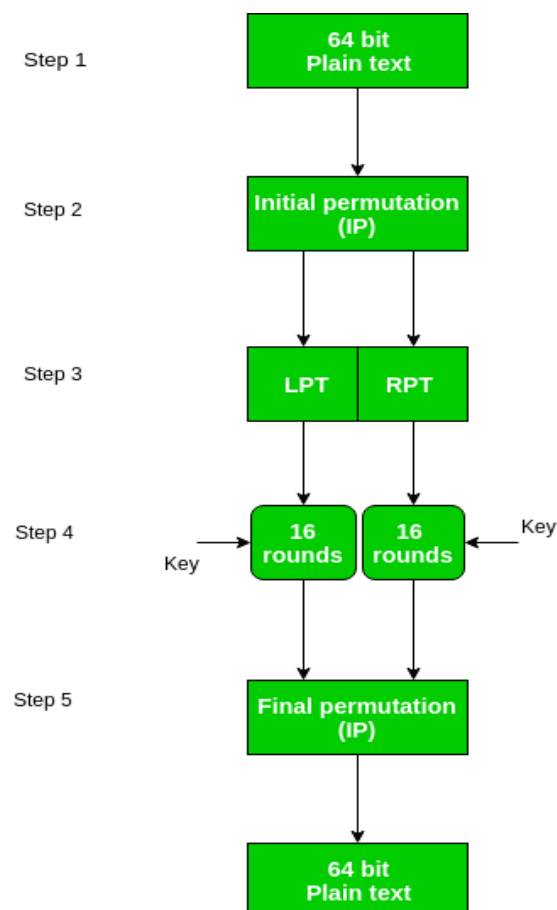
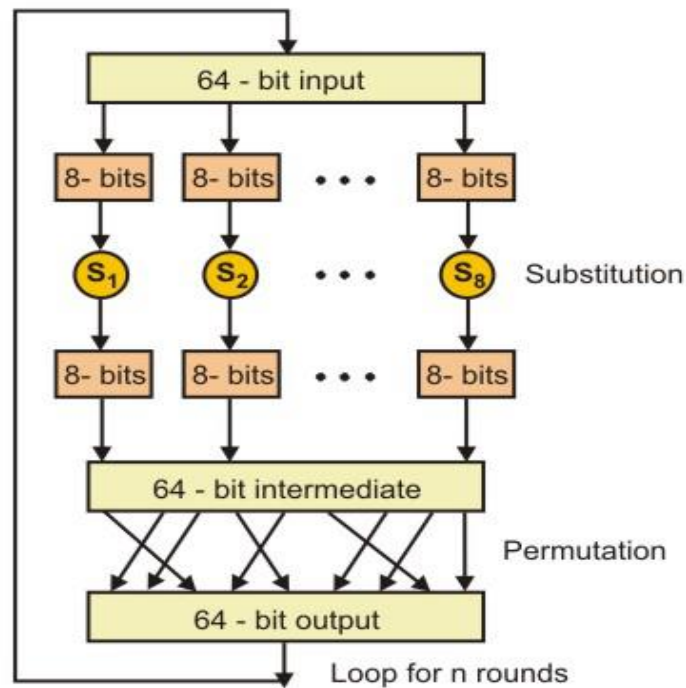
|    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 | 11 | 12 | 13 | 14 | 15 | 16 |
| 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 |
| 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 |
| 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 |

Figure - discarding of every 8<sup>th</sup> bit of original key

Thus, the discarding of every 8th bit of the key produces a 56-bit key from the original 64-bit key.

DES is based on the two fundamental attributes of cryptography: substitution (also called as confusion) and transposition (also called as diffusion). DES consists of 16 steps, each of which is called as a round. Each round performs the steps of substitution and transposition. The broad-level steps in DES.

1. In the first step, the 64 bit plain text block is handed over to an initial Permutation (IP) function.
2. The initial permutation performed on plain text.
3. Next the initial permutation (IP) produces two halves of the permuted block; says Left Plain Text (LPT) and Right Plain Text (RPT).
4. Now each LPT and RPT to go through 16 rounds of encryption process.
5. In the end, LPT and RPT are rejoined and a Final Permutation (FP) is performed on the combined block
6. The result of this process produces 64 bit cipher text.



### Initial Permutation (IP) –

As we have noted, the Initial permutation (IP) happens only once and it happens before the first round. It suggests how the transposition in IP should proceed, as shown in figure.

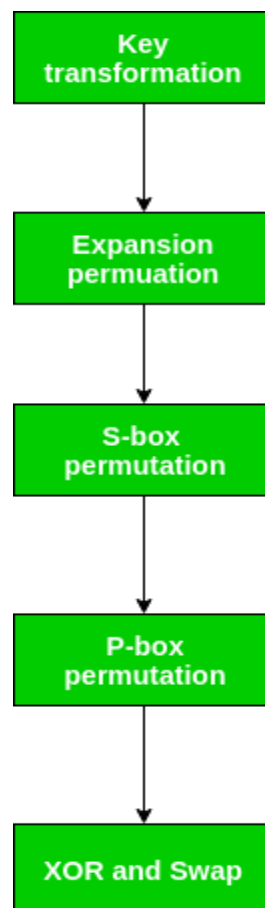
For example, it says that the IP replaces the first bit of the original plain text block with the 58th bit of the original plain text, the second bit with the 50th bit of the original plain text block and so on.

This is nothing but jugglery of bit positions of the original plain text block. the same rule applies for all the other bit positions which shows in the figure.

|    |    |    |    |    |    |    |   |    |    |    |    |    |    |    |   |
|----|----|----|----|----|----|----|---|----|----|----|----|----|----|----|---|
| 58 | 50 | 42 | 34 | 26 | 18 | 10 | 2 | 60 | 52 | 44 | 36 | 28 | 20 | 12 | 4 |
| 62 | 54 | 46 | 38 | 30 | 22 | 14 | 6 | 64 | 56 | 48 | 40 | 32 | 24 | 16 | 8 |
| 57 | 49 | 41 | 33 | 25 | 17 | 9  | 1 | 59 | 51 | 43 | 35 | 27 | 19 | 11 | 3 |
| 61 | 33 | 45 | 37 | 29 | 21 | 13 | 5 | 63 | 55 | 47 | 39 | 31 | 23 | 15 | 7 |

Figure - Initial permutation table

As we have noted after IP done, the resulting 64-bit permuted text block is divided into two half blocks. Each half block consists of 32 bits, and each of the 16 rounds, in turn, consists of the broad level steps outlined in figure.



#### Step-1: Key transformation –

We have noted initial 64-bit key is transformed into a 56-bit key by discarding every 8th bit of the initial key. Thus, for each a 56-bit key is available. From this 56-bit key, a different 48-bit Sub Key is generated during each round using a process called as key transformation. For this the 56 bit key is divided into two halves, each of 28 bits. These halves are circularly shifted left by one or two positions, depending on the round.

For example, if the round number 1, 2, 9 or 16 the shift is done by only position for other rounds, the circular shift is done by two positions. The number of key bits shifted per round is show in figure.

| Round             | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 |
|-------------------|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|
| #key bits shifted | 1 | 1 | 2 | 2 | 2 | 2 | 2 | 2 | 1 | 2  | 2  | 2  | 2  | 2  | 2  | 1  |

Figure - number of key bits shifted per round

After an appropriate shift, 48 of the 56 bit are selected. for selecting 48 of the 56 bits the table show in figure given below. For instance, after the shift, bit number 14 moves on the first position, bit number 17 moves on the second position and so on. If we observe the table carefully, we will realize that it contains only 48 bit positions. Bit number 18 is discarded (we will not find it in the table), like 7 others, to reduce a 56-bit key to a 48-bit key. Since the key transformation process involves permutation as well as selection of a 48-bit sub set of the original 56-bit key it is called Compression Permutation.

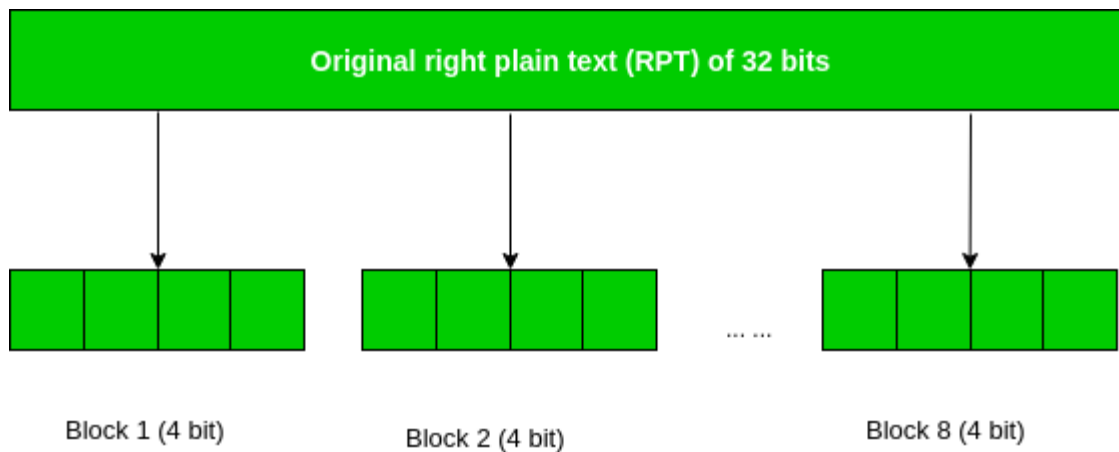
|    |    |    |    |    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|----|----|----|----|
| 14 | 17 | 11 | 24 | 1  | 5  | 3  | 28 | 15 | 6  | 21 | 10 |
| 23 | 19 | 12 | 4  | 26 | 8  | 16 | 7  | 27 | 20 | 13 | 2  |
| 41 | 52 | 31 | 37 | 47 | 55 | 30 | 40 | 51 | 45 | 33 | 48 |
| 44 | 49 | 39 | 56 | 34 | 53 | 46 | 42 | 50 | 36 | 29 | 32 |

Figure - compression permutation

Because of this compression permutation technique, a different subset of key bits is used in each round. That's make DES not easy to crack.

### Step-2: Expansion Permutation –

Recall that after initial permutation, we had two 32-bit plain text areas called as Left Plain Text (LPT) and Right Plain Text (RPT). During the expansion permutation, the RPT is expanded from 32 bits to 48 bits. Bits are permuted as well hence called as expansion permutation. This happens as the 32 bit RPT is divided into 8 blocks, with each block consisting of 4 bits. Then, each 4 bit block of the previous step is then expanded to a corresponding 6 bit block, i.e., per 4 bit block, 2 more bits are added.



**Figure - division of 32 bit RPT into 8 bit blocks**

This process results into expansion as well as permutation of the input bit while creating output. Key transformation process compresses the 56-bit key to 48 bits. Then the expansion permutation process expands the 32-bit RPT to 48-bits. Now the 48-bit key is XOR with 48-bit RPT and resulting output is given to the next step, which is the S-Box substitution.

### Asymmetric Key- RSA

The RSA algorithm is named after Ron Rivest, Adi Shamir and Len Adleman, who invented it in 1977 [RIVE78]. The basic technique was first discovered in 1973 by Clifford Cocks [COCK73] of CESG (part of the British GCHQ) but this was a secret until 1997. The patent taken out by RSA Labs has expired.

The RSA cryptosystem is the **most widely-used public key cryptography algorithm** in the world. It can be used to **encrypt a message without the need to exchange a secret key separately**.

The RSA algorithm can be used for both public key encryption and digital signatures. Its **security is based on the difficulty of factoring large integers**.

Party A can send an encrypted message to party B without any prior exchange of secret keys. A just uses **B's public key to encrypt the message** and B decrypts it using **the private key**, which only he knows. RSA can also be used to sign a message, so A can sign a message using their private key and B can verify it using A's public key.

### **Algorithm of RSA to generate key**

**This is the original algorithm.**

- 1. Generate two large random primes,  $p$  and  $q$ , of approximately equal size such that their product  $n=p*q$  is of the required bit length, e.g. 1024 bits. Compute  $n= p*q$  and  $\phi=(p-1)*(q-1)$ .**
- 2. Choose an integer  $e$ ,  $1<e<\phi$ , such that  $\gcd(e, \phi)=1$ .**

3. Compute the secret exponent  $d$ ,  $1 < d < \phi$ , such that  $ed \equiv 1 \pmod{\phi}$ . The public key is  $(n, e)$  and the private key  $(d, p, q)$ . Keep all the values  $d$ ,  $p$ ,  $q$  and  $\phi$  secret. [Sometimes the private key is written as  $(n, d)$  because you need the value of  $n$  when using  $d$ . Other times we might write the key pair as  $((N, e), d)$ .]
- $n$  is known as the modulus.
  - $e$  is known as the public exponent or encryption exponent or just the exponent.
  - $d$  is known as the secret exponent or decryption exponent.

### Encryption

Sender A does the following:-

- Obtains the recipient B's public key  $(n, e)$ .
- Represents the plaintext message as a positive integer  $m$  with  $1 < m < n$  [see note 4].
- Computes the cipher text  $c = m^e \pmod{n}$ .
- Sends the cipher text  $c$  to B.

### Decryption

Recipient B does the following:-

- Uses his private key  $(n, d)$  to compute  $m = c^d \pmod{n}$ .
- Extracts the plaintext from the message representative  $m$ .

### Example

- Encrypt Message "LOVE" using RSA algorithm
  - Let  $p = 5$ ,  $q = 7$  (smaller number are selected for simplicity)
  - Then  $n = p * q = 5 * 7 = 35$
  - and  $z = (p-1) * (q-1) = 4 * 6 = 24$
  - selecting  $e = 5$  ( $e < n$  and no common factor with  $z$ ) and  $d = 29$  (such that:  $e * d \pmod{z} = 1$ )
    - Public key  $(n, e) = (35, 5)$
    - Private Key  $(n, d) = (35, 29)$

| Plain Text | $m$ (numeric Representation) | $m^e$   | Cipher Message<br>$c = m^e \pmod{n}$ |
|------------|------------------------------|---------|--------------------------------------|
| L          | 12                           | 248832  | 17 (Q)                               |
| O          | 15                           | 759375  | 15 (O)                               |
| V          | 22                           | 5153632 | 22 (V)                               |
| E          | 5                            | 3125    | 10 (J)                               |

Encryption ( LOVE => QOVJ )

| c      | c <sup>d</sup>   | m = c <sup>d</sup> mod n | Character |
|--------|------------------|--------------------------|-----------|
| 17 (Q) | <sub>17</sub> 29 | 12                       | L         |
| 15 (O) | <sub>15</sub> 29 | 15                       | O         |
| 22 (V) | <sub>22</sub> 29 | 22                       | V         |
| 10 (J) | <sub>5</sub> 29  | 5                        | E         |

### Decryption ( QOVJ => LOVE )

- An important property of RSA algorithm:

$$\underbrace{K_B^-(K_B^+(m))}_{\text{use public key first, followed by private key}} = m = \underbrace{K_B^+(K_B^-(m))}_{\text{use private key first, followed by public key}}$$

Result is the same!

### Advantages and disadvantages of RSA Algorithm

There are advantages and disadvantages of RSA algorithm. The advantages include; RSA algorithm is safe and secure for its users through the use of complex mathematics. RSA algorithm is hard to crack since it involves factorization of prime numbers which are difficult to factorize. Moreover, RSA algorithm uses the public key to encrypt data and the key is known to everyone, therefore, it is easy to share the public key.

The disadvantages include; RSA algorithm can be very slow in cases where large data needs to be encrypted by the same computer. It requires a third party to verify the reliability of public keys. Data transferred through RSA algorithm could be compromised through middlemen who might temper with the public key system. In conclusion, both the symmetric encryption technique and the asymmetric encryption technique are important in encryption of sensitive data

### Key Exchange Protocols

Although symmetric-key and public-key cryptography can be used for privacy and user authentication, question arises about the techniques used for the distribution of keys. Particularly, symmetric-key distribution involves the following three problems:

- For n people to communicate with each other requires  $n(n-1)/2$  keys. The problem is aggravated as n becomes very large.

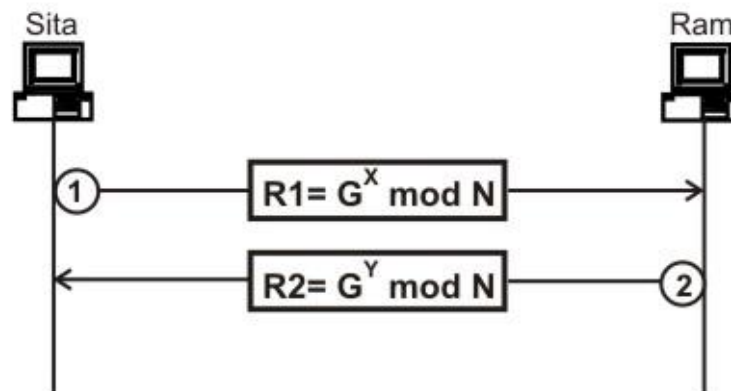
- Each person needs to remember (n-1) keys to communicate with the remaining (n-1) persons.
- How the two parties will acquire the shared key in a secured manner?

In view of the above problems, the concept of session key has emerged. A session key is created for each session and destroyed when the session is over. The Diffie-Hellman protocol is one of the most popular approach for providing one-time session key for both the parties.

### **Diffie-Hellman Protocol**

Key features of the Diffie-Hellman protocol are mentioned below

- Used to **establish a shared secret key**
- Prerequisite: N is a large prime number such that (N-1)/2 is also a prime number. G is also a prime number. Both N and G are known to Ram and Sita..
- Sita chooses a large random number x and calculates  $R1 = G^x \bmod N$  and sends it to Ram
- Ram chooses another large random number y and calculates  $R2 = G^y \bmod N$  and sends it to Sita
- Ram calculates  $K = (R1)^y \bmod N$
- Sita calculates  $K = (R2)^x \bmod N$



*Figure 84 Diffie- Hellman Protocol*

### **Key Management using KDC**

It may be noted that both R1 and R2 are sent as plaintext, which may be intercepted by an intruder. This is a serious flaw of the Diffie-Hellman Protocol. Another approach is to use a trusted third party to assign a symmetric key to both the parties. This is the basic idea behind the use of key distribution center (KDC).

### **Kerberos**

Another popular authentication protocol known as Kerberos. It uses an authentication server (AS), which performs the role of KDC and a ticket-granting server (TGS), which provides the

session key ( $K_{AB}$ ) between the sender and receiver parties. Apart from these servers, there is the real data server say Ram that provides services to the user Sita. The operation of Kerberos is depicted with the help of Fig. 8.2.12. The client process (Sita) can get a service from a process running in the real server Ram after six steps as shown in the figure. The steps are as follows:

**Step 1.** Sita uses her registered identity to send her message in plaintext.

**Step 2.** The AS server sends a message encrypted with Sita's symmetric key  $K_S$ . The message contains a session key  $K_{SE}$ , which is used by Sita to contact the TGS and a ticket for TGS that is encrypted with the TGS symmetric key  $K_{TG}$ .

**Step 3.** Sita sends three items to the TGS; the ticket received from the AS, the name of the real server, and a timestamp encrypted by  $K_{SE}$ . The timestamp prevents replay by Ram.

**Step 4.** The TGS sends two tickets to Sita. The ticket for Sita encrypted with  $K_{SE}$  and the ticket for Ram encrypted with Ram's key. Each of the tickets contains the session key  $K_{SR}$  between Sita and Ram.

**Step 5.** Sita sends Ram's ticket encrypted by  $K_{SR}$ .

**Step 6.** Ram sends a message to Sita by adding 1 to the timestamp confirming the receipt of the message using  $K_{SR}$  as the key for encryption.

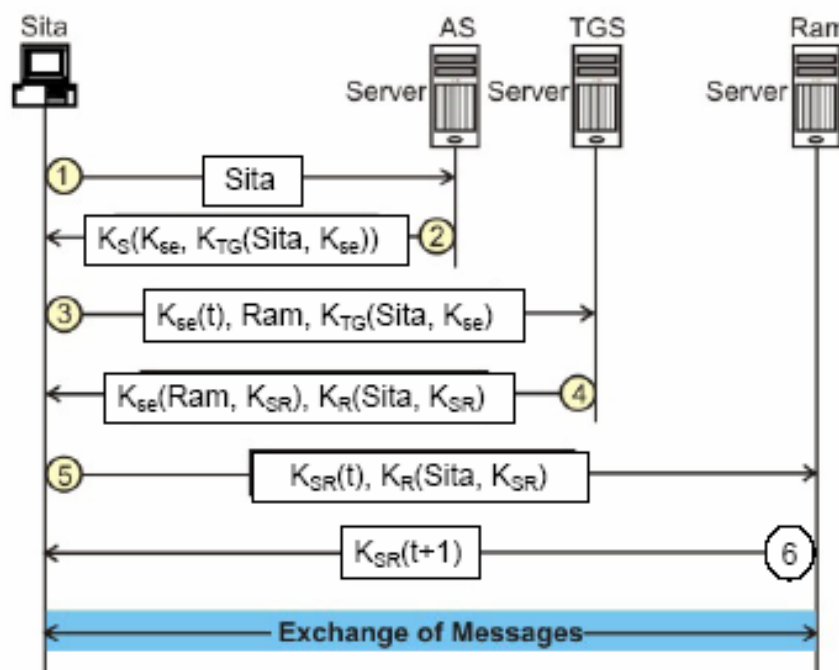


Figure 85 Kerberos Protocol

Based on the encryption techniques we have discussed so far, security measures can be applied to different layers such as network, transport or application layers. However, implementation of security features in the application layer is far simpler and feasible compared to implementing at the other two lower layers. In this subsection, a protocol known as **Pretty Good Privacy (PGP)**, invented by Phil Zimmermann, that is used in the application layer to provide all the four aspects of security for sending an email is briefly discussed. PGP uses a combination of private-key and public key for privacy. For integrity, authentication and nonrepudiation, it uses a combination of hashing to create digital signature and public-key encryption as shown in Fig.

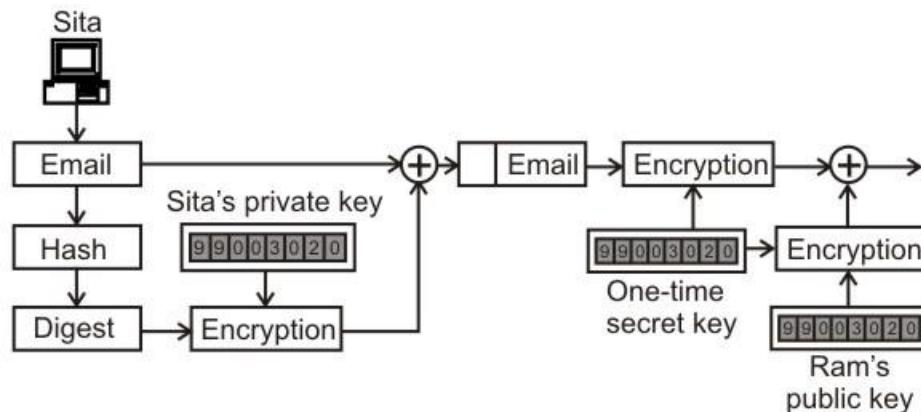


Figure 86 Sender Site of the PGP

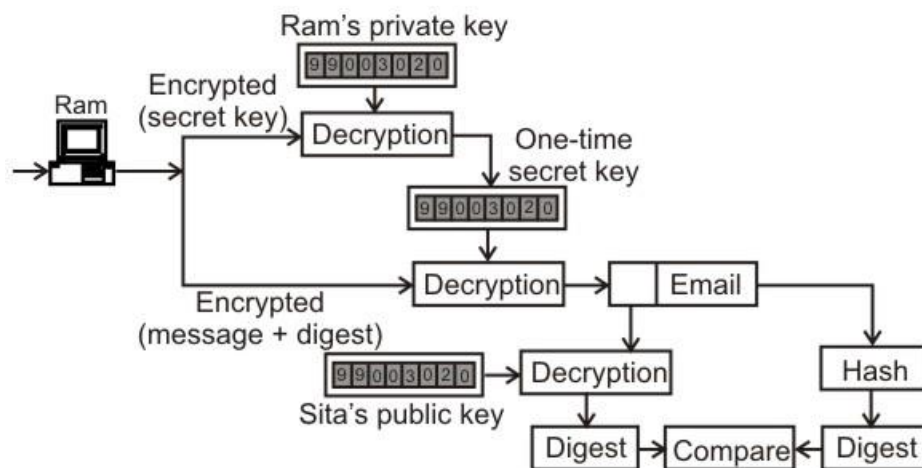


Figure 87 Receiver site of the PGP

### Virtual Private Network (VPN)

With the availability of huge infrastructure of public networks, the Virtual Private Network (VPN) technology is gaining popularity among enterprises having offices distributed throughout the country. Before we discuss about the VPN technology, let us first discuss about two related terms: intranet and extranet.

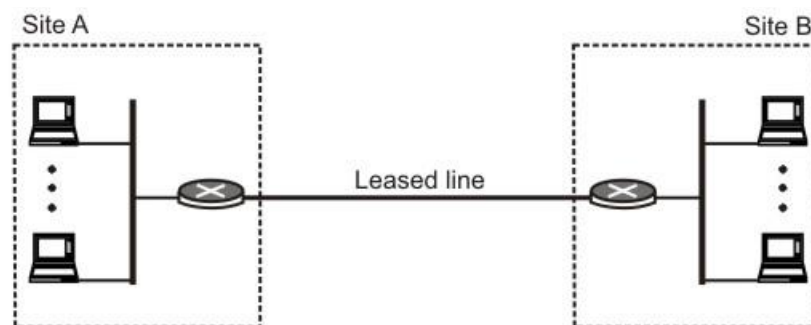
**Intranet** is a private network (typically a LAN) that uses the internet model for exchange of information. A private network has the following features:

- It has limited applicability because access is limited to the users inside the network
- Isolated network ensures privacy
- Can use private IP addresses within the private network

**Extranet** is same as the intranet with the exception that some resources can be allowed to access by some specific groups under the control of network administrator.

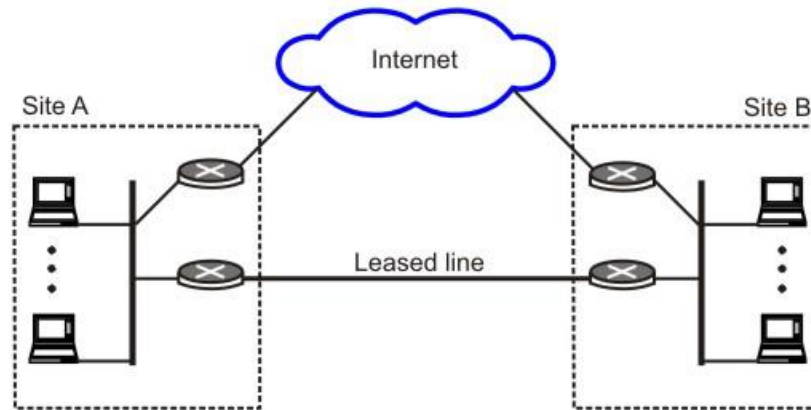
Privacy can be achieved by using one of the three models: Private networks, Hybrid Networks and Virtual Private Networks.

**Private networks:** A small organization with a single site can have a single LAN whereas an organization with several sites geographically distributed can have several LANs connected by leased lines and routers as shown in Fig. 8.2.14. In this scenario, people inside the organization can communicate with each other securely through a private internet, which is totally isolated from the global internet.



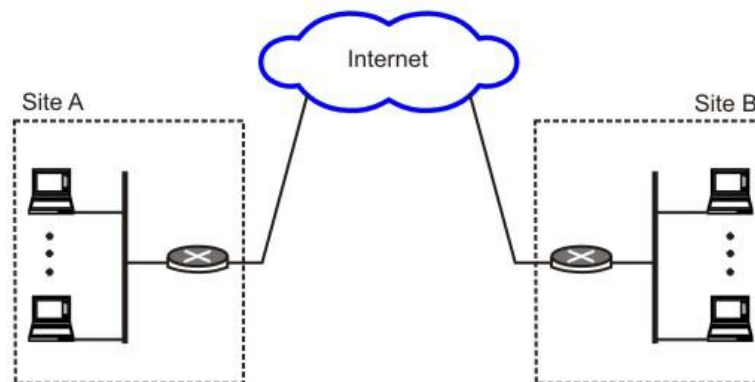
*Figure 88 Private network with two LAN sites*

**Hybrid Networks:** Many organizations want privacy for inter-organization level data exchange, at same time they want to communicate with others through the global internet. One solution to achieve this is to implement a hybrid network as shown in Fig. 8.2.15. In this case, both private and hybrid networks have high cost of implementation, particularly private WANs are expensive to implement.



*Figure 89 Hybrid network with two LAN Sites*

**Virtual Private Networks (VPN):** VPN technology allows both private communication and public communications through the global internet as shown in Fig. 8.2.16. VPN uses IPsec in the tunnel mode to provide authentication, integrity and privacy. In the IPsec tunnel mode the datagram to be sent is encapsulated in another datagram as payload. It requires two sets of addressing as shown in Fig.



*Figure 90VPN Linking two LANs*

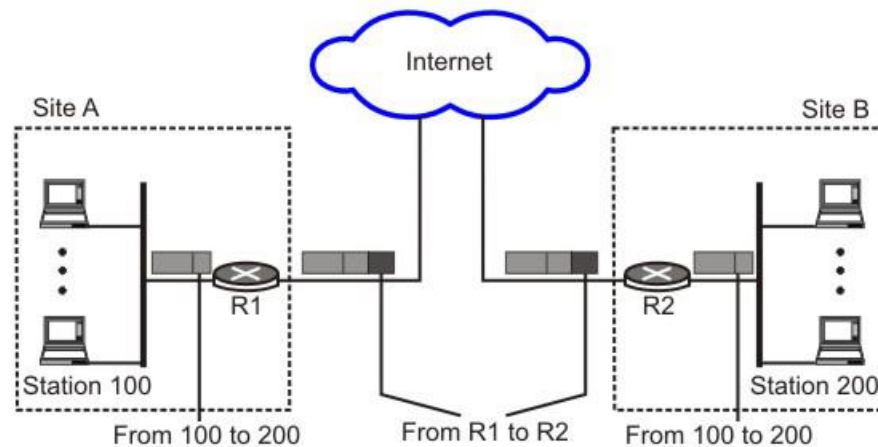


Figure 91 VPN linking two LANs

### Overview of IPSec

The **IP security (IPsec)** is an Internet Engineering Task Force (IETF) standard suite of protocols between 2 communication points across the IP network that provide data authentication, integrity, and confidentiality. It also defines the encrypted, decrypted and authenticated packets. The protocols needed for secure key exchange and key management are defined in it.

#### Uses of IP Security –

IPsec can be used to do the following things:

- To encrypt application layer data.
- To provide security for routers sending routing data across the public internet.
- To provide authentication without encryption, like to authenticate that the data originates from a known sender.
- To protect network data by setting up circuits using IPsec tunneling in which all data is being sent between the two endpoints is encrypted, as with a Virtual Private Network (VPN) connection.

#### Components of IP Security –

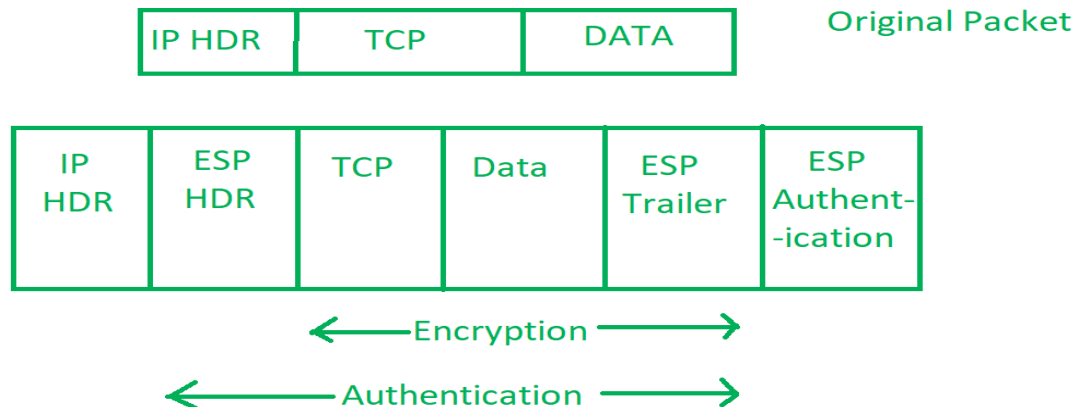
It has the following components:

1. **Encapsulating Security Payload (ESP)** –It provides data integrity, encryption, authentication and anti-replay. It also provides authentication for payload.
2. **Authentication Header (AH)** –It also provides data integrity, authentication and anti-replay and it does not provide encryption. The anti-replay protection, protects against unauthorized transmission of packets. It does not protect data's confidentiality.



3. **Internet Key Exchange (IKE)** – It is a network security protocol designed to dynamically exchange encryption keys and find a way over Security Association (SA) between 2 devices. The Security Association (SA) establishes shared security attributes between 2

network entities to support secure communication. The Key Management Protocol (ISAKMP) and Internet Security Association which provides a framework for authentication and key exchange. ISAKMP tells how the set up of the Security Associations (SAs) and how direct connections between two hosts that are using IPsec. Internet Key Exchange (IKE) provides message content protection and also an open frame for implementing standard algorithms such as SHA and MD5. The algorithm's IP sec users produces a unique identifier for each packet. This identifier then allows a device to determine whether a packet has been correct or not. Packets which are not authorized are discarded and not given to receiver.



### Working of IP Security –

1. The host checks if the packet should be transmitted using IPsec or not. These packet traffic triggers the security policy for themselves. This is done when the system sending the packet apply an appropriate encryption. The incoming packets are also checked by the host that they are encrypted properly or not.
2. Then the **IKE Phase 1** starts in which the 2 hosts (using IPsec) authenticate themselves to each other to start a secure channel. It has 2 modes. The **Main mode** which provides the greater security and the **Aggressive mode** which enables the host to establish an IPsec circuit more quickly.
3. The channel created in the last step is then used to securely negotiate the way the IP circuit will encrypt data across the IP circuit.
4. Now, the **IKE Phase 2** is conducted over the secure channel in which the two hosts negotiate the type of cryptographic algorithms to use on the session and agreeing on secret keying material to be used with those algorithms.
5. Then the data is exchanged across the newly created IPsec encrypted tunnel. These packets are encrypted and decrypted by the hosts using IPsec SAs.
6. When the communication between the hosts is completed or the session times out then the IPsec tunnel is terminated by discarding the keys by both the hosts.

### Firewall

**Firewall** is a barrier between Local Area Network (LAN) and the Internet. It allows keeping private resources confidential and minimizes the security risks. It controls network traffic, in both directions.

Many organizations have confidential or proprietary information, such as trade secrets, product development plans, marketing strategies, etc., which should be protected from unauthorized access and modification. One possible approach is to use suitable encryption/decryption technique for transfer of data between two secure sites, as we have discussed in the previous lesson. Although these techniques can be used to protect data in transit, it does not protect data from digital pests and hackers. To accomplish this it is necessary to perform user authentication and access control to protect the networks from unauthorized traffic. This is known as firewalls. A firewall system is an electronic security guard and electronic barrier at the same time.

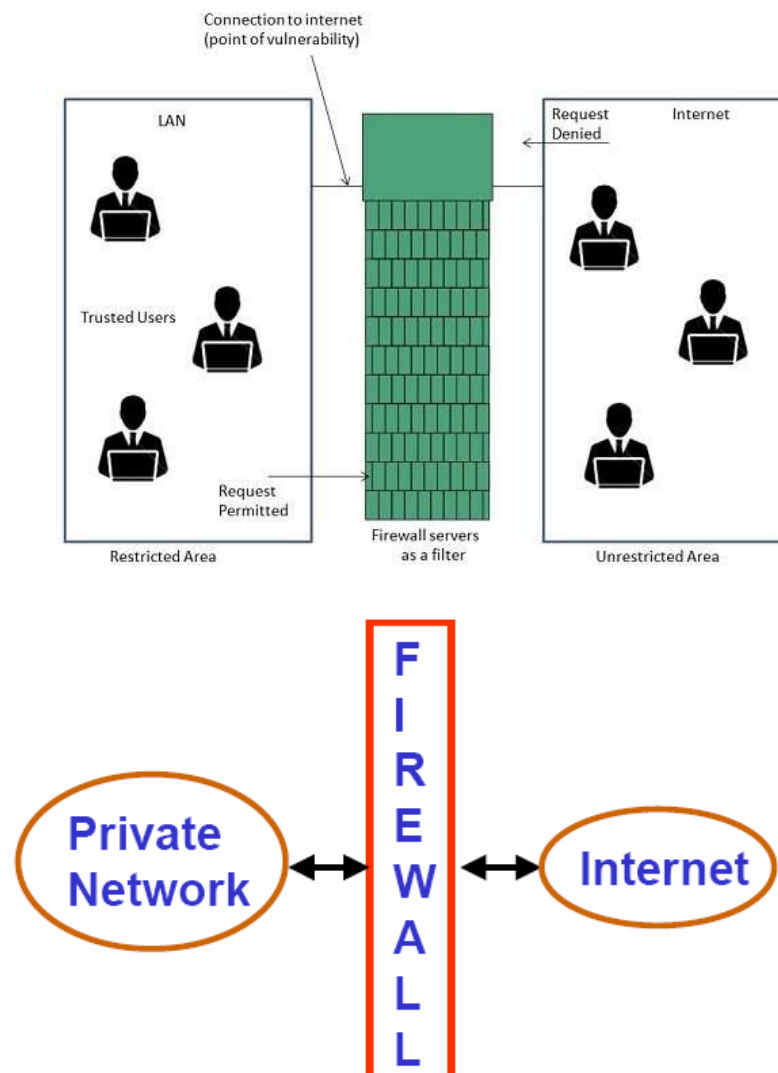


Figure 92 Schematic Diagram of firewall

### Why a Firewall is needed?

There is no need for a firewall if each and every host of a private network is properly secured. Unfortunately, in practice the situation is different. A private network may consist of different platforms with diverse OS and applications running on them. Many of the applications were designed and developed for an ideal environment, without considering the

possibility of the existence of bad guys. Moreover, most of the corporate networks are not designed for security. Therefore, it is essential to deploy a firewall to protect the vulnerable infrastructure of an enterprise.

### Access Control Policies

Access control policies play an important role in the operation of a firewall. The policies can be broadly categorized in to the following four types:

#### Service Control:

- Determines the types of internet services to be accessed
- Filters traffic based on IP addresses and TCP port numbers
- Provides Proxy servers that receives and interprets service requests before it is passed on

#### Direction Control:

Determines the direction in which a particular service request may be initiated and allowed to flow through the firewall

#### User Control:

- Controls access to a service according to which user is attempting to access it
- Typically applied to the users inside the firewall perimeter
- Can be applied to the external users too by using secure authentication technique

#### Behavioral Control:

- Controls how a particular service is used
- For example, a firewall may filter email to eliminate spam
- Firewall may allow only a portion of the information on a local web server to an external user

### Firewall Capabilities

Important capabilities of a firewall system are listed below:

- It defines a single choke point to keep unauthorized users out of protected network
- It prohibits potentially vulnerable services from entering or leaving the network
- It provides protection from various kinds of IP spoofing
- It provides a location for monitoring security-related events
- Audits and alarms can be implemented on the firewall systems
- A firewall is a convenient platform for several internet functions that are not security related
- A firewall can serve as the platform for IPSec using the tunnel mode capability and can be used to implement VPNs

### Limitations of a Firewall

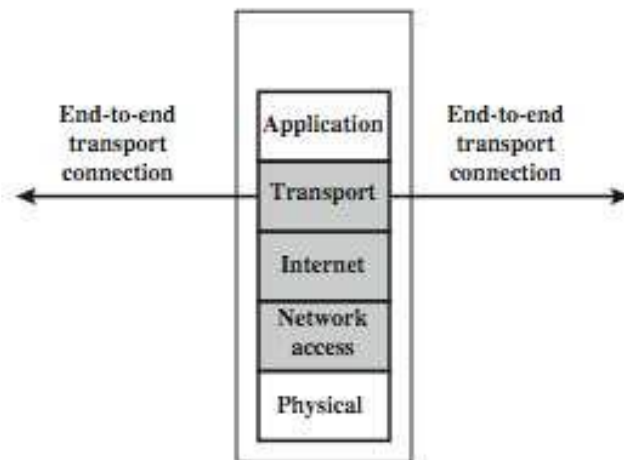
Main limitations of a firewall system are given below:

- A firewall cannot protect against any attacks that bypass the firewall. Many organizations buy expensive firewalls but neglect numerous other back-doors into their network
- A firewall does not protect against the internal threats from traitors. An attacker may be able to break into network by completely bypassing the firewall, if he can find a ``helpful" insider who can be fooled into giving access to a modem pool
- Firewalls can't protect against tunneling over most application protocols. For example, firewall cannot protect against the transfer of virus-infected programs or files

### Types of Firewalls

The firewalls can be broadly categorized into the following three types:

- Packet Filters
  - Application-level Gateways
  - Circuit-level Gateways
  - Stateful inspection Firewall
- **Packet Filters:** Packet filtering router applies a set of rules to each incoming IP packet and then forwards or discards it. Packet filter is typically set up as a list of rules based on matches of fields in the IP or TCP header. An example table of telnet filter rules is given in Fig. The packet filter operates with positive filter rules. It is necessary to specify what should be permitted, and everything that is explicitly not permitted is automatically forbidden.



*Figure 93 Packet filtering*

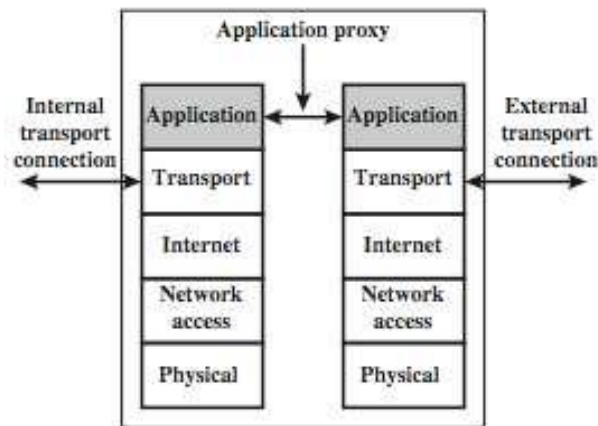
Advantage:

- Cost
- Low resource usage
- Best suited for smaller network

Disadvantage:

- Can work only on the network layer

- Do not support complex rule based support
  - Vulnerable to spoofing
- **Application-level Gateway:** Application level gateway, also called a Proxy Server acts as a relay of application level traffic. Users contact gateways using an application and the request is successful after authentication. The application gateway is service specific such as FTP, TELNET, SMTP or HTTP.



*Figure 94 Application Proxy Firewall*

Advantages:

- More Secure than Packet filter firewall
- Easy to log and audit incoming traffic

Disadvantage

- Additional Processing overhead on each connection
- **Circuit Level Gateway:** Circuit-level gateway can be a standalone or a specialized system. It does not allow end-to-end TCP connection; the gateway sets up two TCP connections. Once the TCP connections are established, the gateway relays TCP segments from one connection to the other without examining the contents. The security.

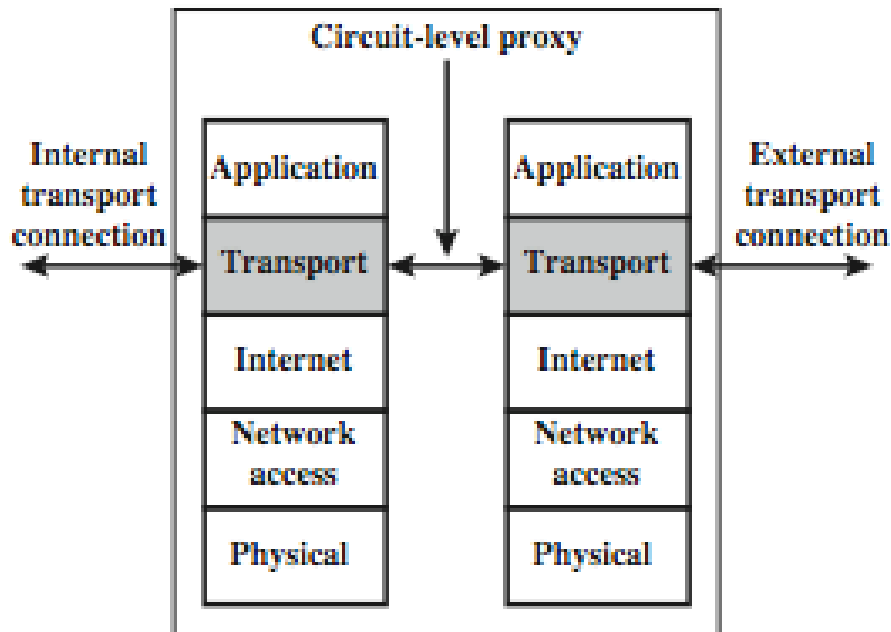


Figure 95 Circuit- level Firewall

Advantage:

- Comparatively inexpensive and provide anonymity to the private network

Disadvantage:

- Do not filter Individual Packets

- **Stateful Inspection Firewall-** A stateful inspection packet firewall tightens up the rules for TCP traffic by creating a directory of outbound TCP Connections there is an entry for each currently established connection. The packet filter now allow incoming traffic to high- numbered ports only for those packets that fits the profile of one of the entries in this directory. A stateful packet inspection firewall reviews the same packet information as a packet filtering, but also records information about TCP connections.

Advantages

- Can Work on a transparent mode allowing direct connections between the client and the server.
- Can also implement algorithm and complex security models which are protocol specific, making the connections and data transfer more secure.